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Ansys Fluid Dynamics Verification Manual



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Release 2026 R1
March 2026

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Published in the U.S.A.

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Part 1: Introduction to the Fluid Dynamics Verification Test Cases

Chapter 1: Overview

The *Fluid Dynamics Verification Manual* presents a collection of test cases that demonstrate a representative set of the capabilities of the ANSYS Fluid Dynamics product suite. The primary purpose of this manual is to demonstrate a wide range of capabilities in straightforward problems that have "classical" or readily-obtainable theoretical solutions and in some cases have experimental data for comparison. The close agreement of the ANSYS solutions to the theoretical or experimental results in this manual is intended to provide user confidence in the ANSYS solutions. These problems may then serve as the basis for your additional validation and qualification of ANSYS capabilities for specific applications that may be of interest to you.

Although these test cases allow comparison against theoretical solutions and experimental data, we have selected relatively simple problems that run quickly and can be conveniently used to test for consistency across multiple platforms. They are not intended to provide a rigorous validation of the model, which generally requires a greater number of much longer running problems. This manual represents a small subset of the Quality Assurance test case library that is used in full when testing new versions of Ansys Fluent, Ansys CFX, and Ansys Forte. Furthermore, because these programs are capable of solving very complicated practical engineering problems that have no closed-form theoretical solutions, the test cases in this manual do not illustrate their full capability.

The Ansys software suite is continually being verified by the developers at ANSYS as new capabilities are added to the software. Verification of Ansys products is conducted in accordance with written procedures that form a part of an overall Quality Assurance program at Ansys.

Note:

To solve the fluid dynamics test cases, you must have a product license that supports the relevant solver (the Ansys Fluent, Ansys CFX, or Ansys Forte solver). A complete list of the products and license features can be found in the ANSYS, Inc. Product to License Feature Mapping table. For instructions on how to access this table, see [Product to License Feature Mapping in the Ansys, Inc. License Management Guide](#).

Additional Topics Covered in this Introduction:

- [1.1. Using the Verification Manual and Test Cases](#)
- [1.2. Expected Results](#)
- [1.3. References](#)
- [1.4. Verification and Validation](#)
- [1.5. Quality Assurance Services](#)
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- [1.8. Index to the Ansys Fluent and CFX Test Cases](#)
- [1.9. Ansys Fluent GPU Supported Test Cases](#)

1.10. Index to the Ansys Forte Test Cases

1.1. Using the Verification Manual and Test Cases

You are encouraged to use these tests as starting points when exploring features in these products. Geometries, material properties, loads, and output results can easily be changed and the solution repeated. As a result, the tests offer a quick introduction to new features with which you may be unfamiliar.

The test cases in this manual are primarily intended for verification of the ANSYS programs. An attempt has been made to include most significant analysis capabilities of the ANSYS products in this manual. Although they are valuable as demonstration problems, the test cases are not presented as step-by-step examples with lengthy data input instructions and printouts. The reader should refer to the online help for complete input data instructions.

Users desiring more detailed instructions for solving problems or in-depth treatment of specific topics should refer to the Ansys Fluent, Ansys CFX, or the Ansys Forte documentation. Tutorials are also available for various specific topics. These publications focus on particular features or program areas, supplementing other ANSYS reference documents with theory, procedures, guidelines, examples, and references.

1.2. Expected Results

The test cases in this manual have been modeled to give reasonably accurate comparisons with experimental data wherever applicable, with a low number of elements and iterations. In some cases, even fewer elements and/or iterations will still yield an acceptable accuracy for the purpose of verifying the solution. The test cases employ a balance between accuracy and solution time. An attempt has been made to present a test case and results that are grid-independent. If test results are not grid-independent, it is due to the need to limit the run time for the test to be in the manual. Improved results can be obtained in some cases by refining the mesh, but this requires longer solution times.

Important:

It should be noted that these are not validation cases of the models presented. The test cases are single instance simulations using one mesh, one turbulence model, and one scenario of the model.

The ANSYS solutions in this manual are compared with solutions or experimental data from textbooks or technical publications. Fluid-dynamics simulations have to make use of data available from experimental measurements for their verification primarily because closed-form theoretical solutions are not available for modeling the related phenomena. In this manual, test cases make use of experimental data published in reputable journals or conference proceedings for verification of the computational results. The experimental measurements for fluid-flow systems are often presented in the form of plots of the relevant parameters. The published experimental data for those cases and the corresponding simulation results are presented in graphical format to ease comparison.

Different computers and operating systems may yield slightly different results for some of the test cases in this manual due to numerical precision variation from machine to machine. Solutions that are non-linear, iterative, or have convergence options activated are among the most likely to exhibit machine-dependent numerical differences. Because of this, an effort has been made to report an appropriate

and consistent number of significant digits in both the target and the ANSYS solution. If you run these test cases on your own computer hardware, be advised that an ANSYS result reported in this manual as 0.01234 may very well show up in your printout as 0.012335271. Slightly different results may be obtained when different processor types or operating systems are used.

1.3. References

The goal for the test cases contained in this manual was to have results accuracy within 3% of the target solution. The solutions for the test cases have been verified; however, certain differences may exist with regard to the references. These differences have been examined and are considered acceptable.

It should be noted that only those items corresponding to the given theoretical solution values are reported for each problem. The same solution may also contain other useful numerical solution data.

1.4. Verification and Validation

The test cases provided in this manual are a single instance, using one mesh, one turbulence model, and one scenario for the model. They are not validation cases of the models presented since they do not provide, nor attempt to provide, the necessary methodology on how to arrive at the presented results. The intent of these cases is to provide a means to verify that you are obtaining the same results in your computing environment as ANSYS obtained in its computing environment. You are encouraged to do a complete study to validate the ANSYS solution.

1.5. Quality Assurance Services

For customers who may have further need for formal verification of the ANSYS products on their computers, Ansys, Inc. offers the Quality Assurance Testing Agreement. You are provided with input data, output data, comparator software, and software tools for automating the testing and reporting process. If you are interested in contracting for such services, contact the Ansys, Inc. Quality Assurance Group.

For additional information, refer to the [Ansys Quality Assurance Effort in the Ansys Workbench Verification Manual](#).

1.6. Downloading the Project Files

All project files for the test cases are available for download at the Customer Portal. Ansys publishes these files annually for each product release. You can find the zipped archives at: **ANSYS Customer Portal > Downloads > Current Release > Tools**, or you can use the download links below.

Project File Release Schedule

The latest verification files become available for download several weeks after each release of the Ansys software.

If you click the download link for the latest project files, you might receive a message that the files are not yet available. If that happens, check back again later. Meanwhile, you can download the project files for the previous release at any time. (You'll find the download link below as well.) The download package for the *previous* release contains project files for all the test cases, except for the most recent

ones added for the current release. And those earlier files are fully compatible with the latest version of your Ansys software.

How to Download the Fluids Verification Case Files

1. Using the **Current Release 2026 R1** link below, downloads the .zip package containing the project files for all the test cases in the Ansys CFX, Ansys Fluent, and Ansys Forte Product's verification manual.

- Download link:

If you receive a message that the project files are not yet available, you may download the package for the previous **2025 R2** release for the time being.

- Download link:

2. Select the .zip file you are interested in and extract the contents to a folder of your choice.

Note:

Typically, the project files appear on the Portal several weeks after the release of the software package. If the downloads are not yet available, check back again later. Meanwhile, you can download the project files of earlier releases any time. They are fully compatible with the latest version of your Ansys software. Direct download links are provided below.

1.7. CFX Supported Test Cases

Due to differences between the Fluent and CFX solvers, only a subset of the test cases in this manual are supported by CFX. The following is a list of test cases which are supported by CFX:

- VMFL001 (p. 15) - Flow Between Rotating and Stationary Concentric Cylinders
- VMFL002 (p. 17) - Laminar Flow Through a Pipe with Uniform Heat Flux
- VMFL003 (p. 19) - Pressure Drop in Turbulent Flow Through a Pipe
- VMFL004 (p. 21) - Plain Couette Flow with Pressure Gradient
- VMFL005 (p. 25) - Poiseuille Flow in a Pipe
- VMFL007 (p. 29) - Non-Newtonian Flow in a Pipe
- VMFL008 (p. 31) - Flow Inside a Rotating Cavity
- VMFL009 (p. 35) - Natural Convection in a Concentric Annulus
- VMFL010 (p. 39) - Laminar Flow in a 90° Tee-Junction
- VMFL011 (p. 41) - Laminar Flow in a Triangular Cavity
- VMFL012 (p. 45) - Turbulent Flow in a Wavy Channel
- VMFL013 (p. 51) - Turbulent Flow with Heat Transfer in a Backward-Facing Step
- VMFL014 (p. 55) - Species Mixing in Co-axial Turbulent Jets
- VMFL015 (p. 61) - Flow Through an Engine Inlet Valve
- VMFL016 (p. 65) - Turbulent Flow in a Transition Duct

- VMFL017 (p. 69) - Transonic Flow over an RAE 2822 Airfoil
- VMFL018 (p. 71) - Shock Reflection in Supersonic Flow
- VMFL019 (p. 77) - Transient Flow Near a Wall Set in Motion
- VMFL020 (p. 79) - Adiabatic Compression of Air in Cylinder by a Reciprocating Piston
- VMFL021 (p. 85) - Cavitation over a Sharp-Edged Orifice Case A: High Inlet Pressure
- VMFL023 (p. 89) - Oscillating Laminar Flow Around a Circular Cylinder
- VMFL024 (p. 91) - Interface of Two Immiscible Liquids in a Rotating Cylinder
- VMFL026 (p. 99) - Supersonic Flow with Real Gas Effects inside a Shock Tube
- VMFL027 (p. 103) - Turbulent Flow Over a Backward-Facing Step
- VMFL032 (p. 115) - Turbulent Flow with Separation Along an Axisymmetric Afterbody
- VMFL037 (p. 127) - Turbulent Flow Over a Forward Facing Step
- VMFL039 (p. 133) - Boiling in a Pipe with Heated Wall
- VMFL040 (p. 137) - Separated Turbulent Flow in a Diffuser
- VMFL042 (p. 143) - Turbulent Mixing of Two Streams with Different Densities
- VMFL044 (p. 149) - Supersonic Nozzle Flow
- VMFL045 (p. 153) - Oblique Shock Over an Inclined Ramp
- VMFL047 (p. 157) - Turbulent Flow with Separation in an Asymmetric Diffuser
- VMFL051 (p. 165) - Isentropic Expansion of Supersonic Flow Over a Convex Corner
- VMFL052 (p. 167) - Turbulent Natural Convection Inside a Tall Cavity
- VMFL054 (p. 173) - Laminar flow in a Trapezoidal Cavity
- VMFL059 (p. 185) - Conduction in a Composite Solid Block
- VMFL063 (p. 193) - Separated Laminar Flow Over a Blunt Plate

1.8. Index to the Ansys Fluent and CFX Test Cases

Dimensionality Column Key:

- **2**: 2D
- **3**: 3D
- **A**: 2D Axisymmetric

	Dimensionality	Unsteady	Turbulent	Trans. Turbulence	Inviscid	Compressible	Non-Isothermal	Radiation	Multi-Species	Reacting	Multi-phase	Phase-Change	External Forces	Moving Frame
VMFL001 (p. 15)	2													
VMFL002 (p. 17)	A						X							
VMFL003 (p. 19)	A		X											

	Dimensionality	Unsteady	Turbulent	Trans. Turbulence	Inviscid	Compressible	Non-Isothermal	Radiation	Multi-Species	Reacting	Multi-phase	Phase-Change	External Forces	Moving Frame
VMFL004 (p. 21)	2													
VMFL005 (p. 25)	A													
VMFL006 (p. 27)	A								X					
VMFL007 (p. 29)	A													
VMFL008 (p. 31)	A													X
VMFL009 (p. 35)	2						X						X	
VMFL010 (p. 39)	2													
VMFL011 (p. 41)	2													
VMFL012 (p. 45)	2		X											
VMFL013 (p. 51)	2		X				X							
VMFL014 (p. 55)	A		X						X					
VMFL015 (p. 61)	3		X											
VMFL016 (p. 65)	3		X											
VMFL017 (p. 69)	2		X			X	X							
VMFL018 (p. 71)	2		X			X	X							
VMFL019 (p. 77)	2	X												
VMFL020 (p. 79)	2	X					X							X
VMFL021 (p. 85)	A		X								X	X		
VMFL022 (p. 87)	A		X								X	X		
VMFL023 (p. 89)	2	X												
VMFL024 (p. 91)	A	X									X		X	
VMFL025 (p. 93)	A		X				X		X	X				
VMFL026 (p. 99)	3	X			X	X	X							
VMFL027 (p. 103)	2		X											
VMFL028 (p. 107)	A		X				X							
VMFL029 (p. 109)	2				X		X							
VMFL030 (p. 111)	3		X											
VMFL031 (p. 113)	2		X											
VMFL032 (p. 115)	A		X											
VMFL033 (p. 119)	2						X							
VMFL034 (p. 121)	2		X								X			
VMFL035 (p. 123)	3		X			X	X							X
VMFL036 (p. 141)	A		X											
VMFL037 (p. 127)	2		X											

	Dimensionality	Unsteady	Turbulent	Trans. Turbulence	Inviscid	Compressible	Non-Isothermal	Radiation	Multi-Species	Reacting	Multi-phase	Phase-Change	External Forces	Moving Frame
VMFL038 (p. 131)	2	X											X	
VMFL039 (p. 133)	A		X				X				X	X	X	
VMFL040 (p. 137)	A		X											
VMFL041 (p. 141)	2		X			X	X							
VMFL042 (p. 143)	2		X								X		X	
VMFL043 (p. 147)	2		X	X										
VMFL044 (p. 149)	A		X			X	X							
VMFL045 (p. 153)	2					X	X							
VMFL046 (p. 155)	2				X	X	X							
VMFL047 (p. 157)	2		X											
VMFL048 (p. 159)	3		X											
VMFL049 (p. 161)	A		X				X		X	X				
VMFL050 (p. 163)	2	X					X							
VMFL051 (p. 165)	2				X	X	X							
VMFL052 (p. 167)	2		X										X	
VMFL053 (p. 171)	2		X			X	X							
VMFL054 (p. 173)	2													
VMFL055 (p. 177)	2		X	X									X	
VMFL056 (p. 179)	2						X	X						
VMFL057 (p. 181)	2						X	X						
VMFL058 (p. 183)	A		X											
VMFL059 (p. 185)	2						X							
VMFL060 (p. 187)	2		X	X		X	X							
VMFL061 (p. 189)	2						X	X						
VMFL062 (p. 191)	2		X											
VMFL063 (p. 193)	2													
VMFL064 (p. 195)	2													
VMFL065 (p. 197)	A		X											
VMFL066 (p. 199)	2						X	X						
VMFL067 (p. 201)	A		X				X				X			
VMFL068 (p. 203)	3		X											
VMFL069 (p. 205)	3				X						X			
VMFL070 (p. 207)	2						X	X						
VMFL071 (p. 209)	2		X				X							

	Dimensionality	Unsteady	Turbulent	Trans. Turbulence	Inviscid	Compressible	Non-Isothermal	Radiation	Multi-Species	Reacting	Multi-phase	Phase-Change	External Forces	Moving Frame
VMFL072 (p. 211)	3				X									
VMFL073 (p. 213)	A		X											
VMFL074 (p. 215)	2		X								X			
VMFL075 (p. 217)	2				X	X	X							
VMFL076 (p. 219)	2				X		X							
VMFL077 (p. 221)	3		X								X			
VMFL078 (p. 223)	3				X									

1.9. Ansys Fluent GPU Supported Test Cases

The following is a list of test cases which are supported by Ansys Fluent GPU Solver:

- [VMFLGPU001 \(p. 225\)](#) - Flow Between Rotating and Stationary Concentric Cylinders
- [VMFLGPU002 \(p. 227\)](#) - Laminar Flow in a 90° Tee-Junction
- [VMFLGPU003 \(p. 229\)](#) - Laminar Flow in a Triangular Cavity
- [VMFLGPU004 \(p. 233\)](#) - Anisotropic Conduction Heat Transfer
- [VMFLGPU005 \(p. 235\)](#) - Turbulent Natural Convection Inside a Tall Cavity
- [VMFLGPU006 \(p. 239\)](#) - Mid-Span Flow Over a Goldman Stator Blade
- [VMFLGPU007 \(p. 243\)](#) - Turbulent Flow with Heat Transfer in a Backward-Facing Step
- [VMFLGPU008 \(p. 247\)](#) - Radiative Heat Transfer in a Rectangular Enclosure with Participating Medium
- [VMFLGPU009 \(p. 249\)](#) - Two Phase Poiseuille Flow
- [VMFLGPU010 \(p. 251\)](#) - Surface to Surface Radiative Heat Transfer Between Two Concentric Cylinders

1.10. Index to the Ansys Forte Test Cases

- [VMFRT001 \(p. 255\)](#): Large Eddy Simulation in Internal Combustion Engine Case
- [VMFRT002 \(p. 259\)](#): Engine Combustion Network Nonreacting Flow Case - bklraAL4
- [VMFRT003 \(p. 261\)](#): Engine Combustion Network Nonreacting Flow Case - bklfaAL4
- [VMFRT004 \(p. 263\)](#): Engine Combustion Network Nonreacting Flow Case - bkldaAL4
- [VMFRT005 \(p. 265\)](#): Engine Combustion Network Reacting Flow Case - jkldaAL4
- [VMFRT006 \(p. 269\)](#): Adiabatic Compression of Air in Cylinder by a Reciprocating Piston

- [VMFRT007 \(p. 273\)](#): Small-Bore Direct Injection Diesel Engine

Part 2: Ansys Fluent and CFX Test Cases

VMFL001: Flow Between Rotating and Stationary Concentric Cylinders

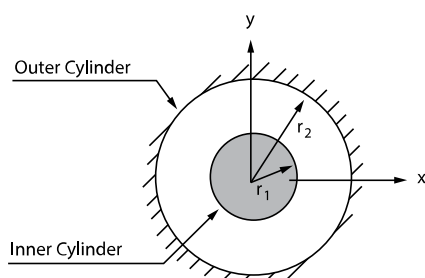
Overview

Reference	F. M. White. <i>Viscous Fluid Flow</i> . Section 3-2.3. McGraw-Hill Book Co., Inc.. New York, NY. 1991.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar flow, rotating wall
Input Files	rot_conc_cyl.cas for Ansys Fluent rotating_cylinder.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Steady laminar flow between two concentric cylinders is modeled. The flow is induced by rotation of the inner cylinder with a constant angular velocity, while the outer cylinder is held stationary. Due to periodicity only a section of the domain needs to be modeled. In the present simulation a 180° segment (half of the domain shown in [Figure .01.1: Flow Domain \(p. 15\)](#)) is modeled. The sketch is not to scale.

Figure .01.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³ Viscosity = 0.0002 kg/m-s	Radius of the Inner Cylinder = 17.8 mm Radius of the Outer Cylinder = 46.28 mm	Angular velocity of the inner wall = 1 rad/s

Analysis Assumptions and Modeling Notes

The flow is steady. The tangential velocity at various sections can be calculated using analytical equations for laminar flow. These values are used for comparison with simulation results.

Results Comparison for Ansys Fluent

Table .01.1: Comparison of Tangential Velocity in the Annulus at Various Radial Locations

Tangential Velocity at	Target, m/s	Ansys Fluent, m/s	Ratio
r = 20 mm	0.0151	0.0151	1.000
r = 25 mm	0.0105	0.0105	1.000
r = 30 mm	0.0072	0.0072	1.000
r = 35 mm	0.0046	0.0045	0.978

Results Comparison for Ansys CFX

Table .01.2: Comparison of Tangential Velocity in the Annulus at Various Radial Locations

Location	Target, m/s	Ansys CFX, m/s	Ratio
r = 20 mm	0.0151	0.0150	0.991
r = 25 mm	0.0105	0.0105	0.998
r = 30 mm	0.0072	0.0071	0.988
r = 35 mm	0.0046	0.0045	0.976

VMFL002: Laminar Flow Through a Pipe with Uniform Heat Flux

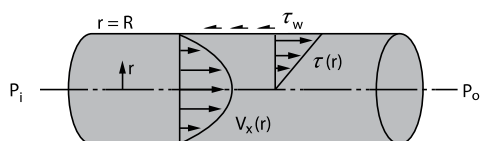
Overview

Reference	F.M. White. <i>Fluid Mechanics</i> . 3rd Edition. McGraw-Hill Book Co., New York, NY. 1994. F.P. Incropera and D.P. DeWitt. <i>Fundamentals of Heat Transfer</i> . John Wiley & Sons. 1981.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar flow with heat transfer
Input File	laminar-pipe-hotflow.cas for Ansys Fluent VMFL002B_VV002CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar flow of Mercury through a circular pipe is modeled, with uniform heat flux across the wall. A fully developed laminar velocity profile is prescribed at the inlet. The resulting pressure drop and exit temperature are compared with analytical calculations for Laminar flow. Only half of the 2-D domain is modeled due to symmetry.

Figure .02.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Fluid: Mercury Density = 13529 kg/m^3 Viscosity = 0.001523 kg/m-s Specific Heat = 139.3 J/kg-K Thermal Conductivity = 8.54 W/m-K	Length of the pipe = 0.1 m Radius of the pipe = 0.0025 m	Fully developed velocity profile at inlet. Inlet temperature = 300 K Heat Flux across wall = 5000 W/m^2

Analysis Assumptions and Modeling Notes

The flow is steady and incompressible. Pressure drop can be calculated from the theoretical expression for laminar flow given in Ref. 1. Correlations for temperature calculations are given in Ref. 2.

Results Comparison for Ansys Fluent

Table .02.1: Comparison of Pressure Drop and Outlet Temperature

	Target	Ansys Fluent	Ratio
Pressure Drop, Pa	1.000	0.999	0.999
Centerline Temperature at the Outlet, K	341.00	340.50	0.999

Results Comparison for Ansys CFX

Table .02.2: Comparison of Pressure Drop and Outlet Temperature

	Target	Ansys CFX	Ratio
Pressure Drop, Pa	1.000	1.019	1.019
Centerline Temperature at the Outlet, K	341.00	340.8	0.9994

VMFL003: Pressure Drop in Turbulent Flow Through a Pipe

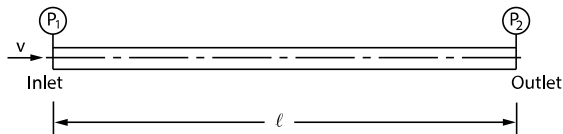
Overview

Reference	F.M. White. <i>Fluid Mechanics</i> . 3rd Edition. McGraw-Hill Co., New York, NY. 1994.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Turbulent flow, standard k- ϵ Model
Input File	turb_pipe_flow.cas for Ansys Fluent VMFL003B_VV003CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Air flows through a horizontal pipe with smooth walls. The flow Reynolds number is 1.37×10^4 . Only half of the axisymmetrical domain is modeled.

Figure .03.1: Flow Domain



The figure is not to scale.

Material Properties	Geometry	Boundary Conditions
Density = 1.225 kg/m^3 Viscosity = $1.7894 \times 10^{-5} \text{ kg/m-s}$	Length of the pipe = 2 m Radius of the pipe = 0.002 m	Inlet velocity = 50 m/s Outlet pressure = 0 Pa

Analysis Assumptions and Modeling Notes

The flow is steady. Pressure drop can be calculated from analytical formula using friction factor f which can be determined for the given Reynolds number from Moody chart. The calculated pressure drop is compared with the simulation results (pressure difference between inlet and outlet).

Results Comparison for Ansys Fluent

Table .03.1: Comparison of Pressure Drop in the Pipe

	Target	Ansys Fluent	Ratio
Pressure Drop, Pa	21744	21480	0.988

Results Comparison for Ansys CFX

Table .03.2: Comparison of Pressure Drop in the Pipe

	Target	Ansys CFX	Ratio
Pressure Drop, Pa	21744	21740	1.000

VMFL004: Plain Couette Flow with Pressure Gradient

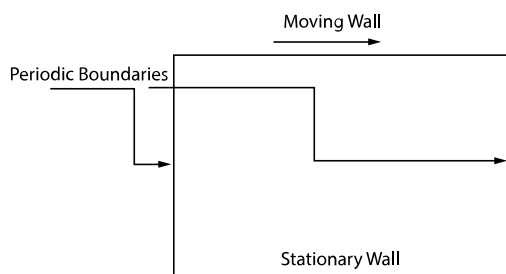
Overview

Reference	B.R. Munson, T.H. Okiishi, W.W. Huebsch, <i>Fundamentals of Fluid Mechanics</i> , 5th Edition, Wiley, 2006
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar flow, moving wall, periodic boundaries
Input Files	couette_flow.cas for Ansys Fluent Couette_Flow.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Viscous flow between two parallel plates is modeled. The top plate moves with a uniform velocity while the lower plate is fixed. A pressure gradient is imposed in a direction parallel to the plates.

Figure .04.1: Flow Domain



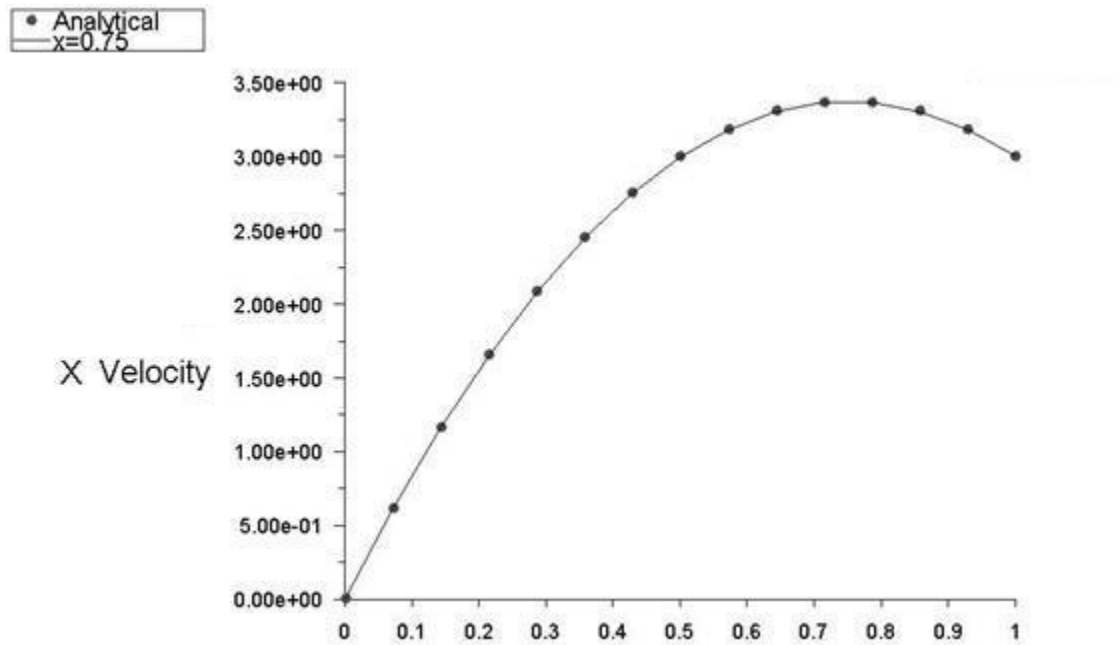
Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³ Viscosity = 1 kg/m-s	Length of the domain = 1.5 m Width of the domain = 1 m	Velocity of the moving wall = 3 m/s in X-direction For Ansys Fluent, pressure gradient across periodic boundaries = -12 Pa/m For Ansys CFX, pressure gradient across periodic boundaries = -12 Pa/m (pressure change = -18 Pa)

Analysis Assumptions and Modeling Notes

The flow is steady and laminar. Periodic conditions with specified pressure drop are applied across the flux boundaries.

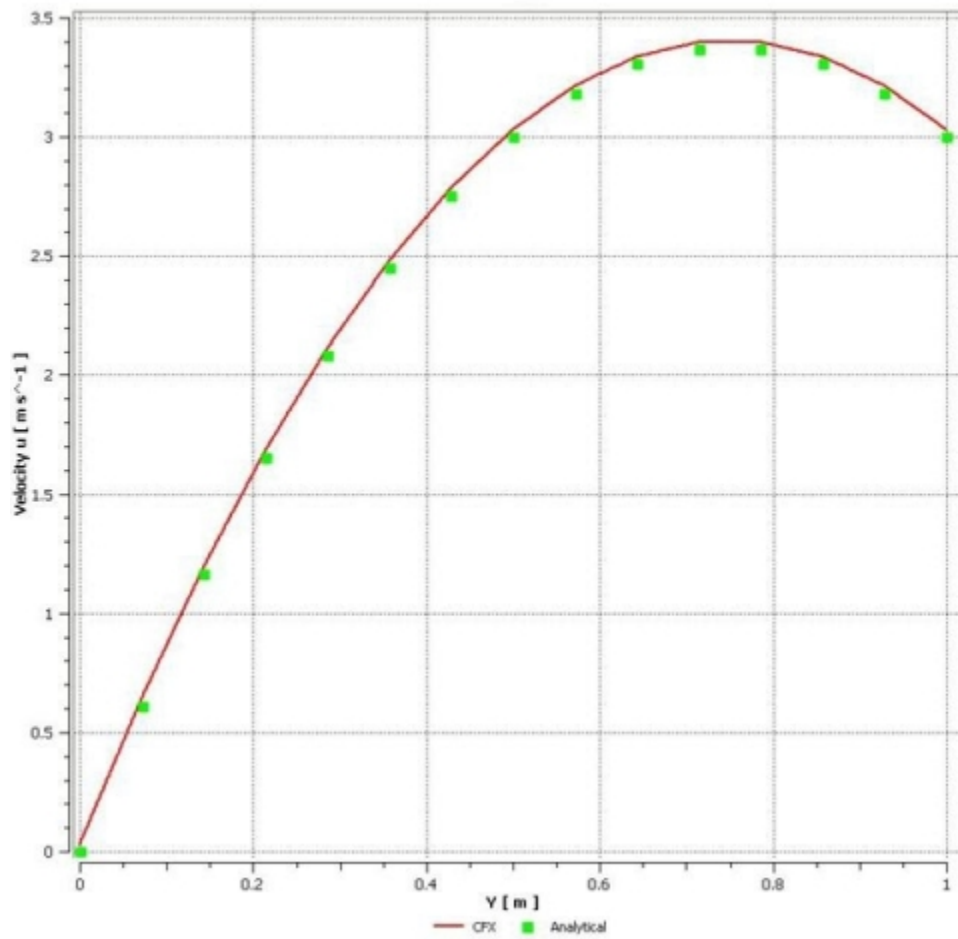
Results Comparison for Ansys Fluent

Figure .04.2: Comparison of X-Velocity (m/s) at a Section Where X = 0.75 m



Results Comparison for Ansys CFX

Figure .04.3: Comparison of X-Velocity (m/s) at a Section Where $X = 0.75$ m



VMFL005: Poiseuille Flow in a Pipe

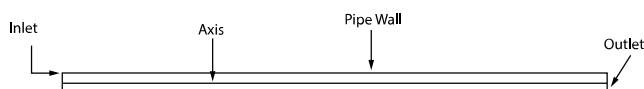
Overview

Reference	F.M. White. <i>Fluid Mechanics</i> . 3rd Edition. McGraw-Hill Book Co., New York, NY. 1994.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Steady laminar flow
Input File	poiseuille-flow.cas for Ansys Fluent VMFL005B_VV005CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Fully developed laminar flow in a circular tube is modeled. Reynolds number based on the tube diameter is 500. Only half of the axisymmetric domain is modeled.

Figure .05.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³ Viscosity = 1e-5 kg/m-s	Length of the pipe = 0.1 m Radius of the pipe = 0.00125 m	Fully developed laminar velocity profile at inlet with an average velocity of 2.00 m/s

Analysis Assumptions and Modeling Notes

The flow is steady. A fully developed laminar velocity profile is prescribed at the inlet. Hagen-Poiseuille equation is used to determine the pressure drop analytically.

Results Comparison for Ansys Fluent

Table .05.1: Comparison of Pressure Drop in the Pipe

	Target	Ansys Fluent	Ratio
Pressure Drop, Pa	10.24	10.22	0.998

Results Comparison for Ansys CFX

Table .05.2: Comparison of Pressure Drop in the Pipe

	Target	Ansys CFX	Ratio
Pressure Drop, Pa	10. 24	10.49	1.024

VMFL006: Multicomponent Species Transport in Pipe Flow

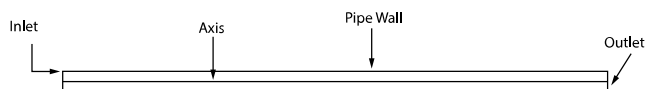
Overview

Reference	W.M. Kays and M.E. Crawford. <i>Convective Heat and Mass Transfer</i> . 3rd Edition. McGraw-Hill Book Co., Inc., New York, NY. 126-134. 1993.
Solver	Ansys Fluent
Physics/Models	Steady laminar flow, species transport
Input File	Species-diffusion.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Fully developed laminar flow in a circular tube, with two species is modeled. Species A enters at the inlet and species B enters from the wall. Uniform and dissimilar mass fractions are specified at the pipe inlet and wall. Fluid properties are assumed to be the same for both species, so that computed results can be compared with analytical solution.

Figure .06.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Species A Density = 1 kg/m ³ Viscosity = 1.0 x 10 ⁻⁵ Pa-s Diffusivity $_{BA} = 1.43 \times 10^{-5}$ m ² /s	Length of the pipe = 0.1 m Radius of the pipe = 0.0025 m	Fully developed laminar velocity profile at inlet with an average velocity of 1 m/s Mass fraction of species A at pipe inlet = 0.5 Mass fraction of species B at pipe inlet = 0.5
Species B Density = 1 kg/m ³ Viscosity = 1.0 x 10 ⁻⁵ Pa-s Diffusivity $_{AB} = 1.43 \times 10^{-5}$ m ² /s		Mass fraction of species A at pipe wall = 0.9 Mass fraction of species B at pipe wall = 0.1

Analysis Assumptions and Modeling Notes

The flow is steady. A fully developed laminar velocity profile is prescribed at the inlet. Species transport model is used.

Results Comparison

In the following table, the value of species A is the mass-weighted average of the normalized species mass fraction A at x-locations.

Table .06.1: Comparison of Mass Fraction of Species A Along the Axis

Axial location (m)	Target	Ansys Fluent	Ratio
0.01	0.8225	0.8227	1.002
0.02	0.7308	0.7309	1.001
0.03	0.6593	0.6594	1.002
0.04	0.5992	0.5993	1.002
0.05	0.5469	0.5471	1.004
0.06	0.5006	0.5007	1.002
0.07	0.4589	0.4591	1.004
0.08	0.4212	0.4215	1.007
0.09	0.3869	0.3872	1.008
0.10	0.3555	0.3557	1.006

VMFL007: Non-Newtonian Flow in a Pipe

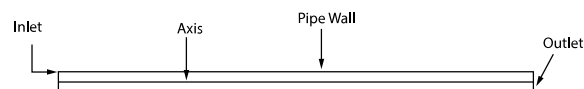
Overview

Reference	W.F. Hughes and J.A. Brighton. <i>Schaum's Outline of Theory and Problems of Fluid Dynamics</i> . McGraw-Hill Book Co., Inc., New York, NY. 1991.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Steady laminar flow, power law for viscosity
Input File	power law-visc.cas for Ansys Fluent VMFL007B_vv007CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow of a non-Newtonian fluid in a circular pipe is modeled. Viscosity is specified by power law equation.

Figure .07.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1000 kg/m ³ Viscosity: Power law Parameters: k = 10 n = 0.4	Pipe length = 0.1 m Pipe diameter = 0.0025 m	Fully developed velocity profile at inlet with an average velocity of 2 m/s

Analysis Assumptions and Modeling Notes

The flow is steady. Viscosity is specified using non-Newtonian power law equation.

Results Comparison for Ansys Fluent

Table .07.1: Comparison of Pressure Drop in the Pipe

	Target	Ansys Fluent	Ratio
Pressure Drop, kPa	60.52	60.41	0.998

Results Comparison for Ansys CFX

Table .07.2: Comparison of Pressure Drop in the Pipe

	Target	Ansys CFX	Ratio
Pressure Drop, kPa	60.52	61.52	1.0165

VMFL008: Flow Inside a Rotating Cavity

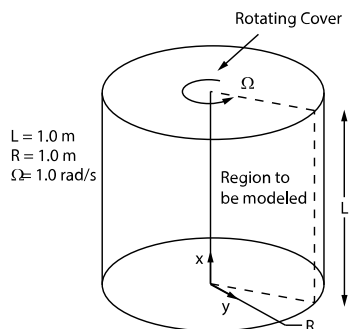
Overview

Reference	J.A. Michelsen. "Modeling of Laminar Incompressible Rotating Fluid Flow". AFM 86-05., Ph.D. thesis. Department of Fluid Mechanics, Technical University of Denmark. 1986.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar flow, Rotating reference frame
Input File	rotcv_RRF.cas for Ansys Fluent VMFL008B_rot_cyl.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow in a cylindrical cavity enclosed with a lid that spins at $\Omega = 1.0$ rad/s. The flow field is 2-D axisymmetric, so only the region bounded by the dashed lines in [Figure .08.1: Flow Domain \(p. 31\)](#) needs to be modeled. The Reynolds number of the flow based on the cavity radius **R** and the tip-speed of the disk is **1800**.

Figure .08.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m^3 Viscosity: 0.000556 kg/m-s	Height of the cavity = 1 m Radius of cavity = 1 m	Speed of rotation of the moving wall = 1 rad/s Rotational velocity for cell zone = -1 rad/s

Analysis Assumptions and Modeling Notes

The flow is laminar. The problem is solved using rotating reference frame.

Results Comparison for Ansys Fluent

Figure .08.2: Comparison of Distribution of Radial Velocity Along a Section at X= 0.6 m

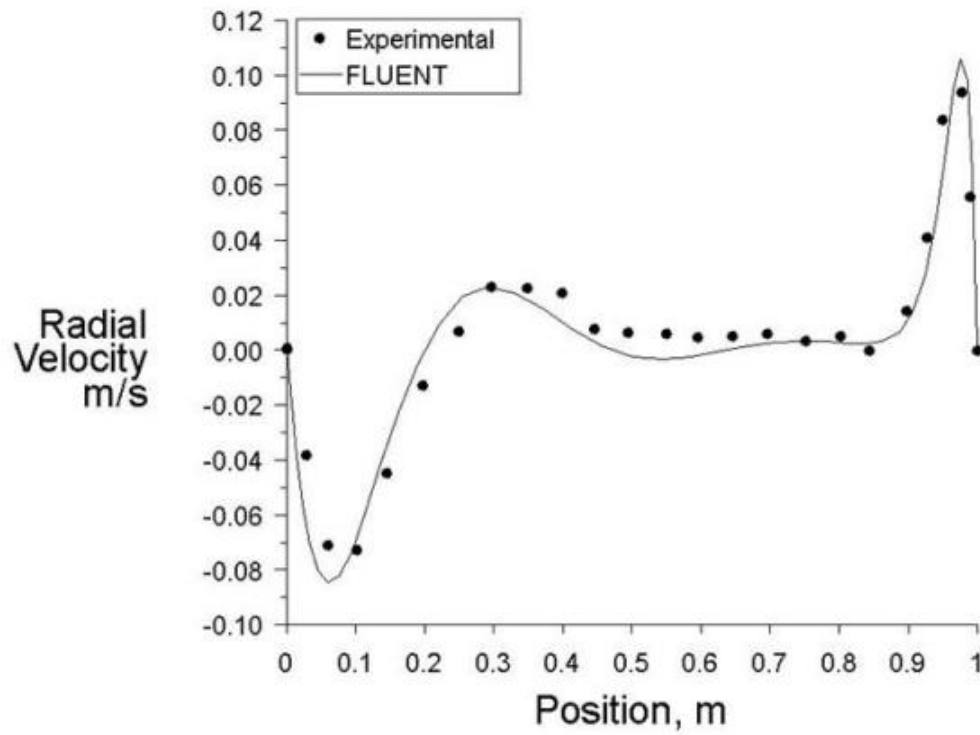
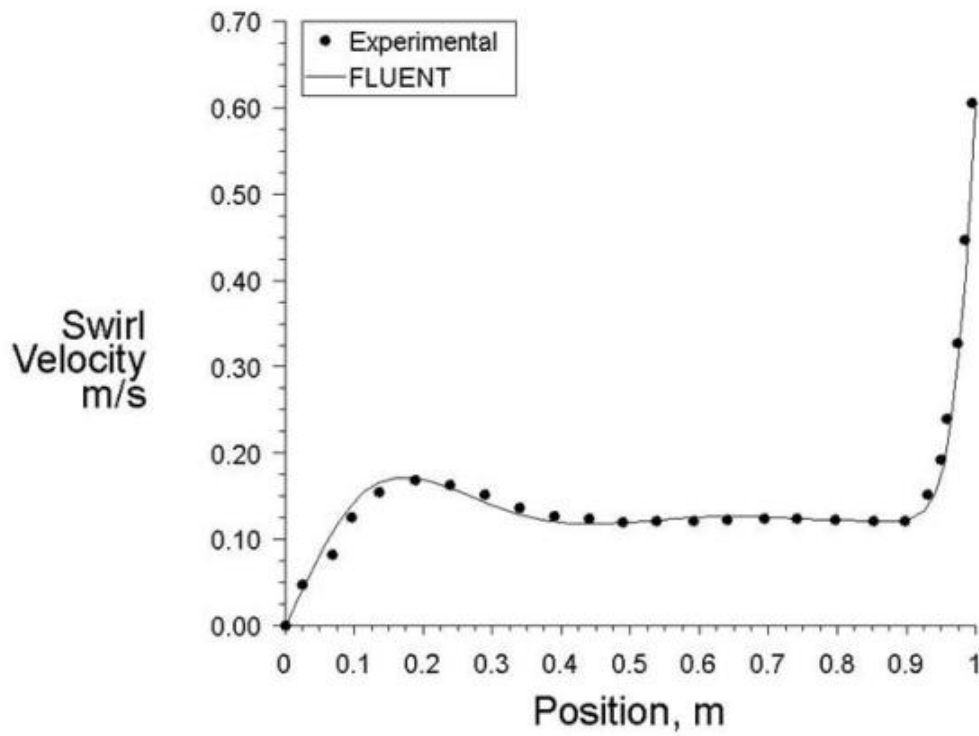


Figure .08.3: Comparison of Distribution of Swirl Velocity Along a Section at X= 0.6 m



Results Comparison for Ansys CFX

Figure .08.4: Comparison of Distribution of Radial Velocity Along a Section at X= 0.6 m

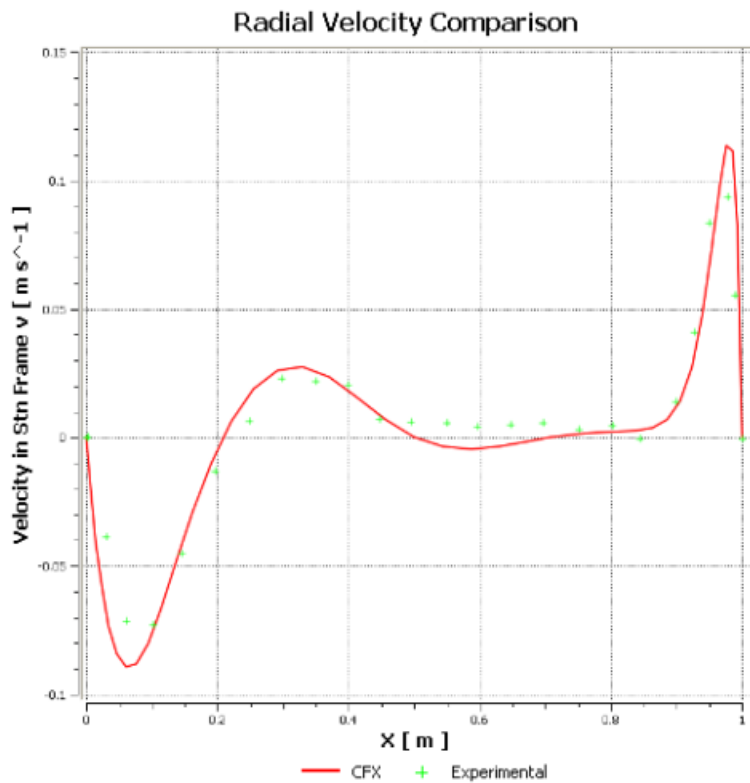
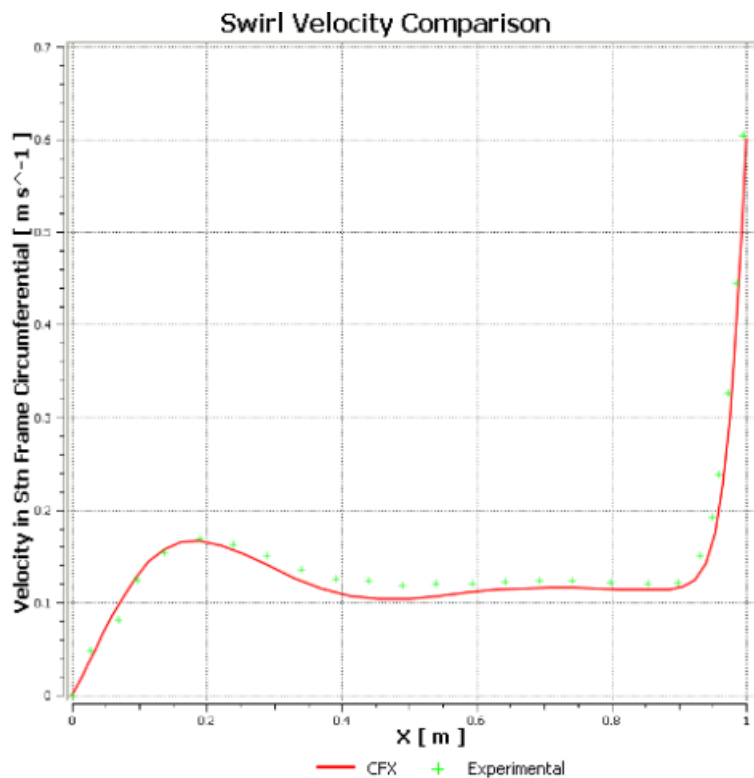


Figure .08.5: Comparison of Distribution of Swirl Velocity Along a Section at X= 0.6 m



VMFL009: Natural Convection in a Concentric Annulus

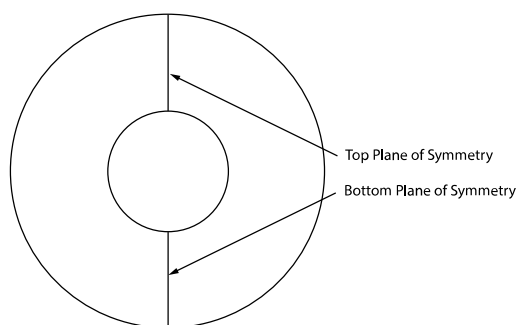
Overview

Reference	T.H. Kuehn, R.J. Goldstein, "An Experimental Study of Natural Convection Heat Transfer in Concentric and Eccentric Horizontal Cylindrical Annuli", <i>Journal of Heat Transfer</i> , Vol 100, pp. 635-640, 1978.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Heat transfer, natural convection, laminar flow
Input Files	concn.cas for Ansys Fluent ecc_cfx.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Natural convection inside a concentric annular domain. The inner wall is maintained at a higher temperature than the outer wall, thereby causing buoyancy induced circulation.

Figure .09.1: Flow Domain



Only half of the domain is modeled due to symmetry.

Material Properties	Geometry	Boundary Conditions
Density: Incompressible ideal gas	Radius of outer cylinder = 46.25 mm	Inner wall temperature = 373 K
Viscosity: 2.081×10^{-5} kg/m-s	Radius of inner cylinder = 17.8 mm	Outer wall temperature = 327 K
Specific Heat: 1008 J/kg-K		
Thermal Conductivity: 0.02967 W/m-K		

Analysis Assumptions and Modeling Notes

The flow is symmetric and only half of the domain is modeled. Density is calculated based on incompressible ideal gas assumption. The flow is laminar.

Results Comparison for Ansys Fluent

Figure .09.2: Comparison of Static Temperature Distribution on the Bottom Wall of Symmetry

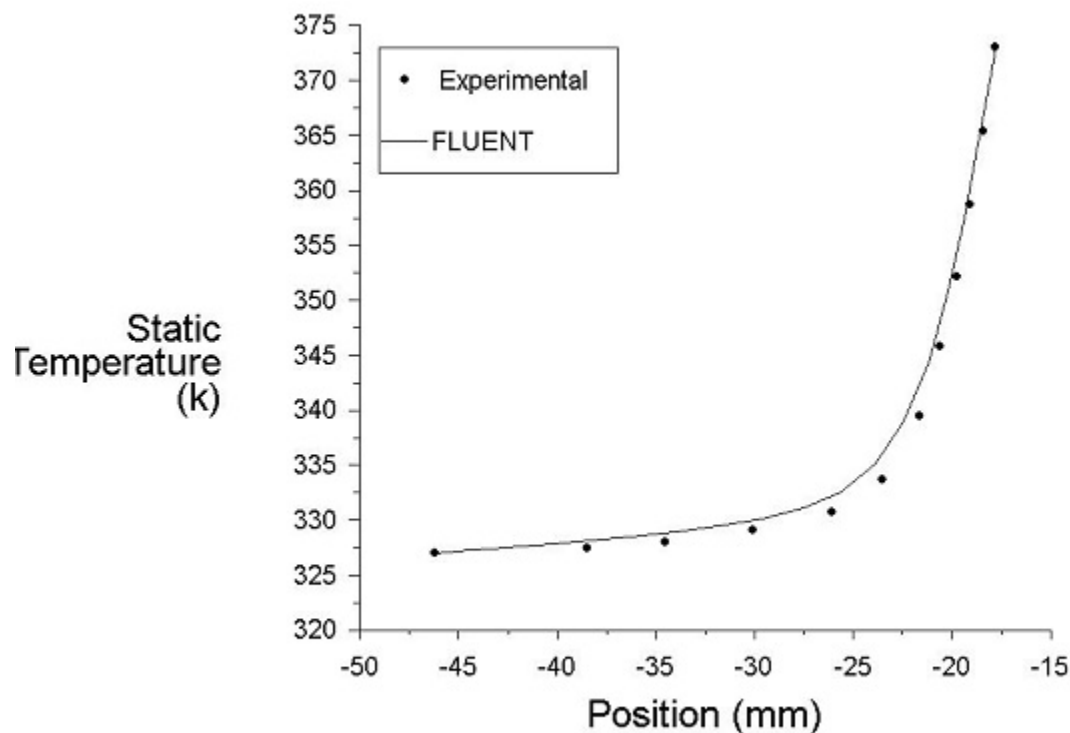
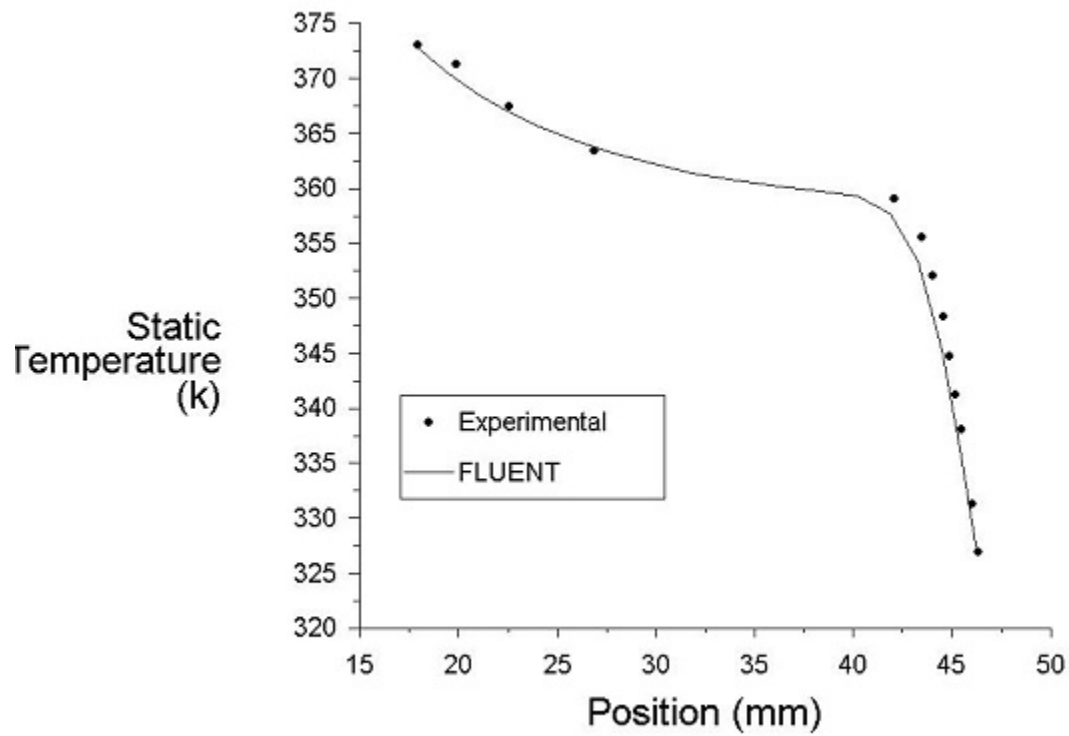
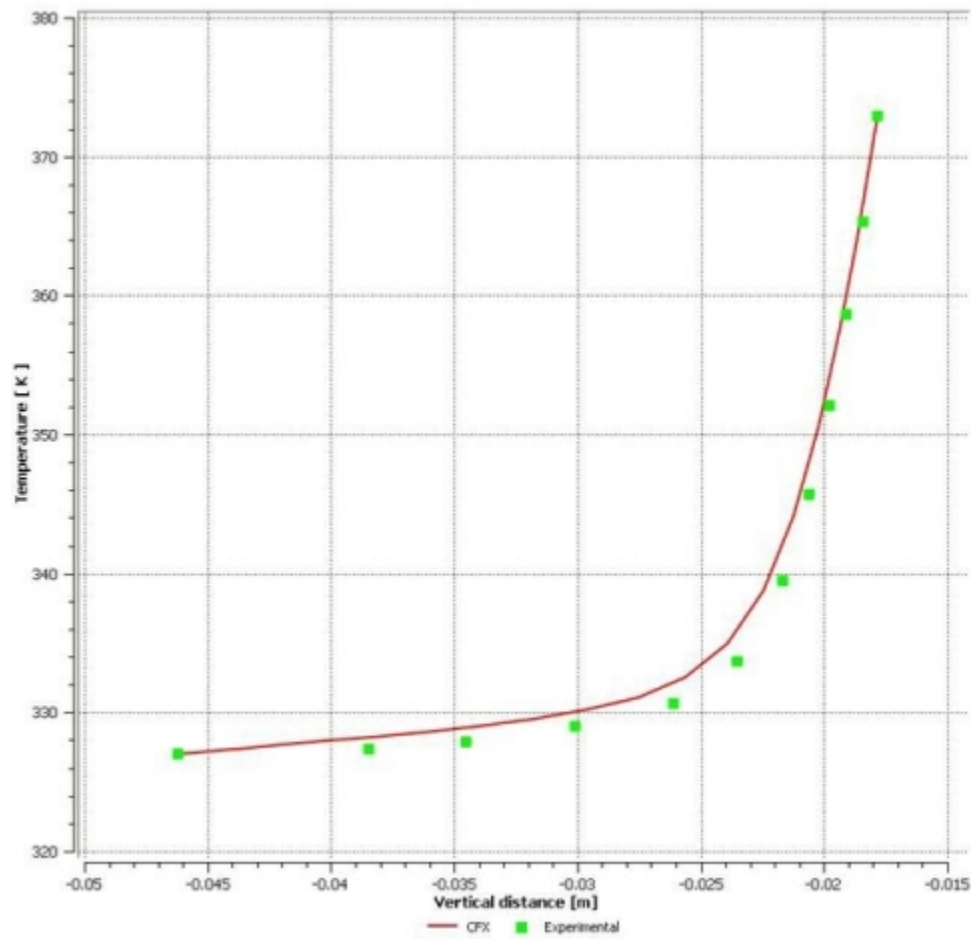


Figure .09.3: Comparison of Static Temperature Distribution on the Top Wall of Symmetry

Results Comparison for Ansys CFX

Figure .09.4: Comparison of Static Temperature Distribution on the Bottom Wall of Symmetry



VMFL010: Laminar Flow in a 90° Tee-Junction.

Overview

Reference	R.E. Hayes, K. Nandkumar, H. Nasr-El-Din, "Steady Laminar Flow in a 90 Degree Planar Branch". <i>Computers and Fluids</i> , Vol 17, pp. 537-553, 1989.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar flow
Input File	p1arb_r4.cas for Ansys Fluent VMFL010B_p1arb.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The purpose of this test is to compare prediction of the fractional flow in a dividing tee-junction with experimental results. The fluid enters through the bottom branch and divides into the two channels whose exit planes are held at the same static pressure.

Figure .10.1: Flow Domain

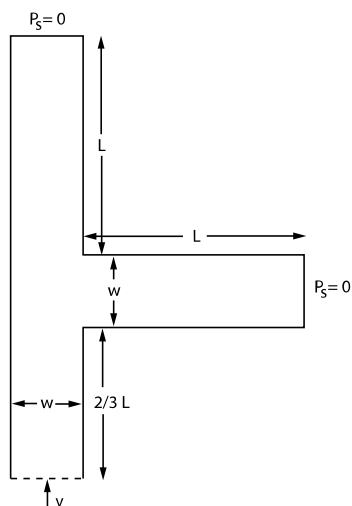


Table .10.1: Comparison of Flow Split from Tee

Material Properties	Geometry	Boundary Conditions
Fluid: Air Density : 1 kg/m ³ Viscosity: 0.003333 kg/m-s	L=3.0 m W=1.0 m	Fully developed inlet velocity profile for: $Re = \frac{\rho V_c W}{\mu} = 300$ where V_c is the inlet centerline velocity. $P_s = 0$

Analysis Assumptions and Modeling Notes

The flow is steady and incompressible. Pressure based solver is used. It is seen that with increasing flow rate in the main channel, less fluid escapes through the secondary (right) branch. For analysis of results, we calculate and compare the fractional flow in the upper branch.

Results Comparison for Ansys Fluent

Table .10.2: Comparison of Flow Split from Tee

	Target	Ansys Fluent	Ratio
Flow split	0.887	0.884	0.997

Results Comparison for Ansys CFX

Table .10.3: Comparison of Flow Split from Tee

	Target	Ansys CFX	Ratio
Flow split	0.887	0.8837	0.9962

VMFL011: Laminar Flow in a Triangular Cavity

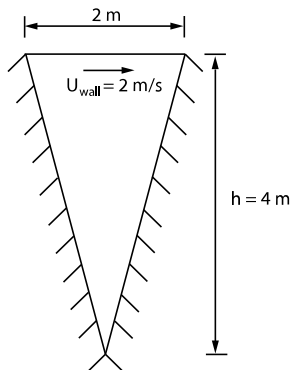
Overview

Reference	R. Jyotsna, S.P. Vanka. "Multigrid Calculation of Steady, Viscous Flow in a Triangular Cavity". <i>J. Comp. Phys.</i> , Vol 122, pp. 107-117, 1995.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Viscous flow, driven by a moving wall
Input Files	driv.cas for FLUENT driven_cavity.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar flow induced by the motion of the top wall of a triangular cavity (Figure .11.1: Flow Domain (p. 41)). The side walls are stationary.

Figure .11.1: Flow Domain



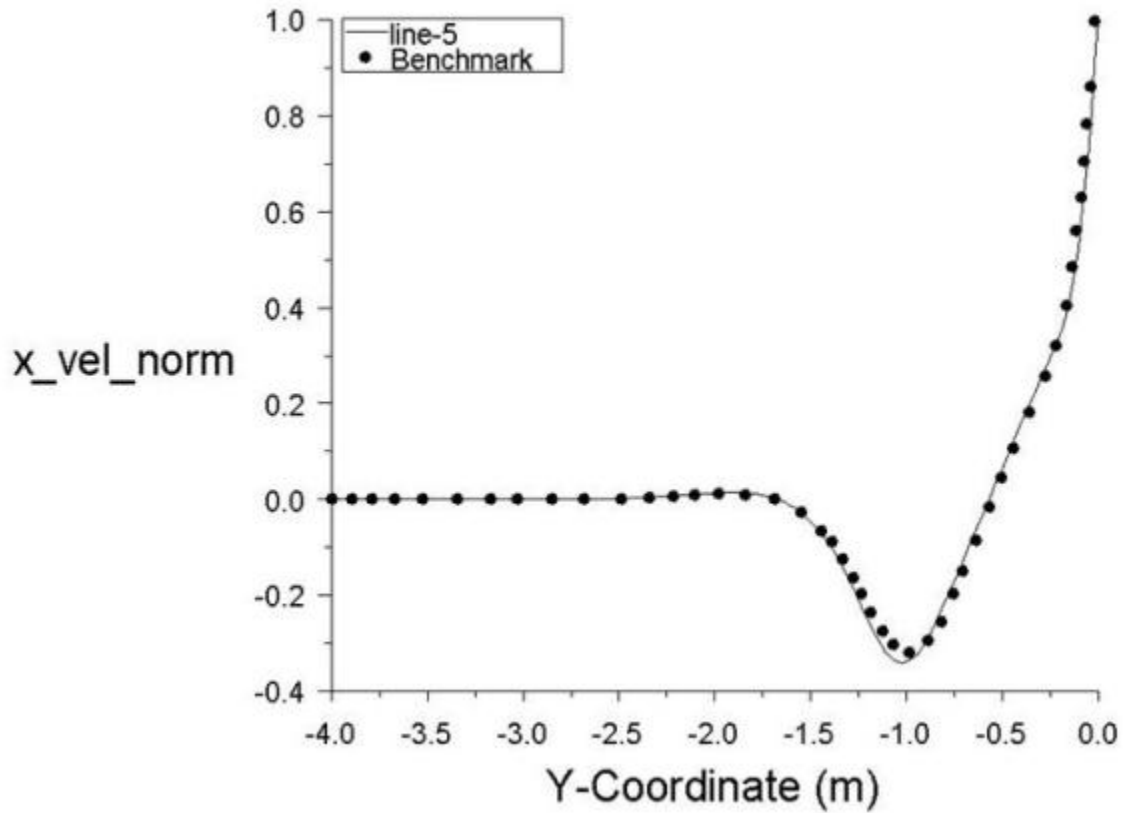
Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m^3 Viscosity = 0.01 kg/m-s	Height of the triangular cavity = 4 m Width of the base = 2 m	Velocity of the top (base) wall = 2 m/s Other walls are stationary

Analysis Assumptions and Modeling Notes

The flow is steady. Pressure based solver is used. A hybrid mesh with triangular and quadrilateral cells is used to discretize the domain.

Results Comparison for FLUENT

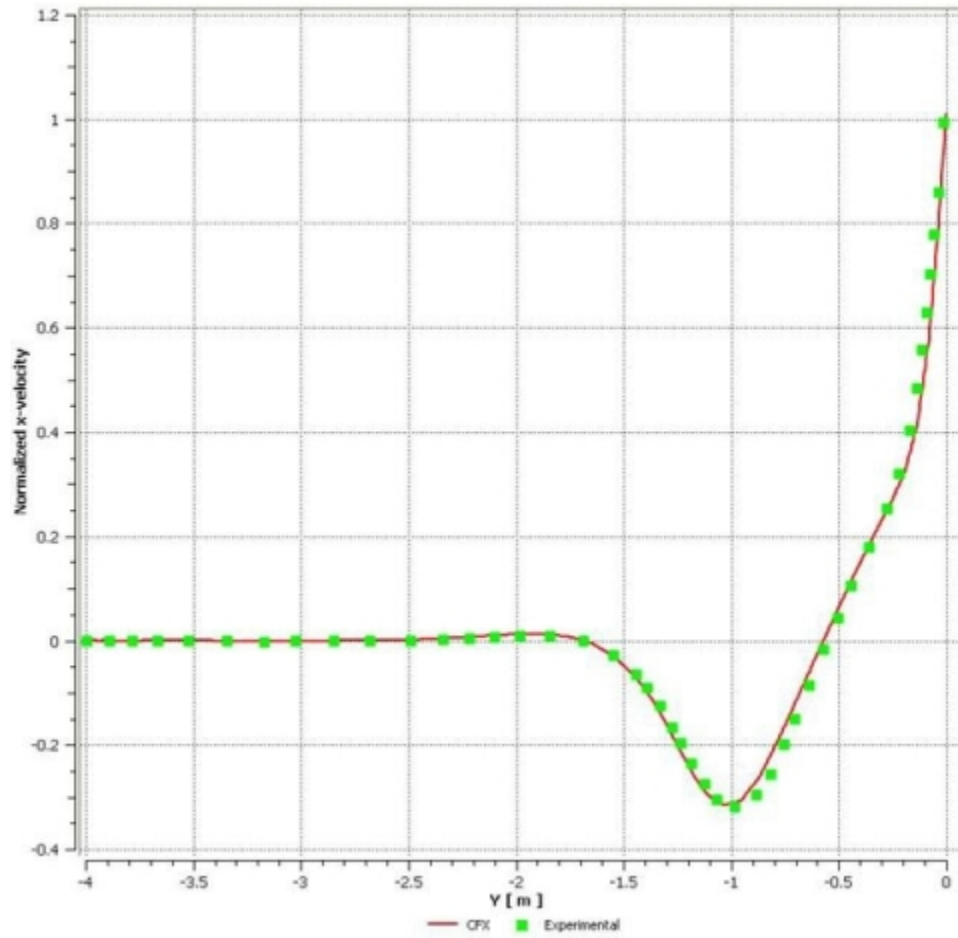
Figure .11.2: Comparison of Distribution of Normalized X-Velocity Along a Vertical Line that Bisects the Base of the Cavity



In this figure, X-velocity is normalized by the velocity of the moving wall.

Results Comparison for Ansys CFX

Figure .11.3: Comparison of Distribution of Normalized X-Velocity Along a Vertical Line that Bisects the Base of the Cavity



In this figure also the X-velocity is normalized by the velocity of the moving wall.

VMFL012: Turbulent Flow in a Wavy Channel

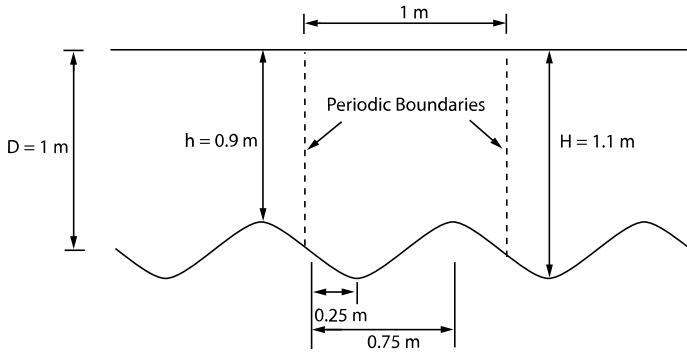
Overview

Reference	J.D. Kuzan, "Velocity Measurements for Turbulent Separated and Near-Separated Flows Over Solid Waves". Ph.D. thesis. Department of Chemical Engineering, University of Illinois, Urbana, IL, 1986.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Turbulent internal flow with separation and recirculation, periodic boundaries
Input File	wavy . cas for Ansys Fluent VMFL012B_VV012 . def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A periodic flow domain bounded on one side by a sinusoidal wavy wall and with a straight wall on the other side. Due to periodicity only a part of the channel needs to be modeled. [Figure .12.1: Flow Domain \(p. 45\)](#) depicts the channel geometry. Flow direction is from left to right.

Figure .12.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m^3 Viscosity = 0.0001 kg/m-s	Amplitude of the sinusoidal wave = 0.1 m Wave length = 1 m Length of the periodic segment = 1 m	Periodic Conditions: Mass flow rate = 0.816 kg/s Pressure Gradient = -0.01687141 Pa/m

Analysis Assumptions and Modeling Notes

The flow is steady. Pressure based solver is used. Periodic boundaries are used. For analysis of results, velocity in the x –direction is normalized by the mean mainstream velocity, $U = 0.816$ m/s, at mean channel height. Turbulence model used is Realizable k-E in Fluent and Shear Stress Transport in CFX.

Results Comparison for Ansys Fluent

Figure .12.2: Comparison of Distribution of Normalized X-Velocity along Transverse Direction at the Wave Crest

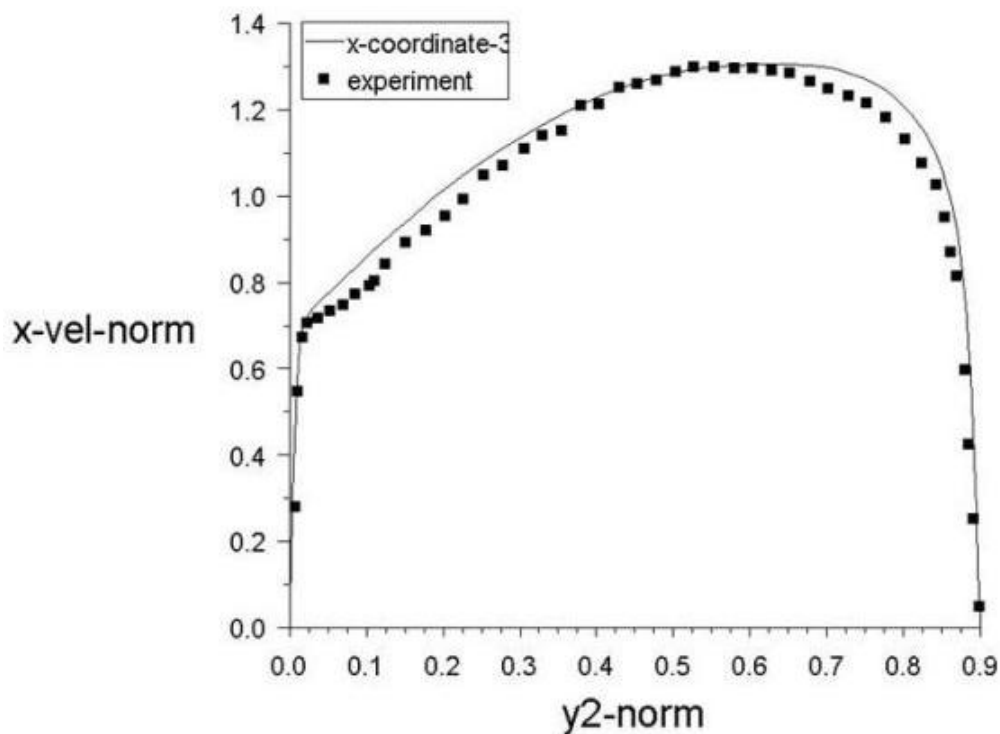
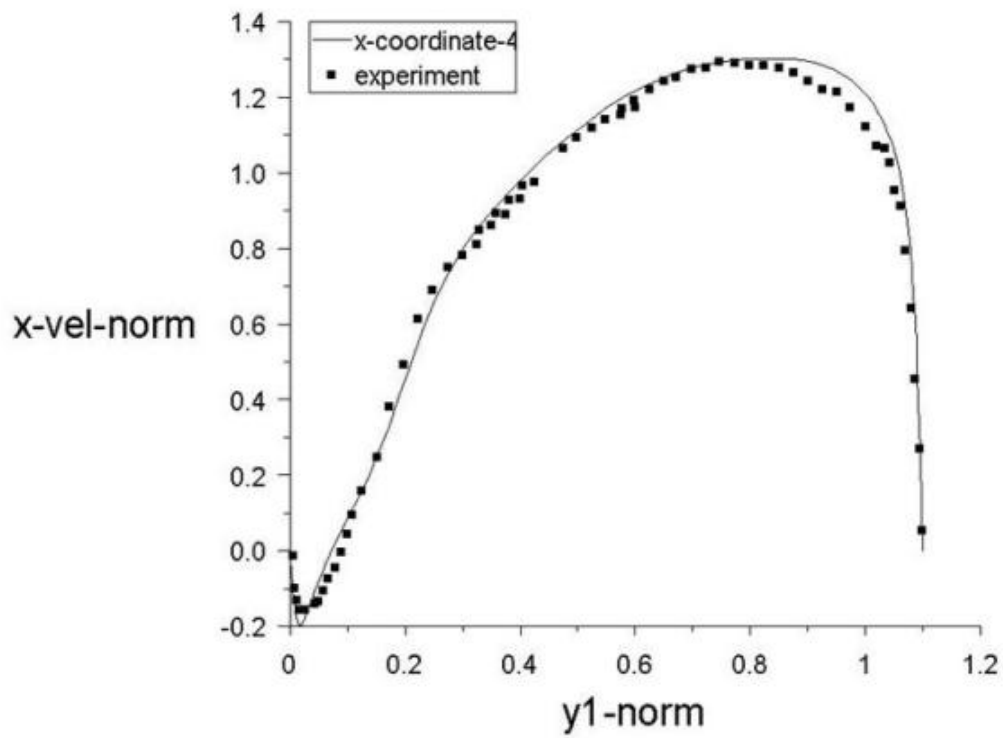


Figure .12.3: Comparison of Predicted Normalized X-Velocity along Transverse Direction at the Wave Trough



Results Comparison for Ansys CFX

Figure .12.4: Comparison of Distribution of Normalized X-Velocity along Transverse Direction at the Wave Crest

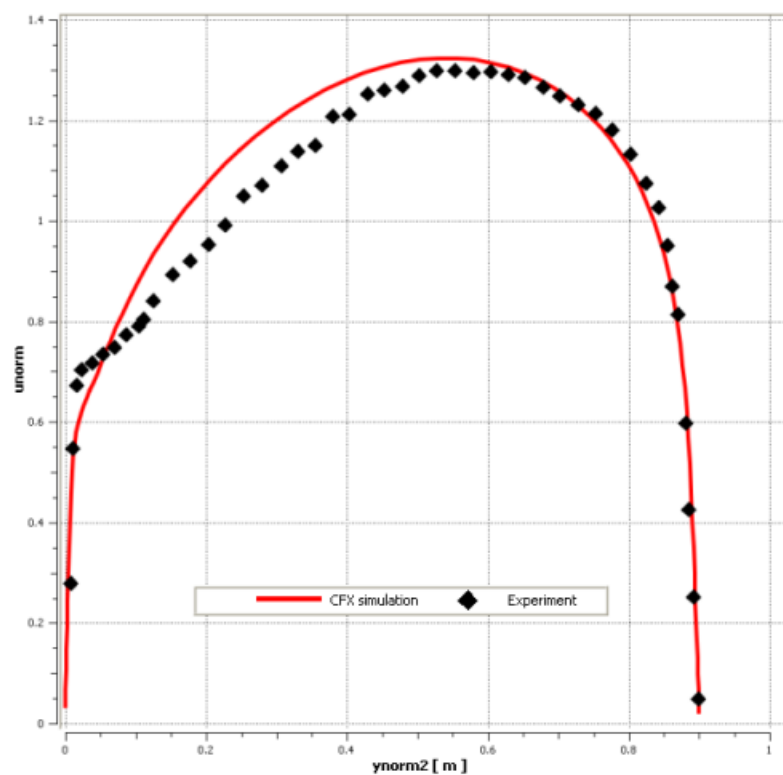
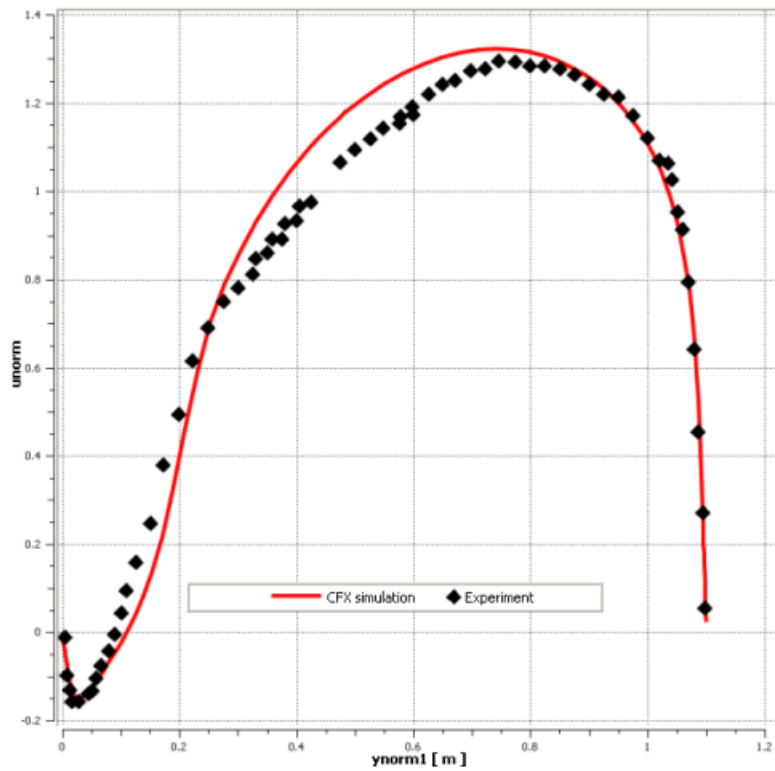


Figure .12.5: Comparison of Predicted Normalized X-Velocity along Transverse Direction at the Wave Trough



VMFL013: Turbulent Flow with Heat Transfer in a Backward-Facing Step

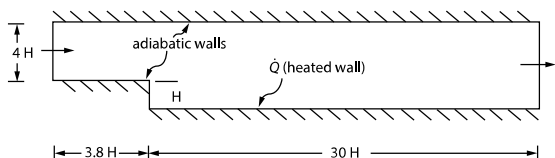
Overview

Reference	J.C. Vogel, J.K. Eaton, "Combined Heat Transfer and Fluid Dynamic Measurements Downstream of a Backward-Facing Step". <i>Journal of Heat Transfer</i> , Vol. 107, pp. 922-929, 1985.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Incompressible, turbulent flow with heat convection and reattachment.
Input File	step_ve.cas for Ansys Fluent VMFL013_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The fluid flow and convective heat transfer over a 2-D backward-facing step is modeled. A constant heat-flux surface behind the sudden expansion leads to a separated and reattaching boundary layer that disturbs local heat transfer. Measured values of the distribution of the local Nusselt number along the heated wall are used to validate the CFD simulation.

Figure .13.1: Flow Domain



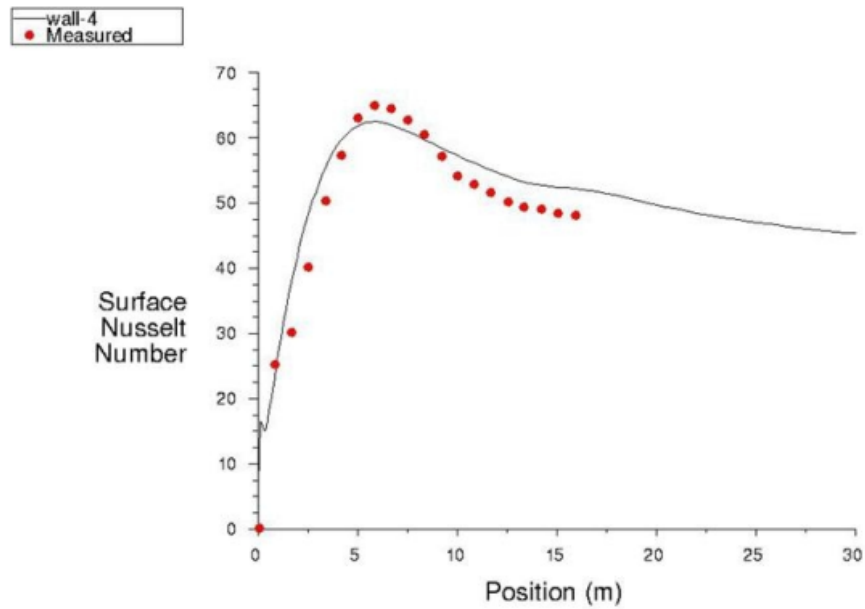
Material Properties for Dry Air	Geometry	Boundary Conditions
Density = 1 kg/m^3 Viscosity = 0.0001 kg/m-s Conductivity = 1.408 W/m-K Specific Heat = $10,000 \text{ J/kg-K}$	$H = 1 \text{ m}$	Velocity profile at inlet corresponding to $Re_H = 28,000$ Wall heat transfer, $Q' = 1,000 \text{ W/m}^2$

Analysis Assumptions and Modeling Notes

A Cartesian non-uniform 121×61 mesh is used. The flow is steady and incompressible. Fluid properties are considered constant. Pressure based solver is used. The inlet boundary conditions are specified using the fully-developed profiles for the U-velocity, k, and epsilon. The incoming boundary layer thickness is $1.1 H$. Under the given pressure conditions, the Reynolds number, Re_H is about 28,000. The RNG $k-\epsilon$ model with standard wall functions is used for accounting turbulence.

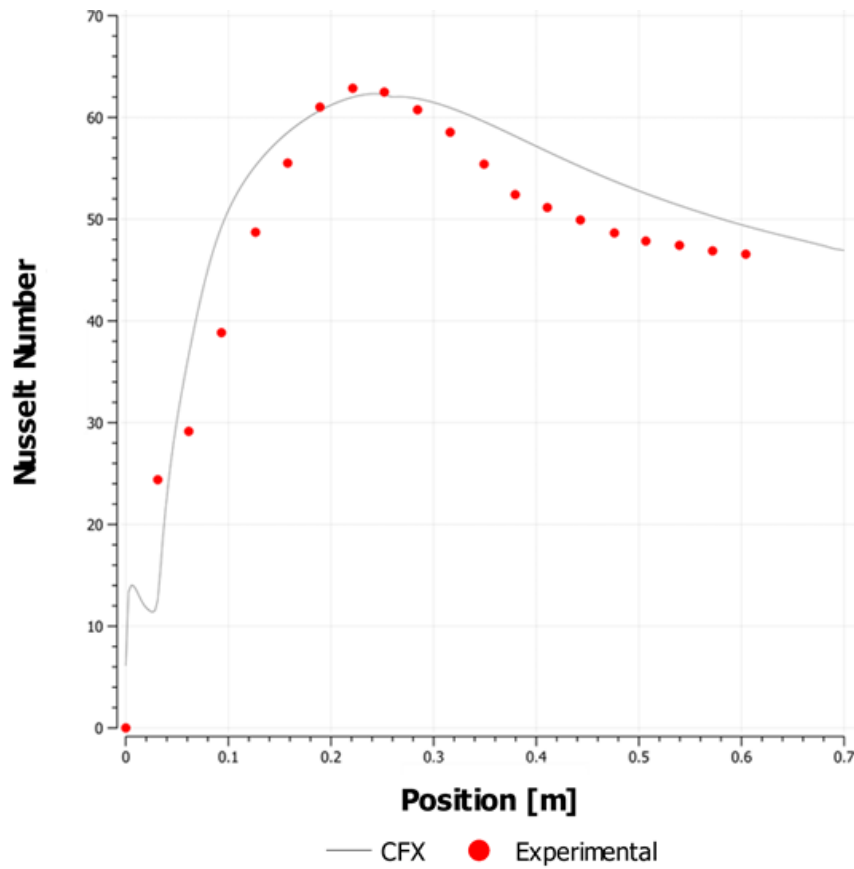
Results Comparison for Ansys Fluent

Figure .13.2: Comparison of Predicted Local Nusselt Number Distribution Along the Heated Wall with Experimental Data



Results Comparison for Ansys CFX

Figure .13.3: Comparison of Predicted Local Nusselt Number Distribution Along the Heated Wall with Experimental Data



VMFL014: Species Mixing in Co-axial Turbulent Jets

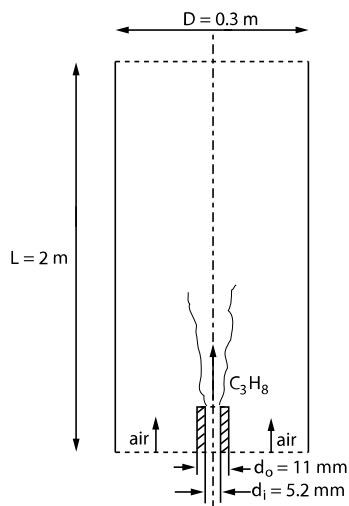
Overview

Reference	R.W. Schefer, R.W. Dibble, "Simultaneous Measurements of Velocity and Density in a Turbulent Non-premixed Flame". <i>AIAA Journal</i>, Vol 23, pp. 1070-1078, 1985. R.W., Schefer, "Data Base for a Turbulent, Nonpremixed, Nonreacting Propane-Jet Flow". http://www.sandia.gov/TNF/DataArch/ProJet.html
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Multi-Species flow, turbulent, jet mixing
Input File	san_jet.cas for Ansys Fluent VMFL014.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A propane jet issues into a co-axial stream of air. There is turbulent mixing between the species in the axisymmetric tunnel. Only half of the domain is considered due to axial symmetry.

Figure .14.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: Incompressible ideal gas law	Tunnel length = 2 m	Inlet velocity of air = 9.2 m/s
Viscosity: 1.72×10^{-5} kg/m-s	Tunnel diameter = 0.3 m	Inlet velocity of Propane – Specified as fully developed profile
	Propane jet tube:	

Material Properties	Geometry	Boundary Conditions
	Inner diameter = 5.2 mm	Inlet temperature (both streams) = 300 K
	Outer diameter = 11 mm	Temperature at the wall = 300 K

Analysis Assumptions and Modeling Notes

The flow is steady. Species mixing is modeled with the three species; propane, oxygen, and nitrogen. There is no reaction.

Results Comparison for Ansys Fluent

Figure .14.2: Comparison of Distribution of Propane Along Axis of the Jets

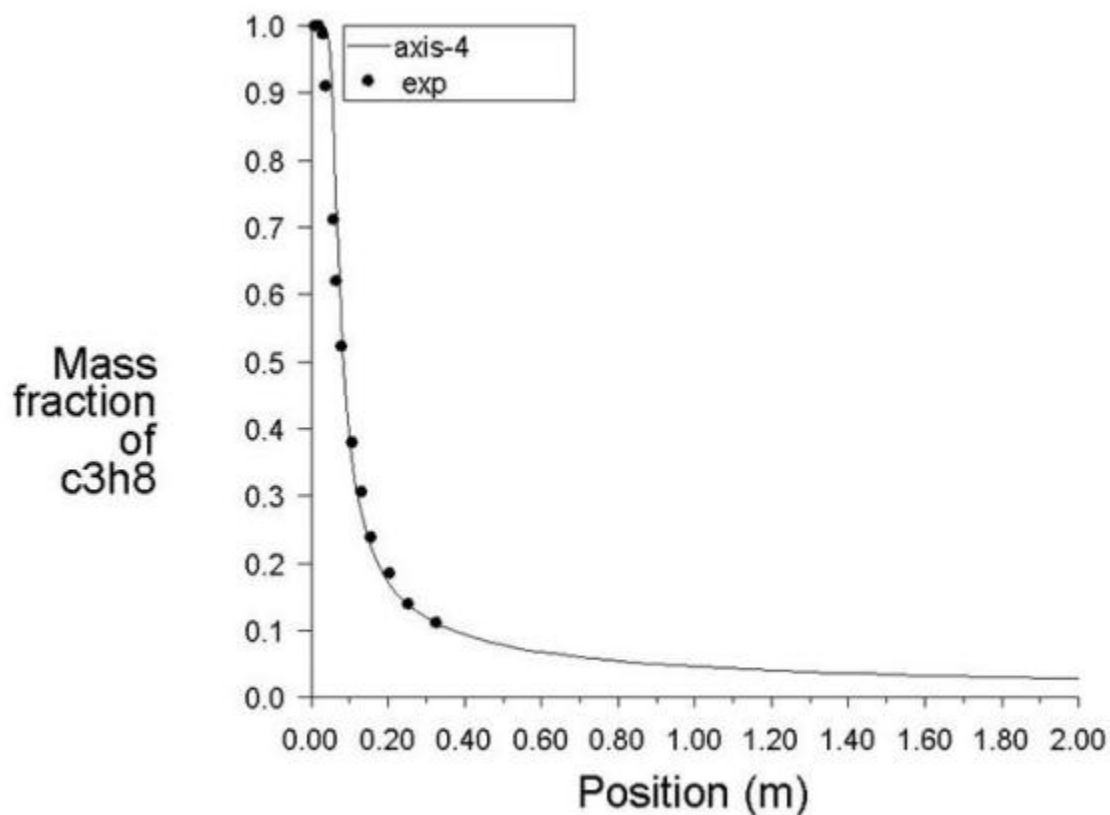
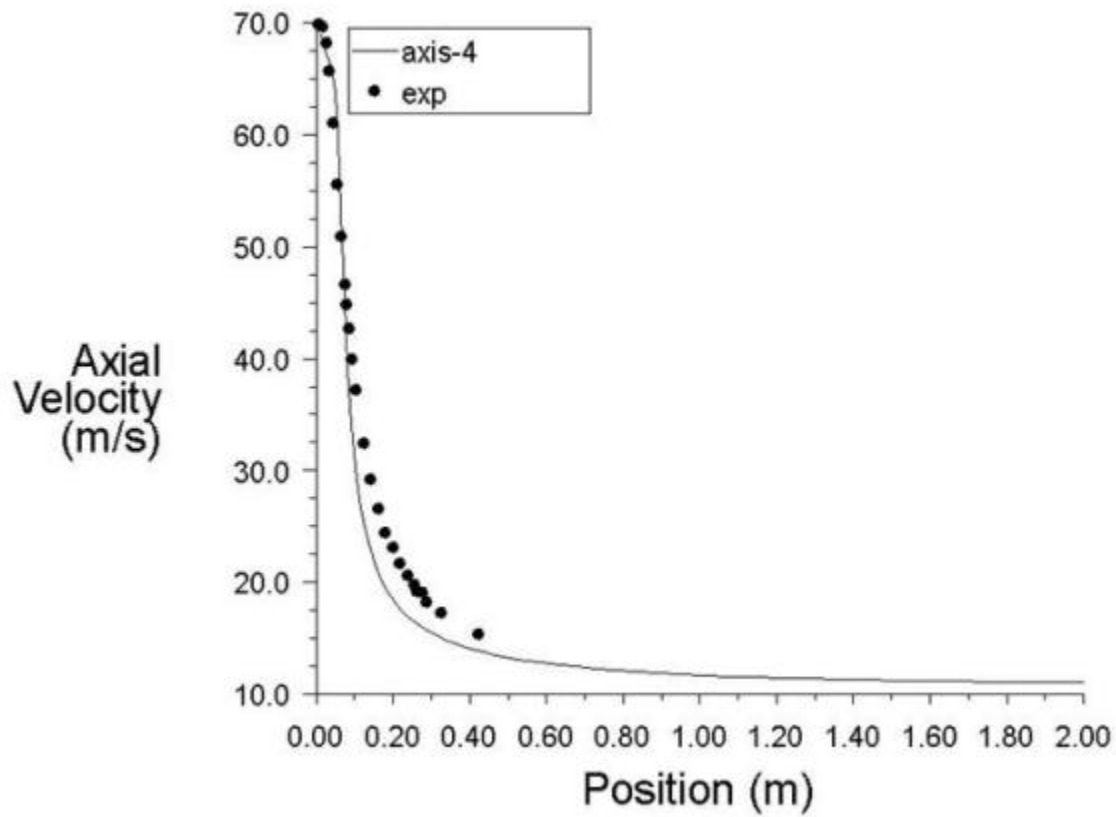


Figure .14.3: Comparison of Distribution of X-Velocity Along Axis of the Jets



Results Comparison for CFX

Figure .14.4: Comparison of Distribution of Propane Along Axis of the Jets

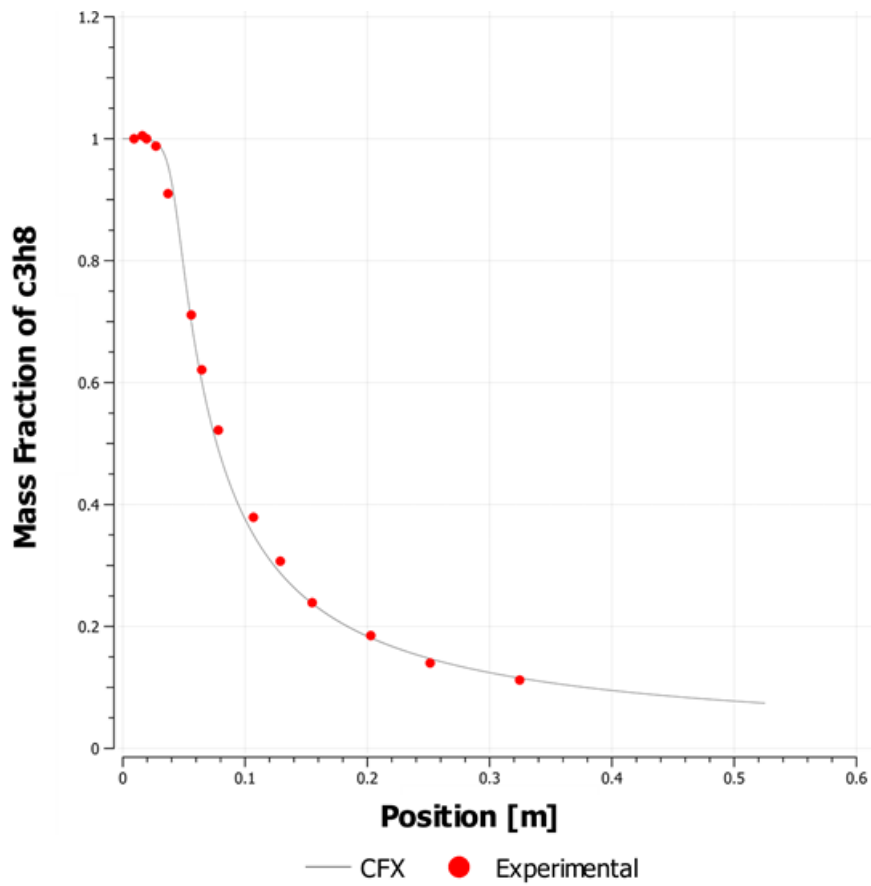
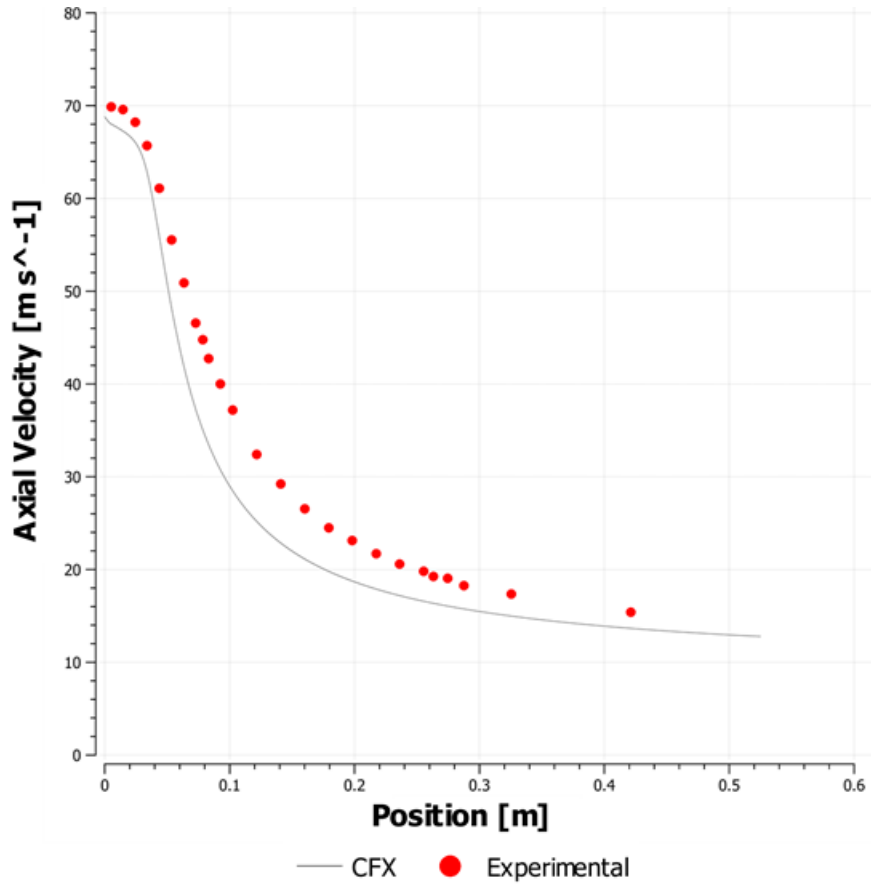


Figure .14.5: Comparison of Distribution of X-Velocity Along Axis of the Jets

VMFL015: Flow Through an Engine Inlet Valve

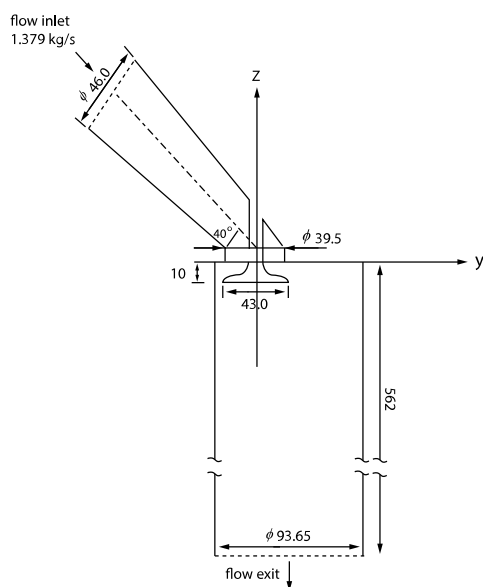
Overview

Reference	A. Chen, K.C. Lee, M. Yianneskis, G. Ganti, "Velocity Characteristics of Steady Flow Through a Straight Generic Inlet Port." <i>International Journal for Numerical Methods in Fluids</i> , Vol 21, pp. 571-590, 1995.
Solver	Ansys Fluent
Physics/Models	3-D turbulent flow
Input File	valve10.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow in an idealized engine cylinder with a straight inlet port and a valve lift of 10 mm (the distance from the top of the cylinder to the bottom of the valve). The configuration of the inlet port, valve, and cylinder is shown in [Figure .15.1: Flow Domain \(p. 61\)](#).

Figure .15.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density : 894 kg/m ³ Viscosity: 0.001529 kg/m-s	All dimensions shown in Figure .15.1: Flow Domain (p. 61) are in mm.	Inlet velocity = 0.9282 m/s Inlet turbulent intensity = 10%

Material Properties	Geometry	Boundary Conditions
		Inlet turbulent length scale = 0.046m Outlet gauge pressure = 0 Pa

Analysis Assumptions and Modeling Notes

The flow is steady, isothermal and incompressible. The standard k- ϵ model with standard wall functions is used. The length of the cylinder is chosen to be large enough that it will not affect the flow in the cylinder.

Results Comparison for Ansys Fluent

Figure .15.2: Z-Velocity Component at Z= -5mm (p. 62) and Figure .15.3: Z-Velocity Component at Z = +10mm (p. 63) compare Ansys Fluent's results with the experimental data (z-component of velocity at different heights).

Figure .15.2: Z-Velocity Component at Z= -5mm

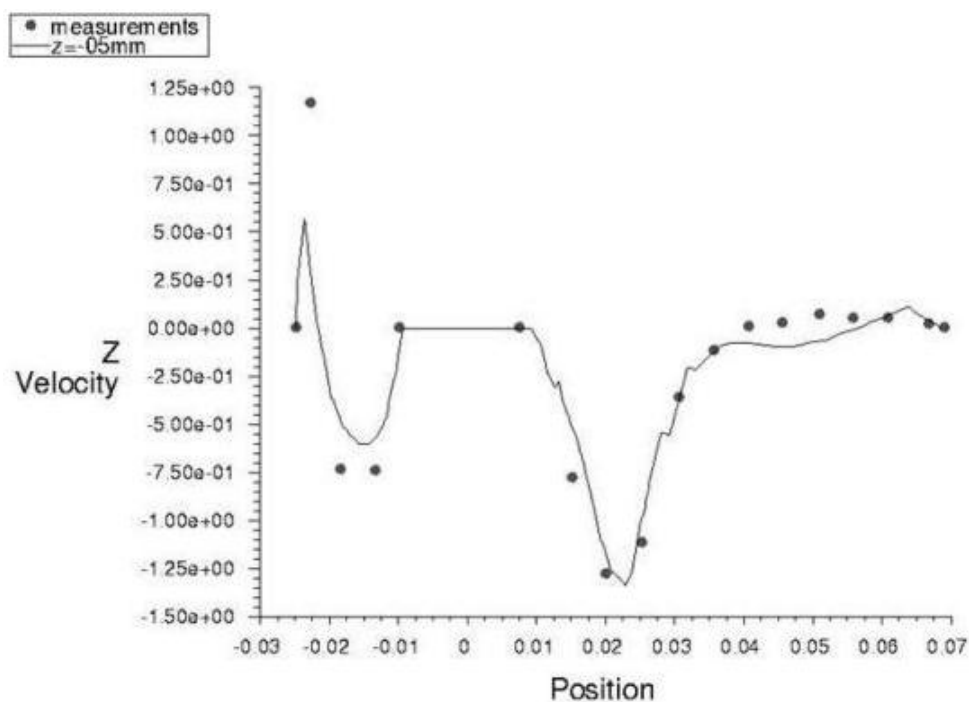
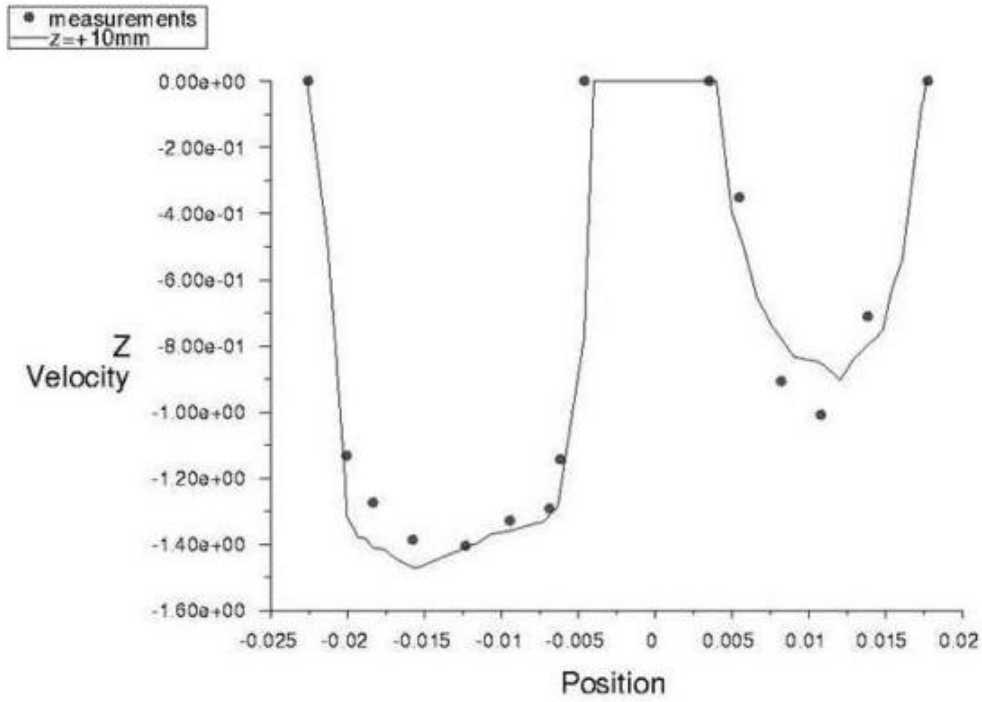


Figure .15.3: Z-Velocity Component at Z = +10mm

VMFL016: Turbulent Flow in a Transition Duct

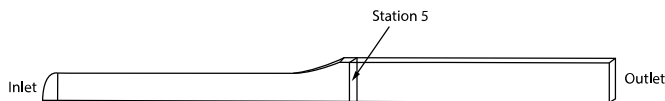
Overview

Reference	D.O. Davis, F.B. Gessner, "Experimental Investigation of Turbulent Flow Through a Circular-to-Rectangular Transition Duct". <i>AIAA Journal</i> , Vol 30, pp. 367-375, 1992
Solver	Ansys Fluent
Physics/Models	3-D Turbulent flow with separation, Reynolds stress model
Input Files	tranduct-rsm-1.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Turbulent flow through a circular-to-rectangular transition duct having the same inlet and outlet cross-sectional areas is modeled. The curvature of the duct walls induces a strong pressure-driven cross-flow that develops into a counter-rotating vortex pair near the short side walls of the duct. Due to symmetry of the flow field, only one fourth of the duct is modeled (as shown in [Figure .16.1: Flow Domain \(p. 65\)](#)). Station 5 is located 23 m downstream of the inlet.

Figure .16.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 5.13X10 ⁻⁶ kg/m-s	Inlet radius = 1 m Length of duct = 35 m	Inlet velocity: 1 m/s

Analysis Assumptions and Modeling Notes

The flow is steady. Reynolds Stress Model (RSM) is used to model turbulence.

Results Comparison for Ansys Fluent

Figure .16.2: Comparison of Pressure Coefficient at Station 5

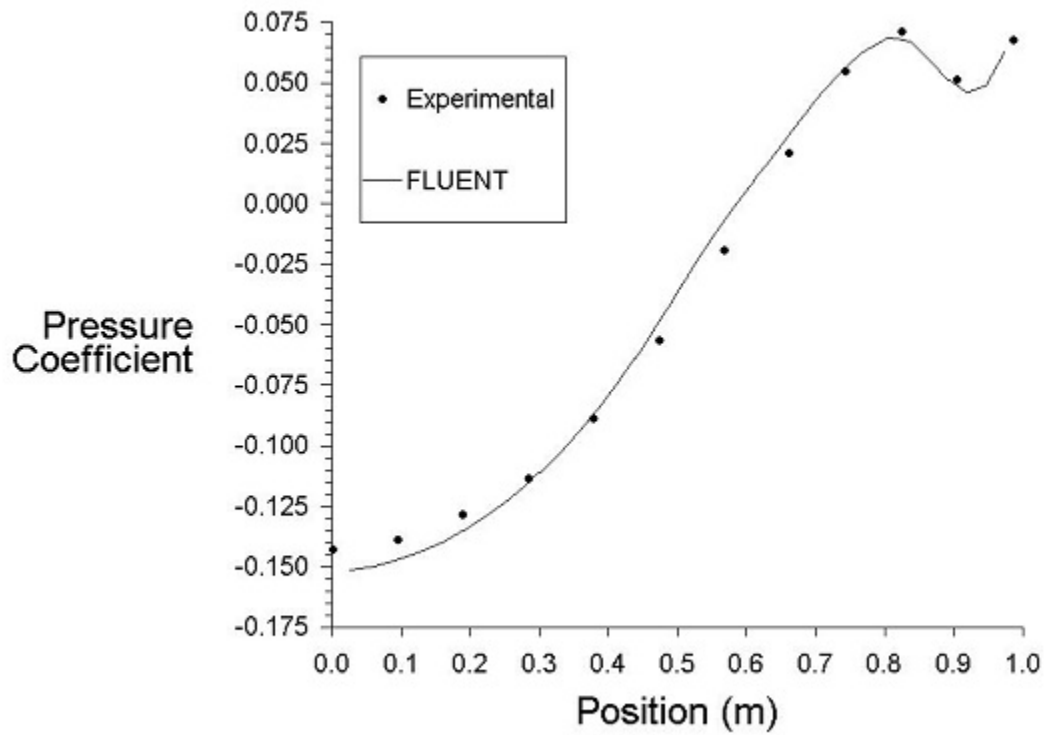
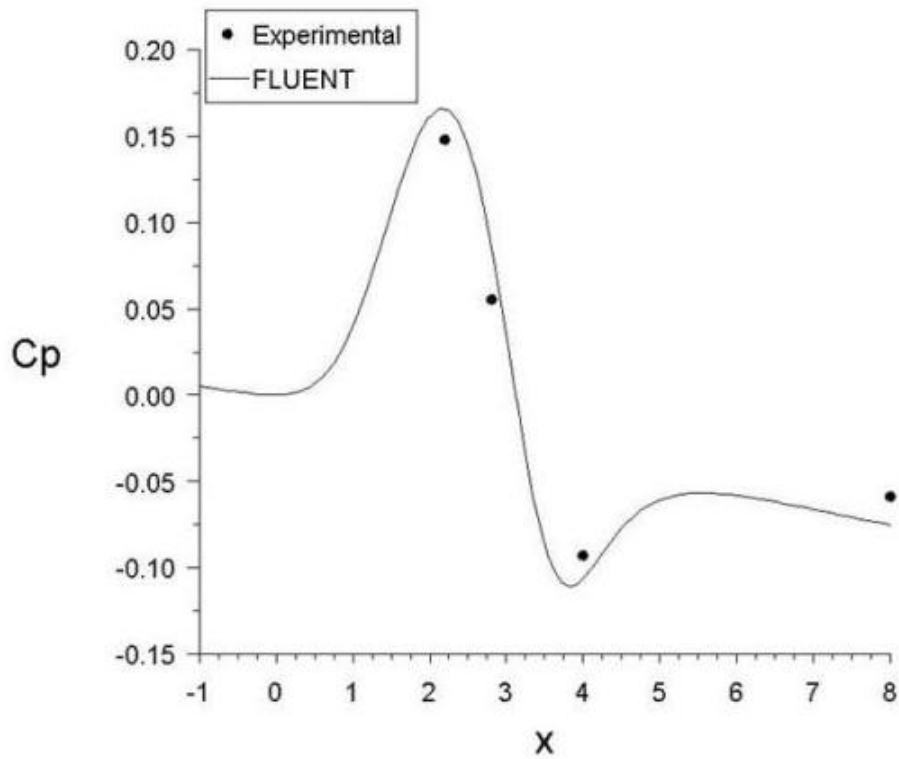


Figure .16.3: Comparison of Pressure Coefficient Along Centerline of the Duct

VMFL017: Transonic Flow over an RAE 2822 Airfoil

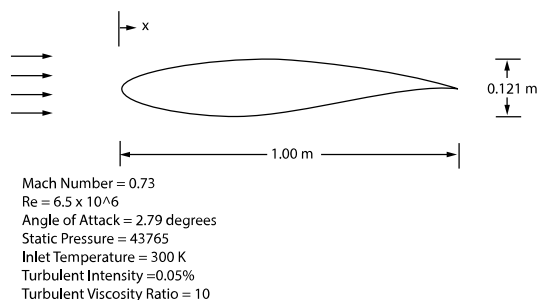
Overview

Reference	P.H. Cook, M.A. McDonald, M.C.P. Firmin, "AEROFOIL RAE 2822 Pressure Distribution and Boundary Layer and Wake Measurements". <i>AGARD Advisory Report</i>. No. 138. 1979. S.J. Kline, B.J. Cantwell, G.M. Lilley, "1980-81 AFOSR-HTTM-Stanford Conference on Complex Turbulent Flows: Comparison of Computation and Experiment", Stanford University, Stanford, Calif., 1982.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Compressible, turbulent flow
Input File	r2822.cas for Ansys Fluent VMFL017.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow over an RAE 2822 airfoil at a free-stream Mach number of 0.73. The angle of attack is 2.79°. The flow field is 2D, compressible (transonic), and turbulent. The geometry of the RAE 2822 airfoil is shown in [Figure .17.1: Geometry of the RAE 2822 Airfoil \(p. 69\)](#). It is a thick airfoil with a chord length, c , of 1.00 m and a maximum thickness, d , of 0.121 m. The flow domain spans over 100 Chord lengths in both streamwise and transverse directions, so that the presence of the airfoil is not felt at the outer boundary.

Figure .17.1: Geometry of the RAE 2822 Airfoil



Material Properties	Geometry	Boundary Conditions
Fluid: Air - Density: Ideal Gas - Viscosity: 1.983×10^{-5} kg/m-s	Chord length = 1 m Maximum thickness = 0.121 m	The inlet conditions are: Mach number = 0.73 $Re = 6.5 \times 10^6$ Static pressure = 43765 Pa

Material Properties	Geometry	Boundary Conditions
- Thermal conductivity: 0.0242 W/m-K - Molecular Weight: 28.966 g/mol - Specific Heat: 1006.43 J/kg-K		Inlet temperature = 300 K Turbulent intensity = 0.05 % Turbulent viscosity ratio = 10

Analysis Assumptions and Modeling Notes

The implicit formulation of the density-based solver is used. The SST k- ω turbulence model is used to account for turbulence effects. The problem is solved in steady state mode. The experimental procedure included uses a transition trip to fix the boundary layer transition.

Results Comparison for Ansys Fluent

Table .17.1: Comparison of Coefficients

Coefficients	Target	Ansys Fluent	Ratio
Drag	0.0168	0.016	0.952
Lift	0.803	0.78	0.971

Results Comparison for Ansys CFX

Table .17.2: Comparison of Coefficients

Coefficients	Target	Ansys Fluent	Ratio
Drag	0.0168	0.0162	0.9662
Lift	0.803	0.7981	0.9339

VMFL018: Shock Reflection in Supersonic Flow

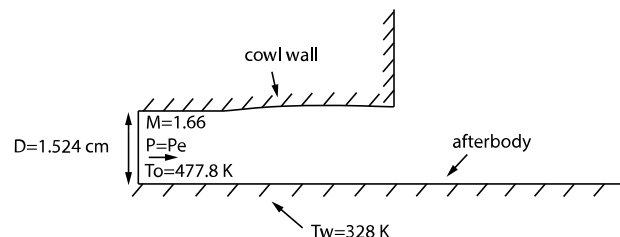
Overview

Reference	H.B. Hopkins, W. Konopka, J. Leng, "Validation of scramjet exhaust simulation technique at Mach 6", <i>NASA Contractor Report 3003</i> , 1979.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Reflecting shocks in supersonic flow; Compressible turbulent flow
Input File	scram-nozzle-flow.cas for Ansys Fluent VMFL018_VV018.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Supersonic flow from a nozzle that represents the exhaust nozzle of a supersonic combustion ramjet (SCRAMJET) is modeled. Jet from the nozzle is issued into a domain which is bounded on one side by an afterbody wall which is parallel to the centerline of the nozzle. Shocks propagating from the nozzle exit reflect from the afterbody. Measured values of (i) the distribution of wall pressure and (ii) heat transfer rate along the afterbody are used to validate the CFD simulation.

Figure .18.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: Ideal Gas	$D = 1.524\text{ cm}$	Inlet Total Pressure (gauge) = 551600 Pa
Molecular Weight: 113.2	Length of cowl = 3.5 D	Inlet Static Pressure (gauge) = 127100 Pa
Viscosity: $1.7894 \times 10^{-5}\text{ kg/m-s}$		Inlet Total Temperature = 477.8 K
Thermal Conductivity: 0.0242 W/m-K		Inlet Turbulent Intensity = 2%
Specific Heat: Temperature Dependent		Wall temperature = 328 K
		Outlet Pressure (gauge) = 2780 Pa

Analysis Assumptions and Modeling Notes

The flow is steady. Specific heat is defined as a linear function of temperature. Density based solver is used. Under the given pressure conditions, the inlet Mach number is about 1.66.

Results Comparison for Ansys Fluent

Figure .18.2: Comparison of Predicted Static Pressure Distribution on the Afterbody with Experimental Data

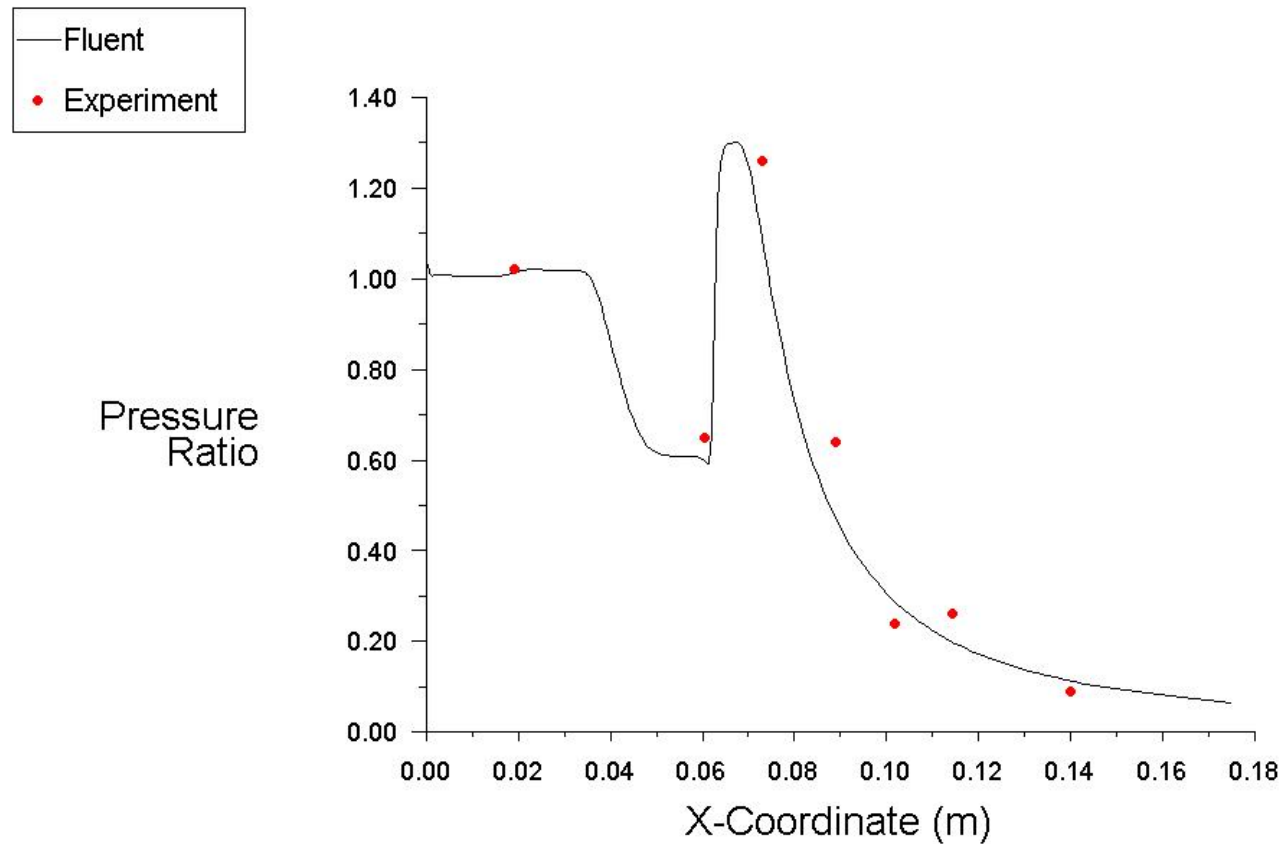
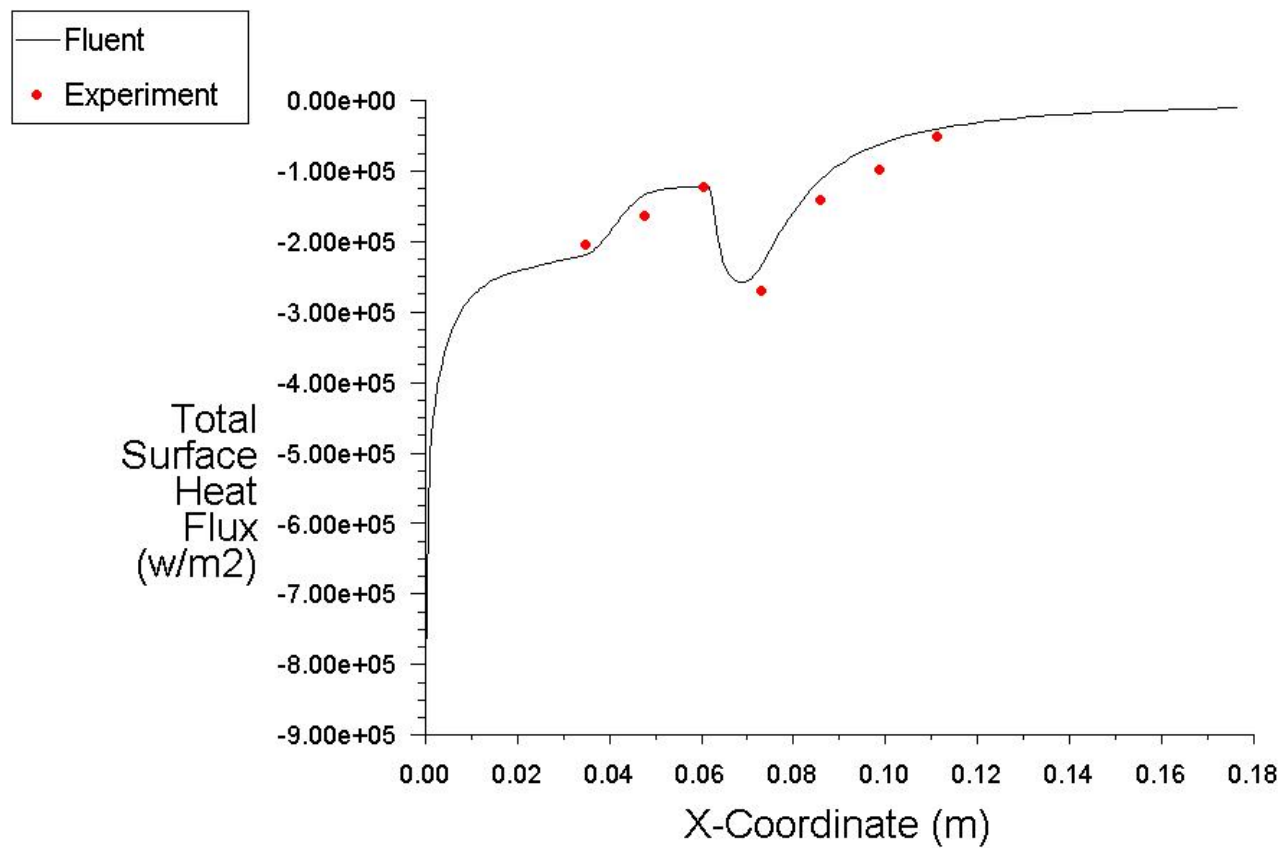


Figure .18.3: Comparison of Predicted Total Heat Flux Along the Afterbody with Experimental Data



Results Comparison for Ansys CFX

Figure .18.4: Comparison of Predicted Static Pressure Distribution on the Afterbody with Experimental Data

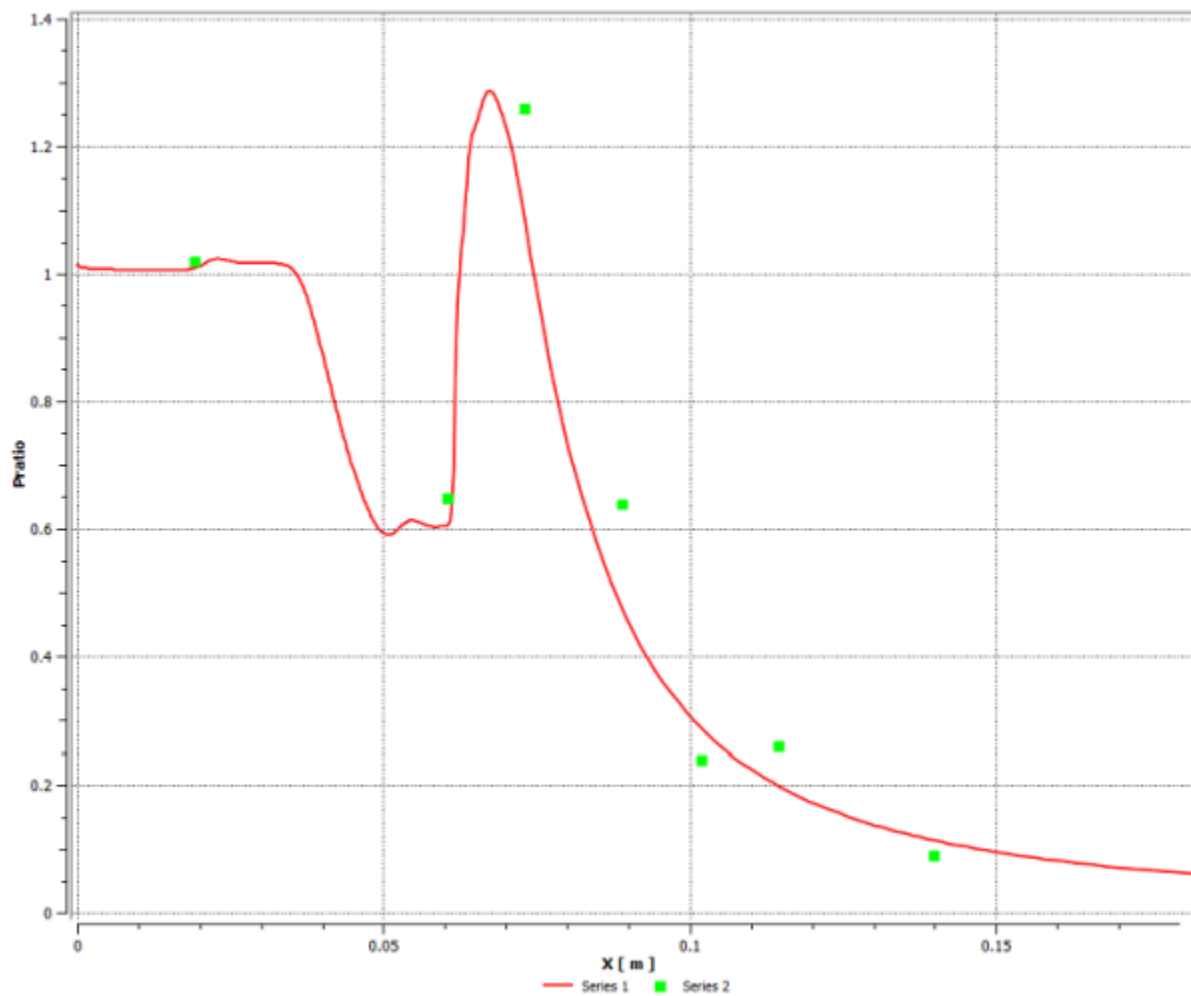
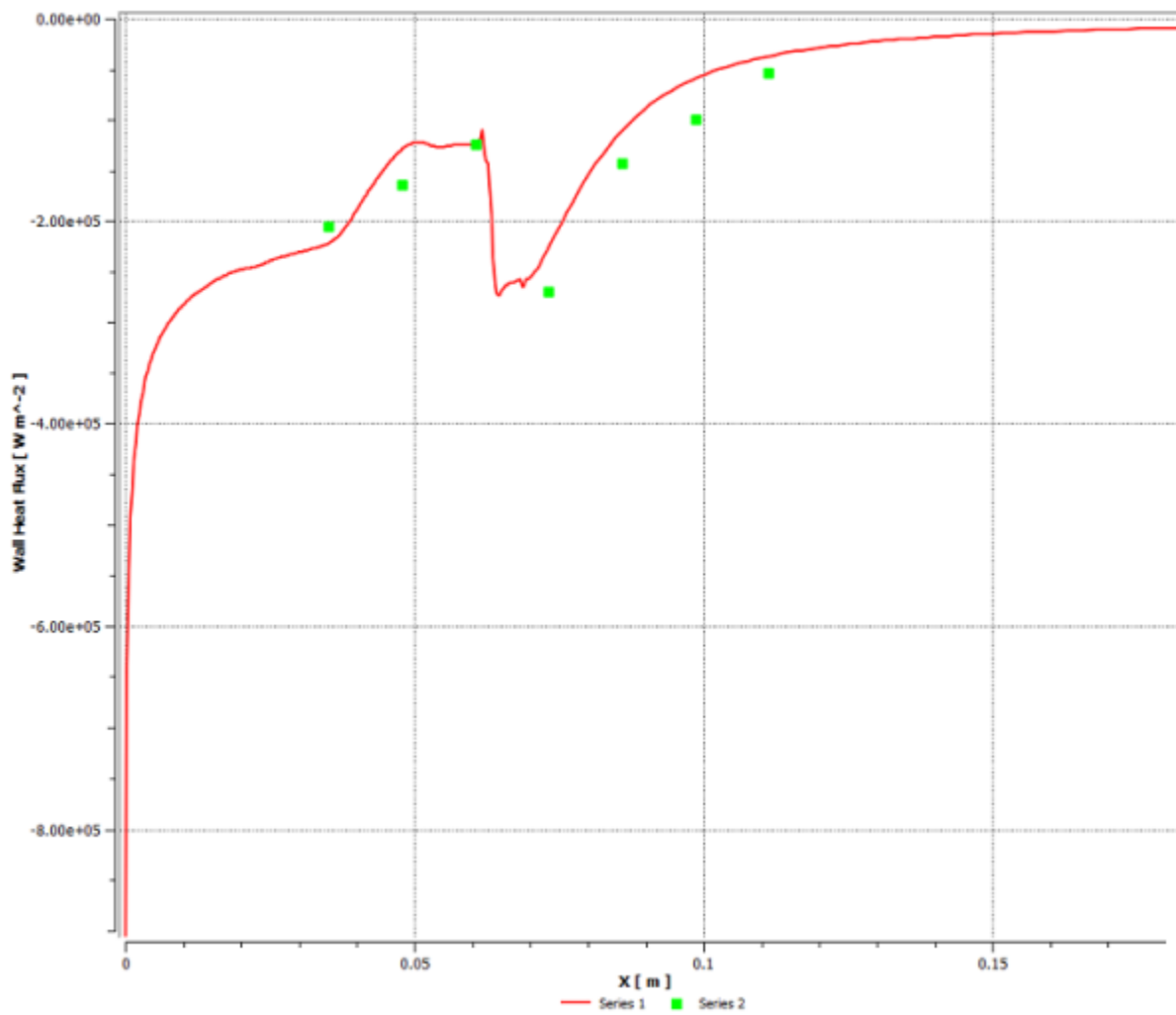


Figure .18.5: Comparison of Predicted Total Heat Flux Along the Afterbody with Experimental Data



VMFL019: Transient Flow Near a Wall Set in Motion

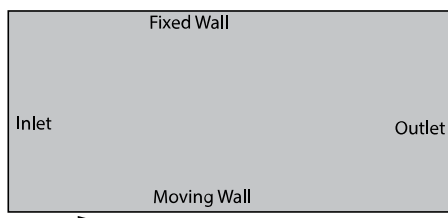
Overview

Reference	H. Schlichting, K. Gersten, <i>Boundary Layer Theory</i> , 8th Edition, pp. 126-127, 2000
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Unsteady flow, moving wall
Input File	VMFL019_FLUENT.cas for Ansys Fluent VMFL019_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow near a wall suddenly set into motion is modeled. The start up flow is modeled as a transient problem with a constant wall-velocity at t (time) > 0 . The flow is highly viscous and the velocity is 0 at $t = 0$.

Figure .19.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1000 kg/m ³ Viscosity = 1 kg/m-s	Dimensions of the domain: 0.75 m X 0.3 m	Velocity of the moving wall = 0.01 m/s Gauge Pressure at Inlet = 0 N/m ² Gauge Pressure at Outlet = 0 N/m ²

Analysis Assumptions and Modeling Notes

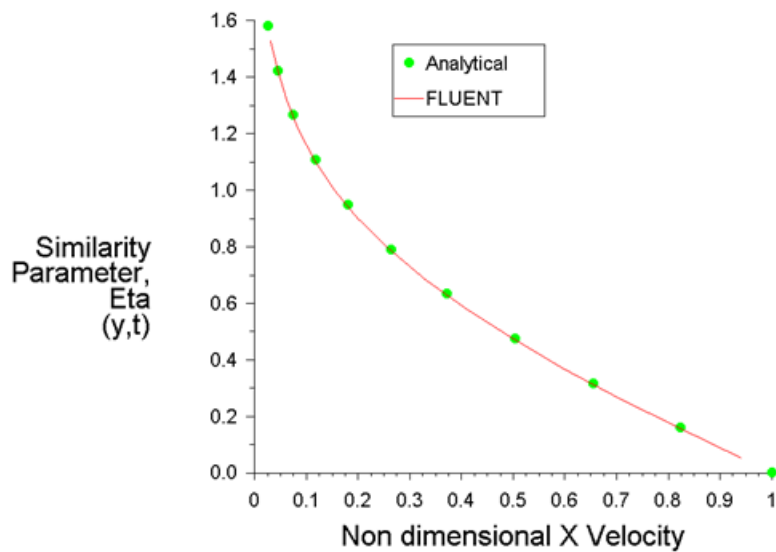
The pressure based solver is used in Ansys Fluent. Pressure boundaries are specified to model the driving head in the direction of flow. The fluid is at rest initially ($t = 0$). The similarity parameter is defined as:

$$\eta(y, t) = y / (2\sqrt{\nu t})$$

Where ν is the kinematic viscosity.

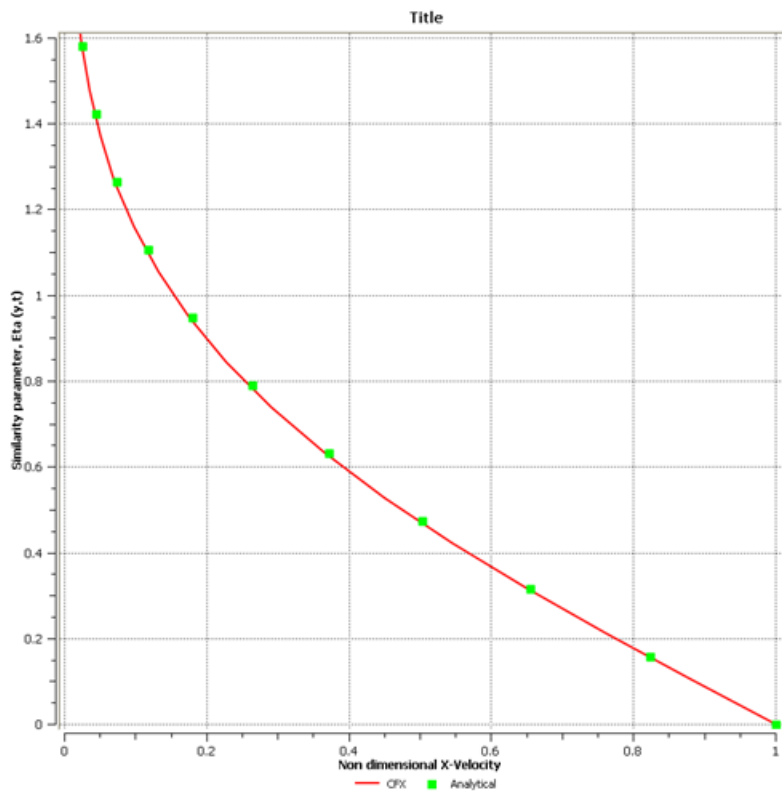
Results Comparison using Ansys Fluent

Figure .19.2: Comparison of Velocity Profile Near the Wall at Outlet



Results Comparison using Ansys CFX

Figure .19.3: Comparison of Velocity Profile Near the Wall at Outlet



VMFL020: Adiabatic Compression of Air in Cylinder by a Reciprocating Piston

Overview

Reference	L.D. Russell, G.A. Adebiyi, <i>Classical Thermodynamics</i> , Saunders College Publishing, Philadelphia, PA, 1993
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Dynamic Mesh, Transient flow with ideal gas effects
Input File	box2d_remesh.cas for Ansys Fluent VMFL020_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Air undergoes adiabatic compression due to the movement of a piston inside a rectangular box, representing a cylinder geometry in 2-D as shown in [Figure .20.1: In-Cylinder Piston Description \(p. 79\)](#). The Top Dead Center (TDC) corresponds to a crank angle of 360° . The piston moves back after reaching TDC.

Figure .20.1: In-Cylinder Piston Description

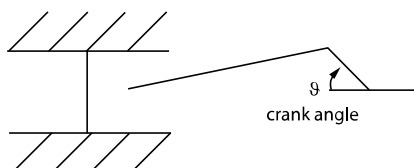
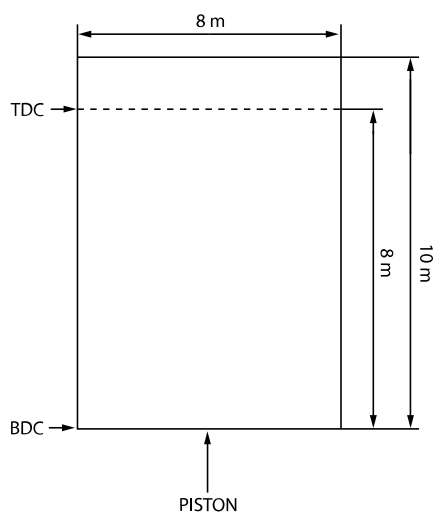


Figure .20.2: Flow Domain



Material Properties	Geometry	Boundary Conditions
Ideal gas law for density Viscosity = 1.7894×10^{-5} kg/m-s	Length of the block = 10 m Width of the block = 8 m	Movement of the piston is modeled using deforming mesh

Analysis Assumptions and Modeling Notes

The compression within the cylinder is assumed to be adiabatic. The spring-based smoothing method with local remeshing is used for modeling the dynamic mesh motion.

Results Comparison Ansys Fluent

Figure .20.3: Comparison of Static Temperature Variation with Time

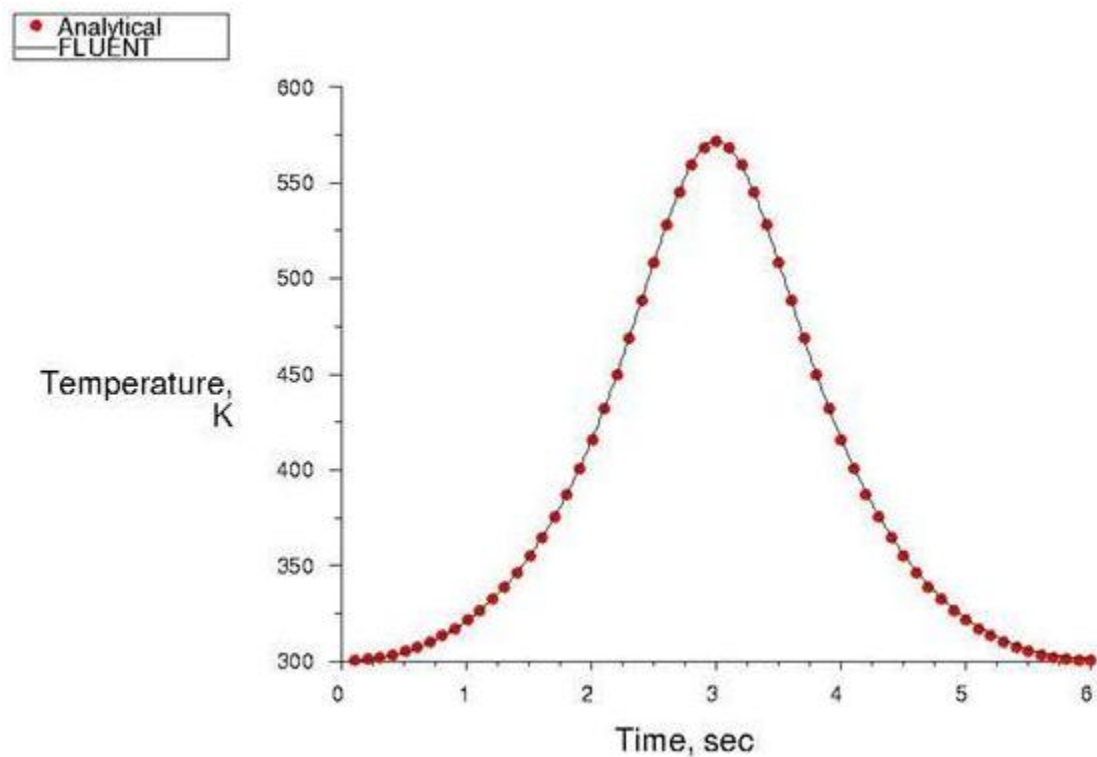
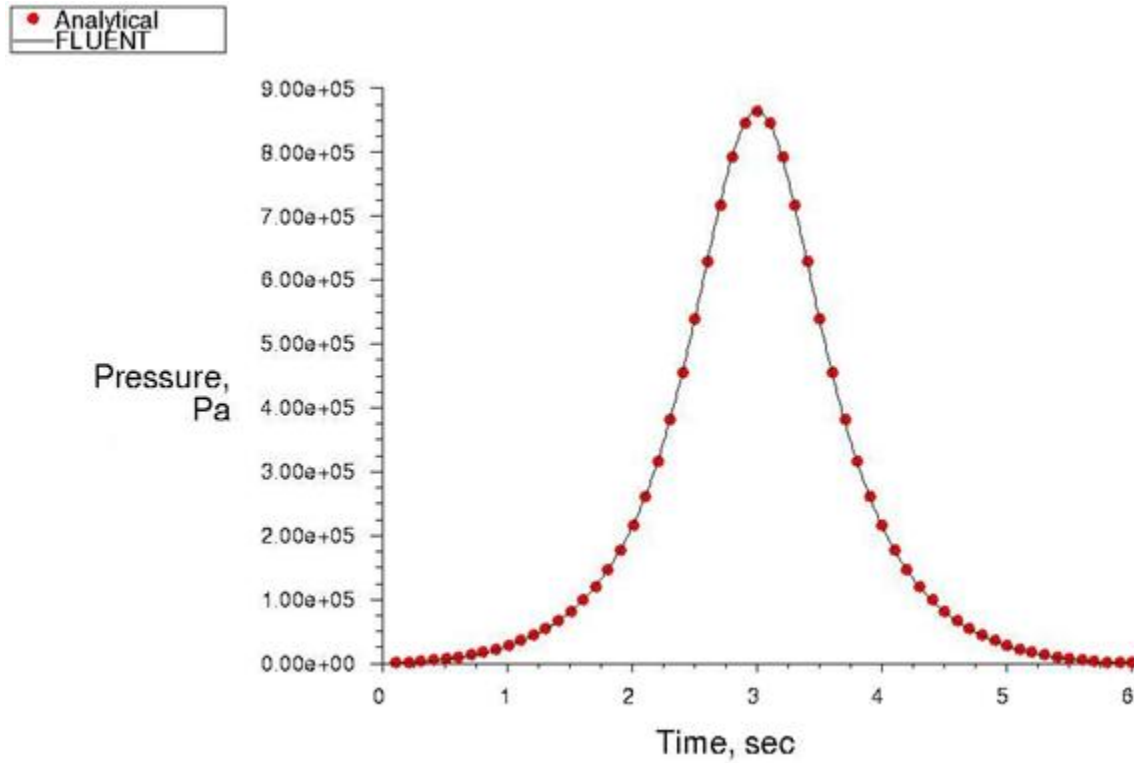


Figure .20.4: Comparison of Static Pressure Variation with Time

Results Comparison Ansys CFX

Figure .20.5: Comparison of Static Temperature Variation with Time

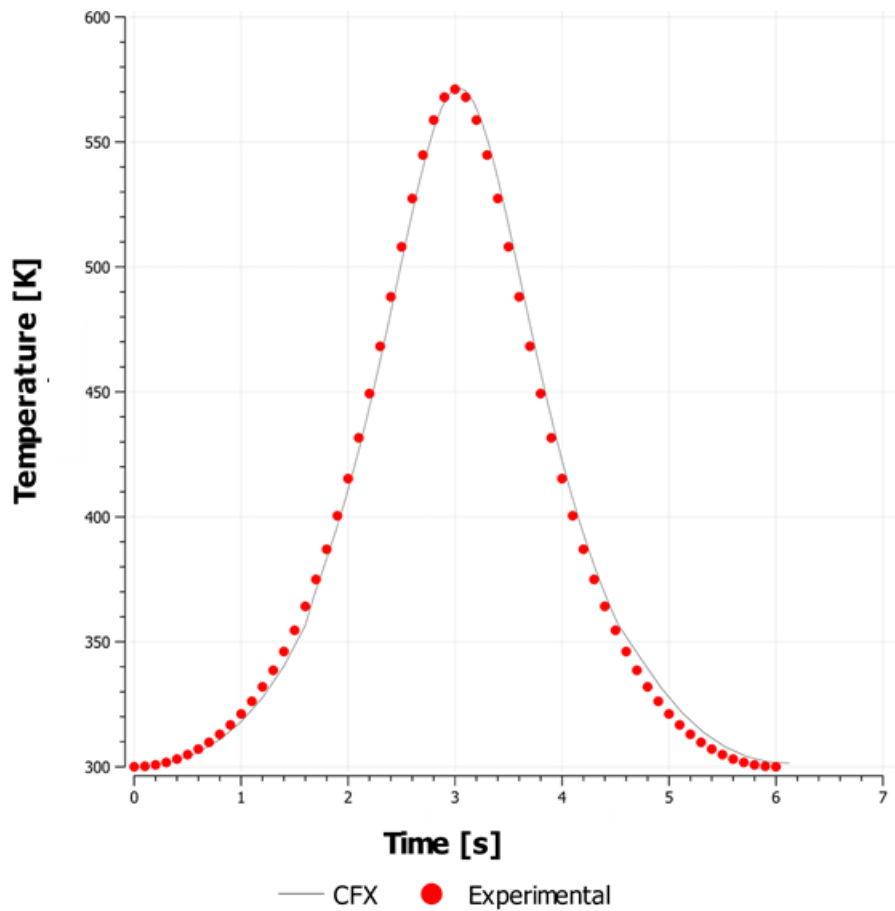
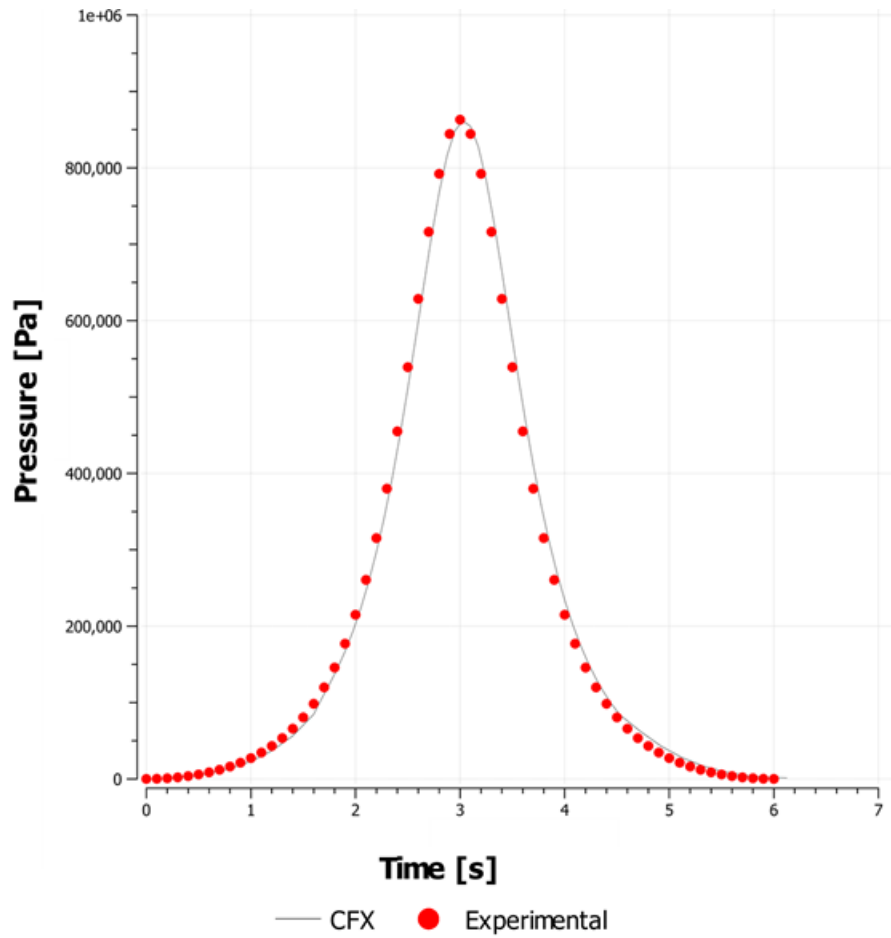


Figure .20.6: Comparison of Static Pressure Variation with Time

VMFL021: Cavitation Over a Sharp-Edged Orifice Case A: High Inlet Pressure

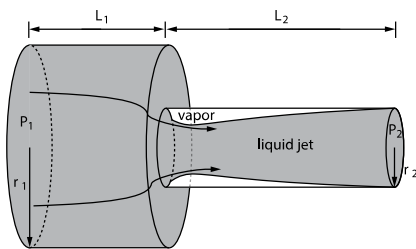
Overview

Reference	W.H. Nurick, "Orifice Cavitation and Its Effects on Spray Mixing". <i>Journal of Fluids Engineering</i> , Vol.98, pp. 681-687, 1976
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Turbulent multiphase flow with cavitation and phase change
Input File	cav_orifice_HP.cas for Ansys Fluent VMFL021B_VV021.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A steady, axisymmetric, multiphase (water/steam) flow, with phase change taking place. Due to sudden contraction a low pressure region occurs near the sharp edge which results in cavitation. [Figure .21.1: Flow Domain \(p. 85\)](#) depicts the orifice geometry. Flow direction is from left to right.

Figure .21.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Liquid: Water	$L_1 = 1.60$ cm	$P_1 = 250,000,000$ Pa
Density : 1000 kg/m ³	$L_2 = 3.20$ cm	$P_2 = 95,000$ Pa
Viscosity: 0.001 kg/m-s	$r_1 = 1.15$ cm	$T = 300$ K
Gas: Water-Vapor	$r_2 = 0.40$ cm	$P_{\text{vapor}} = 3,540$ Pa
Density: 0.02558 kg/m ³		
Viscosity: 1.26×10^{-6} kg/m-s		

Analysis Assumptions and Modeling Notes

The flow is steady and incompressible. Pressure based solver is used. Fluent uses standard $k-\epsilon$ model with standard wall functions for the turbulence model while CFX uses SST. The Zwart-Gerber-Belamri cavitation model is applied together with mixture multiphase model.

For analysis of results, we calculate and compare the discharge coefficient with the experimental data.

The coefficient of discharge, C_d , is the ratio of the mass flow rate through the nozzle to the theoretical maximum mass flow rate:

$$C_d = \frac{\dot{m}}{A\sqrt{2\rho(P_1 - P_2)}}$$

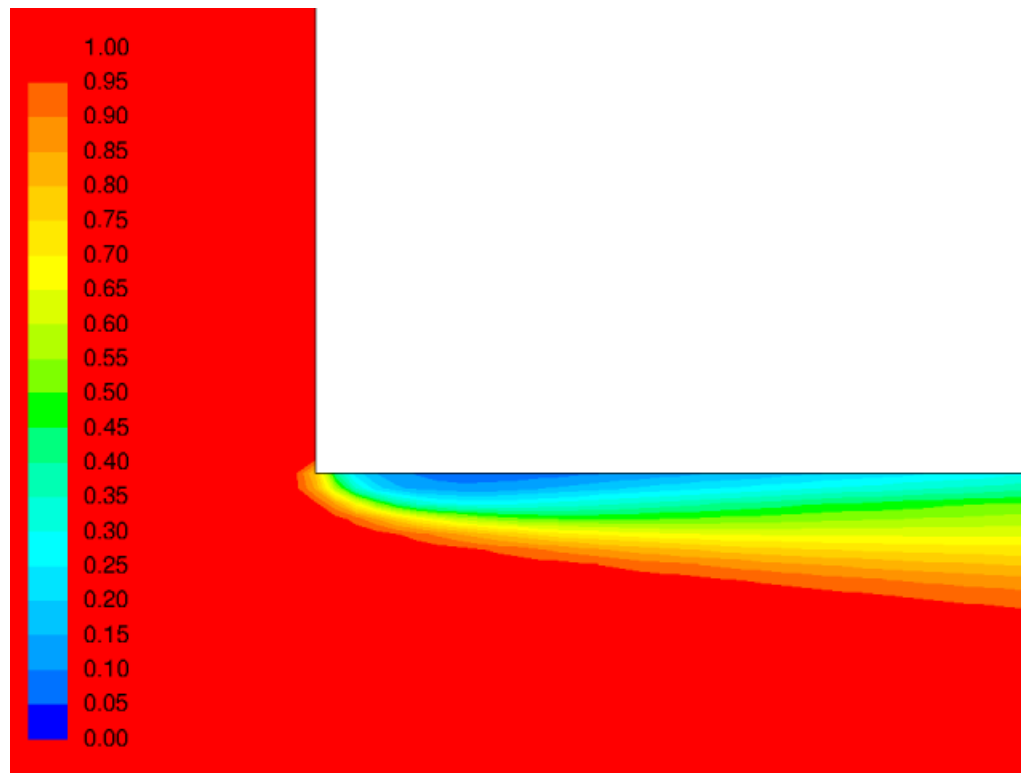
In the above equation, $\dot{m}$ is the mass flow rate as calculated by the CFD solver.

Results Comparison for Ansys Fluent

Table .21.1: Comparison of Discharge Coefficient

	Target	Ansys Fluent	Ratio
Coefficient of Discharge	0.620	0.631	1.018

Figure .21.2: Contours of Liquid (Water) Volume Fraction



Results Comparison for Ansys CFX

Table .21.2: Comparison of Discharge Coefficient

	Target	Ansys CFX	Ratio
Coefficient of Discharge	0.620	0.637	1.03

VMFL022: Cavitation Over a Sharp-Edged Orifice Case B: Low Inlet Pressure

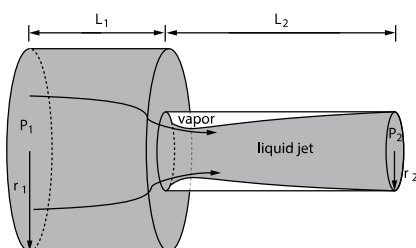
Overview

Reference	W.H. Nurick, "Orifice Cavitation and Its Effects on Spray Mixing", <i>Journal of Fluids Engineering</i> , Vol 98, pp. 681-687, 1976
Solver	Ansys Fluent
Physics/Models	Turbulent multiphase flow with cavitation and phase change
Input File	cav_orifice_LP.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A steady, axisymmetric, multiphase (water/steam) flow, with phase change taking place. Due to sudden contraction a low pressure region occurs near the sharp edge which results in a weak cavitation. [Figure .22.1: Flow Domain \(p. 87\)](#) depicts the orifice geometry. Flow direction is from left to right.

Figure .22.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Liquid: Water	$L_1 = 1.60$ cm	$P_1 = 250,000$ Pa
Density: 1000 kg/m ³	$L_2 = 3.20$ cm	$P_2 = 95,000$ Pa
Viscosity: 0.001 kg/m-s	$r_1 = 1.15$ cm	$T = 300$ K
Gas: Water-Vapor	$r_2 = 0.40$ cm	$P_{\text{vapor}} = 3,540$ Pa
Density: 0.02558 kg/m ³		
Viscosity: 1.26×10^{-6} kg/m-s		

Analysis Assumptions and Modeling Notes

The flow is steady and incompressible. Pressure based solver is used. Standard $k-\epsilon$ model with standard wall functions is used for turbulence. The Zwart-Gerber-Belamri cavitation model is applied together with mixture multiphase model.

For analysis of results, we calculate and compare the discharge coefficient with the experimental data.

The coefficient of discharge, C_d , is the ratio of the mass flow rate through the nozzle to the theoretical maximum mass flow rate:

$$C_d = \frac{\dot{m}}{A\sqrt{2\rho(P_1 - P_2)}}$$

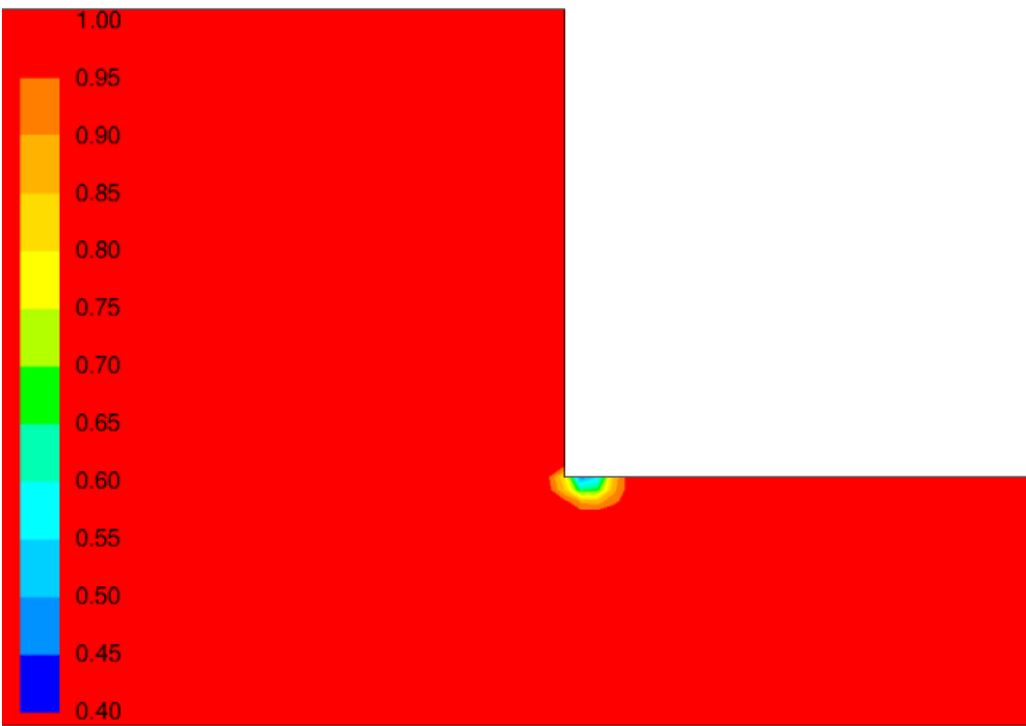
In the above equation, $\dot{m}$ is the mass flow rate as calculated by the CFD solver.

Results Comparison for Ansys Fluent

Table .22.1: Comparison of Discharge Coefficient

	Target	Ansys Fluent	Ratio
Coefficient of Discharge	0.780	0.777	0.996

Figure .22.2: Contours of Liquid (Water) Volume Fraction



VMFL023: Oscillating Laminar Flow Around a Circular Cylinder

Overview

Reference	F.M. White, <i>Fluid Mechanics</i> , 3rd Edition. McGraw-Hill Book Co., New York, NY. 1994 S.J. Kim, C.M. Lee, "Numerical Investigation of Cross-Flow Around a Circular Cylinder at a Low-Reynolds Number Flow Under an Electromagnetic Force". <i>KSME International Journal</i> . Vol 16, pp. 363-375, 2002
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar, transient flow
Input File	cyl_2d.cas for Ansys Fluent VMFL023B_osc_cyl.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The purpose of this case is to validate the ability of Ansys Fluent and Ansys CFX to predict the flow structure as well as the reattachment length and Strouhal number against experimental results. The present calculations are confined to the low-Reynolds-number regime ($Re = 100$), which encompasses unsteady asymmetric flow.

Figure .23.1: Flow Domain

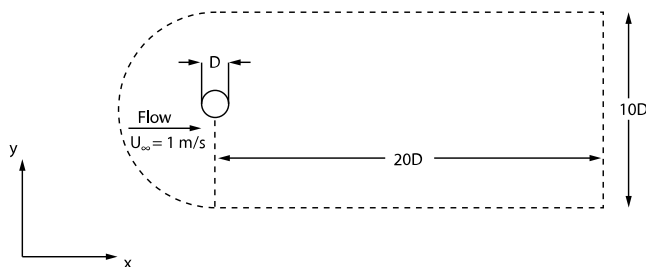


Table .23.1: Materials, Geometry, and Boundary Conditions

Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m^3 Viscosity: 0.02 kg/m-s	Diameter of the cylinder = 2 m	$U_{\infty} = 1 \text{ m/s}$

Analysis Assumptions and Modeling Notes

The flow is laminar, and unsteady. An FFT analysis of the lift coefficient on the cylinder wall is presented to determine the frequency of oscillations. The Strouhal number corresponding to the maximum mag-

nitude of oscillations is presented in the Table below. The formula for the Strouhal number is $S = (N * D)/U_{\infty}$, where N is the frequency, D is the diameter of the cylinder, and U_{∞} is the freestream velocity.

CFX time duration was chosen to accommodate FFT calculations. These were chosen based on

$\Delta f = \frac{1}{(\Delta t \cdot N)}$, where Δf is the size of the frequency bins and N is an even divisor of 720720, as described in FFT documentation.

Results Comparison for Ansys Fluent

Table .23.2: Predicted Strouhal Number for Re = 100

	Target	Ansys Fluent	Ratio
Strouhal Number	0.165	0.178	1.1

Results Comparison for Ansys CFX

Table .23.3: Predicted Strouhal Number for Re = 100

	Target	Ansys CFX	Ratio
Strouhal Number	0.165	0.167	1.01

VMFL024: Interface of Two Immiscible Liquids in a Rotating Cylinder

Overview

Reference	T. Sugimoto, M. Iguchi, "Behavior of Immiscible Two Liquid Layers Contained in Cylindrical Vessel Suddenly Set in Rotation". <i>ISIJ International</i> , Vol 42, pp. 338-343, 2002
Solver	Ansys Fluent
Physics/Models	Multiphase (Volume of Fluid), transient flow, body force
Input File	rot-cyl_2liq_vof.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar interface between two immiscible liquids, water and silicon oil, inside a vertical cylinder which is set in rotation starting from a state of rest. The silicone oil layer rests on top of the water due to its lower density. The cylinder is sealed at the top. The vessel is set to rotate with a constant angular velocity.

Figure .24.1: Flow Domain

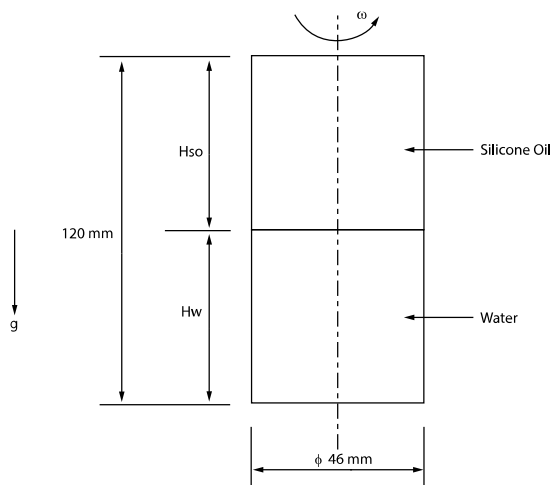


Table .24.1: Materials, Geometry, and Boundary Conditions

Material Properties	Geometry	Boundary Conditions
Water : Density: 1030 kg/m ³ Viscosity: 0.00103 kg/m-s	Diameter of the cylinder = 46 mm Height of the cylinder = 120 mm	All walls are set up at rotational speed of 2.39577 rad/s
Silicon Oil: Density: 935 kg/m ³		

Material Properties	Geometry	Boundary Conditions
Viscosity: 0.0103 kg/m-s		

Analysis Assumptions and Modeling Notes

The flow is laminar, unsteady and axisymmetric. Non-dimensionalized swirl velocity, defined as **Swirl velocity / (Rotational speed X Cylinder radius)** is used to validate the results.

Results Comparison for Ansys Fluent

Table .24.2: Comparison of the Non-Dimensional Swirl Velocity at Various Radial Locations (for a Given Axial Location, X = 20mm) at Time t = 80 s

Radial locations (at x = 20 mm)	Target	Ansys Fluent	Ratio
4.83 mm	0.21	0.2093	0.997
9.43 mm	0.41	0.4109	1.002
14.26 mm	0.62	0.6221	1.003

VMFL025: Turbulent Non-Premixed Methane Combustion with Swirling Air

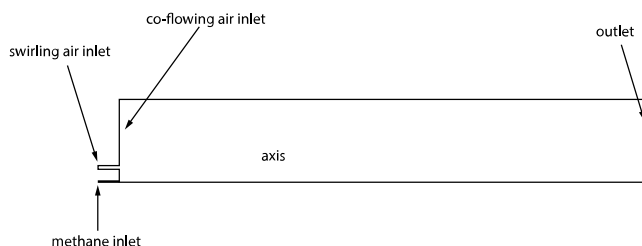
Overview

Reference	P.A.M. Kalt, Y.M. Al-Abdeli, A.R. Masri, R.S. Barlow, "Swirling turbulent non-premixed flames of methane: Flow field and compositional structure". <i>Proceedings of the Combustion Institute</i> , Vol 29, pp. 1913-1919, 2002 Y.M. Al-Abdeli, A.R. Masri, "Stability Characteristics and Flow Fields of Turbulent Swirling Jet Flows". <i>Combustion Theory and Modeling</i> , Vol 7, pp. 731-766, 2003
Solver	Ansys Fluent
Physics/Models	Turbulent swirling flow with reaction, non-premixed combustion model, Radiation heat transfer, Discrete ordinates method
Input File	non-premix_17k-final.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Air and Methane enter as separate streams into an annular chamber. Air issues as a swirling jet and also as a separate co-flowing stream with axial velocity. Both the air streams are free of methane. Species mixing and combustion take place in the axisymmetric chamber. Radiative heat transfer is taken into account.

Figure .25.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Species mixture properties specified through PDF file Viscosity: 1.72×10^{-05} kg/m-s Refractive Index = 1	The fuel (methane only) inlet has a diameter of 3.6 mm. The air inlet for the annular shroud has an inner diameter of 50mm and an outer diameter of 60 mm.	Methane inlet velocity: 32.7 m/s Axial velocity of swirling air: 38.2 m/s Swirl velocity of air: 19.1 m/s Co-flowing air velocity: 20 m/s

Material Properties	Geometry	Boundary Conditions
	Co-flowing air inlet has an outer diameter of 310 mm.	Walls are adiabatic

Analysis Assumptions and Modeling Notes

The flow is steady. Realizable $k-\epsilon$ is used to model turbulence. Discrete ordinates method used to model radiation. The walls are treated as adiabatic. Non-premixed combustion model is used to model reactions.

Results Comparison Ansys Fluent

Figure .25.2: Comparison of Axial Velocity at X = 40mm

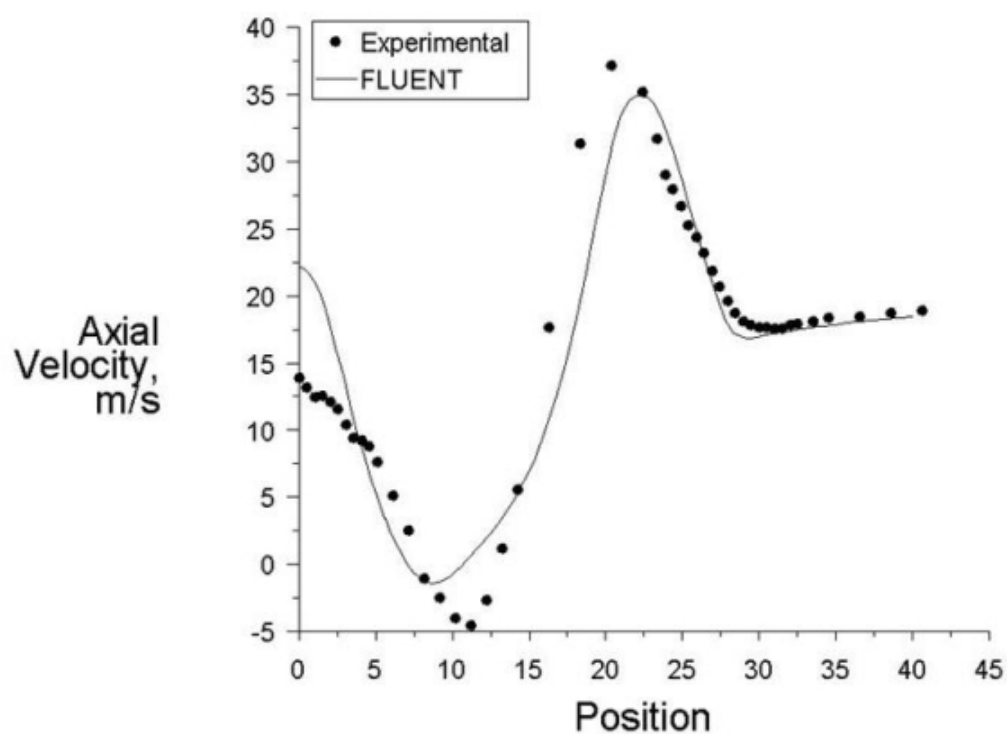


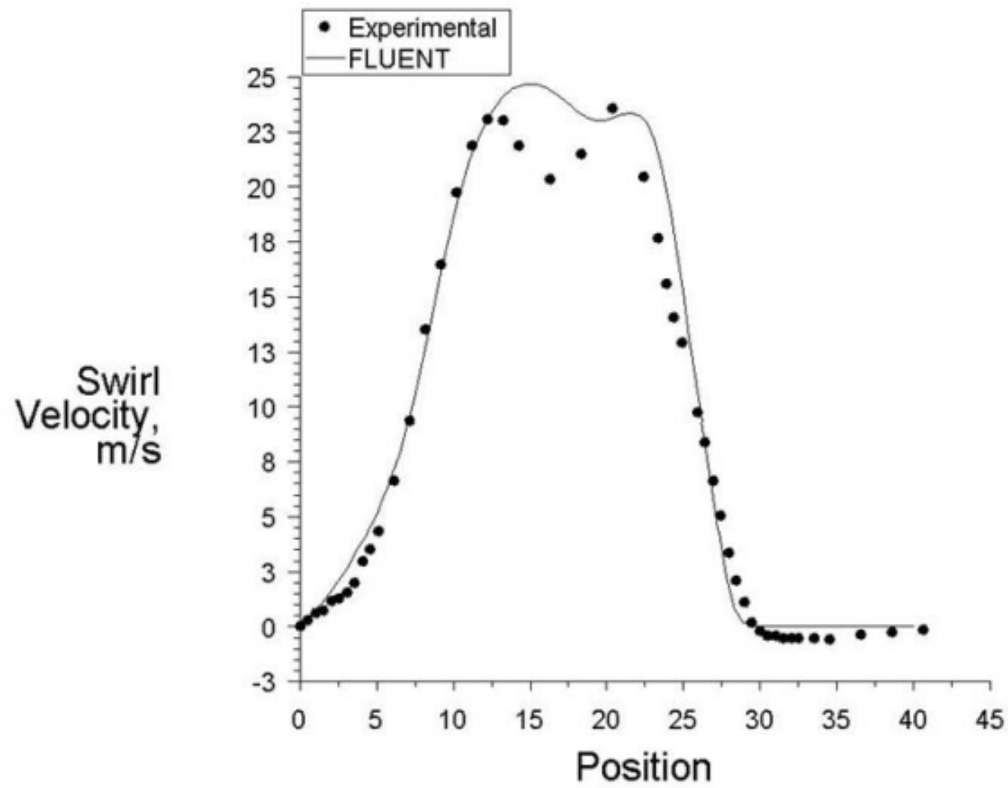
Figure .25.3: Comparison of Swirl Velocity at X = 40mm

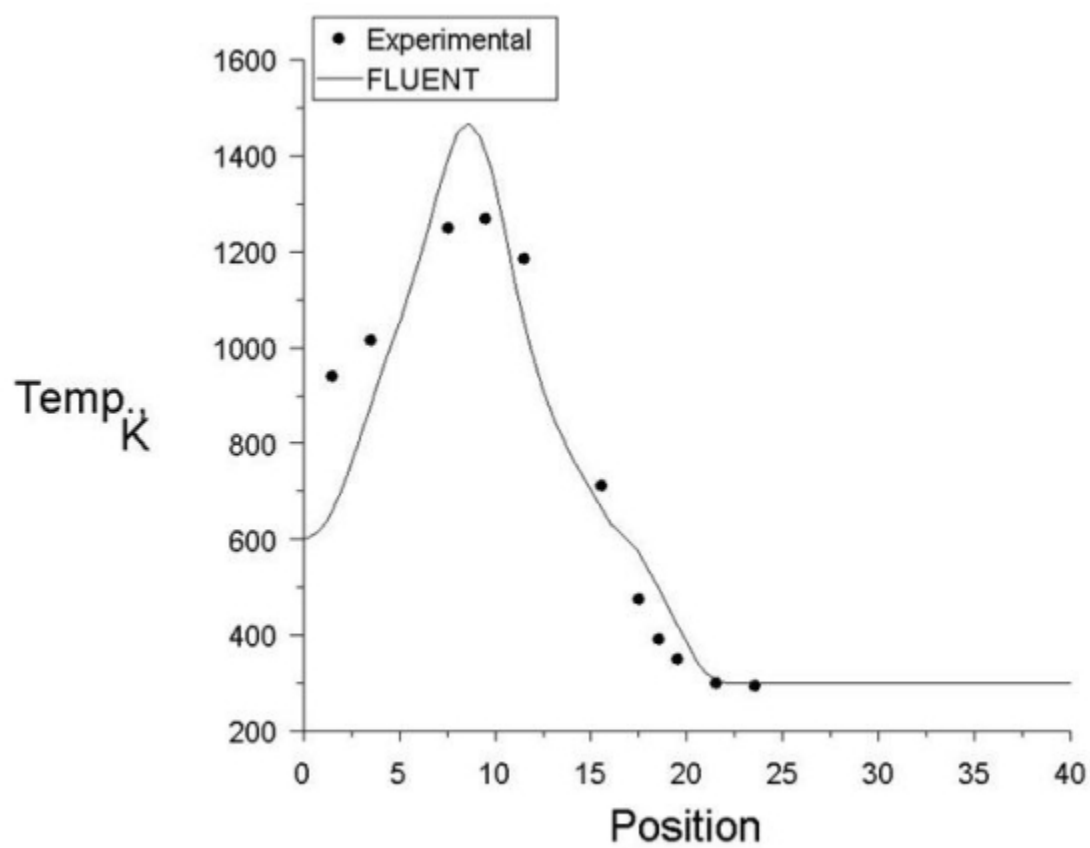
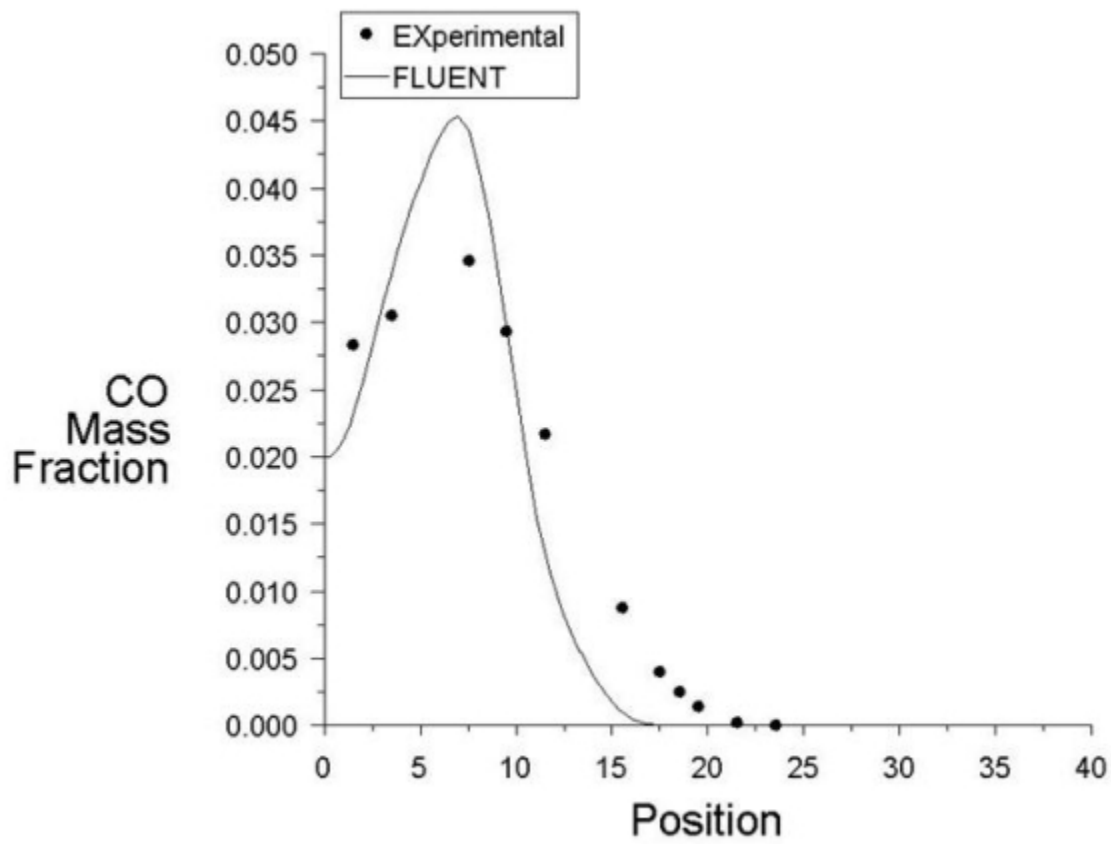
Figure .25.4: Comparison of Temperature at X = 40mm

Figure .25.5: Comparison of Mass Fraction of CO at X = 40mm



VMFL026: Supersonic Flow with Real Gas Effects inside a Shock Tube

Overview

Reference	K. Mohamed, M. Paraschivoiu, "Real Gas Numerical Simulation of Hydrogen Flow". <i>2nd International Energy Conversion Engineering Conference</i> , Providence, Rhode Island, Aug. 16-19, 2004
Solver	Ansys Fluent
Physics/Models	Transient Compressible flow, Real Gas effects, Shock
Input File	realgas_shock-tube.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Transient flow inside a hydrogen filled shock tube is modeled. A diaphragm separating regions of high and low pressures ruptures at $t = 0$ thereby creating a shock wave in the tube.

Figure .26.1: Flow Domain

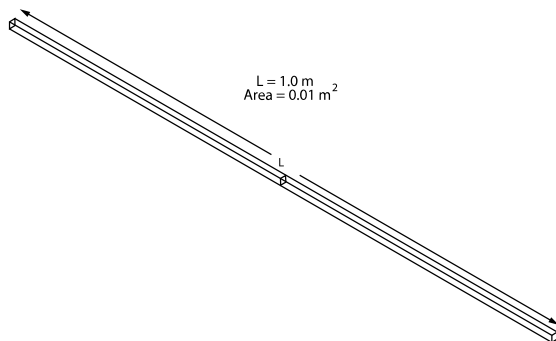


Table .26.1: Materials, Geometry, and Boundary Conditions

Material Properties	Geometry	Boundary Conditions
Density is specified using the Aungier-Redlich-Kwong real gas model	Length of the tube = 1 m Area of cross section = 0.01 m^2	Cell zone conditions are specified with high pressure and low pressure properties of hydrogen

Analysis Assumptions and Modeling Notes

The flow is compressible and unsteady by nature. Real gas effects are significant in the pressure range considered here.

Results Comparison for Ansys Fluent

Figure .26.2: Comparison of Static Temperature Along Centerline of the Tube

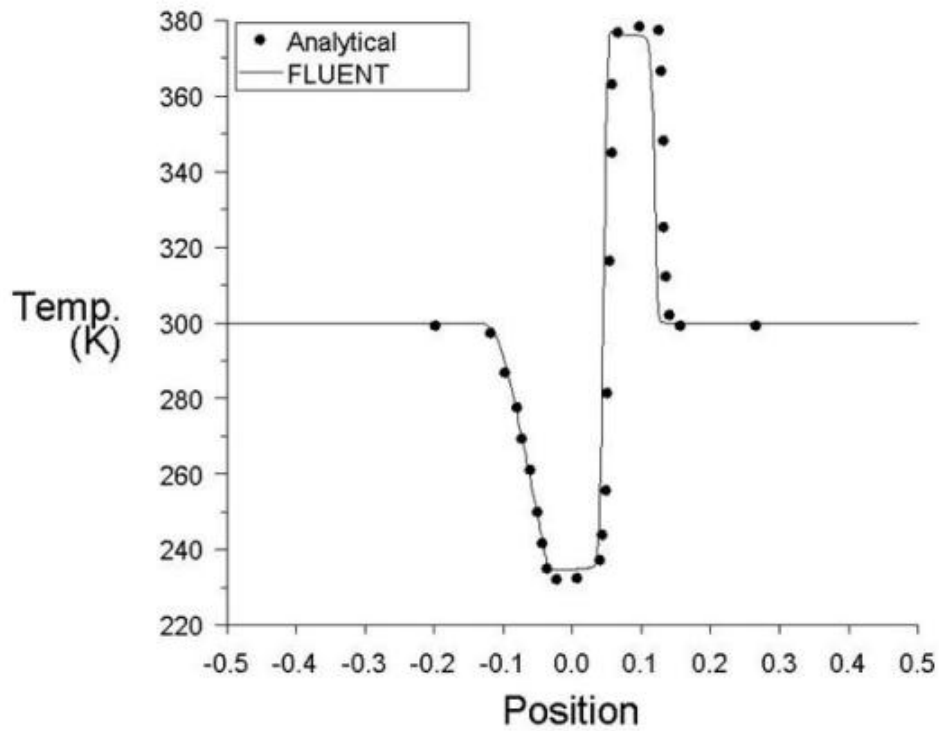
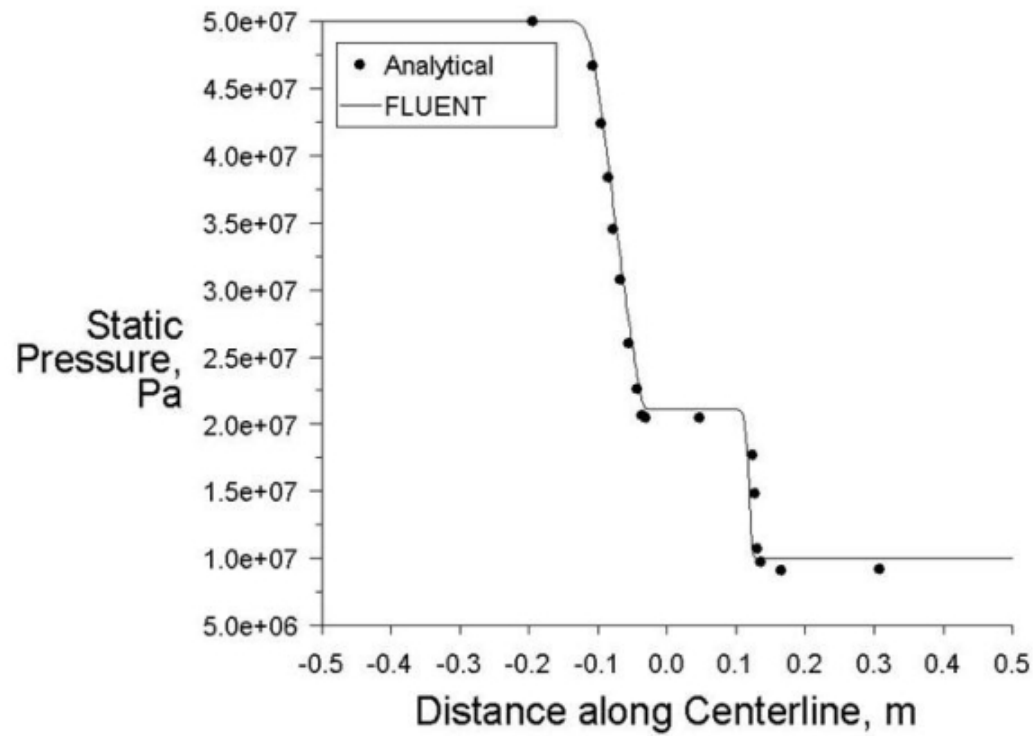


Figure .26.3: Comparison of Static Pressure Along Centerline of the Tube

VMFL027: Turbulent Flow Over a Backward-Facing Step

Overview

Reference	D.M. Driver, H.L. Seegmiller, "Features of a Reattaching Turbulent Shear Layer in Divergent Channel Flow". <i>AIAA Journal</i> , Vol 23, pp. 163-171, 1985
Solver	Ansys Fluent, Ansys CFX
Physics/Models	2-D turbulent flow with separation and reattachment, realizable k- ϵ model
Input File	drivseeg-rke-neqwf.cas for Ansys Fluent VMFL020_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Turbulent flow over a backward facing step is modeled. The flow separates at the step and reattaches on the wall downstream, enclosing a region of recirculation. The inlet is at 4 H upstream and the outlet at 30 H downstream from the location of the step, where H is the step height. Reynolds number based on the step-height is about 37,400.

Figure .27.1: Flow Domain



Table .27.1: Materials, Geometry, and Boundary Conditions

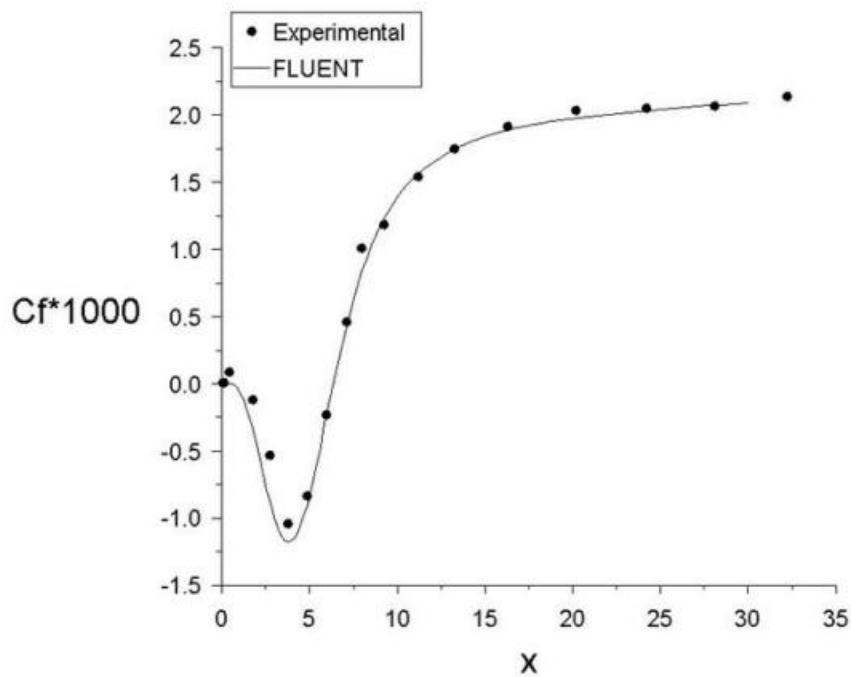
Material Properties	Geometry	Boundary Conditions
Density : 1 kg/m ³ Viscosity: 0.0001 kg/m-s	Step height = 1 m Total length of the channel = 34 m Height of the channel = 9 m	Inlet velocity specified as fully developed turbulent velocity profile

Analysis Assumptions and Modeling Notes

The flow is steady. Realizable k- ϵ model was used to model turbulence.

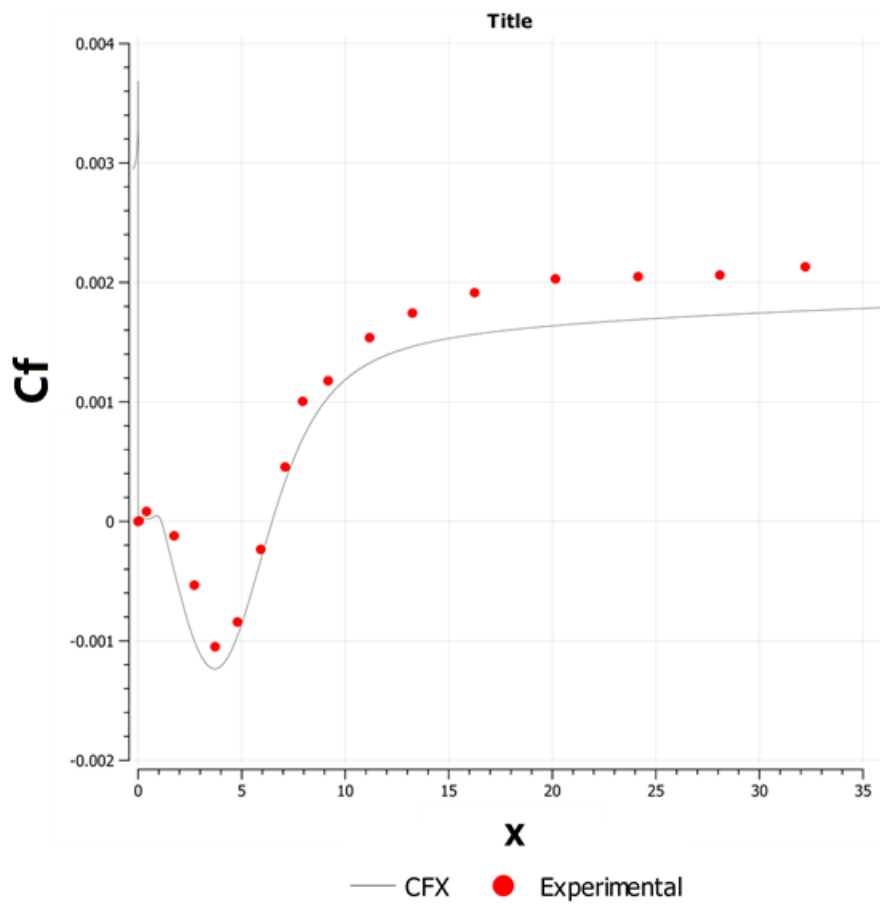
Results Comparison for Ansys Fluent

Figure .27.2: Comparison of Skin Friction Coefficient Along the Wall



Results Comparison for Ansys CFX

Figure .27.3: Comparison of Skin Friction Coefficient Along the Wall



VMFL028: Turbulent Heat Transfer in a Pipe Expansion

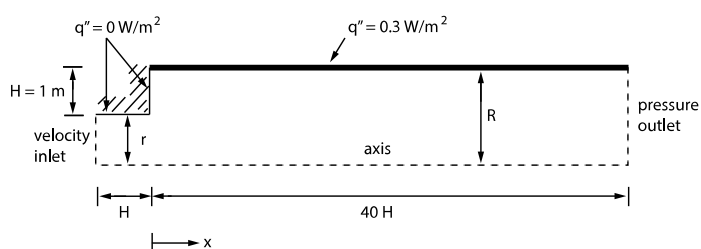
Overview

Reference	J.W. Baughn, B.E. Launder, M.A. Hoffman, R.K. Takahashi, "Local Heat Transfer Downstream of an Abrupt Expansion in a Circular Channel With Constant Wall Heat Flux", <i>Journal of Heat Transfer</i> , Vol 106, pp. 789-796, 1984
Solver	Ansys Fluent
Physics/Models	Heat transfer, turbulent flow with recirculation and reattachment
Input File	bghnexp.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Fully developed turbulent flow through an axisymmetric pipe expansion is modeled. The flow reattaches to the pipe wall downstream of the expansion, enclosing a zone of recirculation. The pipe wall downstream of the expansion is heated at a constant rate. Inlet to the computational domain is placed at 1 step height upstream of the expansion and the outlet at 40 step-heights downstream.

Figure .28.1: Flow Domain



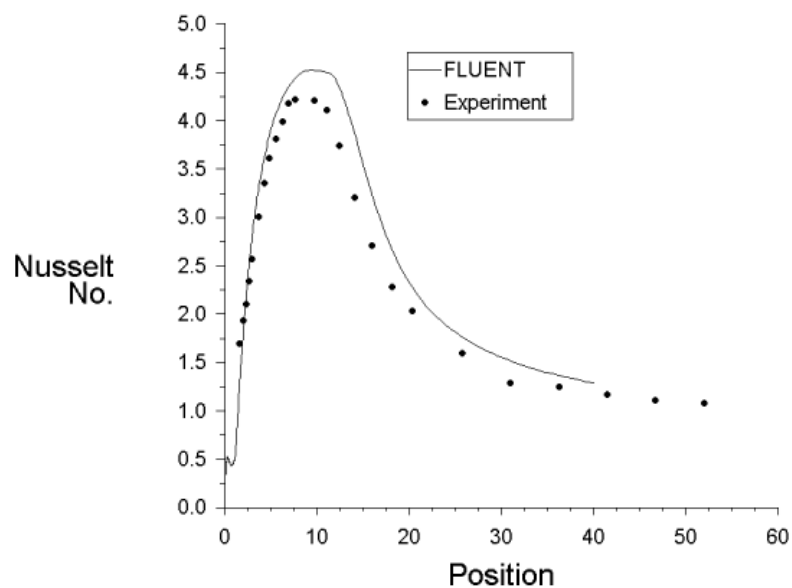
Material Properties	Geometry	Boundary Conditions
Density: 1.225 kg/m ³ Viscosity: 1.68318e-5 kg/m-s Specific Heat: 1006.43 J/kg-K Thermal Conductivity: 0.0242 W/m-K	Pipe radius before expansion = 0.667 m Pipe radius after expansion = 1.6667 m	Inlet velocity: Specified by fully developed turbulent velocity profile Inlet temperature = 273 K Heat flux across the wall after expansion = 0.3 W/m ²

Analysis Assumptions and Modeling Notes

Steady flow in axisymmetric domain. The wall upstream of expansion is adiabatic.

Results Comparison for Ansys Fluent

Figure .28.2: Nusselts Number Variation along the Heated Wall



VMFL029: Anisotropic Conduction Heat Transfer

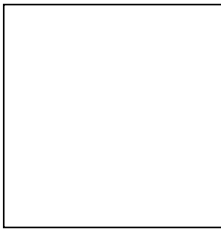
Overview

Reference	
Solver	Ansys Fluent
Physics/Models	Heat conduction, anisotropic conductivity
Input File	aniso.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Heat conduction in a solid with anisotropic thermal conductivity is modeled. A square domain is considered. Two opposite walls are maintained at uniform temperatures. Conductivity of the solid material is specified using matrix components to account for the anisotropy. The simulation results are compared with analytical solution for temperature distribution.

Figure .29.1: Domain



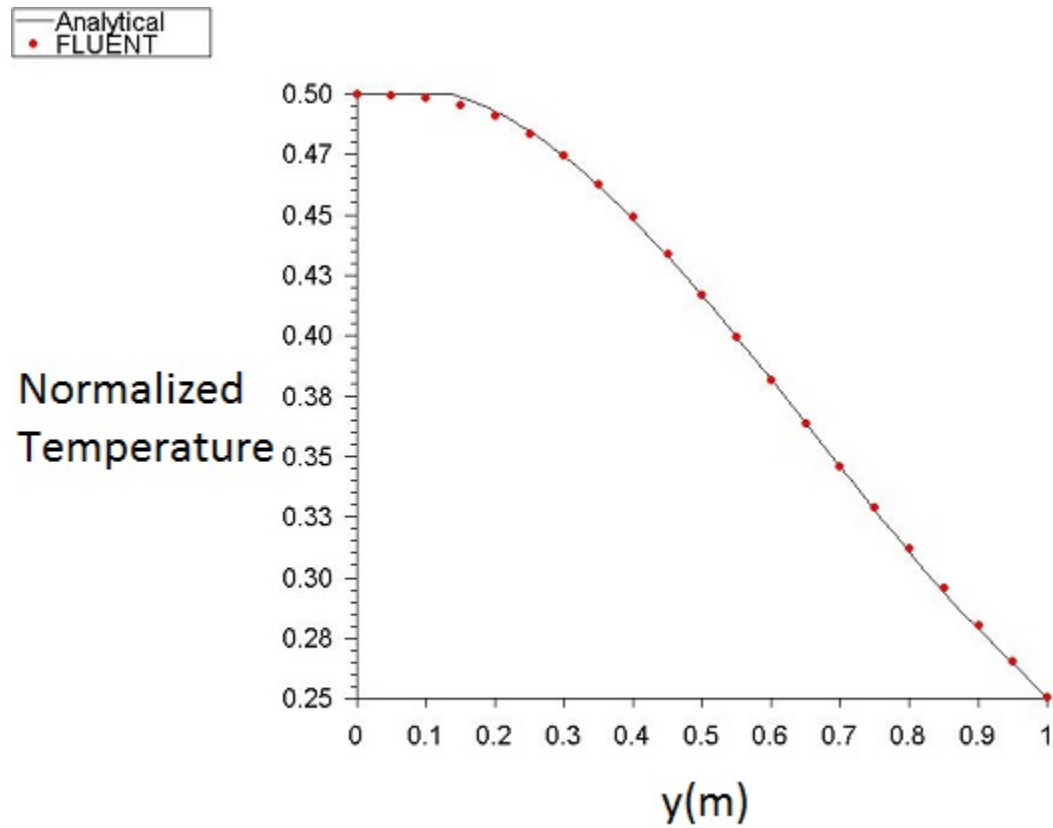
Material Properties	Geometry	Boundary Conditions
Density of solid = 2719 kg/m^3 Specific heat = 871 J/kg-K Thermal conductivity: Anisotropic	Dimensions of the domain: $1 \text{ m} \times 1 \text{ m}$	Fixed wall temperatures = 100 K and 200 K respectively User-defined profile for temperature distribution on the other two walls

Analysis Assumptions and Modeling Notes

Steady state conduction. Anisotropic conductivity modeled by specified matrix components for the solid conductivity.

Results Comparison

Figure .29.2: Comparison of Normalized Temperature Distribution at X = 0.5 m



VMFL030: Turbulent Flow in a 90° Pipe-Bend

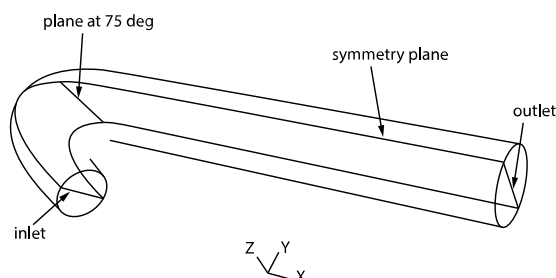
Overview

Reference	M.M. Enayet, M.M. Gibson, A.M.K.P. Taylor, M. Yianneskis, "Laser-Doppler Measurements of Laminar and Turbulent Flow in a Pipe Bend". <i>Znt. J. Heat & Fluid Flow</i> , Vol 3. pp. 213-219, 1982
Solver	Ansys Fluent
Physics/Models	3-D Turbulent flow with separation, RNG k- ϵ model with non-equilibrium wall functions
Input File	pipebnd - rng - noneq .cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Turbulent flow through a 90° circular pipe bend is modeled. The flow separates and reattaches around the bend. Due to symmetry of the flow field only half of the domain is modeled. Velocity profile at an angle of 75° (as measured from the inlet) is used to validate the simulation.

Figure .30.1: Flow Domain



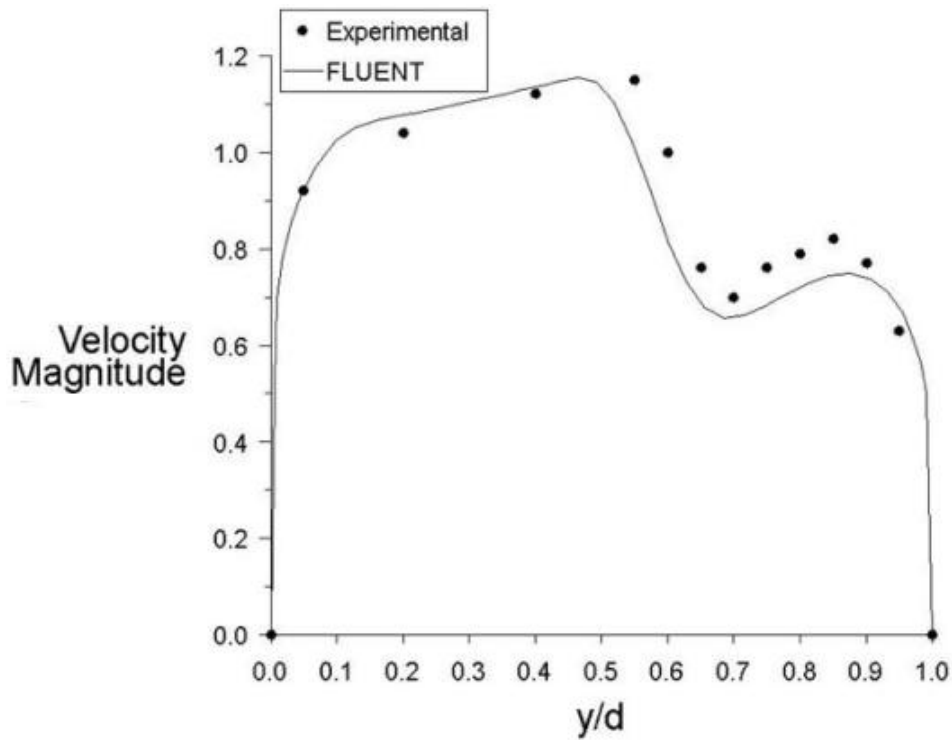
Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 2.3256 x 10 ⁻⁰⁵ kg/m-s	Radius of the pipe = 0.5 m	Inlet velocity: Fully developed turbulent profile for z-velocity. Non components in other directions

Analysis Assumptions and Modeling Notes

The flow is steady. RNG k- ϵ is used to model turbulence along with non-equilibrium wall functions.

Results Comparison for Ansys Fluent

Figure .30.2: Comparison of Velocity Magnitude (m/s) at 75° Along the Bend



VMFL031: Turbulent Flow Behind an Open-Slit V Gutter

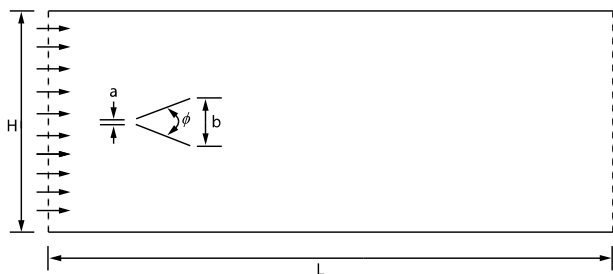
Overview

Reference	J.T. Yang, G.L. Tsai, "Near-wake flow of a v-gutter with slit bleed". <i>Journal of Fluid Engineering</i> . Vol 115. pp. 13-20, 1993
Solver	Ansys Fluent
Physics/Models	Turbulent flow
Input File	spltvee.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The near-wake flow structure behind an open-slit V gutter at airflow speed of 20 m/s is modeled. The interaction between the flow penetrating through the open slit and the shear layer results in an asymmetric wake flow structure. The size of the entire recirculation zone shifts toward one of the two wings due to the Coandă effect.

Figure .31.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 1.8333 X 10 ⁻⁰⁵ kg/m-s	L = 40 cm H = 10 cm a = 2 mm b = 22 mm φ = 45°	u _{inlet} = 20 m/s k _{inlet} = 0.04335 m ² /s ² ε _{inlet} = 0.2119 m ² /s ³

Analysis Assumptions and Modeling Notes

Steady, turbulent, incompressible flow. The standard k-ε model is used for turbulence.

Results Comparison for Ansys Fluent

The x-velocity at $x = 22$ mm downstream of the split-V-gutter, is compared with experimental data.

Figure .31.2: X-Velocity at $x = 22$ mm Downstream of the V-Gutter

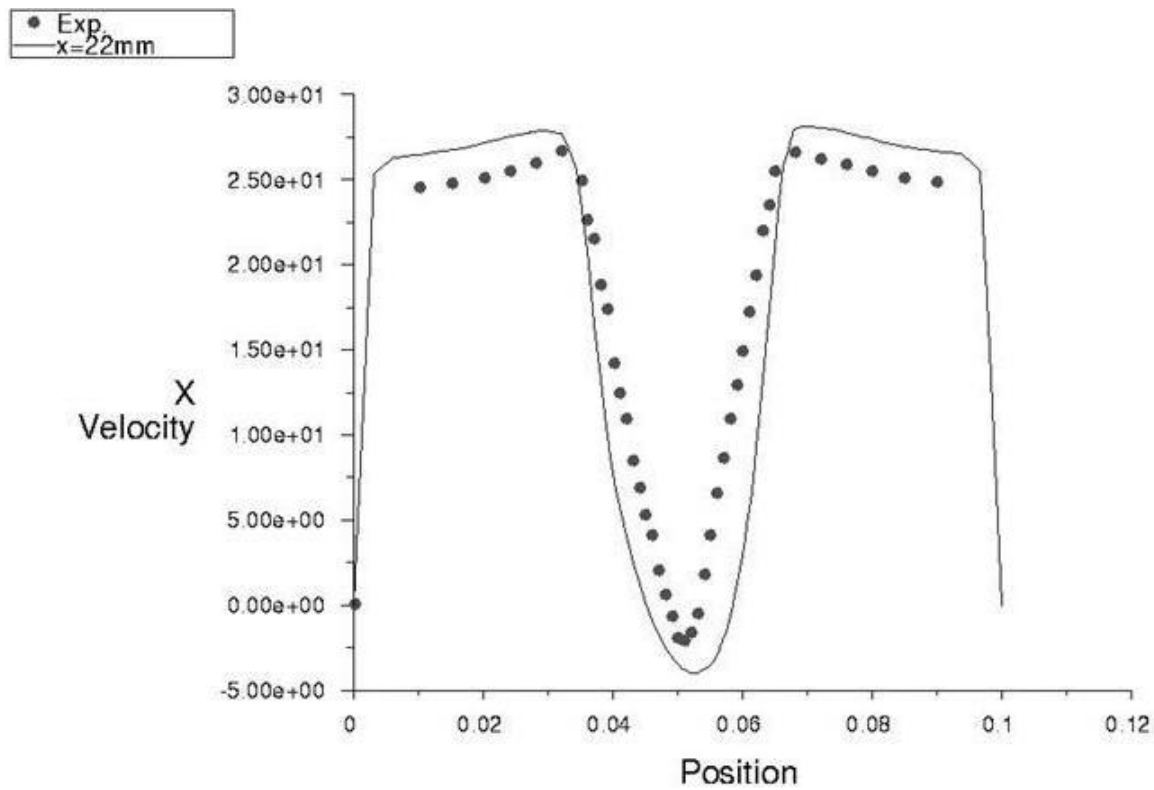
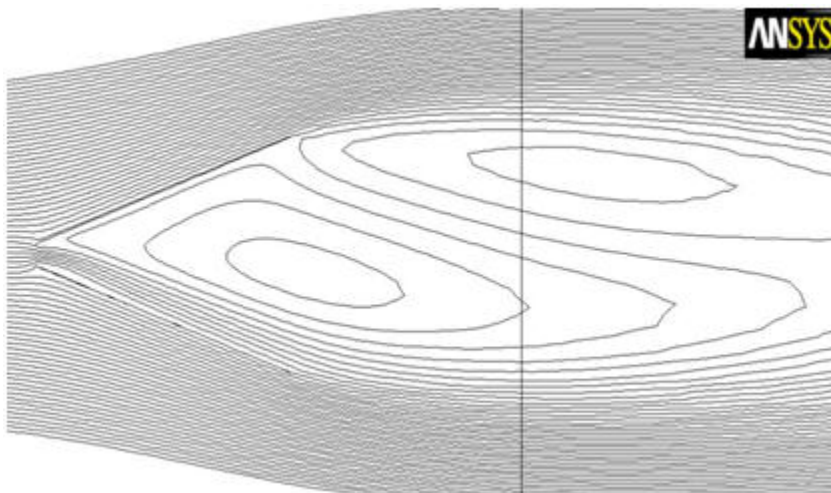


Figure .31.3: The Coand Effect



VMFL032: Turbulent Flow with Separation Along an Axisymmetric Afterbody

Overview

Reference	T.T. Huang, N.C. Groves, "Propeller/stern boundary layer interaction on axisymmetric bodies: Theory and experiment". <i>David W. Taylor Naval Ship Research and Development Center Report</i> . 76-0113. 1976.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Turbulent flow
Input File	axiaft.cas for Ansys Fluent VMFL032B_afterbody.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow past an axisymmetric afterbody, representing the hull of ship. The flow separates on the rear face of the body.

Figure .32.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 1 X 10 ⁻⁶ kg/m-s	Length of the afterbody = 1.0 m Maximum radius of the afterbody = 0.04556 m	Fully developed turbulent velocity profile on the inlet normal to axis Axial velocity = 5.9 m/s on the inlet parallel to axis

Analysis Assumptions and Modeling Notes

The far-field boundary of the domain is set parallel to the axis and is modeled as velocity inlet. Fully developed profile is specified at the transverse velocity inlet.

Results Comparison for Ansys Fluent

Figure .32.2: Comparison of Pressure Coefficient Along the Afterbody Wall

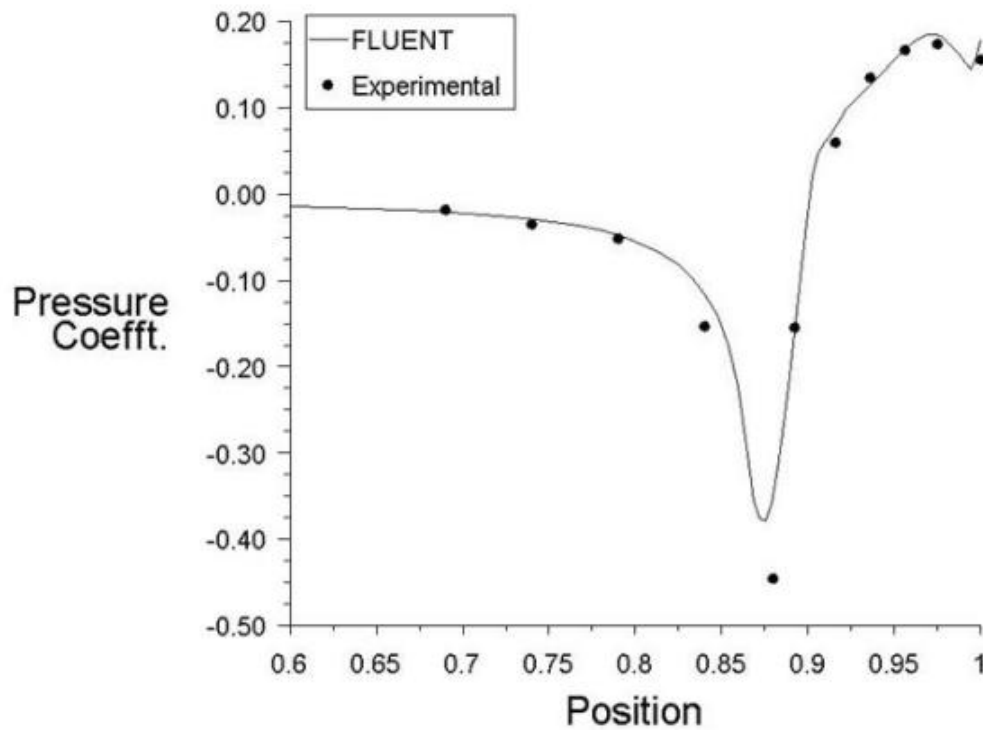
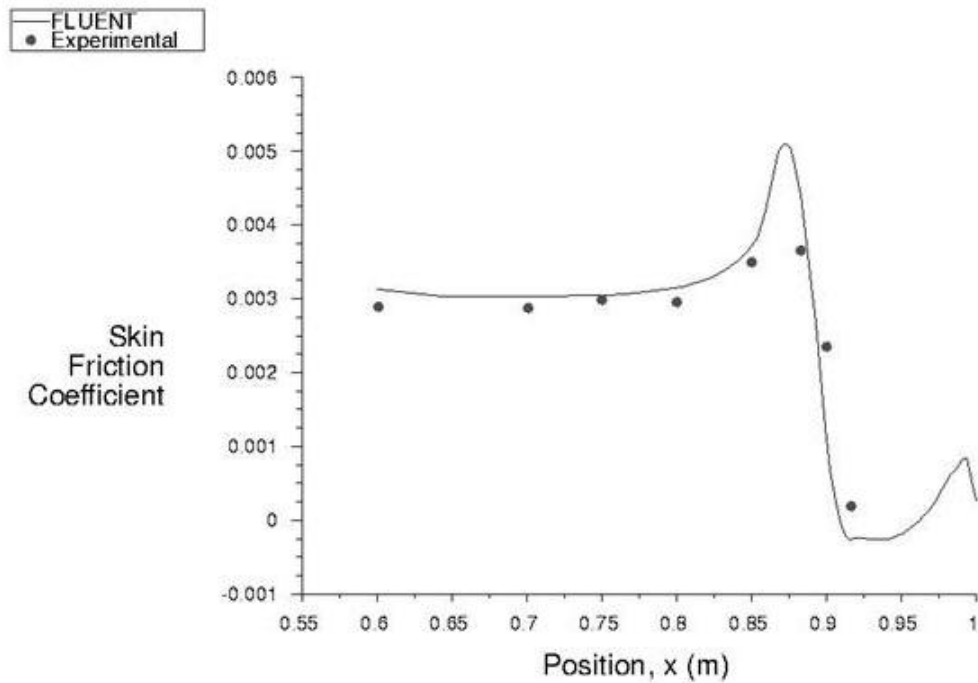


Figure .32.3: Comparison of Skin Friction Coefficient Along the Afterbody Wall



Results Comparison for Ansys CFX

Figure .32.4: Comparison of Pressure Coefficient Along the Afterbody Wall

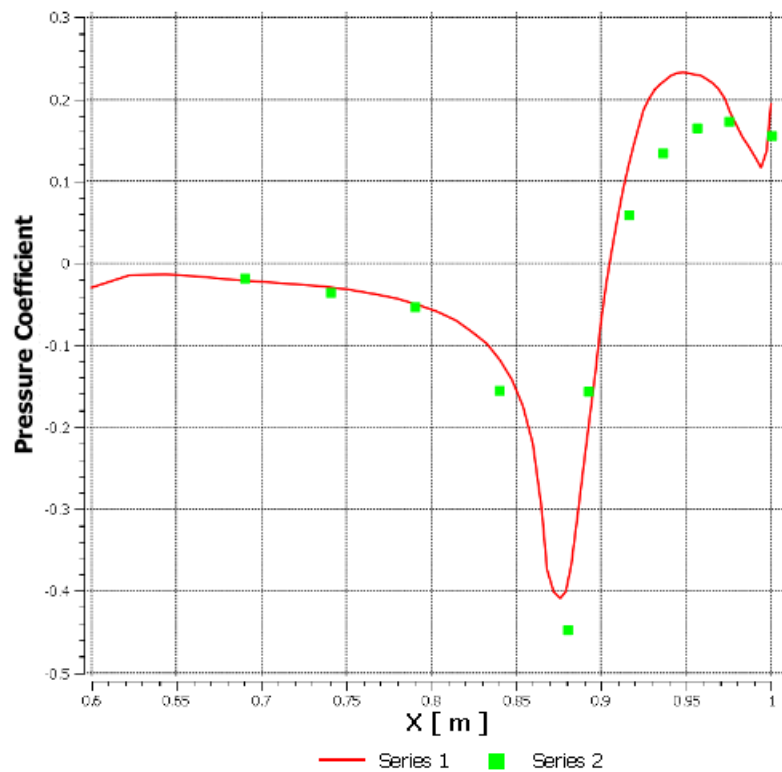
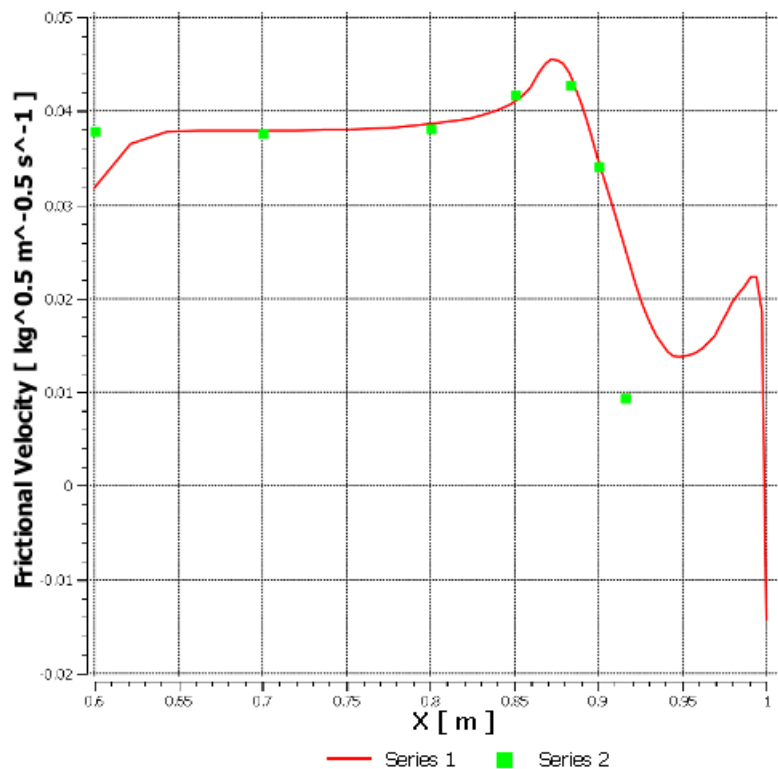


Figure .32.5: Comparison of Skin Friction Coefficient Along the Afterbody Wall



VMFL033: Viscous Heating in an Annulus

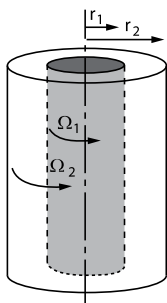
Overview

Reference	R.B. Bird, W.E. Stewart, E.N. Lightfoot, <i>Transport phenomena</i> , John Wiley and Sons, New York, 1960
Solver	Ansys Fluent
Physics/Models	Viscous flow and heating driven by a moving wall
Input File	plate_polar.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

In this problem, we model the viscous heating and mass flow in a 2-D annulus induced by the rotation of one of the two walls ([Figure .33.1: Geometry \(p. 119\)](#)). This problem can be solved analytically.

Figure .33.1: Geometry



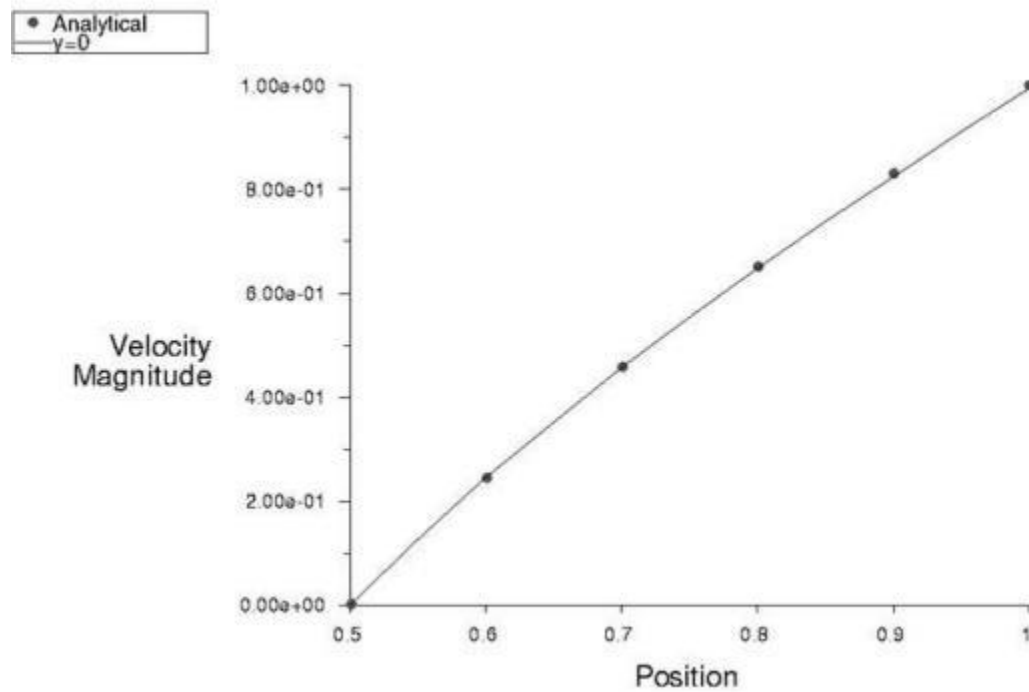
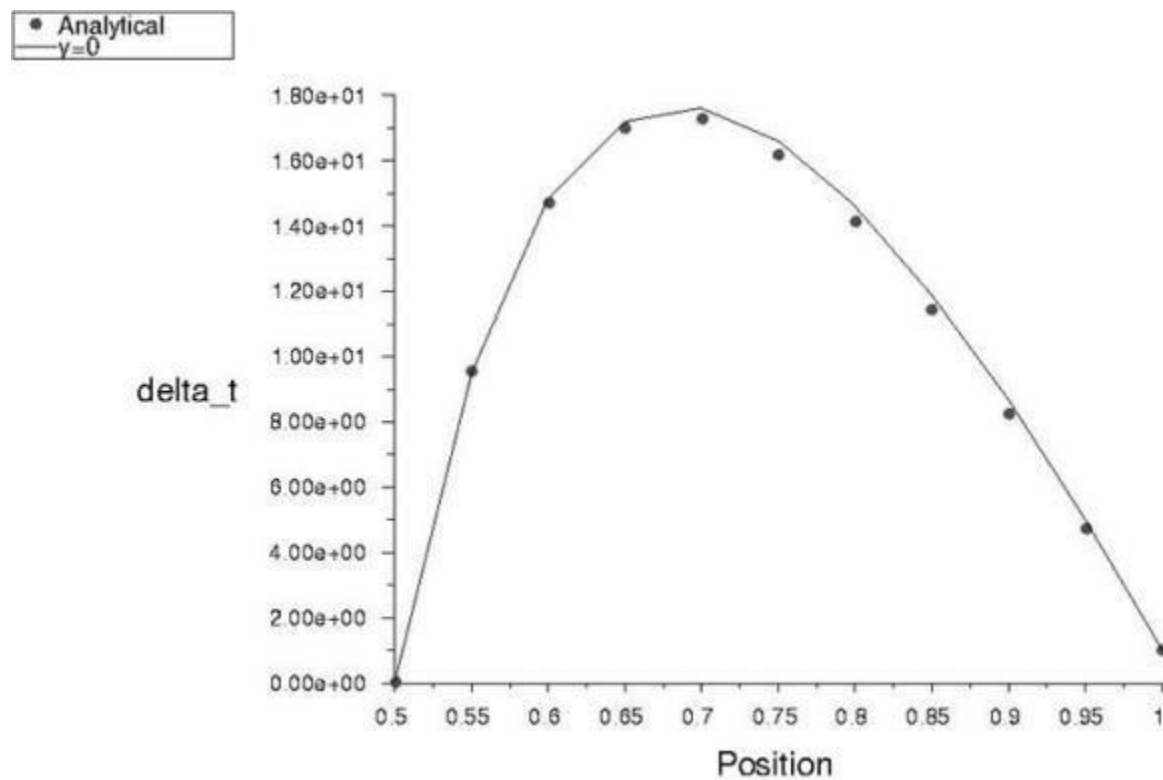
Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³	$r_1 = 1 \text{ m}$	$\Omega_1 = 0.0 \text{ rad/s}$
Specific heat: 1 J/kg-K	$r_2 = 2 \text{ m}$	$\Omega_2 = 0.5 \text{ rad/s}$
Thermal conductivity: 1 W/m-K		$T_1 = 273 \text{ K}$
Viscosity: 300 kg/m-s		$T_2 = 274 \text{ K}$

Analysis Assumptions and Modeling Notes

The flow is laminar and steady. Pressure based solver is used. A 2-D mesh with quadrilateral cells is used to discretize the domain.

Results Comparison for Ansys Fluent

Normalized velocity and temperature profiles are compared with the analytical solution provided by Bird et al (1960).

Figure .33.2: Comparison of Velocity Profile**Figure .33.3: Comparison of Temperature Profile**

VMFL034: Particle Aggregation inside a Turbulent Stirred Tank

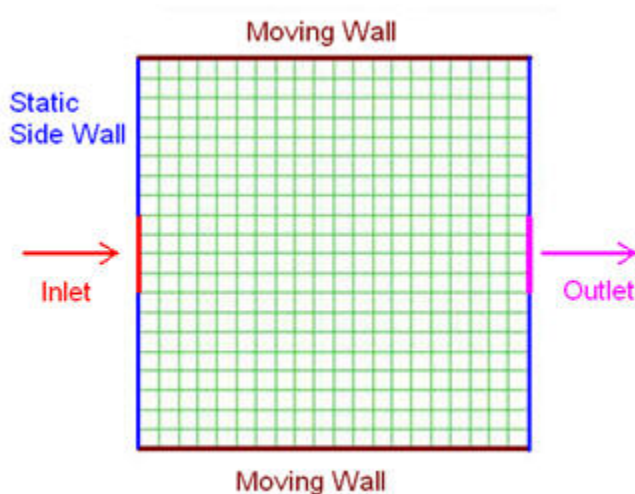
Overview

Reference	B. Wan, T.A. Ring, K. Dhanasekharan, J. Sanyal, "Comparison of Analytical Solutions for CMSMPR Crystallizer with QMOM Population Balance Modeling in Ansys Fluent". <i>China Particuology</i> , Vol. 3, pp. 213-218, 2005
Solver	Ansys Fluent
Physics/Models	Multi-phase, Population balance model, turbulent flow
Input File	agglomeration.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A 2-D approximation of a stirred tank is simulated in order to verify the population balance model that operates in conjunction with its multiphase calculations to predict the particle size distribution within the flow field. The flow rate at the inlet is equal to that at the outlet, allowing the mean residence time to be calculated from the inlet flow rate (velocity x inlet area) and the "volume" (box area x unit depth) of the box. To simulate the agitation in the tank the top and bottom walls are assumed to move in the direction of the outlet. The flow is turbulent, steady, and incompressible. Multi-phase, with QMOM population balance model is used for particle aggregation. The standard $k-\epsilon$ model is used for turbulence. Moments are solved on a frozen flow field.

Figure .34.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 998.2 kg/m ³	Square box side = 0.1 m	Top wall velocity: 101 m/s
Viscosity: 0.00103 kg/m-s	Inlet/Outlet openings = 0.02 m	Bottom wall velocity: 100 m/s
		Inlet velocity = 0.005 m/s

Material Properties	Geometry	Boundary Conditions
		Outlet gauge pressure = 0 Pa

Analysis Assumptions and Modeling Notes

The results of the Ansys Fluent simulation are compared to steady a state analytical solution for the population balance in a stirred tank where aggregation takes place.

To optimize the calculations, the operations shown in the journal file below were performed. These steps must be followed to replicate the test case results.

```
rc VMFL034_aggregation.cas
solve set equations mixture flow yes ke yes mp no
solve set equations phase-2 moment-0 no moment-1 no moment-2 no moment-3 no moment-4 no moment-5 no
solve initialize initialize-flow
solve iterate 700
solve set equations mixture flow no ke no mp no
solve set equations phase-2 moment-0 yes moment-1 yes moment-2 yes moment-3 yes moment-4 yes moment-5 yes
solve patch phase-2 fluid ( ) mp 1e-06
solve set under-relaxation phase-2 moment-0 0.9 moment-1 0.9 moment-2 0.9 moment-3 0.9 moment-4 0.9 moment-5 0.9
solve iterate 200
solve set under-relaxation phase-2 moment-0 1.0 moment-1 1.0 moment-2 1.0 moment-3 1.0 moment-4 1.0 moment-5 1.0
solve iterate 2000
report population-balance moments outlet ( ) ( ) 6 no
```

Results Comparison

In this table, moment of PBE for Ansys Fluent turbulent simulations is compared with analytical solution for aggregation alone at the outlet of the tank.

Table .34.1: Comparison of Moment of PBE

Moment	Target	Ansys Fluent	Ratio
m_0	0.132	0.132	1.000
m_1	0.225	0.226	1.004
m_2	0.547	0.548	1.002
m_3	1.910	1.910	1.000
m_4	9.073	9.133	1.007
m_5	53.797	53.816	1.000

VMFL035: 3-Dimensional Single-Stage Axial Compressor

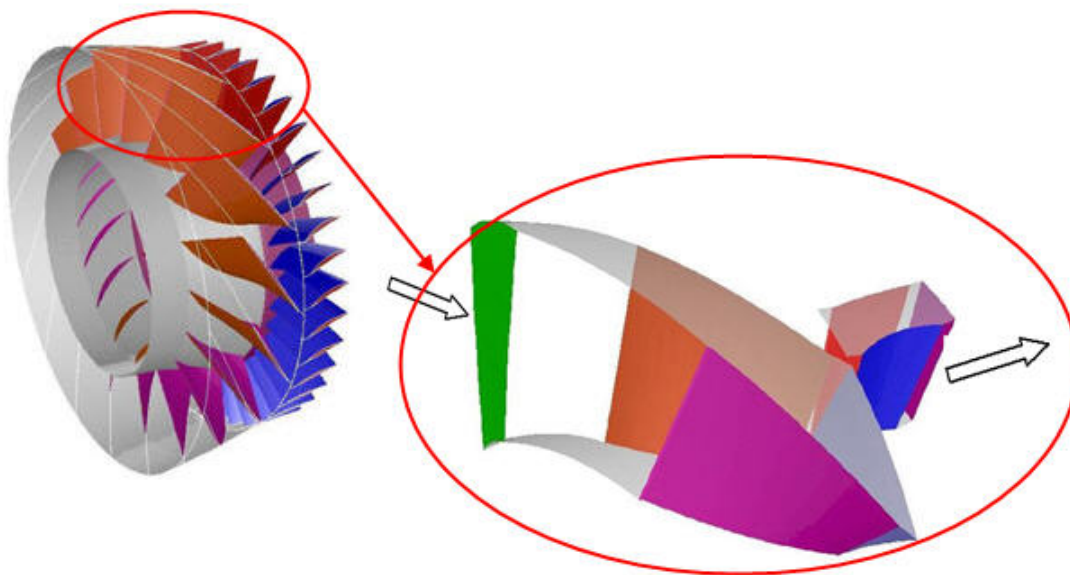
Overview

Reference	Density-based solver (Ansys Fluent)
Solver	Ansys Fluent
Physics/Models	Compressible (transonic), turbulent flow, moving reference frame
Input File	axial-compressor.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A 3-D model of a single-stage axial compressor is simulated. The flow through the rotor blades is computed in a rotating reference frame, while the flow in the stator blades in a stationary frame. The purpose of this case is to validate the performance of the pressure-based coupled solver for a compressible turbomachinery problem with a mixing plane. The flow is compressible, turbulent and steady.

Figure .35.1: Flow Domain



Material Properties (for Air)	Geometry	Boundary Conditions
Density = Ideal – Gas Molecular weight = 28.966 Specific heat = 1006.43 J/kg-K Viscosity - Sutherland's law Conductivity - Kinetic theory	Geometry is as shown in in Figure .34.1: Flow Domain (p. 121) Number of rotor blades = 16 Number of stator blades = 40	Rotational speed = 37,500 rpm For Inlet: • $P_{\text{total}} = 1 \text{ atm}$ • $T_{\text{total}} = 288 \text{ K}$

Analysis Assumptions and Modeling Notes

Steady, turbulent, compressible flow. Ideal-gas law is used for density calculations and kinetic theory for fluid viscosity and thermal conductivity. The standard k- ϵ model is used for turbulence. Pressure-based coupled solver with a mixing plane at the rotor-outlet/stator-inlet interface.

Results Comparison for Ansys Fluent

The results of the pressure-based Ansys Fluent simulation are compared to the steady state solution from the density-based solver.

Table .35.1: Comparison of Pressure and Mass Flow Rate

	Target	Ansys Fluent	Ratio
Pressure at Stator-Outlet (atm)	1.4725	1.48	1.005
Mass-Flow Rate at Stator-Outlet (kg/s)	0.1049	0.1076	1.026

VMFL036: Laminar Flow Past Sphere

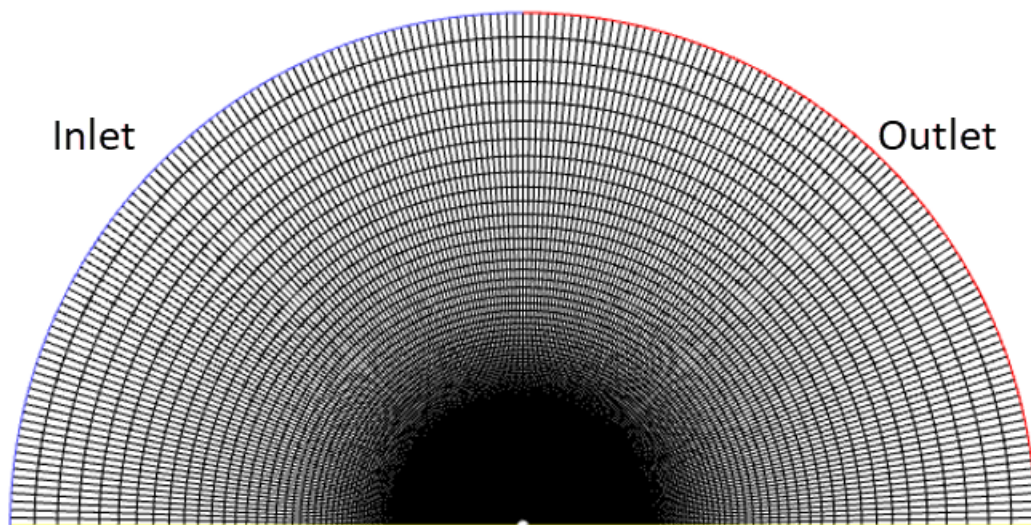
Overview

Reference	Mittal, R. (1999). A Fourier-Chebyshev spectral collocation method for simulating flow past spheres and spheroids. <i>International Journal for Numerical Methods in Fluids</i> , 30 (7). Tabata, M. & Itakura, K. (1998). A precise computation of drag coefficients of a sphere. <i>International Journal of Computational Fluid Dynamics</i> , 9 (3-4).
Solver	Ansys Fluent
Physics/Models	Laminar Flow
Input File	VMFL036_FLUENT.cas.h5
Project Files	Link to Project Files Download Page (p. 5)

Test Case

To verify that FLUENT can capture the flow regime in wake and flow separation where a sphere is immersed in the flow. The forces computed on the sphere wall surface are compared with the literature.

Figure .36.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 0.02 kg/m-s	A section of the spherical geometry is modeled in 2D with axisymmetric boundary conditions. The sphere diameter is 1 m, and the circular fluid domain around the sphere has a radius of 50 D.	Inlet Velocity = 1 m/s

Analysis Assumptions and Modeling Notes

Flow is laminar and steady.

Results Comparison for Ansys Fluent

Table .36.1: Comparison of Drag Coefficient

	Target	Ansys Fluent	Ratio
Drag Coefficient	1.0895	1.0875	0.998

VMFL037: Turbulent Flow Over a Forward Facing Step

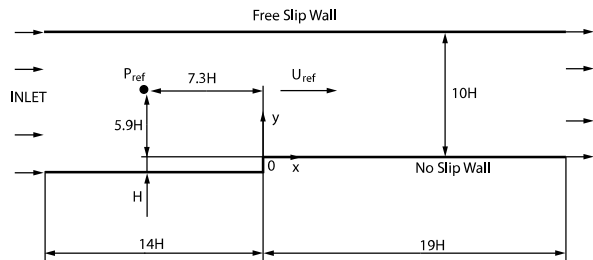
Overview

Reference	S. Baker. "Regions of Recirculating Flow Associated with Two-Dimensional Steps" Ph.D. Thesis. Department of Civil Engineering, University of Surrey. UK. 1977.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	SST model, turbulent flow with separation and reattachment
Input File	VMFL037_ffstep.cas for Ansys Fluent ffstep.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Turbulent flow over a forward facing step is modeled. The flow undergoes separation and reattachment.

Figure .37.1: Flow Domain



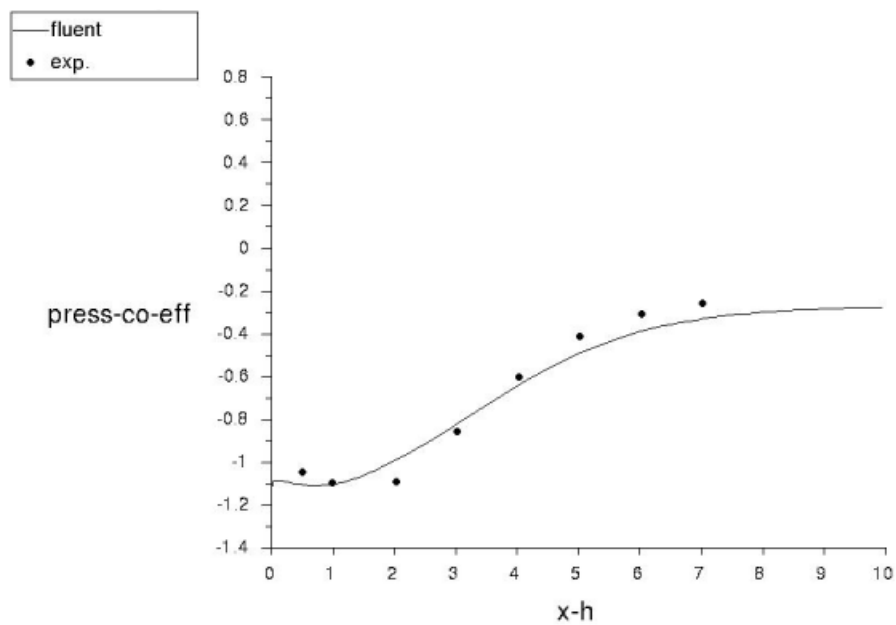
Material Properties	Geometry	Boundary Conditions
Density = 1.02 kg/m^3 Viscosity = $1.5 \times 10^{-5} \text{ kg/m-s}$	Step height $H = 0.0758 \text{ m}$	Inlet Velocity = 9.7 m/s Outer boundary (in transverse direction) is modeled as slip wall

Analysis Assumptions and Modeling Notes

The flow is steady. Pressure coefficient, C_p on the wall is calculated with reference to the pressure at point upstream of the step at coordinates as indicated in [Figure .37.1: Flow Domain \(p. 127\)](#).

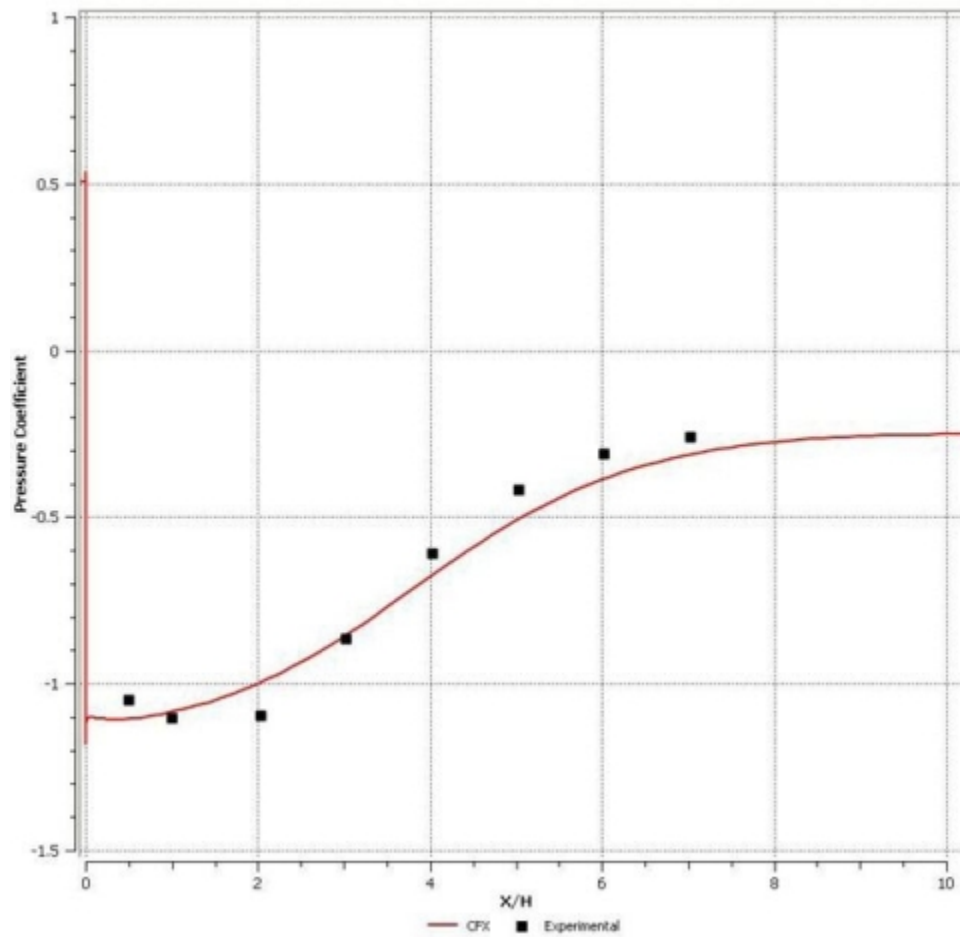
Results Comparison for Ansys Fluent

Figure .37.2: Comparison of Pressure Coefficient Along the Wall



Results Comparison for Ansys CFX

Figure .37.3: Comparison of Pressure Coefficient Along the Wall



VMFL038: Falling Film Over an Inclined Plane

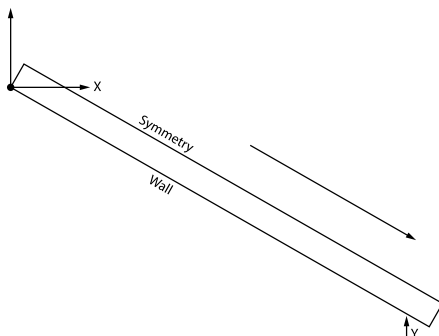
Overview

Reference	R.B. Bird, W.E. Steward, E.N. Lightfoot, <i>Transport Phenomena</i> . John Wiley and Sons, Inc., New York, NY, pg. 45, 2005
Solver	Ansys Fluent
Physics/Models	Laminar Flow, Coupled solver
Input File	VMFL038_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar flow of a fluid over an inclined plane, driven by the pressure difference due to gravity head is modeled. The flow channel is inclined at an angle $\beta = 30^\circ$ with the horizontal direction.

Figure .38.1: Flow Domain



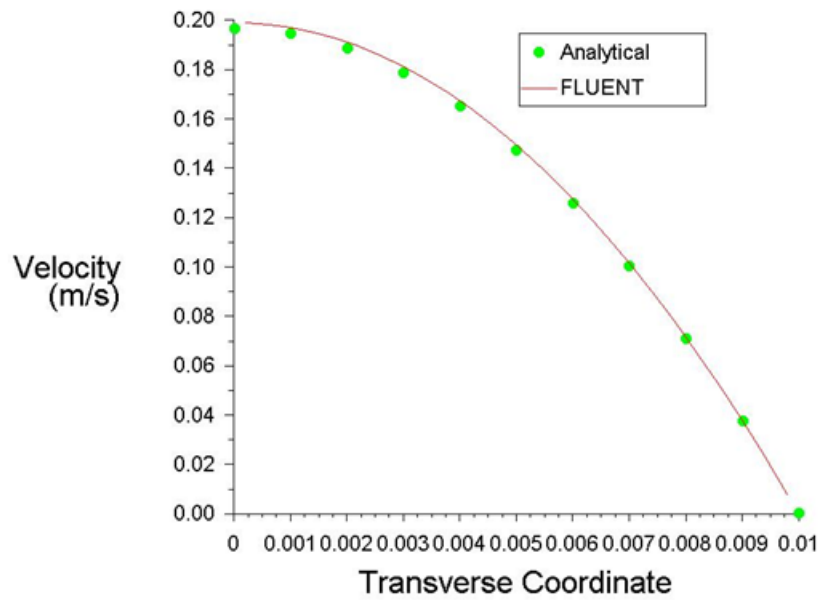
Material Properties	Geometry	Boundary Conditions
Density = 800 kg/m ³ Viscosity = 1 kg/m-s	Dimensions of the domain: 1 m X 18 m Angle with X-axis = 30°	Gauge Pressure at Inlet = 0 N/m ² Gauge Pressure at Outlet = -706.32 N/m ²

Analysis Assumptions and Modeling Notes

The density based solver is used in Ansys Fluent. Pressure boundaries are specified to model the driving head in the direction of flow.

Results Comparison for Ansys Fluent

Figure .38.2: Comparison of Velocity Profile at Outlet



VMFL039: Boiling in a Pipe with Heated Wall

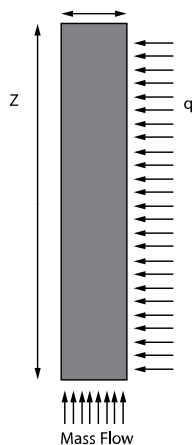
Overview

Reference	G.G. Bartolomei, V.M. Chanturiya "Experimental Study of True Void Fraction When Boiling Subcooled Water in Vertical Tubes". <i>Thermal Engineering</i> . Vol. 14, pp. 123-128, 1967.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Multiphase flow, phase change, RPI Wall boiling Model
Input File	wall-boiling.def for Ansys Fluent VMFL039__wall-boiling.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Bubble formation and boiling near the heated wall of a vertical pipe are modeled. Outer wall of the pipe is heated with a constant heat flux.

Figure .39.1: Flow Domain



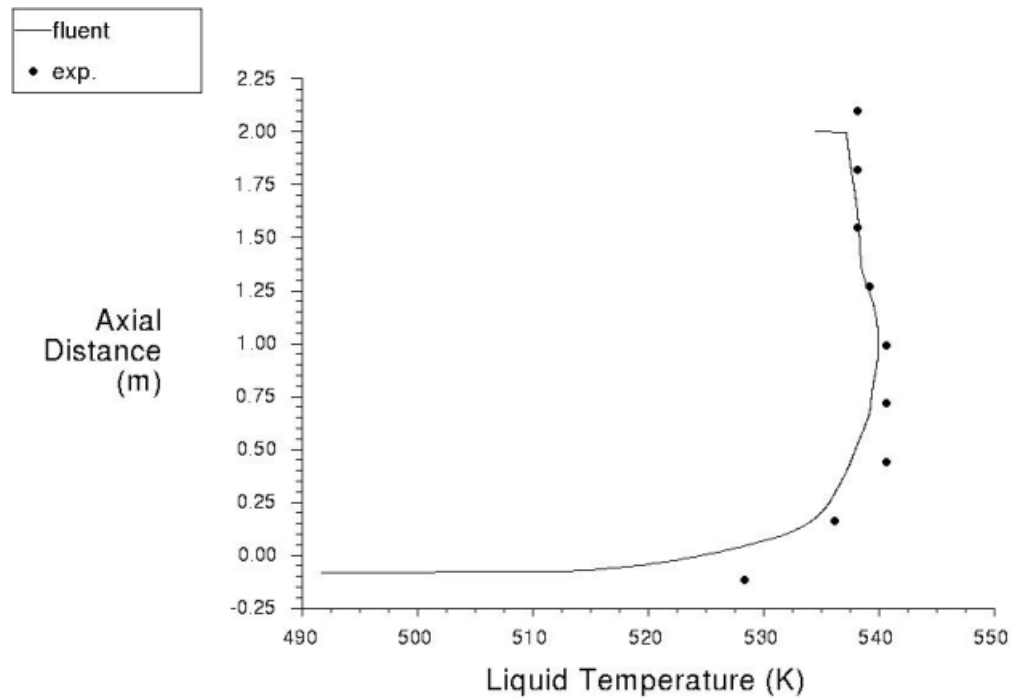
Material Properties	Geometry	Boundary Conditions
Steam-Water 2-phase Flow: - Water: continuous phase - Water Steam: dispersed bubbles Bubble diameter dependent on fluid temperature	Radius of the pipe = 7.7 mm Height of the pipe = 2 m	Mass flux at inlet = 900 kg/m ² /s Inlet pressure = 4.5 X 10 ⁶ N/m ² Heat transfer at the wall = 570000 W/m ²

Analysis Assumptions and Modeling Notes

The flow is steady. SST model is used for turbulence. RPI model for wall boiling is used with a value of 0.8 for the wall area fraction affected by vapor.

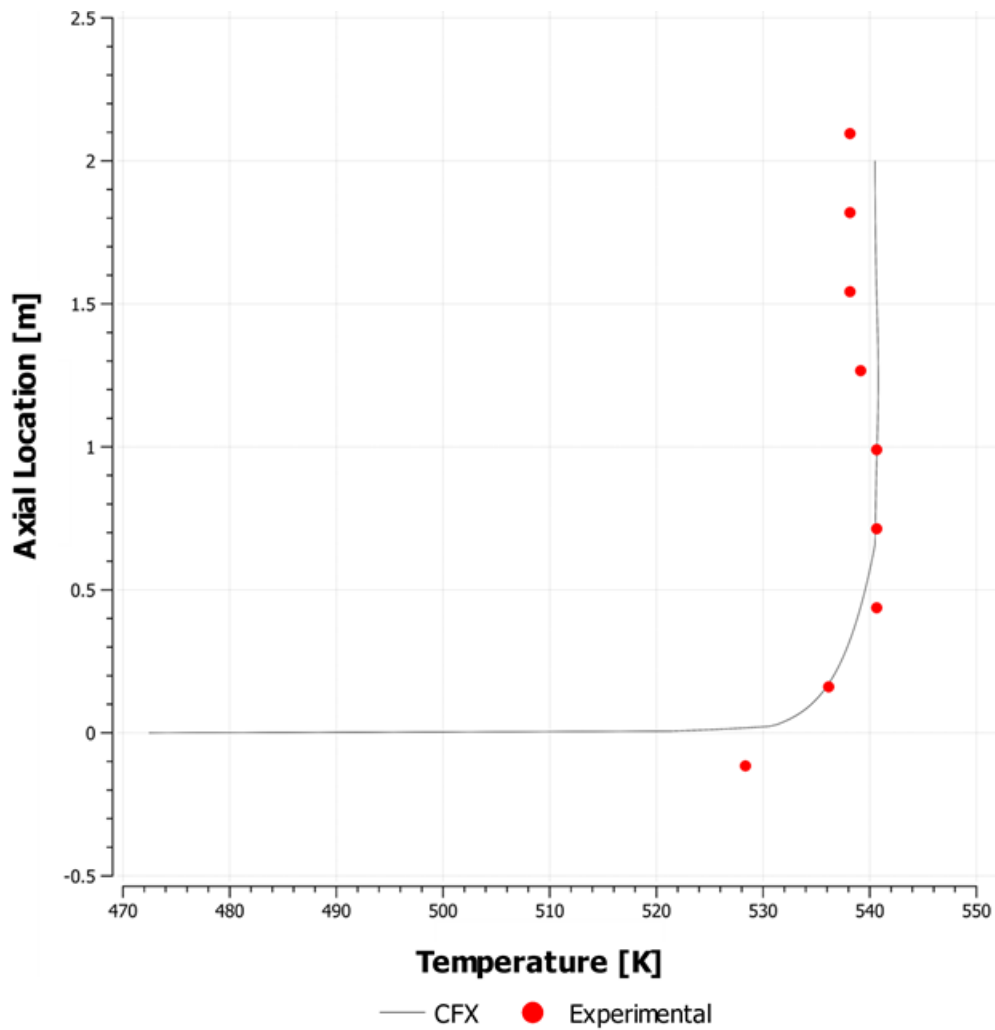
Results Comparison Ansys Fluent

Figure .39.2: Comparison of Temperature Along the Pipe Wall



Results Comparison Ansys CFX

Figure .39.3: Comparison of Temperature Along the Pipe Wall



VMFL040: Separated Turbulent Flow in a Diffuser

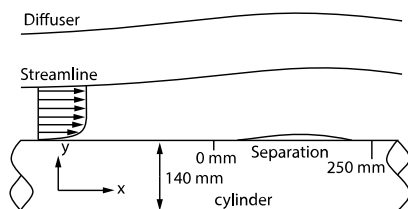
Overview

Reference	D.M. Driver. "Reynolds shear stress measurements in a separated boundary layer flow". <i>AIAA-91-1787</i> . 1991.
Solver	Ansys CFX, Ansys Fluent
Physics/Models	SST model, Adverse pressure gradient, flow separation
Input Files	diffuser-sep.def for Ansys CFX VMFL040A_diffuser-sep.cas for Ansys Fluent
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The test case geometry is shown in [Figure .40.1: Sketch of Flow Domain \(p. 137\)](#). It consists of an axisymmetric diffuser with an internally mounted cylinder along the centre line. The curvature of the diffuser wall results in an adverse pressure gradient. A relatively short separation region was detected in the experiment.

Figure .40.1: Sketch of Flow Domain



This figure is not to scale.

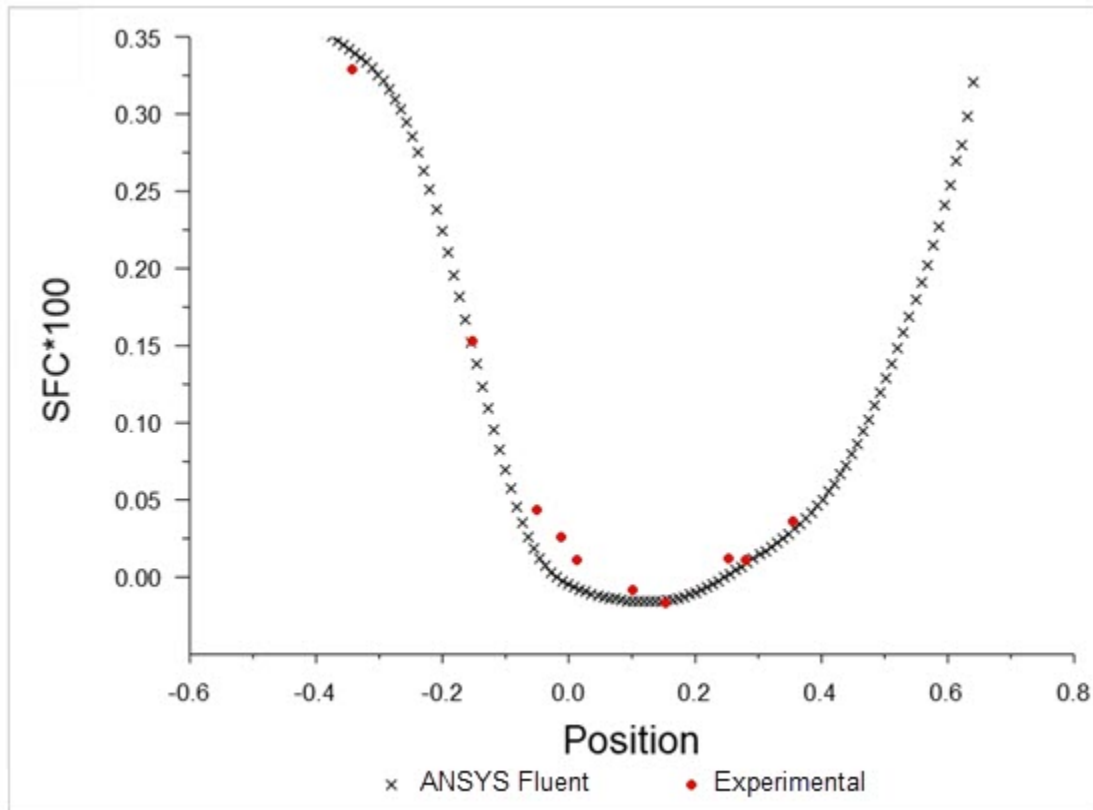
Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 1.5 X 10 ⁻⁵ kg/m-s	Diameter of the cylinder = 140 mm Length of the domain = 1100 mm	Velocity profile at inlet with average velocity = 29 m/s Outer wall modeled as a slip (inviscid) wall

Analysis Assumptions and Modeling Notes

The flow is steady. SST model is used for turbulence.

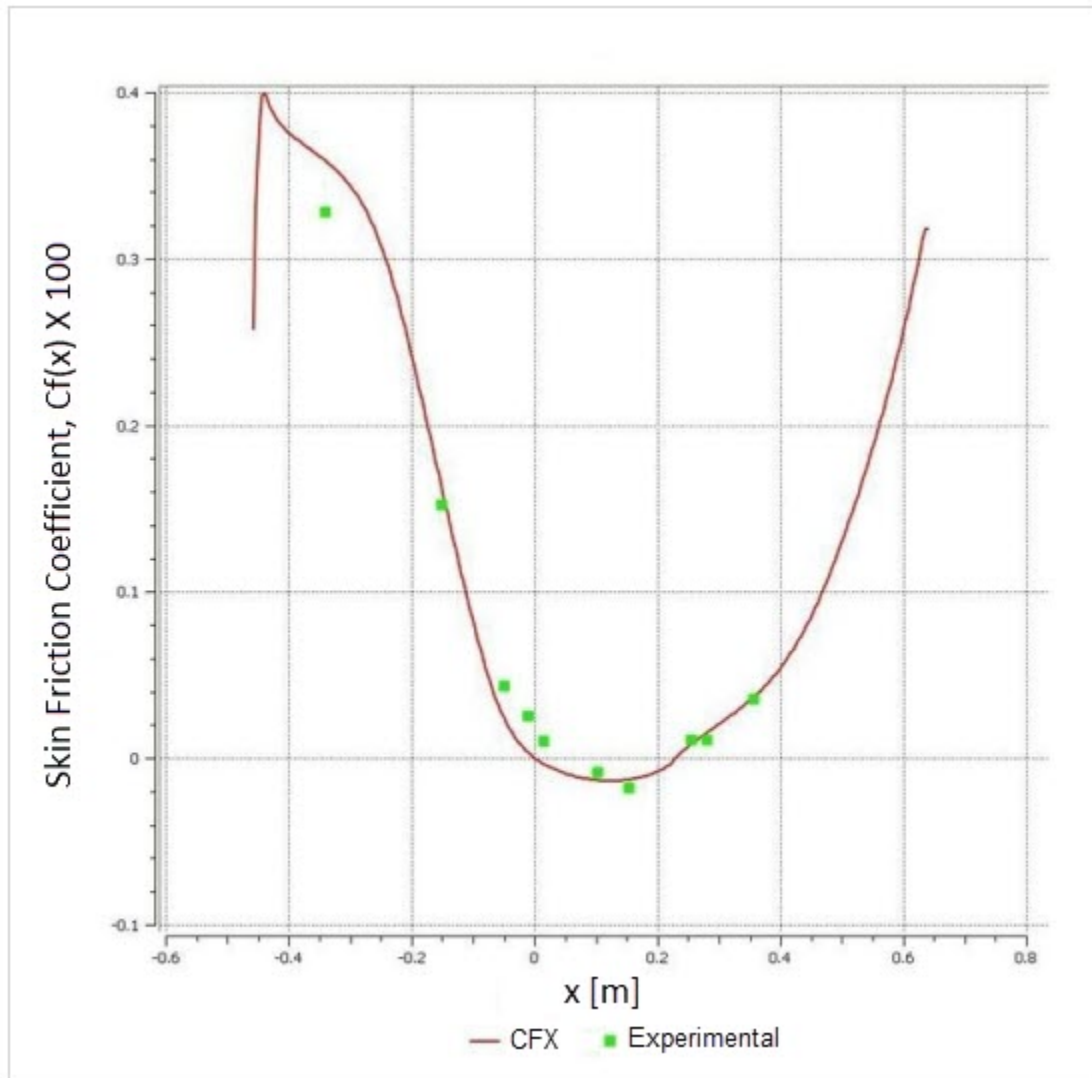
Results Comparison for Ansys Fluent

Figure .40.2: Comparison of Skin Friction Coefficient on the Cylinder Wall



Results Comparison for Ansys CFX

Figure .40.3: Comparison of Skin Friction Coefficient on the Cylinder Wall



VMFL041: Transonic Flow Over an Airfoil

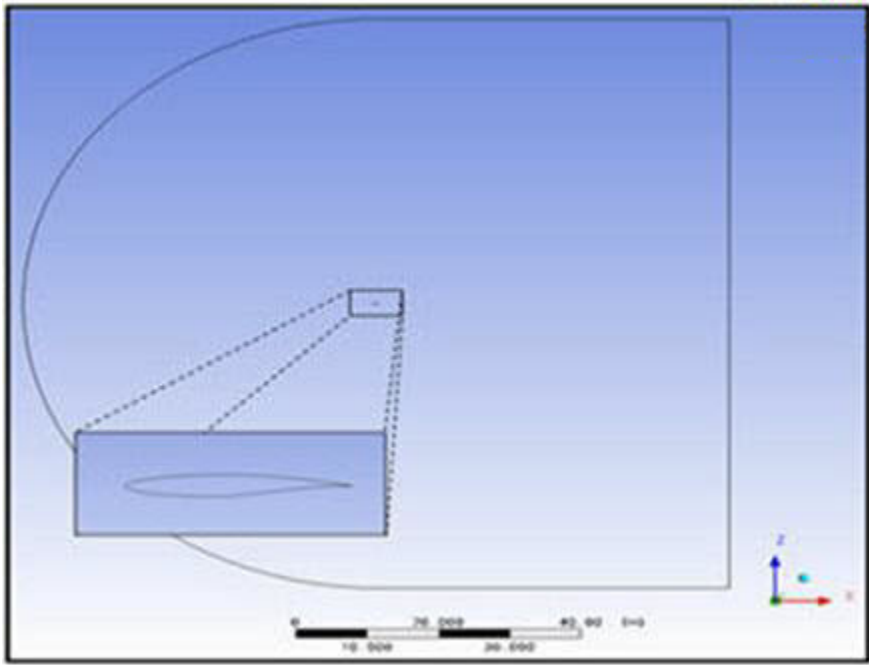
Overview

Reference	P.H. Cook, M.A. McDonald, M.C.P. Firmin. "AEROFOIL RAE 2822 - PRESSURE DISTRIBUTIONS, AND BOUNDARY LAYER AND WAKE MEASUREMENTS." AGARD Advisory Report No. 138.
Solver	Ansys Fluent
Physics/Models	Transonic flow, shock, SST model
Input File	VMFL041_transonic.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Transonic flow over air foil RAE 2822 is modeled for an angle of attack of 3.19°. The flow domain spans over 100 Chord lengths in both stream-wise and transverse directions.

Figure .41.1: Flow Domain



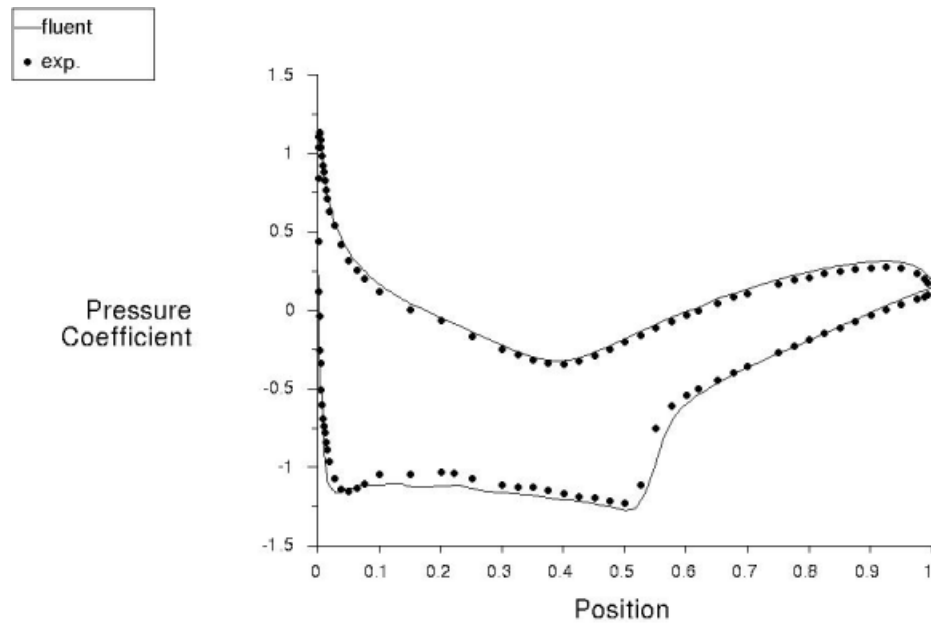
Material Properties	Geometry	Boundary Conditions
Density: Ideal gas law for Air Viscosity: 1.831×10^{-5} kg/m-s	Chord length of the airfoil = 1 m	Velocity profile at inlet with an average velocity of 218 m/s

Analysis Assumptions and Modeling Notes

The flow is steady. The inlet flow Mach number is close to transonic range. Walls are assumed to be adiabatic.

Results Comparison for Ansys Fluent

Figure .41.2: Comparison of Pressure Coefficient on the Airfoil



VMFL042: Turbulent Mixing of Two Streams with Different Densities

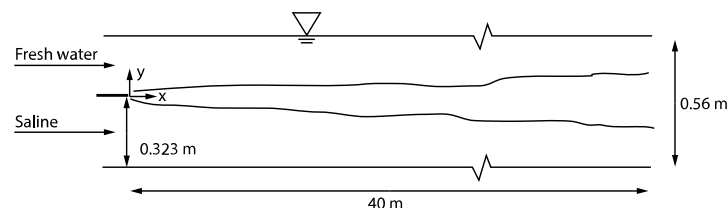
Overview

Reference	R.E. Uittenbogaard. "Stably Stratified Mixing Layer". <i>Data Report for the 14th meeting of the IAHR Working Group on Refined Flow Modeling</i> . 1989. R.E. Uittenbogaard. "The Importance of Internal Waves for Mixing in a Stratified Estuarine Tidal Flow". Manuscript, Delft University of Technology, 1995.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	SST model, mixing layer, density difference, buoyancy
Input File	VMFL042_mixing.cas for Ansys Fluent saline-mixing_layer.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Mixing of two turbulent streams of fresh water and saline water is modeled. The two streams are parallel at the inlet and mixing proceeds downstream.

Figure .42.1: Flow Domain



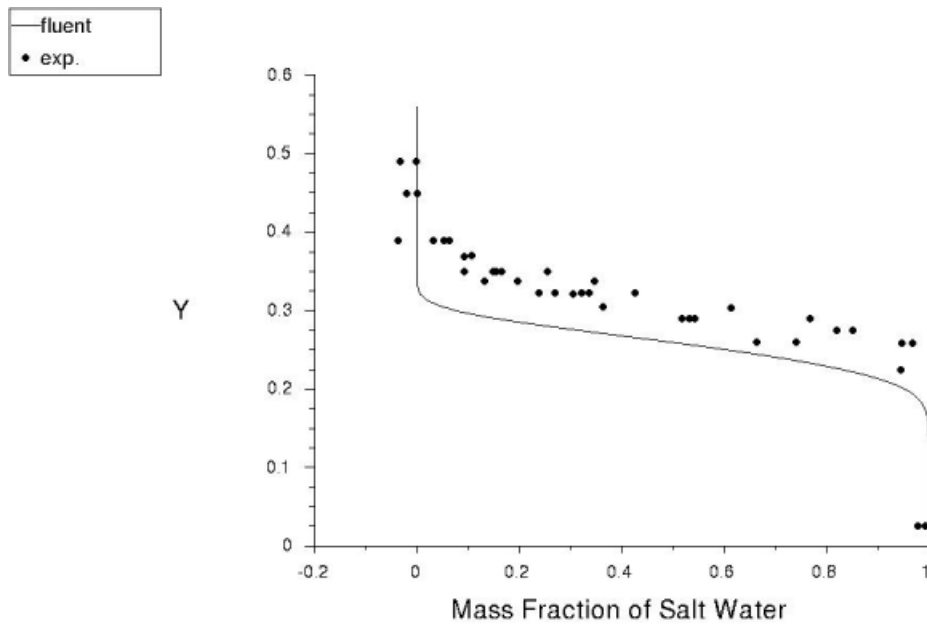
Material Properties	Geometry	Boundary Conditions
Density of fresh water: 1015 kg/m^3 Density of saline water: 1030 kg/m^3 Mixture kinematic diffusivity: $1 \times 10^{-9} \text{ m}^2/\text{s}$	Length of the mixing duct = 40 m	Fresh water inlet velocity = 0.52 m/s Salt water inlet velocity = 0.32 m/s

Analysis Assumptions and Modeling Notes

The flow is steady. SST model is used. Buoyancy turbulence production option is used.

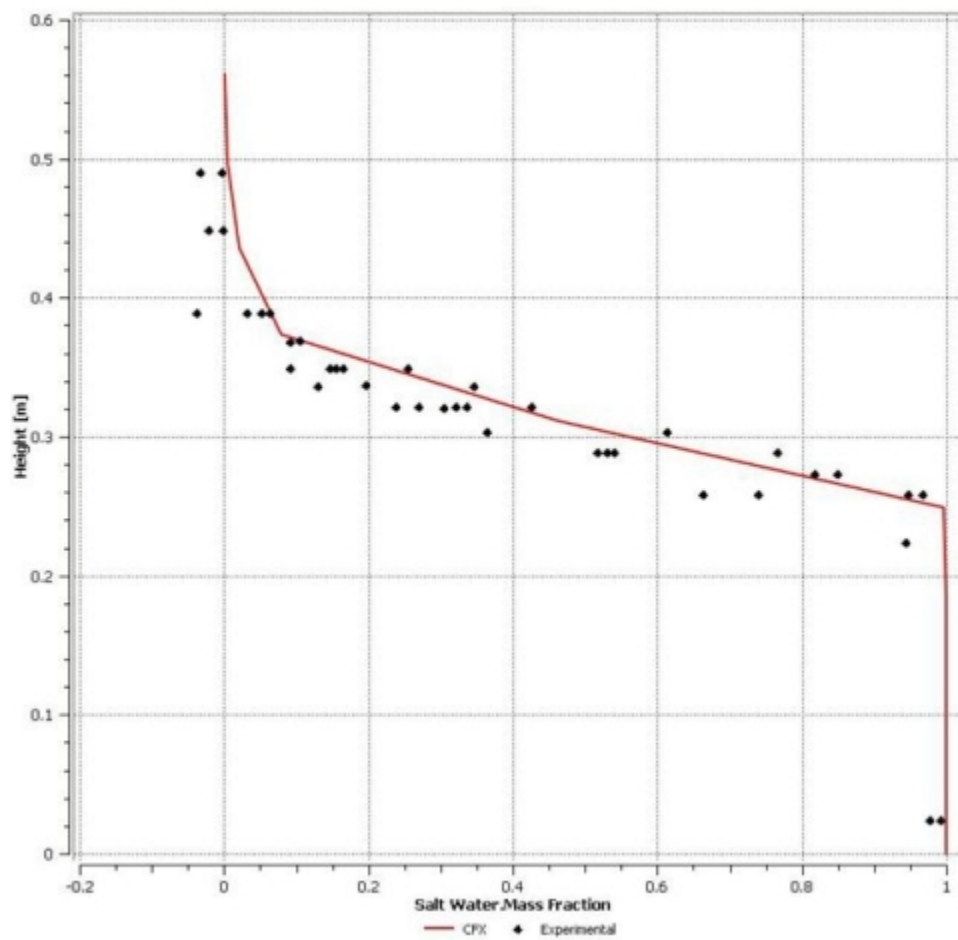
Results Comparison for Ansys Fluent

Figure .42.2: Comparison of Mass Fraction of Salt Water Across the Mixing Layer at $x = 10\text{m}$



Results Comparison for Ansys CFX

Figure .42.3: Comparison of Mass Fraction of Salt Water Across the Mixing Layer at $x = 10\text{m}$



VMFL043: Laminar to Turbulent Transition of Boundary Layer over a Flat Plate

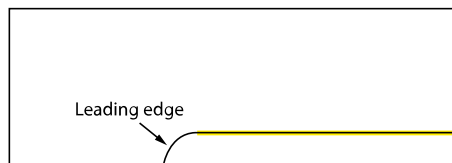
Overview

Reference	A. M. Savill. "Some recent progress in the turbulence modeling of bypass transition". <i>Near-Wall Turbulent Flows</i> . Elsevier Science Publishers, pp. 829-848, 1993. P.E. Roach, D.H. Brierley. "The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development. Part I: Test Cases T3A and T3B". <i>Simulation of Unsteady and Transition to Turbulence</i> . Cambridge University Press. Cambridge, pp. 319-347, 1992.
Solver	Ansys Fluent
Physics/Models	SST model, transitional flow
Input File	VMFL043_transition.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar to turbulent transition of a boundary layer over a flat plate is modeled. The free stream turbulence intensity is 3.3%.

Figure .43.1: Flow Domain



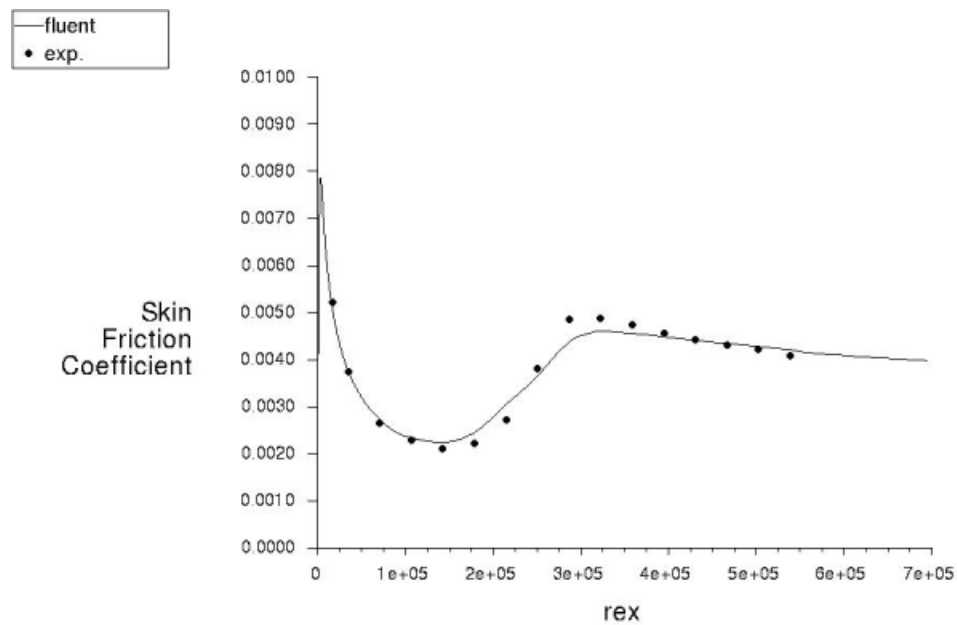
Material Properties	Geometry	Boundary Conditions
Density: 1.2 kg/m ³ Viscosity: 1.831 X 10 ⁻⁵ kg/m-s	Length of the flat plate = 2m	Inlet Velocity = 5.3 m/s Inlet eddy viscosity ratio = 9.7

Analysis Assumptions and Modeling Notes

The flow is steady. SST model with Gamma Theta model for transitional turbulence is used. Langry Menter correlation was used for transition onset.

Results Comparison for Ansys Fluent

Figure .43.2: Comparison of Skin Friction Coefficient on the Plate



VMFL044: Supersonic Nozzle Flow

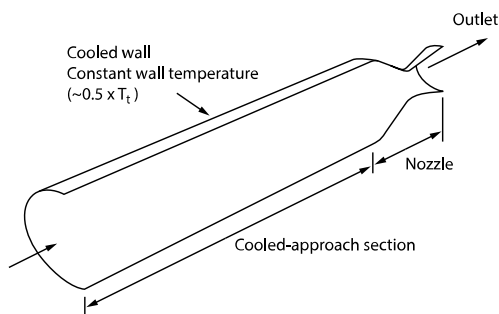
Overview

Reference	L.H. Back, P.F. Massier, H.L. Gier. "Convective Heat Transfer in a Convergent-Divergent Nozzle". <i>Int. J. Heat Mass Transfer</i> , Vol. 7, pp. 549-568, 1964
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Compressible flow in supersonic regime, SST Model
Input File	VMFL044_nozzleflow.cas for Ansys Fluent VMFL044_supersonic_nozzle_flow_serial.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Supersonic flow in a convergent-divergent nozzle is modeled. The flow is supersonic in the entire divergent section of the nozzle.

Figure .44.1: Flow Domain



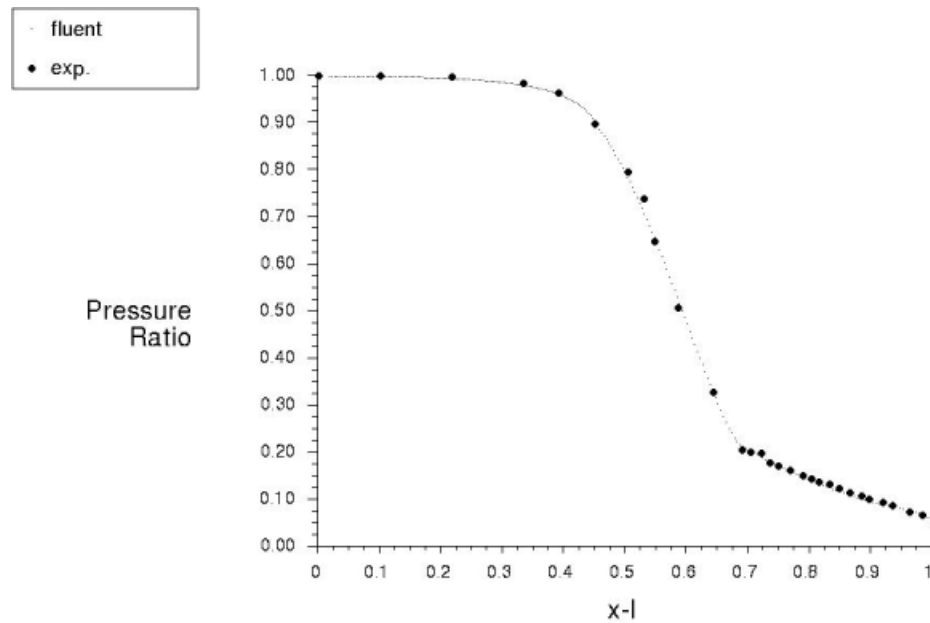
Material Properties	Geometry	Boundary Conditions
Density: Ideal Gas Viscosity: 1.831×10^{-5} kg/m-s	Length of the nozzle = 0.1594 m Exit-to-throat area ratio = 2.68 Half angle of divergence = 15°	Inlet Relative Pressure = 1×10^6 Pa Inlet Total Temperature = 825 K Wall temperature = 413 K

Analysis Assumptions and Modeling Notes

The flow is steady. The walls are assumed to be at constant temperature. Only a 3° sector of the domain is modeled due to symmetry.

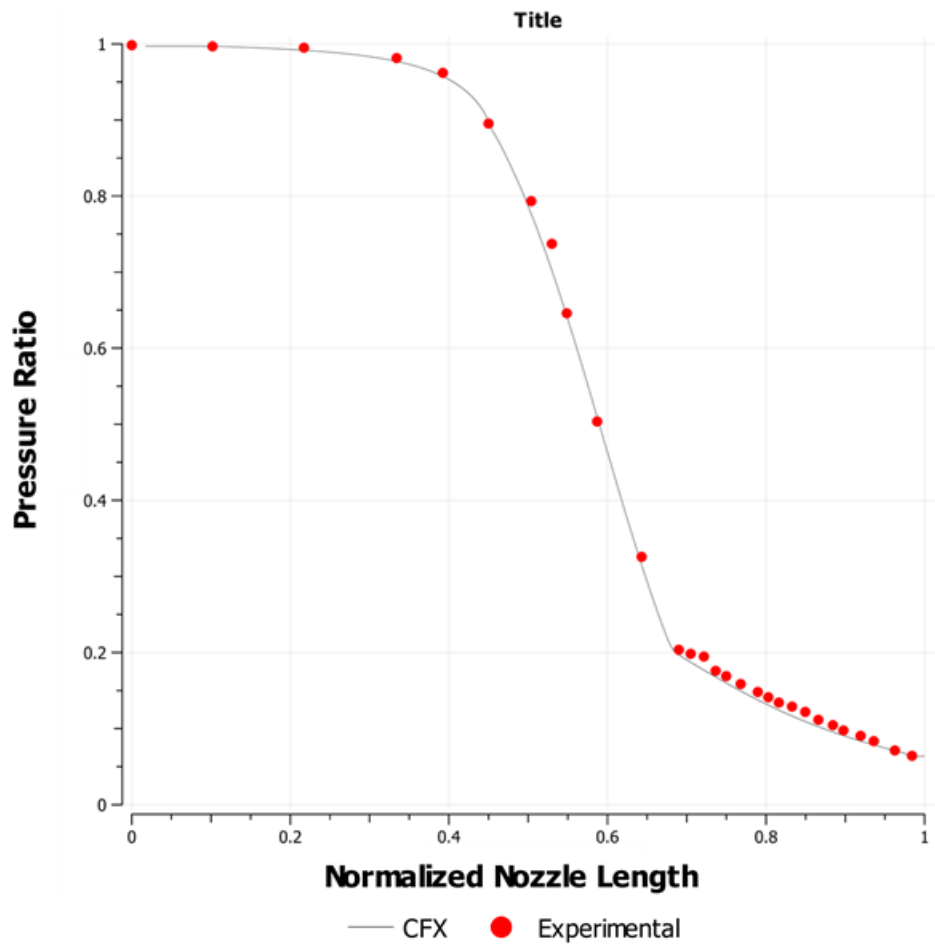
Results Comparison for Ansys Fluent

Figure .44.2: Comparison of Pressure Ratio Along the Nozzle Wall with Experimental Data



Results Comparison for Ansys CFX

Figure .44.3: Comparison of Pressure Ratio Along the Nozzle Wall with Experimental Data



VMFL045: Oblique Shock Over an Inclined Ramp

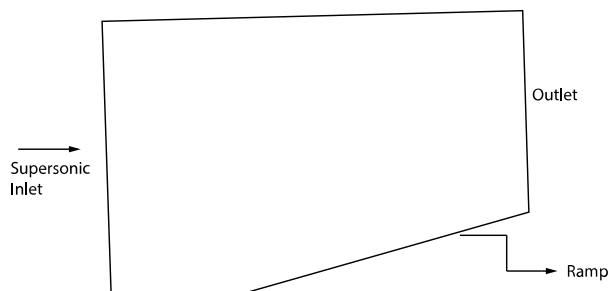
Overview

Reference	F. M. White. <i>Fluid Mechanics</i> . 3rd Edition. McGraw-Hill Book Co., New York, NY. 560-567. 1994.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Compressible flow in supersonic regime, Oblique shock
Input File	VMFL045_obliqueshock.cas for Ansys Fluent ramp_supersonic_tet.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Supersonic flow over a 15° ramp is modeled. The ramp leads to the formation of an oblique shock. Inlet Mach number is about 2.5. The flow is laminar. Inlet density is 1.22 kg/m³ and inlet temperature is 289 K. The simulation values are taken at a Point 1 (x=0.38 m, y=0.14 m).

Figure .45.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: Ideal Gas	Angle of the ramp = 15°	Inlet velocity = 852.68 m/s
Viscosity: 1 X 10 ⁻⁸ kg/m-s	Length = 0.4572 m	Inlet temperature = 289 K
MW: 0.02896 kg/mol		Inlet pressure = 101226.4 Pa
		Wall: Adiabatic

Analysis Assumptions and Modeling Notes

The flow is steady and laminar. The walls are assumed to be adiabatic.

Results Comparison for Ansys Fluent

Table .45.1: Comparison of Properties Downstream of the Oblique Shock

	Target	Ansys Fluent	Ratio
Mach Number	1.874	1.902	1.015
Temperature, K	382.0	377.6	0.9885
Density, kg/m ³	2.277	2.233	0.9807

Results Comparison for Ansys CFX

Table .45.2: Comparison of Properties Downstream of the Oblique Shock

	Target	Ansys CFX	Ratio
Mach Number	1.874	1.871	0.9984
Temperature, K	382.0	382.8	1.002
Density, kg/m ³	2.277	2.278	1.000

VMFL046: Supersonic Flow with Normal Shock in a Converging Diverging Nozzle

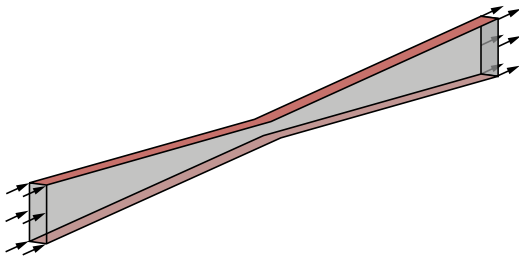
Overview

Reference	F. M. White. <i>Fluid Mechanics</i> . 3rd Edition. McGraw-Hill Book Co., New York, NY. 518-531. 1994.
Solver	Ansys Fluent
Physics/Models	Compressible flow in supersonic regime, Normal shock
Input File	VMFL046_supersonic.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Supersonic flow in a CD nozzle is modeled. The maximum Mach number is 2.2. A normal shock is formed in the divergent section. Mach number distribution in the nozzle is compared with analytical solution for nozzle flow.

Figure .46.1: Flow domain



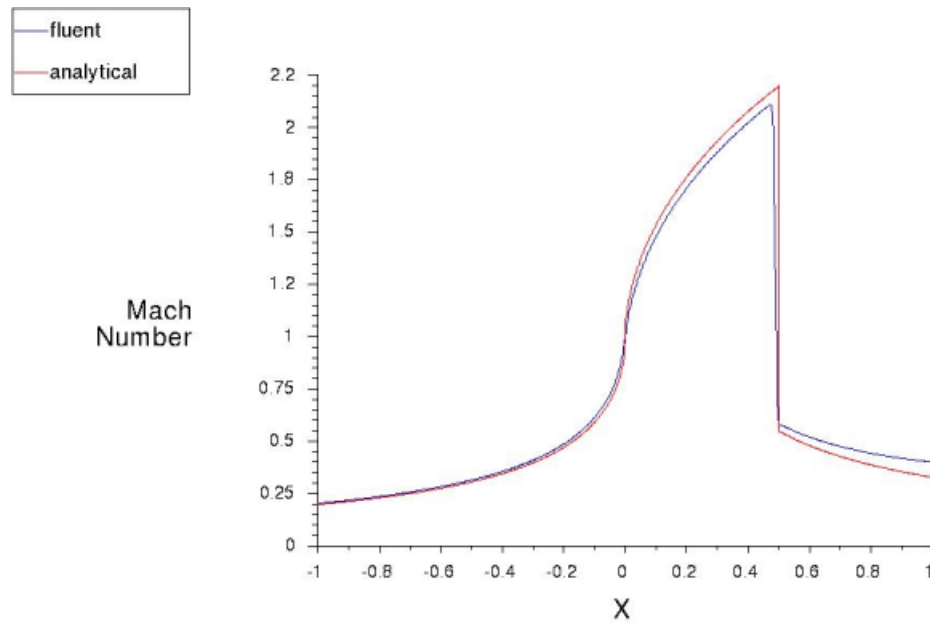
Material Properties	Geometry	Boundary Conditions
Density: Ideal Gas	Length of the nozzle = 2m	Inlet Relative Pressure 200 kPa
Viscosity: 1.7894×10^{-5} kg/m-s	Exit-to-throat area ratio = 3	Inlet Total Temperature = 500 K
		Wall temperature = 328 K
		Outlet Relative Pressure (gauge) = 75 kPa

Analysis Assumptions and Modeling Notes

The flow is steady. The flow is modeled as laminar.

Results Comparison for Ansys Fluent

Figure .46.2: Comparison of Mach Number Along Center Line of the Nozzle With Analytical Solution



VMFL047: Turbulent Flow with Separation in an Asymmetric Diffuser

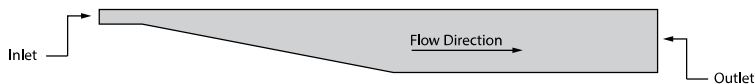
Overview

Reference	C. U. Buice, J. K. Eaton. "Experimental Investigation of Flow Through an Asymmetric Plane Diffuser". <i>Journal of Fluids Engineering</i> . Vol. 122, pp. 433-435, 2000
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Turbulent separation, standard k- ω model
Input File	VMFL047_FLUENT.cas for Ansys Fluent VMFL047_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Turbulent flow with gradual separation and reattachment is modeled in an asymmetric 2-D diffuser. The lower wall of the diffuser is divergent with an angle of 10° and expands to about 4.7 times the inlet height.

Figure .47.1: Flow Domain



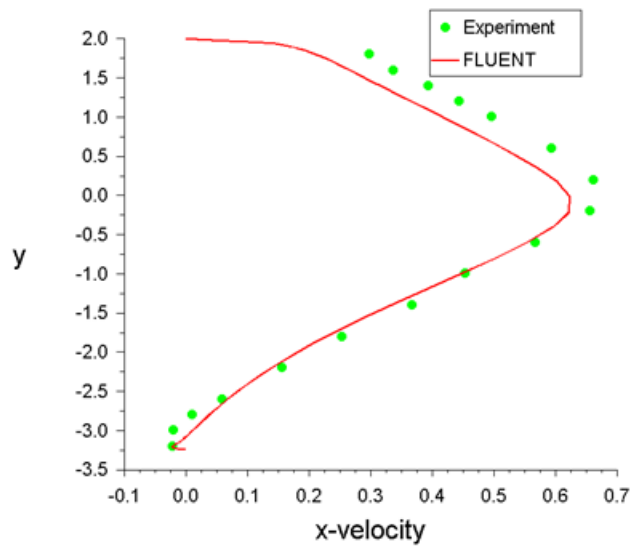
Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m^3 Viscosity: 0.0001 kg/m-s	Inlet height $H = 2 \text{ M}$ Outlet height $= 4.7 \text{ H}$ Angle of the divergent section $= 10^\circ$ Length of the straight section after divergence $= 21 \text{ H}$	Fully developed turbulent profile for velocity at inlet with an average velocity $= 0.7041 \text{ m/s}$

Analysis Assumptions and Modeling Notes

Steady turbulent flow.

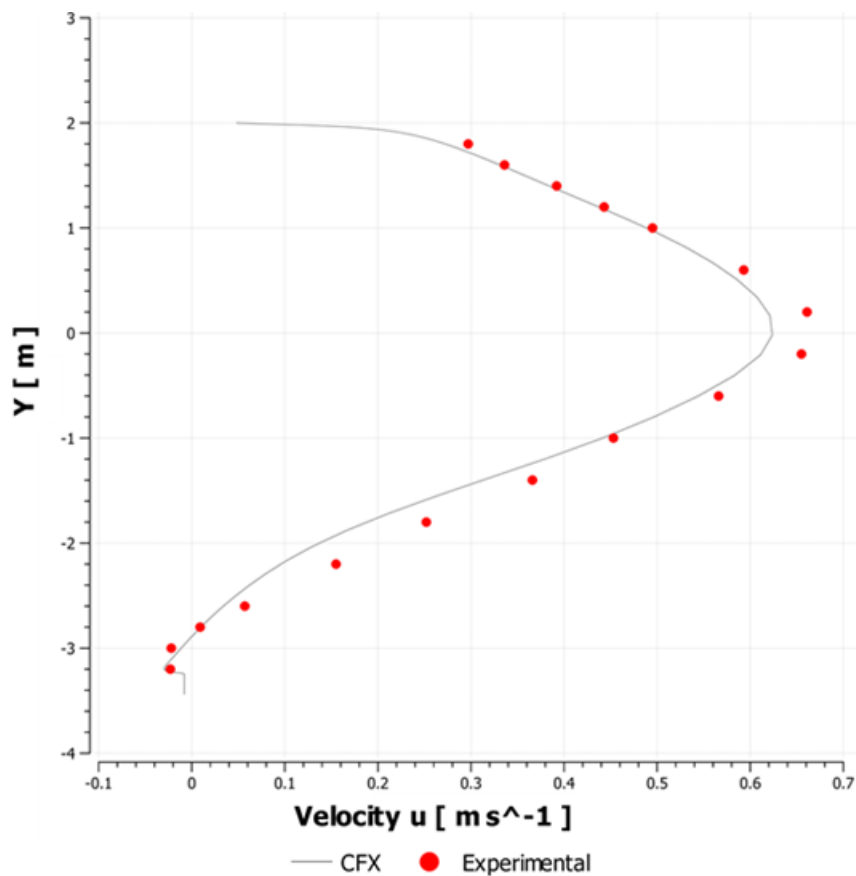
Results Comparison for Ansys Fluent

Figure .47.2: Comparison of X-Velocity at X = 24.4 m



Results Comparison for Ansys CFX

Figure .47.3: Comparison of X-Velocity at X = 24.4 m



VMFL048: Turbulent flow in a 180° Pipe Bend

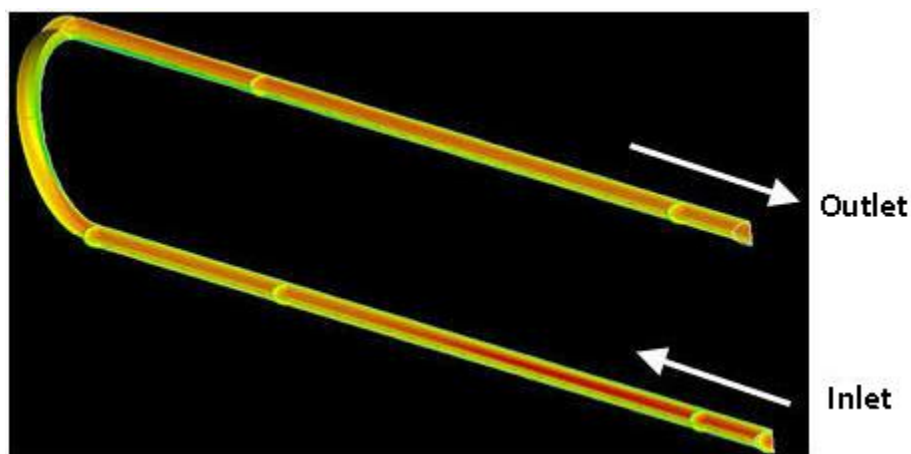
Overview

Reference	T. Takamasa, A. Tomiyama. "Three-dimensional gas-liquid two-phase bubbly flow in a C-shaped tube". <i>NURETH-9</i> . San Francisco, USA. pp. 1-17. 1999.
Solver	Ansys Fluent
Physics/Models	SST model, turbulent flow with separation and reattachment
Input File	VMFL048_pipebend.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow in a 3-D pipe bend as shown in [Figure .48.1: Flow Domain \(p. 159\)](#).

Figure .48.1: Flow Domain



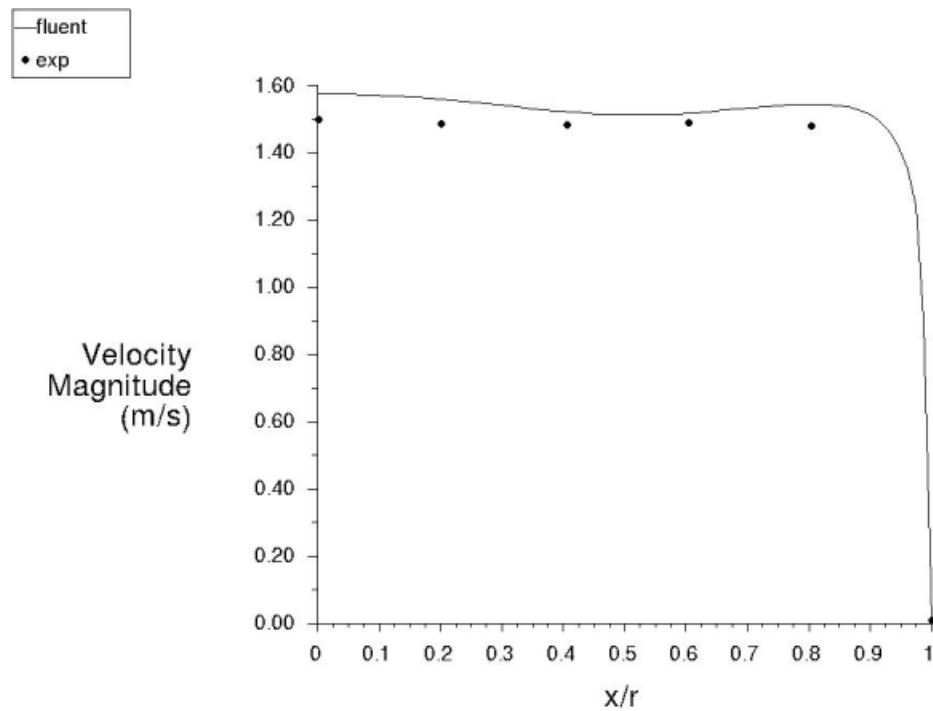
Material Properties	Geometry	Boundary Conditions
Density: 997 kg/m ³ Viscosity: 8.899 X 10 ⁻⁴ kg/m-s	Radius of the pipe = 14 mm Radius of the pipe bend = 125 mm	Velocity profile at inlet with an average velocity of 1.42 m/s

Analysis Assumptions and Modeling Notes

The flow is steady. Symmetry condition is applied on one side of the pipe.

Results Comparison for Ansys Fluent

Figure .48.2: Comparison of Velocity in the Axial Direction at a Section 1.555 m upstream of the Outlet (after the bend)



VMFL049: Combustion in an Axisymmetric Natural Gas Furnace

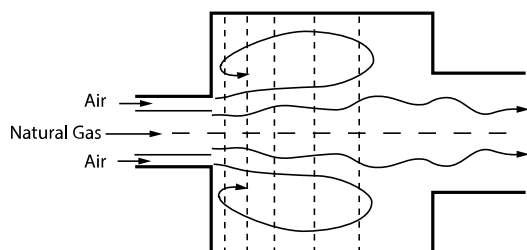
Overview

Reference	K.C. Westbrook, L.F. Dryer, "Simplified reaction mechanism for the oxidation of hydrocarbon fuels in flames", <i>Combustion Science and Technology</i> , Vol 27, pp. 31-43, 1981 T.P. Coffee, "A Lumped Parameter Code for Regenerative Liquid Propellant Guns", <i>BRL-TR-2703</i> , U.S. Army Ballistic Research Laboratory, 1985
Solver	Ansys Fluent
Physics/Models	Turbulent non-premixed combustion, Eddy dissipation model, k- ϵ model
Target File	VMFL049_combustion.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Non-premixed combustion in a natural gas fired furnace is modeled. The axisymmetric flow field is modeled by a 3° cylindrical domain. Fuel jet consists of natural gas modeled as 90% Methane and 10% Nitrogen by mass.

Figure .49.1: Flow Domain



Geometry	Boundary Conditions
Inner diameter of air annulus = 60 mm	Air velocity at inlet = 34 m/s
Outer diameter of air annulus = 100 mm	Fuel velocity at inlet = 7.1 m/s
Diameter of combustion chamber = 500 mm	Wall temperature = 120 °C
Length of chamber = 1700 mm	CH ₄ Mass fraction at inlet = 0.90

Analysis Assumptions and Modeling Notes

The flow is steady. Reactions modeled using Eddy Dissipation Model. The domain is axisymmetric.

Results Comparison for Ansys Fluent

Figure .49.2: Comparison of the Mole Fraction of CH₄ Along the Axis

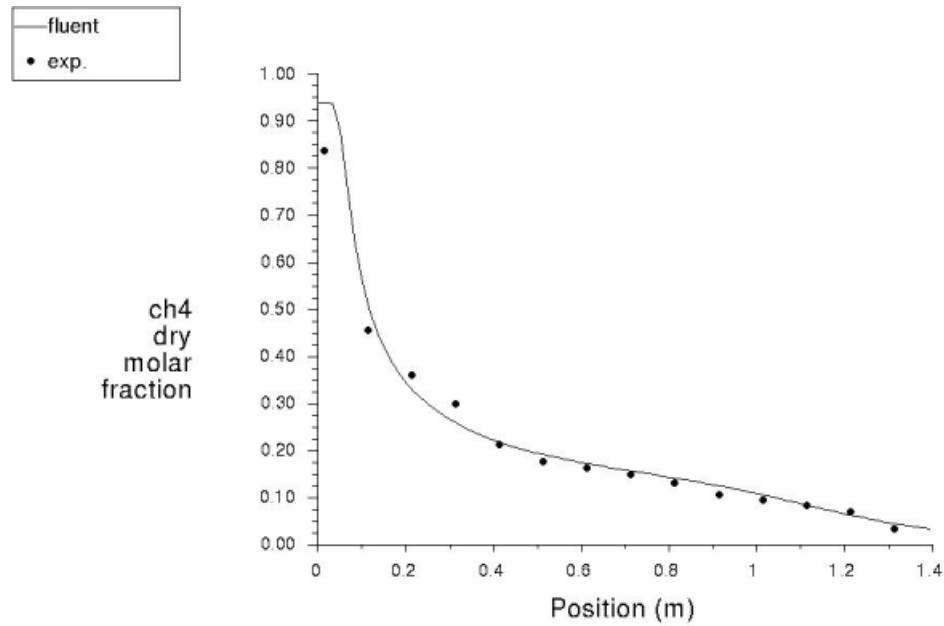
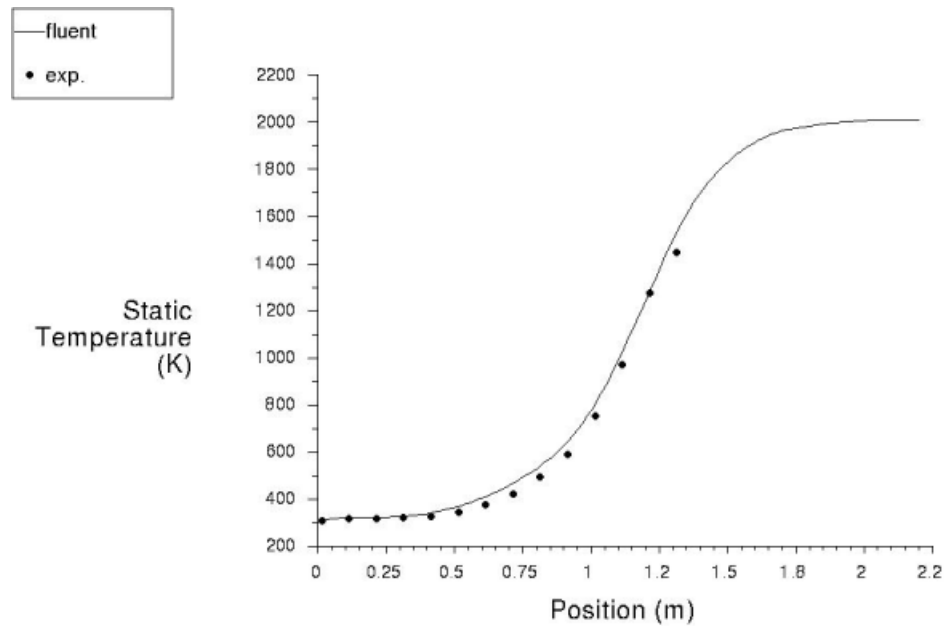


Figure .49.3: Comparison of Temperature Along the Axis



VMFL050: Transient Heat Conduction in a Semi-Infinite Slab

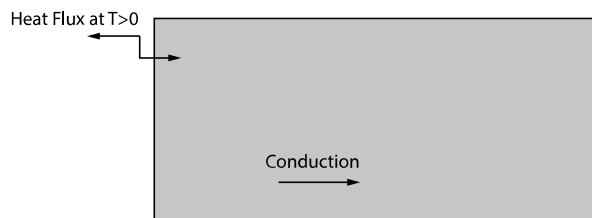
Overview

Reference	F.P. Incropera, D.P. Dewitt, T.L. Bergman, A.S. Lavine, <i>Introduction to Heat Transfer</i> , 5th Edition, Wiley & Sons, p. 287, 2007
Solver	Ansys Fluent
Physics/Models	Transient heat transfer, Conduction
Input File	VMFL050_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Unsteady heat conduction in a thick copper plate is modeled. Initially (at $t = 0$) the plate is at a uniform temperature of 293 K. It is suddenly exposed to a heat transfer at one surface with a constant heat flux of $3 \times 10^5 \text{ W/m}^2$. The temperature distribution after 2 minutes is considered for verification.

Figure .50.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 8995.67 kg/m^3 Specific Heat = 381 J/kg-K Conductivity = 401 W/m-K	Dimensions of the slab: 750 mm X 300 mm	Heat Flux = $3 \times 10^5 \text{ W/m}^2$ on one wall. The opposite wall is adiabatic. Lateral boundaries are modeled as planes of symmetry.

Analysis Assumptions and Modeling Notes

The flow is transient. The dimensions considered here are adequate for the semi-infinite slab assumption. The domain is initialized with a uniform temperature of 293 K corresponding to the condition at time = 0 sec.

Results Comparison for Ansys Fluent

Table .50.1: Comparison of Temperature after 2 Minutes

	Target	Ansys Fluent	Ratio
Temperature of the wall exposed to heat flux at $t = 120$ sec, K	393	392.95	0.9998
Temperature at a point 150 mm from the heat flux wall at $t = 120$ sec, K	318.4	318.41	1.0000

VMFL051: Isentropic Expansion of Supersonic Flow Over a Convex Corner

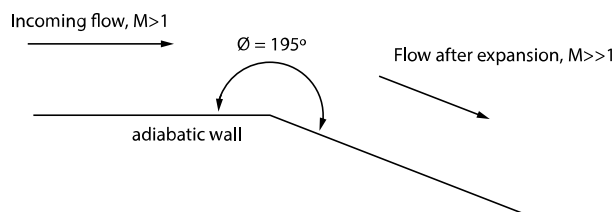
Overview

Reference	John Anderson. <i>Modern Compressible Flow: With Historical Perspective</i> . McGraw-Hill Science/Engineering/Math, 2002
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Compressible, inviscid flow
Input File	VMFL051_FLUENT.cas for Ansys Fluent VMFL051_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Centered expansion of inviscid supersonic flow around a corner is modeled. The expansion results in a change in direction of the flow, a drop in static pressure, and increase in Mach number. The approaching flow is supersonic, with a Mach number of 2.5. The expansion process is reversible and adiabatic.

Figure .51.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: Ideal Gas law Specific Heat = 1006.43 J/kg-K Molecular weight = 28.966	Angle round the corner = 195°	Inlet: Pressure = 202636.9 Pa Mach number = 2.5 Static temperature = 300 K (In CFX, the corresponding velocity is specified). Wall is adiabatic.

Analysis Assumptions and Modeling Notes

The flow is steady, inviscid, and incompressible. Analytic expressions for isentropic expansion can be used to calculate the Mach number downstream of the corner.

Results Comparison for Ansys Fluent

Table .51.1: Comparison of Mach Number Downstream of the Corner, after Expansion

	Target	Ansys Fluent	Ratio
Mach number after expansion	3.2370	3.2316	0.9980

Results Comparison for Ansys CFX

Table .51.2: Comparison of Mach Number Downstream of the Corner, after Expansion

	Target	Ansys Fluent	Ratio
Mach number after expansion	3.237	3.2354	0.9995

VMFL052: Turbulent Natural Convection Inside a Tall Cavity

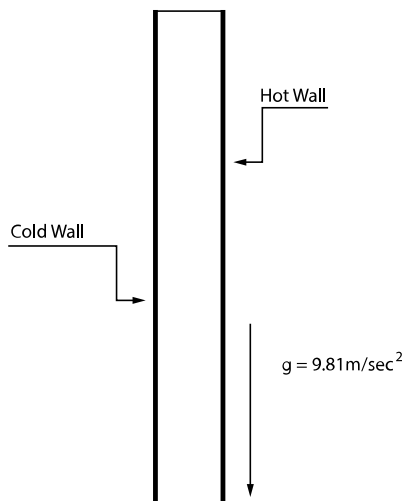
Overview

Reference	P.L. Betts, I.H. Bokhari. "Experiments on turbulent natural convection in an enclosed tall cavity". <i>International Journal of Heat and Fluid Flow</i> , Vol. 21, pp. 675-683, 2000
Solver	Ansys Fluent
Physics/Models	Turbulent flow, buoyancy effects, Boussinesq approximation
Input File	VMFL052_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Natural convection in the turbulent flow field of an enclosed cavity with a length-to-width ratio of 28.6 is modeled. The Rayleigh number is in the turbulent range. The two vertical walls are kept at different temperatures, while the horizontal walls are adiabatic.

Figure .52.1: Flow Domain (not to scale)



Material Properties	Geometry	Boundary Conditions
Density : Boussinesq approximation	Length of the cavity = 2.18 m	Temperature of Cold wall = 288.25 K
Specific Heat = 1005 J/kg-K	Width of the cavity = 0.0762 m	Temperature of Hot wall = 307.85 K
Viscosity = 1.81×10^{-5} kg/m-sec		Top and bottom walls are adiabatic
Molecular weight = 28.966		

Analysis Assumptions and Modeling Notes

The flow is steady and is induced by natural convective heat transfer.

Results Comparison for Ansys Fluent

Figure .52.2: Comparison of Vertical Velocity at $Y/h = 0.05$

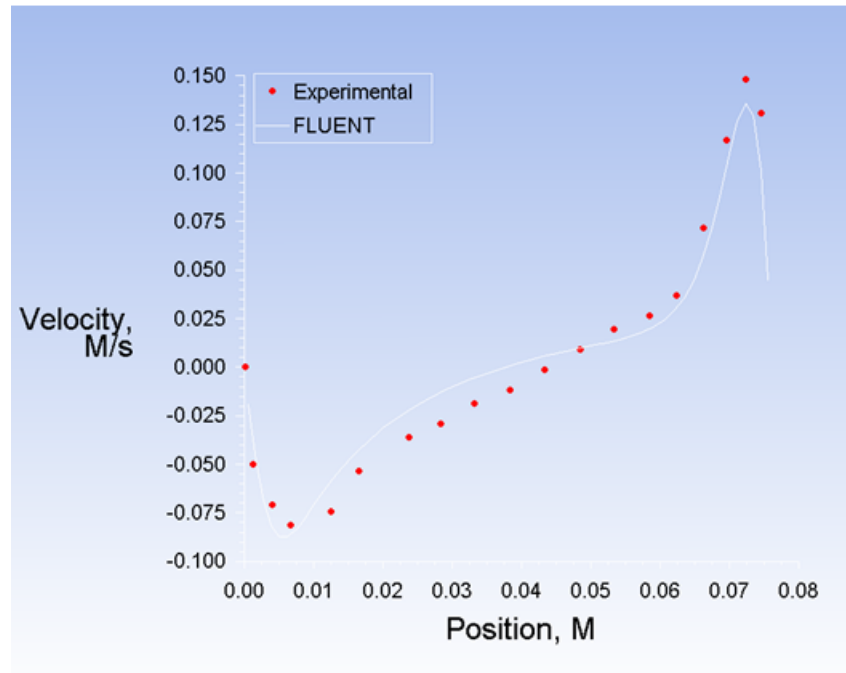
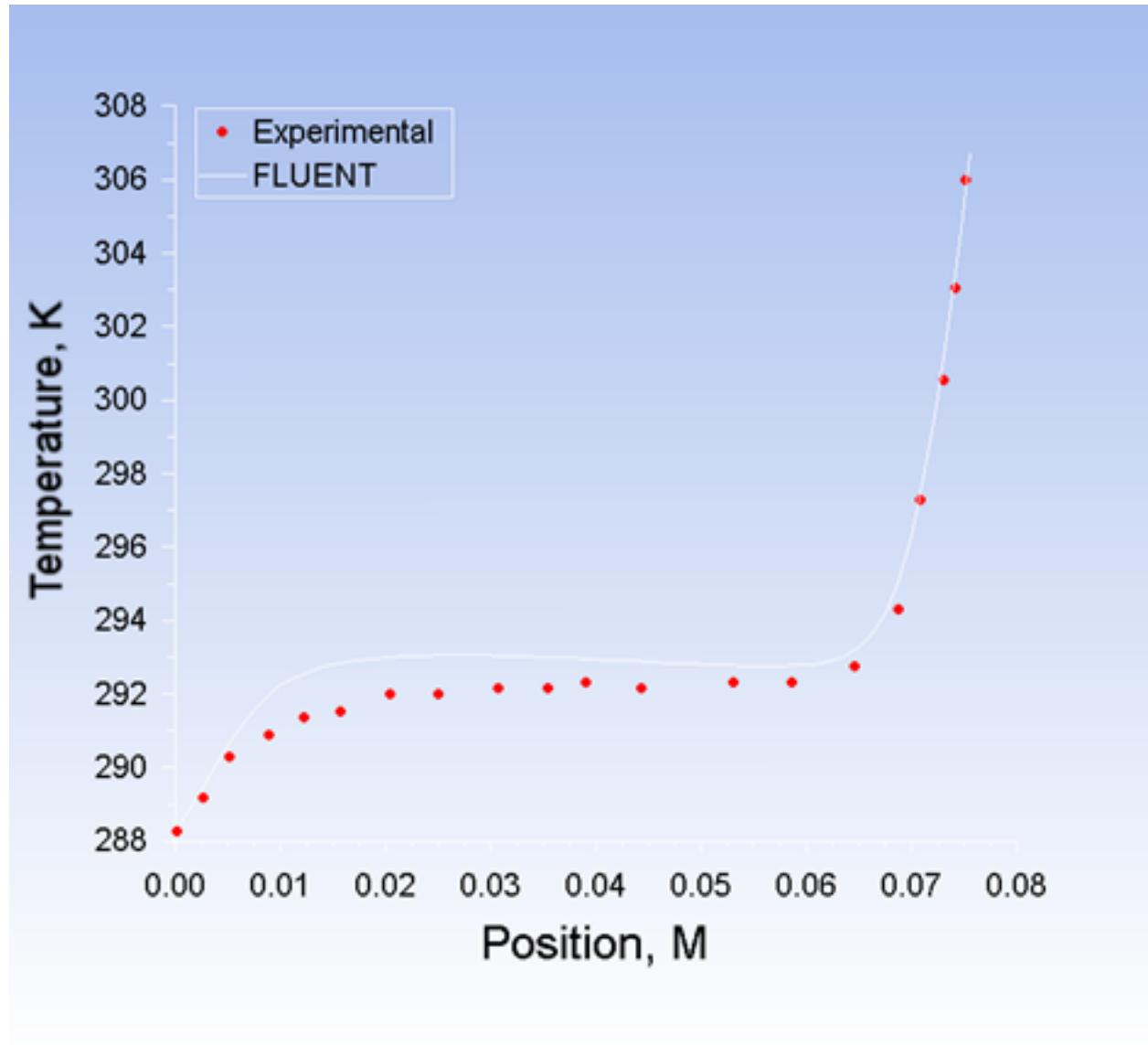


Figure .52.3: Comparison of Temperature at $Y/h = 0.05$ 

VMFL053: Compressible Turbulent Mixing Layer

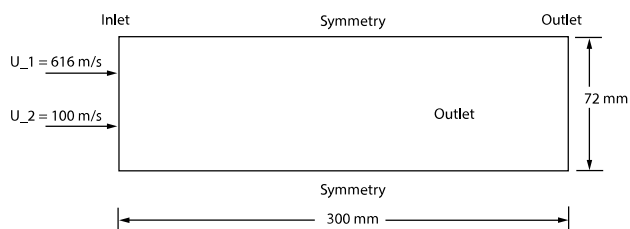
Overview

Reference	S.G. Goebel, J.C. Dutton. "Experimental Study of Compressible Turbulent Mixing Layers". <i>AIAA Journal</i> , Vol. 29, pp. 538-546, 1991
Solver	Ansys Fluent
Physics/Models	Turbulence: RNG k- ϵ model, compressible, energy equation
Input File	VMFL053_FLUENT.cas for Ansys Fluent
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Two streams of air are mixed in a rectangular tunnel. The length of the computational domain is chosen such that the local Reynolds number at the exit of the test section, which is based on the velocity difference between the streams and the mixing layer thickness, is greater than 100,000. This is the Reynolds number needed for the complete development of the mixing layer.

Figure .53.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Air: Density: Ideal Gas Specific Heat: 1006.43 J/kg-K Thermal Conductivity: 0.0242 W/m-K Viscosity: 1.4399e-05 kg/m-s	Dimensions of the domain: 300 mm X 72 mm	Primary Stream (1): Total Pressure = 487 kPa Static Pressure = 36 kPa Total Temperature = 360 K Mach Number = 2.35 Turbulent Kinetic Energy = 74 m ² /s ² Turbulent Dissipation Rate = 62,300 m ² /s ³ Secondary Stream (2): Total Pressure = 37.6 kPa Static Pressure = 36 kPa

Material Properties	Geometry	Boundary Conditions
		Total Temperature = 290 K Mach Number = 0.36 Turbulent Kinetic Energy = $226 \text{ m}^2/\text{s}^2$ Turbulent Dissipation Rate = $332,000 \text{ m}^2/\text{s}^3$

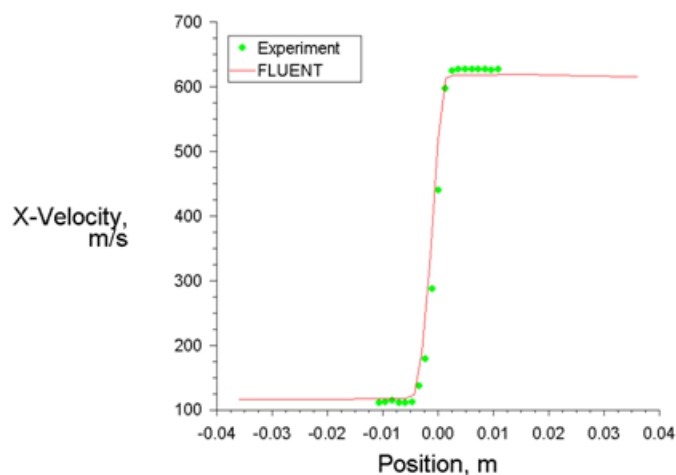
Analysis Assumptions and Modeling Notes

The flow is steady, turbulent, and compressible. The RNG k - ϵ model is used for turbulence.

Results Comparison

The velocity profiles as the mixing layer evolves are compared with the experimental data.

Figure .53.2: X Velocity Profiles at x = 50 mm



VMFL054: Laminar flow in a Trapezoidal Cavity

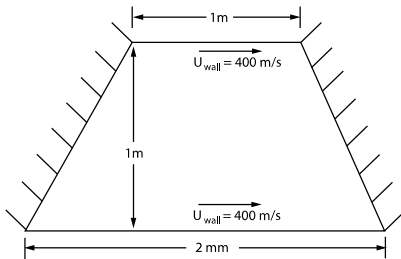
Overview

Reference	J.H. Darr, S.P. Vanka. "Separated Flow in a Driven Trapezoidal Cavity". <i>Phys. Fluids A</i> , Vol. 3, pp. 385-392, 1991.
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Viscous flow, driven by a moving walls
Input File	VMFL054_FLUENT.cas for Ansys Fluent VMFL054_CFX.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar flow induced by the motion of the top wall of a trapezoidal cavity. The top and bottom walls move but the side walls are stationary. The height of the cavity h is 1 m. The widths of the top and bottom walls are 1 m and 2 m, respectively (Figure .38.1: Flow Domain (p. 173)).

Figure .38.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m^3 Viscosity = 1 kg/m-s	Height of cavity = 1 m Width of the bottom base = 2 m Width of the top base = 1 m	Velocity of the base walls = 400 m/s Other walls are stationary

Analysis Assumptions and Modeling Notes

The flow is steady. A pressure based solver is used. A triangular mesh of 4016 cells is used to discretize the domain.

Results Comparison for Ansys Fluent

The u -velocity profile at the vertical centerline of the cavity and the v -velocity profile at the horizontal centerline of the cavity are compared to Darr and Vanka results. Velocity is normalized by velocity of the moving wall.

Figure .38.2: Normalized u -Velocity at the Horizontal Centerline of the Cavity

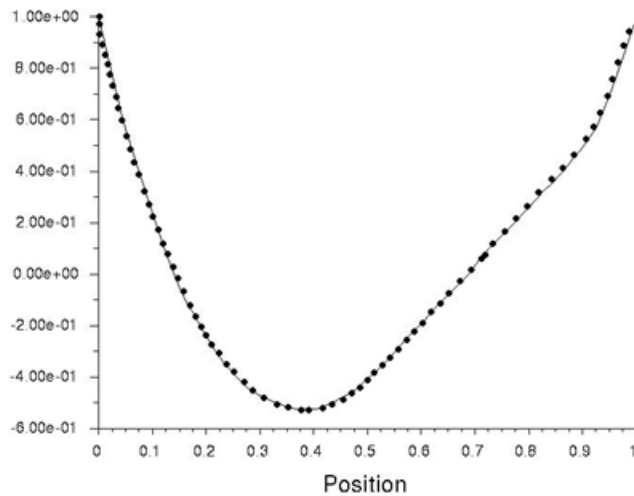
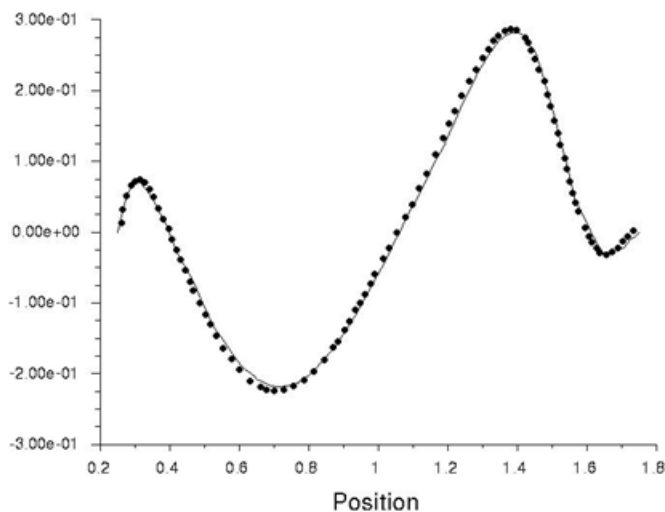


Figure .38.3: Normalized v -Velocity at the Vertical Centerline of the Cavity



Results Comparison for Ansys CFX

The u -velocity profile at the vertical centerline of the cavity and the v -velocity profile at the horizontal centerline of the cavity are compared to Darr and Vanka results. Velocity is normalized by velocity of the moving wall.

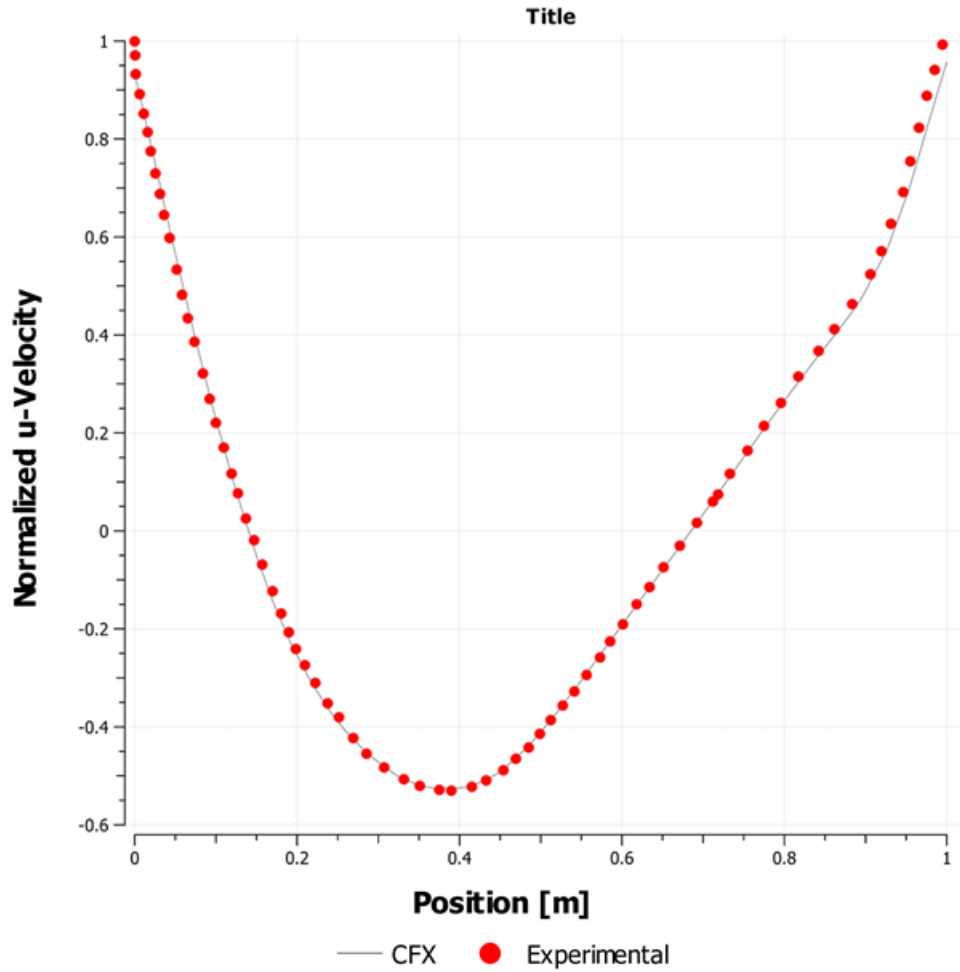
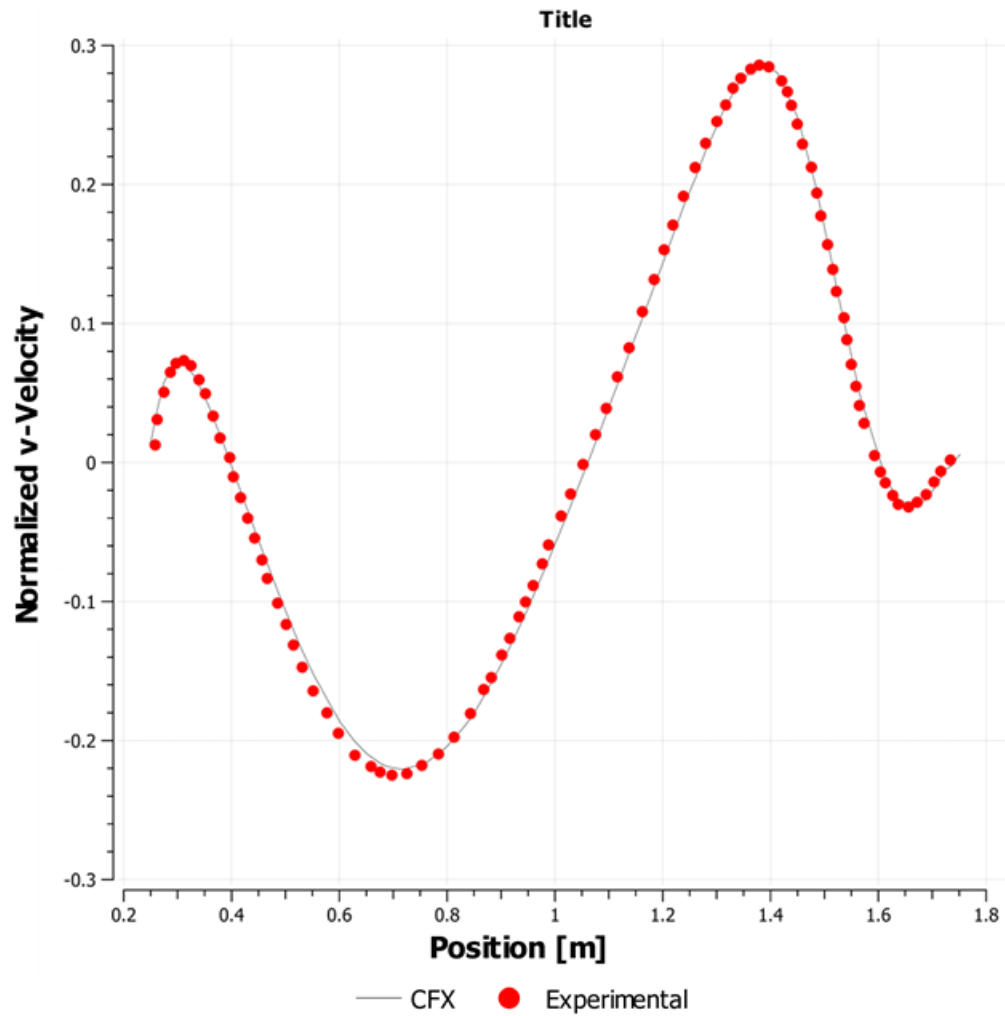
Figure .38.4: Normalized u-Velocity at the Horizontal Centerline of the Cavity

Figure .38.5: Normalized v-Velocity at the Vertical Centerline of the Cavity

VMFL055: Transitional Recirculatory Flow inside a Ventilation Enclosure

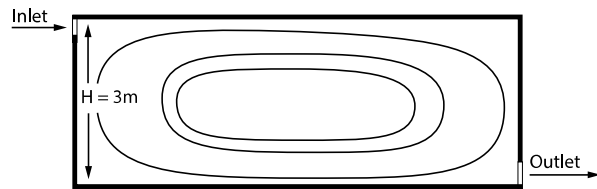
Overview

Reference	A. Restivo, <i>Turbulent Flow in Ventilated Rooms</i> , Ph.D. Thesis, University of London, UK, 1979.
Solver	Ansys Fluent
Physics/Models	Transitional turbulence modeling (k-kl model)
Input File	VMFL055_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow inside an enclosure similar to a ventilated room is modeled. The flow field is transitional and dominated by recirculation. Reynolds number is based on the inlet dimension and is around 5000.

Figure .55.1: Flow Domain



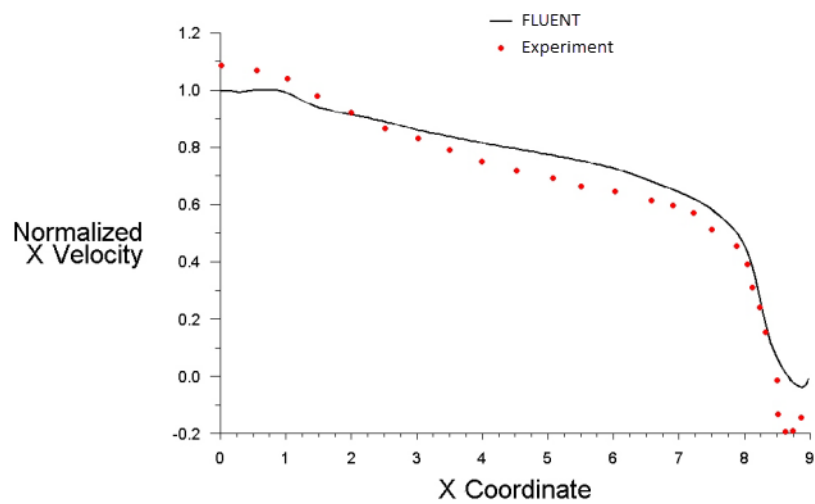
Material Properties	Geometry	Boundary Conditions
Density : 1.225 kg/m ³ Viscosity = 1.7894 X 10 ⁻⁵ kg/m-sec	Height of the enclosure (H) = 3 m Length of the enclosure = 9 m (3 H) Inlet : 0.056 H Outlet : 0.16 H	Inlet velocity = 0.45 m/S

Analysis Assumptions and Modeling Notes

The flow is modeled using transitional turbulence models.

Results Comparison for Ansys Fluent

Figure .55.2: Comparison of X Velocity at Y = 2.916 m



VMFL056: Combined Conduction and Radiation in a Square Cavity

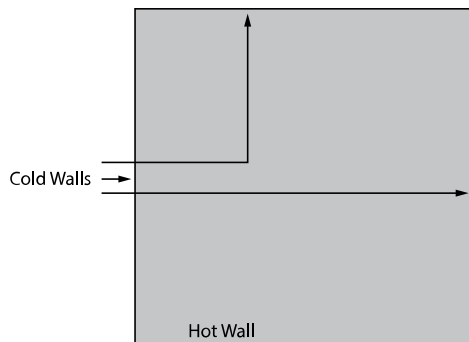
Overview

Reference	D.R. Rousse, G. Gautier, J.F. Sacadura. "Numerical predictions of two-dimensional conduction, convection, and radiation heat transfer. II. Validation", <i>International Journal of Thermal Sciences</i> , Vol 39, pp. 332-353, 2000
Solver	Ansys Fluent
Physics/Models	Radiation modeling; discrete ordinate model
Input File	VMFL056_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Coupled conduction and radiation is modeled in a square enclosure. The material properties are set to model a condition corresponding to the Conduction-Radiation parameter $N = 1.0$. Scattering coefficient of the medium is 0. Steady state heat transfer is modeled. One wall of the square cavity is kept at a higher temperature than the other 3 walls.

Figure .56.1: Flow Domain



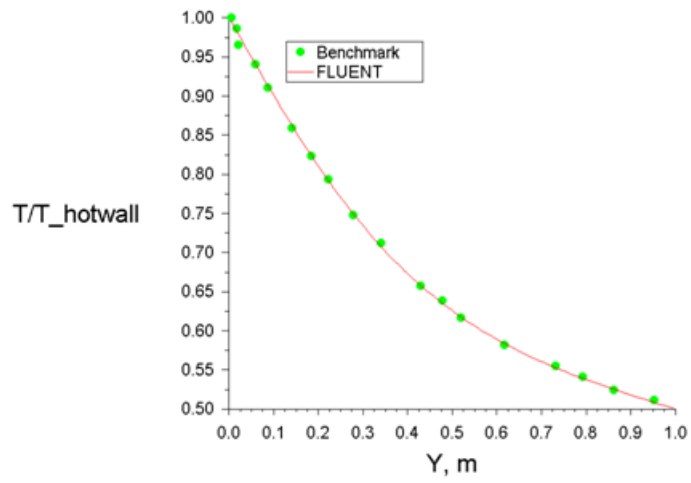
Material Properties	Geometry	Boundary Conditions
Thermal Conductivity = 1 W/m-K	Dimensions of the domain: 1 m X 1 m	Temperature of the hot wall = 100 K
Absorption Coefficient = 0.228/m		Temperature of the cold walls = 50 K

Analysis Assumptions and Modeling Notes

The material properties are set to model the desired conduction-radiation fraction. Radiative heat flux is only a small fraction of the total heat flux.

Results Comparison for Ansys Fluent

Figure .56.2: Comparison of Non-Dimensional Temperature at X = 0.5 m



VMFL057: Radiation and Conduction in Composite Solid Layers

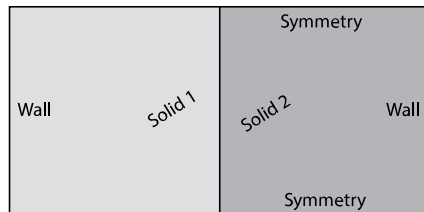
Overview

Reference	C.M. Spuckler, R. Siegel. "Two-Flux and Diffusion Methods for Radiative Transfer in Composite Layers". <i>Journal of Heat Transfer</i> , Vol 118, pp. 218-222, 1996
Solver	Ansys Fluent
Physics/Models	Radiation modeling with DO model, participating medium with gray-band absorption
Input File	VMFL057_FLUENT . cas for Ansys Fluent
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Heat transfer by conduction and radiation is modeled in a composite solid domain consisting of two layers. Both the layers participate in radiation. The two layers are separated by a semi-transparent wall. The upstream and downstream sides of the domain are subjected to convective as well as radiative heat transfer.

Figure .57.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Solid 1: Density = 2719 kg/m ³ Specific Heat = 871 J/kg-k Thermal Conductivity = 5.67 W/m-K Absorption Coefficient: gray-band Refractive Index = 1.5	Dimensions of the domain: 2 m X 1 m (the two solid zones are of equal length)	Left-most wall: Convective Heat Transfer Coefficient = 56.7 W/m ² K free stream temperature = 1000K Semi-transparent
Solid 2:		Right-most wall:

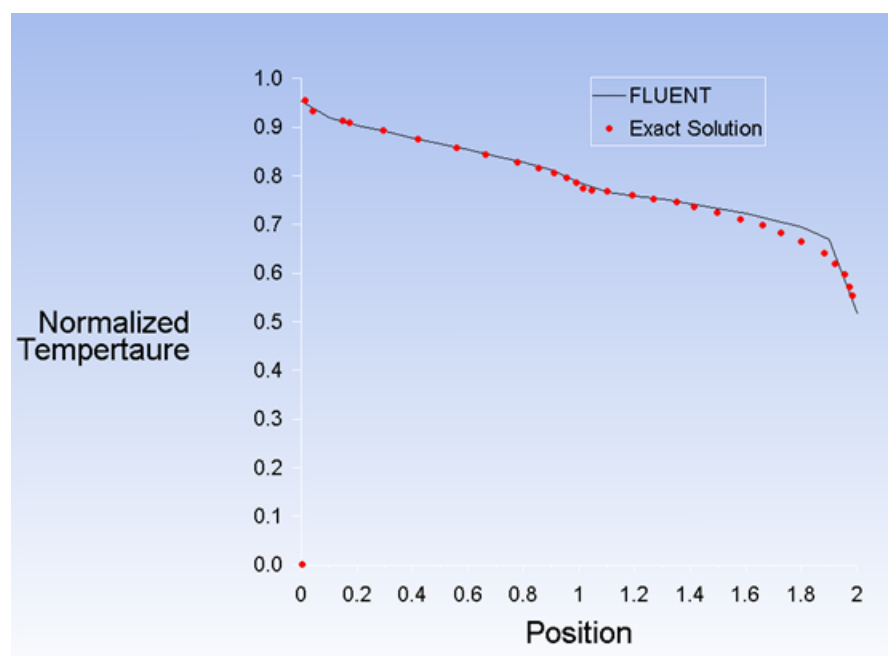
Material Properties	Geometry	Boundary Conditions
Density = 2719 kg/m ³ Specific Heat = 871 J/kg-k Thermal Conductivity = 5.67 W/m-K Absorption Coefficient: gray-band Refractive Index = 3		Convective Heat Transfer Coefficient = 56.7 W/m ² K free stream temperature = 250K Semi-transparent

Analysis Assumptions and Modeling Notes

Transverse boundaries of the domain are modeled as planes of symmetry.

Results Comparison for Ansys Fluent

Figure .57.2: Comparison temperature distribution along Y = 0.5 m



VMFL058: Turbulent Flow in an Axisymmetric Diffuser

Overview

Reference	R. S. Azad, S. Z. Kassab. "Turbulent Flow in a Conical Diffuser: Overview and Implications". <i>Phys. Fluids A</i> Vol 1, pg. 564, 1989
Solver	Ansys Fluent
Physics/Models	Turbulent flow with adverse pressure gradient
Input File	VMFL058_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Fully developed turbulent flow is modeled in an axisymmetric diffuser. The flow is fully developed at the inlet to the diffuser.

Figure .58.1: Flow Domain



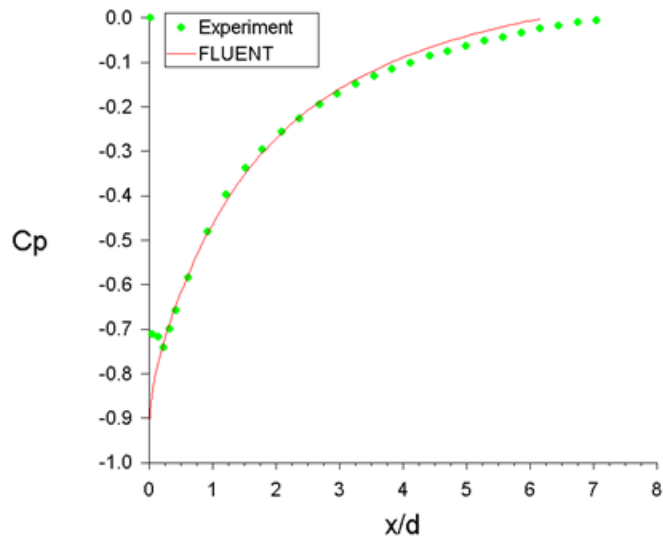
Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³ Viscosity = 1.64 X 10 ⁻⁵ kg/m-s	Included angle for the divergent section = 8° Length of Inlet section (straight) = 6 m Inlet radius = 1 m Outlet radius = 2 m	Fully developed turbulent profile at inlet with an average velocity = 1 m/s

Analysis Assumptions and Modeling Notes

Steady turbulent flow.

Results Comparison for Ansys Fluent

Figure .58.2: Comparison of Pressure Coefficient along the Divergent Diffuser Wall



VMFL059: Conduction in a Composite Solid Block

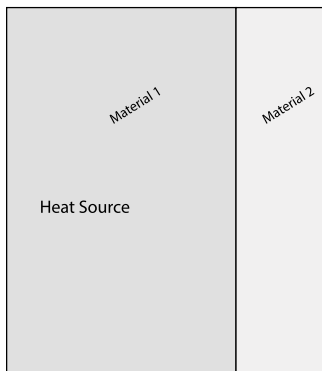
Overview

Reference	F.P. Incropera, D.P. Dewitt. <i>Fundamentals of Heat and Mass Transfer</i> . 5th Edition, pg. 117, 2006.
Solver	Ansys Fluent
Physics/Models	Conduction with heat source
Input File	VMFL059_FLUENT.cas for Ansys Fluent VMFL059_Serial.def for CFX-Solver
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Heat conduction in a plane wall formed as composite of two materials is modeled. One of the materials has a uniform volumetric heat generation source while the other material has an outer surface exposed to convective cooling.

Figure .59.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Material 1: Density = 2702 kg/m ³ Specific Heat = 903 J/kg-K Thermal Conductivity = 75 W/m-K Material 2: Density = 8978 kg/m ³	Dimensions of the domain: 0.07 m X 0.08 m Thickness of slab 1 (material 1) = 0.05 m	Left-most wall: Adiabatic Right-most wall: Convective, with Heat Transfer Coefficient = 1000 W/m ² K and free stream temperature = 303 K Other boundaries are adiabatic walls.

Material Properties	Geometry	Boundary Conditions
Specific Heat = 381 J/kg-k Thermal Conductivity = 150 W/m-K		

Analysis Assumptions and Modeling Notes

Contact resistance between the slabs is neglected.

Results Comparison

Table .59.1: Comparison Temperatures on the Side Walls

	Target	Ansys Fluent	Ratio
Temperature of the cooled wall, K	378	378.14	1.0004
Temperature of the adiabatic wall on extreme left side, K	413	413.17	1.0004

VMFL060: Transitional Supersonic Flow Over a Rearward Facing Step

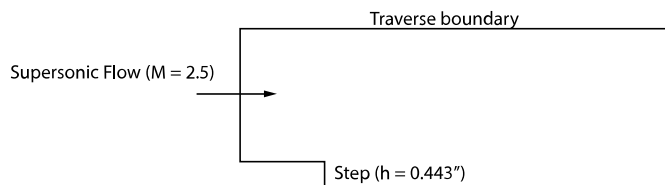
Overview

Reference	H.E. Smith. "The Flow Field and Heat Transfer Downstream of a Rearward Facing Step in Supersonic Flow". ARL 67-0056, Aerospace Research Laboratories. Ohio, USA.
Solver	Ansys Fluent
Physics/Models	Compressible Flow, Transitional turbulence modeling (Transition SST model)
Input File	VMFL060_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Supersonic flow with an inlet Mach number 2.5 past a backward facing step is modeled. Key features of the flow field include sudden expansion, free shear layer, recirculation zone, and oblique shock. Reynolds number of the flow (based on step height) is in the transitional range.

Figure .60.1: Flow Domain



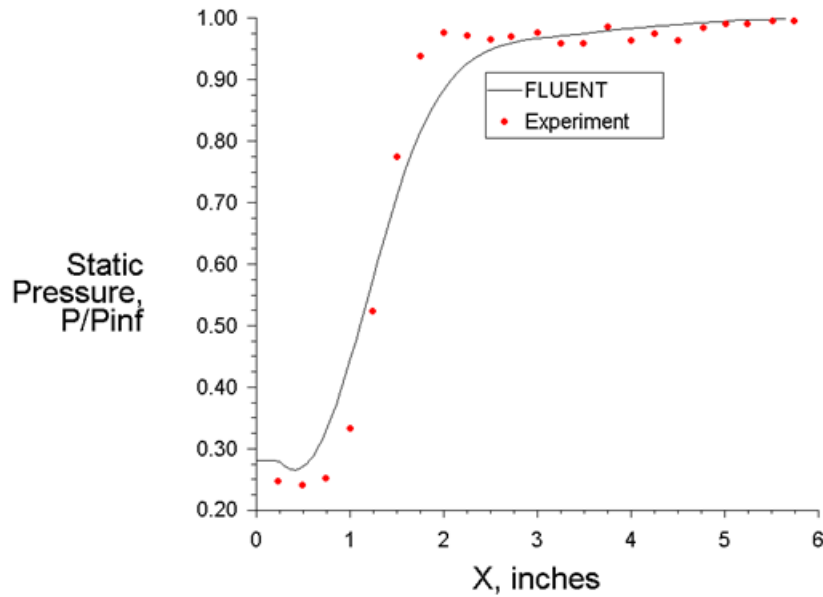
Material Properties	Geometry	Boundary Conditions
Density: Ideal gas Viscosity = 1.7894×10^{-5} kg/m-sec	Step Height = 0.443 in Inlet: 4 in upstream of the step Outlet: 12 in downstream of the step Transverse (top) boundary: 6.25 in above the step.	Total pressure at inlet = 227527 Pa Static Pressure at Inlet = 13316.6 Total temperature at inlet = 344.44 K Transverse boundary modeled as farfield Walls are modeled as adiabatic

Analysis Assumptions and Modeling Notes

The flow is modeled using transitional turbulence models.

Results Comparison for Ansys Fluent

Figure .60.2: Comparison of Non-Dimensionalized Static Pressure along the Stepped Wall Downstream of the Corner



VMFL061: Surface to Surface Radiative Heat Transfer Between Two Concentric Cylinders

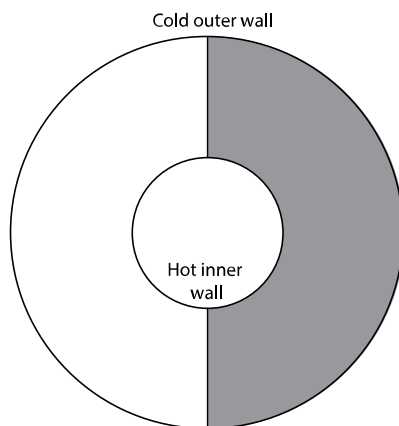
Overview

Reference	F.P. Incropera, D.P. Dewitt. <i>Fundamentals of Heat and Mass Transfer</i> . 4th Edition, John Wiley & Sons, Inc., New York City, New York, 1996.
Solver	Ansys Fluent
Physics/Models	Radiation Modeling (S2S Model)
Input File	VMFL061_FLUENT .cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Radiative heat transfer between two cylindrical surfaces forming a concentric annulus is modeled. There is no participating medium. Due to symmetry only the shaded portion of the domain in [Figure .61.1: Flow Domain \(p. 189\)](#) is modeled.

Figure .61.1: Flow Domain



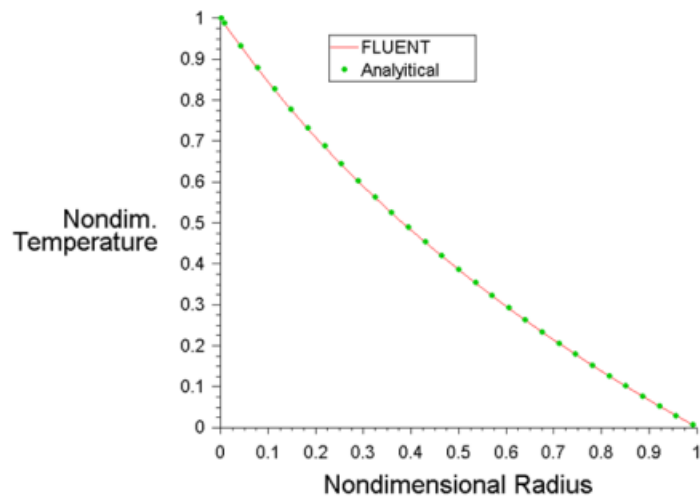
Geometry	Boundary Conditions
Radius of outer wall = 0.04625 m	Temperature of inner wall = 700 K
Radius of inner wall = 0.0178 m	Temperature of outer wall = 300 K

Analysis Assumptions and Modeling Notes

Because there is no flow of mass, only the energy equation is solved. Radiation models are used for the simulation. Heat transfer is purely due to radiation between the two surfaces.

Results Comparison for Ansys Fluent

Figure .61.2: Comparison of Temperature Variation along Radius



VMFL062: Fully Developed Turbulent Flow Over a “Hill”

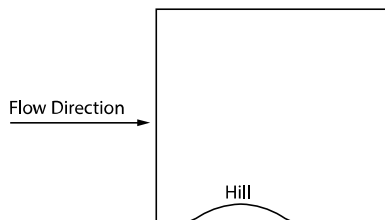
Overview

Reference	V. Baskaran, A.J. Smits, P.N. Joubert, "A turbulent flow over a curved hill Part 1. Growth of an internal boundary layer". <i>Journal of Fluid Mechanics</i> , Vol. 182, pp. 47-83, 1987
Solver	Ansys Fluent
Physics/Models	Low Re k-e Model
Input File	VMFL062_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Flow over a “hill” geometry with separation and reattachment is modeled. Fully developed turbulent profile is specified at the inlet.

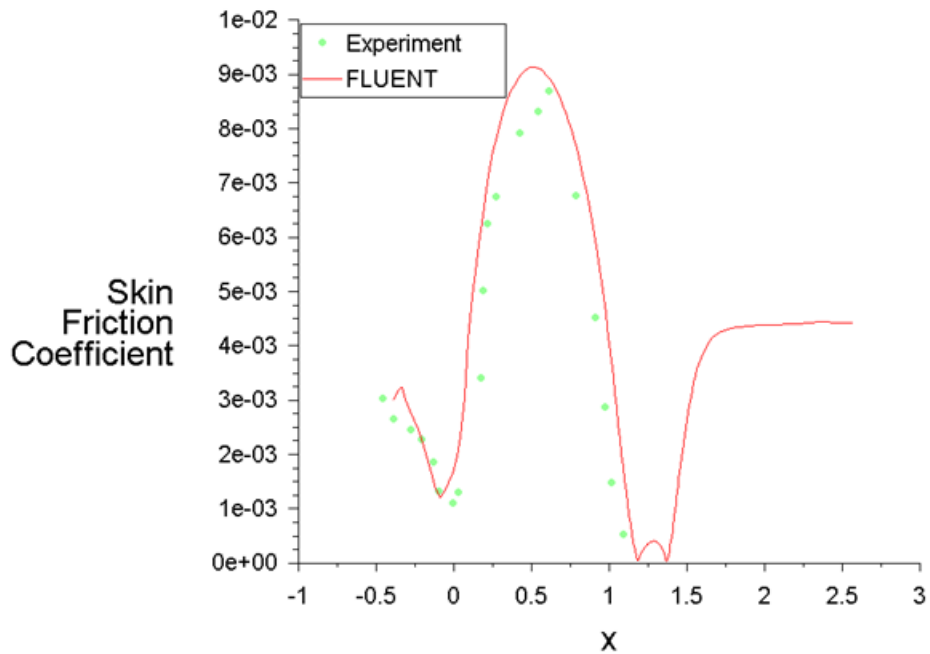
Figure .62.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density : 1 kg/m ³ Viscosity = 7.5188e-07 kg/m-s	Height of the hill = 20.5 mm	Fully developed profiles are specified at the inlet for (i) Velocity, (ii) Turbulent kinetic Energy, and (iii) Eddy dissipation rate. Average velocity at inlet = 1 m/s

Results Comparison for Ansys Fluent

Figure .62.2: Comparison of Skin Friction along the Wall



VMFL063: Separated Laminar Flow Over a Blunt Plate

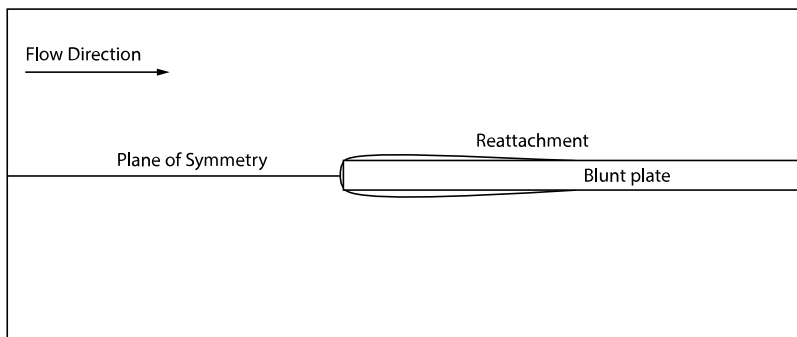
Overview

Reference	J.C. Lane, R.I. Loehrke, "Leading Edge Separation from a Blunt Plate at Low Reynolds Number", <i>Transactions of ASME</i> , Vol. 102, pp. 494-496, 1980
Solver	Ansys Fluent, Ansys CFX
Physics/Models	Laminar flow, high resolution numerical models
Input File	VMFL063_FLUENT.cas for Ansys Fluent VMFL063_Serial.def for Ansys CFX
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The flow separation over a blunt leading edge in laminar flow is modeled. The flow separates and reattaches along the plate. The reattachment length predicted by the solvers is validated against experimental results. Due to symmetry, only half of the domain shown in [Figure .63.1: Flow Domain \(p. 193\)](#) is modeled. The Reynolds number based on plate thickness is 260.

Figure .63.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 1.7894 X 10 ⁻⁵ kg/m-s	Thickness of the plate, 2t = 90 mm Length of the plate = 1500 mm	Velocity at inlet = 0.0517 m/s

Results Comparison for Ansys Fluent

Table .63.1: Comparison of Reattachment Length

	Target	Ansys Fluent	Ratio
Non-dimensionalized Reattachment length ($L_R/2t$)	4.0	4.16	1.04

Results Comparison for Ansys CFX

Table .63.2: Comparison of Reattachment Length

	Target	Ansys Fluent	Ratio
Non-dimensionalized Reattachment length ($L_R/2t$)	4.0	4.05	1.01

VMFL064: Low Reynolds Number Flow in a Channel with Sudden Asymmetric Expansion

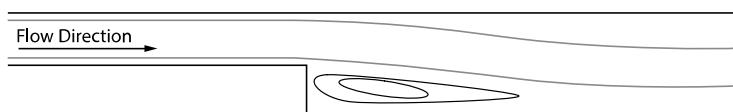
Overview

Reference	B. Armaly, F. Durst, J. Pereira, B. Schöning. "Experimental and theoretical investigation of a backward-facing step". <i>Journal of Fluid Mechanics</i> , Vol 127, pg. 473, 1983 C.J. Freitas. "Perspective: Selected Benchmarks from Commercial CFD Codes". <i>Journal of Fluids Engineering</i> . Vol. 117, pg. 208, 1995
Solver	Ansys Fluent
Physics/Models	Laminar flow, Separation, and Reattachment
Input File	VMFL064_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Laminar flow in a channel with a backward facing step expansion is modeled. The channel section upstream of the expansion is long enough to ensure fully developed laminar profile. The reattachment length predicted by the solvers is validated against experimental results. Reynolds number based on D (equal to twice the channel height at inlet) is 200. The domain extends to about 40 times the step-height upstream and over 20 times the step-height downstream.

Figure .64.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 1.5 X 10 ⁻⁵ kg/m-s	Step height, $s = 4.9$ mm Channel height at inlet = 5.2 mm Length of inlet section = 200 mm Length downstream of the step = 100 mm	Velocity at inlet = 0.288462 m/s No-slip condition at the walls

Analysis Assumptions and Modeling Notes

The flow is fully developed before the step. Reattachment length is measured from the reversal of the sign of the wall shear along the flow direction.

Results Comparison for Ansys Fluent

Table .64.1: Comparison of Reattachment Length

	Target	Ansys Fluent	Ratio
Non-dimensionalized Reattachment length (L_R /Step-height)	5.0	4.91	0.982

VMFL065: Swirling Turbulent Flow Inside a Diffuser

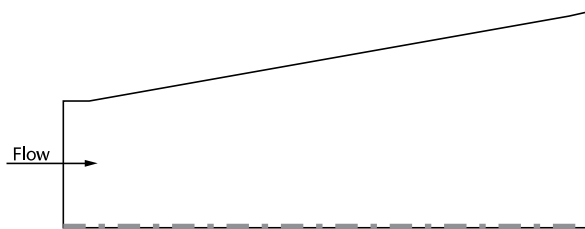
Overview

Reference	P.D. Clausen, S.G. Koh, D.H. Wood. "Measurements of a Swirling Turbulent Boundary Layer Developing in a Conical Diffuser." <i>Experimental Thermal and Fluid Science</i> . Vol. 6, pg. 39-48, 1993
Solver	Ansyst Fluent
Physics/Models	Turbulent flow, swirl velocity, Reynolds stress model for turbulence
Input File	VMFL065_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Turbulent flow with a strong swirl component is modeled in an axisymmetric diffuser. The swirl component of the velocity has a dominant effect on the flow field inside the diffuser.

Figure .65.1: Flow Domain



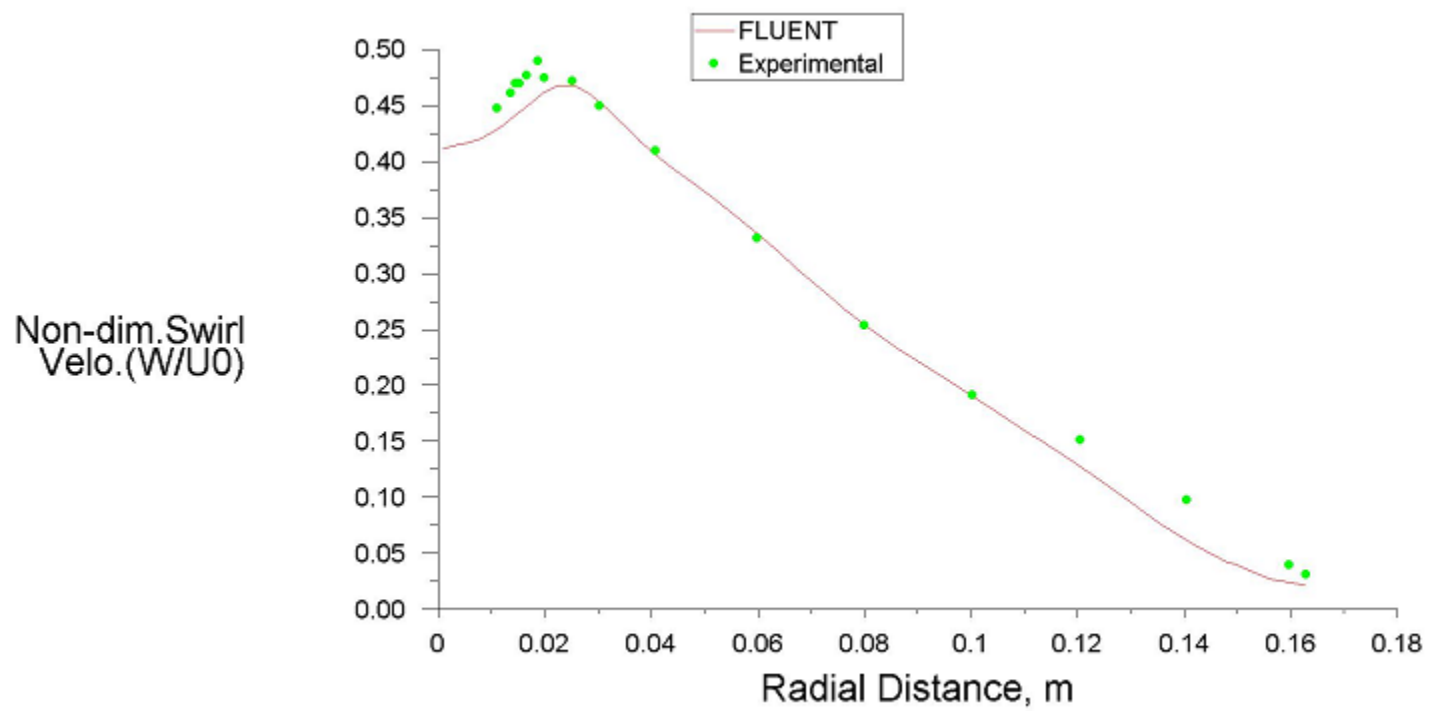
Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 1.293 X 10 ⁻⁶ kg/m-s	Length of the straight inlet section = 25 mm Length of the diffuser (divergent section) = 510 mm Inlet Diameter = 260 mm Outlet Diameter = 440 mm	Fully developed turbulent profile for velocity, k and ϵ at inlet (with average axial inlet velocity = 1 m/s) No-slip condition at the walls

Analysis Assumptions and Modeling Notes

RS model is used for turbulence due to the strong swirl component.

Results Comparison for Ansys Fluent

Figure .65.2: Comparison of Swirl Velocity at $X = 0.175$ m



VMFL066: Radiative Heat Transfer in a Rectangular Enclosure with Participating Medium

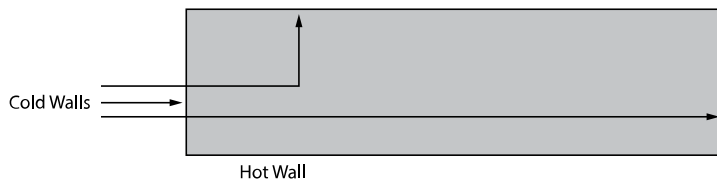
Overview

Reference	G.D Raithby, E.H. Chui. "A Finite Volume Method for Predicting a Radiant Heat Transfer in Enclosures with Participating Media". <i>Journal of Heat Transfer</i> . Volume 112, pp. 415-423, 1990
Solver	Ansys Fluent
Physics/Models	Radiation modeling, discrete ordinate model
Input File	VMFL066_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Two dimensional radiative heat transfer in an enclosure with one hot wall and three cold walls at equal temperature is modeled. The enclosure is a rectangular cavity with a length-to-width ratio of 5. For the problem being considered, $\sigma^s L_y = 1.0$, where σ^s is the scattering coefficient and L_y is the normal distance between the hot wall and the cold wall opposite to it.

Figure .66.1: Flow Domain



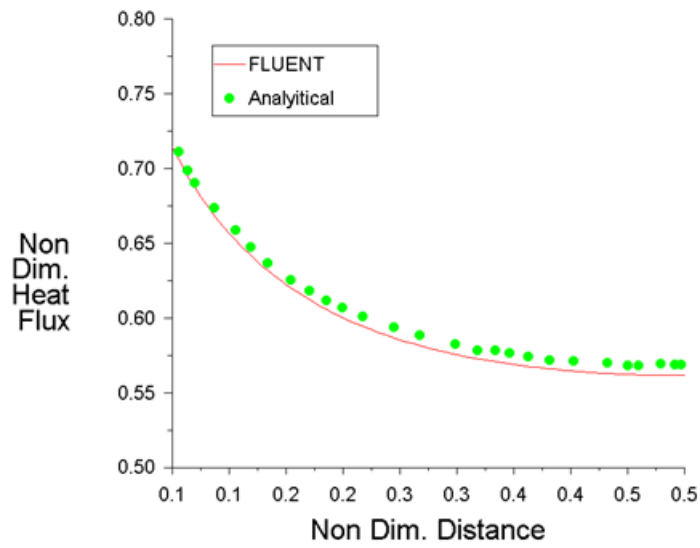
Material Properties	Geometry	Boundary Conditions
Scattering coefficient = 0.5/m	Dimensions of the domain: 10 m x 2 m	Temperature of the hot wall = 200 K Temperature of the cold walls = 100 K

Analysis Assumptions and Modeling Notes

Isotropic scattering and radiative equilibrium are assumed.

Results Comparison for Ansys Fluent

Figure .66.2: Comparison of Non-Dimensional Heat Flux along the Hot Wall



VMFL067: Boiling in a Pipe-Critical Heat Flux

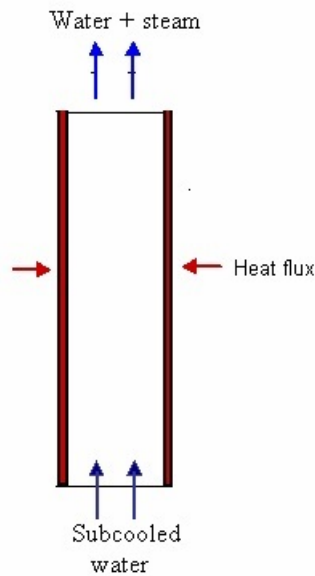
Overview

Reference	N. Hoyer, "Calculation of dryout and post-dryout heat transfer for tube geometry", <i>International Journal of Multiphase Flow</i> , Vol 24, pp. 319-334, 1998
Solver	Ansys Fluent
Physics/Models	Multiphase, heat and mass transfer, boiling
Input File	VMFL067_FLUENT . cas for Ansys Fluent
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test case considers boiling near a heated wall of a vertical pipe. The effects of critical heat flux and post-dryout heat transfer are modeled in a vertical circular pipe. The outer wall of the pipe is heated with a constant heat flux.

Figure .67.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Steam-Water 2-phase Flow: Water: continuous phase	Radius of the pipe = 5 mm Height of the pipe = 7 m	Mass flux at inlet = 1495 kg/m ² /s Heat transfer at the wall =

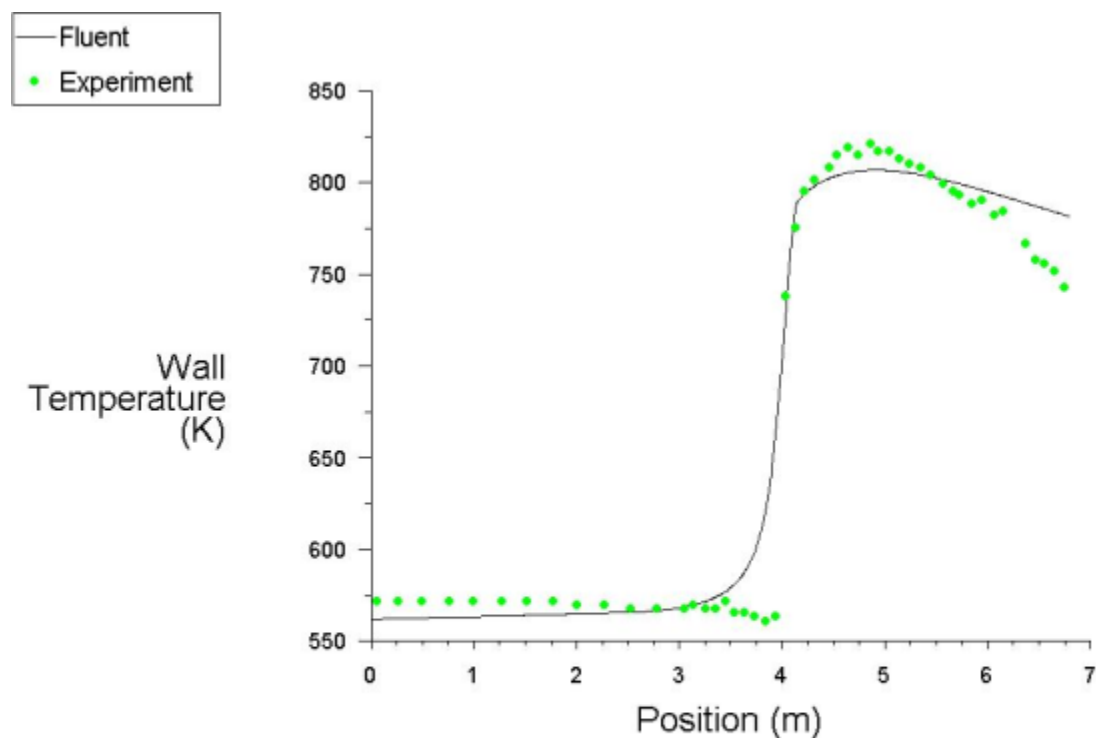
Material Properties	Geometry	Boundary Conditions
Water Steam: dispersed bubbles		797000 W/m^2

Analysis Assumptions and Modeling Notes

The flow is steady. RNG k-epsilon model is used for turbulence.

Results Comparison for Ansys Fluent

Figure .67.2: Comparison of Wall Temperature



VMFL068: Axial Flow in an Eccentric Annulus

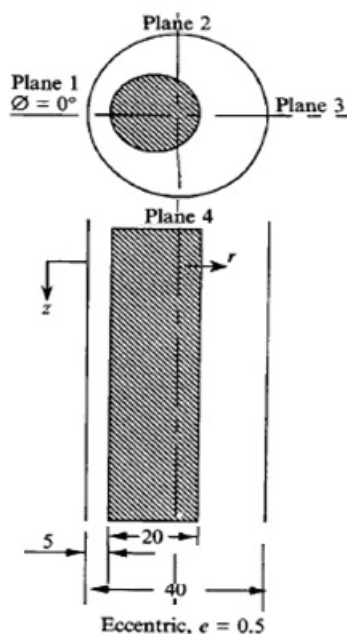
Overview

Reference	J.L.V. Neto, A.L. Martins, A.S. Neto, C.H. Ataide, M.A.S. Barrozo, "Simulated Velocity Profiles for Eccentric Annular Flow with Inner Rotation", Manuscript, Federal University of Uberlandia, Brazil
Solver	Ansys Fluent
Physics/Models	Steady flow, periodic, turbulence, RSM
Input File	VMFL068_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test case considers axial flow in an eccentric annulus without rotation of the inner cylinder. Eccentricity of the channel is 0.5. The Reynolds number is 26,600.

Figure .68.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 896 kg/m^3 Viscosity = $0.00146048 \text{ kg/m-s}$	Diameter, outer cylinder = 40.3 mm Diameter, inner cylinder = 20.1 mm Eccentricity = 0.5	Periodic Boundary is used with a Mass Flow Rate = 1.8368 kg/s.

Analysis Assumptions and Modeling Notes

The flow is steady. The velocity profile at the inlet is fully developed.

Results Comparison for Ansys Fluent

Figure .68.2: Comparison of Normalized Axial Velocity at Plane 2

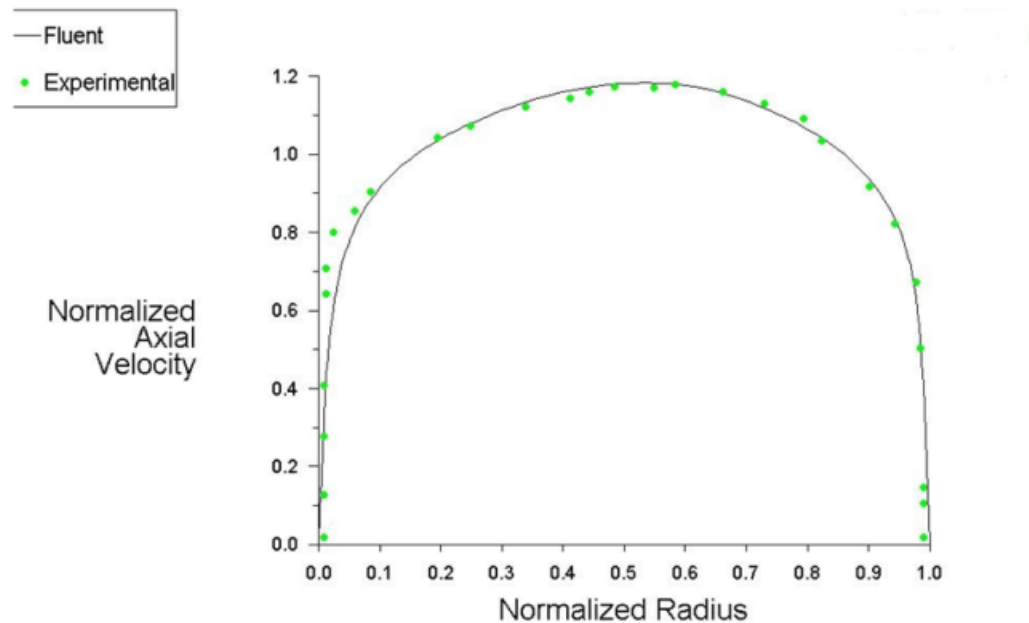
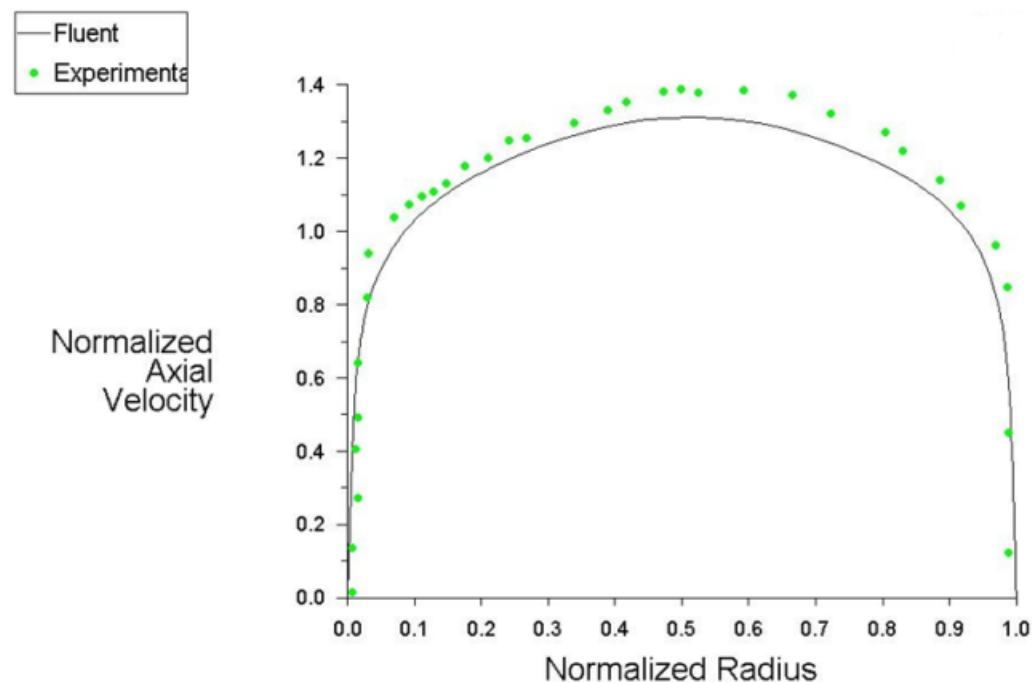


Figure .68.3: Comparison of Normalized Axial Velocity at Plane 3



VMFL069: Two Phase Poiseulle Flow

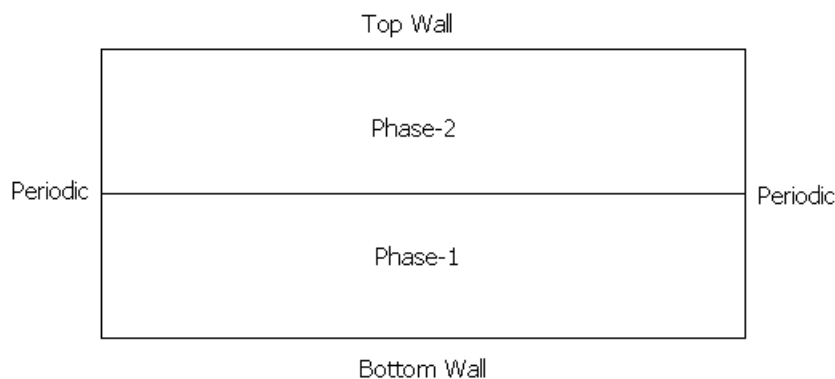
Overview

Reference	E. Marchandise, J.F. Remacle, "A stabilized finite element method using a discontinuous level set approach for solving two phase incompressible flows", <i>Journal of Computational Physics</i> , Vol 219, pp. 780-800, 2006
Solver	Ansys Fluent
Physics/Models	Steady flow, laminar, two phase
Input File	VMFL069_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test case considers the horizontal stratified Poiseulle flow of two fluids between parallel walls. The interface between the two phases is located at half of the height of the channel.

Figure .69.1: Flow Domain



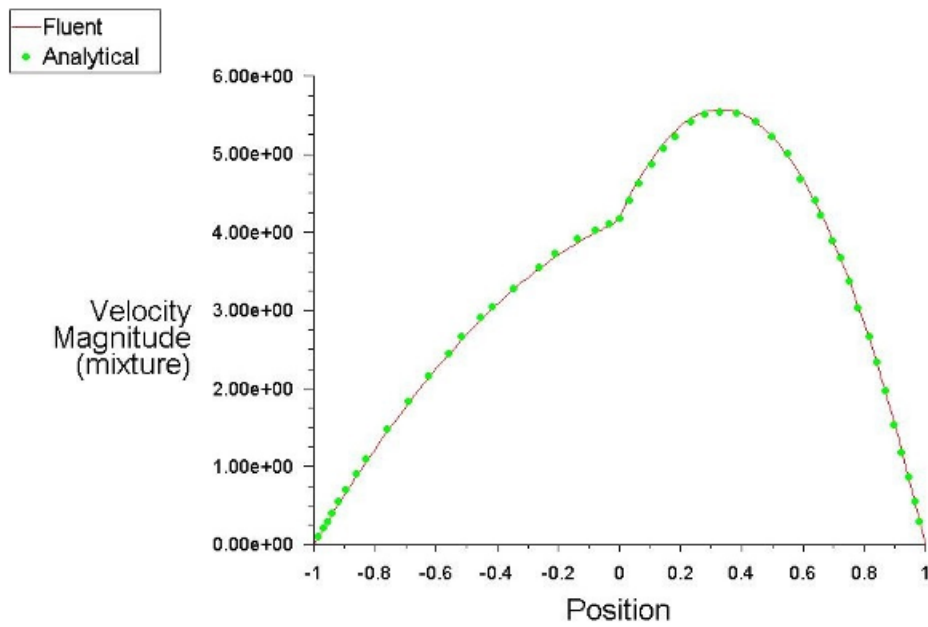
Material Properties	Geometry	Boundary Conditions
The two fluids have the same density. Kinematic Viscosity Fluid-1 = 0.1 Fluid-2 = 0.02	Dimensions of the domain: 2 m X 4 m	Periodic Boundary is used with a Pressure Gradient = -0.5 Pa/m.

Analysis Assumptions and Modeling Notes

The flow is steady. Deformation of the interface is not modeled.

Results Comparison for Ansys Fluent

Figure .69.2: Comparison of Velocity Profile for Two Phase Flow



VMFL070: Radiation Between Two Parallel Surfaces

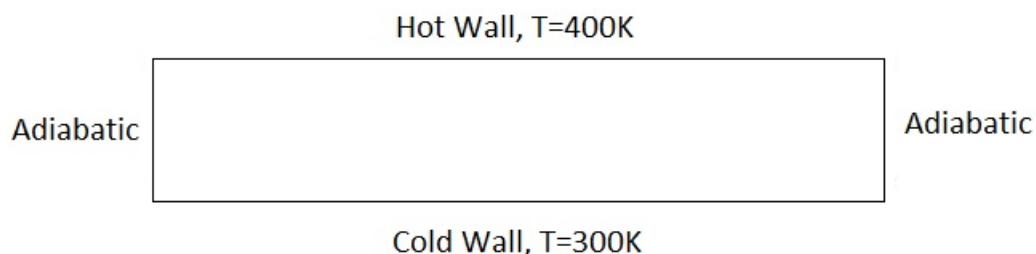
Overview

Reference	F.P. Incropera, D.P. Dewitt, <i>Fundamentals of Heat and Mass Transfer</i> . 4th Edition. New York City, New York: John Wiley & Sons, Inc. 1996.
Solver	Ansys Fluent
Physics/Models	Heat Transfer, radiation
Input File	VMFL070_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test case considers radiation heat transfer between two parallel surfaces. The surfaces are modeled as a rectangle with a length-to-gap aspect ratio of 5 to minimize the edge effect near the mid span. The gap thickness between the upper and lower surfaces is 0.5 m.

Figure .70.1: Flow Domain



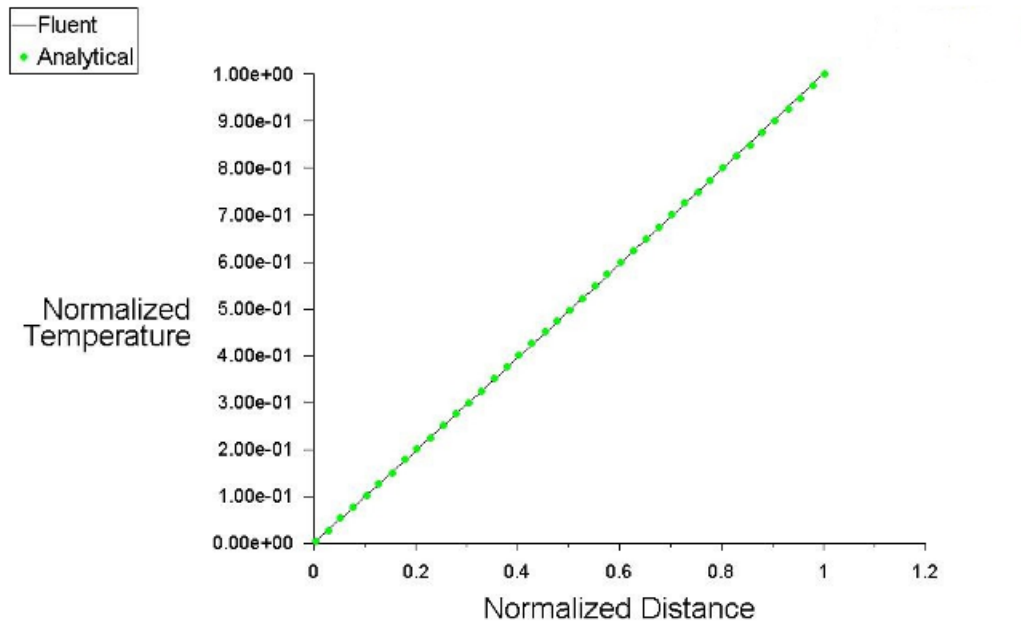
Material Properties	Geometry	Boundary Conditions
Material does not participate in the energy transfer.	Dimensions of the domain: 2.5 m X 0.5 m	Top Wall Temperature = 400 K Bottom Wall Temperature = 300 K

Analysis Assumptions and Modeling Notes

Because there is no flow of mass, only the energy equation is solved. Heat transfer is purely due to radiation between the two surfaces.

Results Comparison for Ansys Fluent

Figure .70.2: Comparison of Normalized Temperature with Analytical Result



VMFL071: Mid-Span Flow Over a Goldman Sator Blade

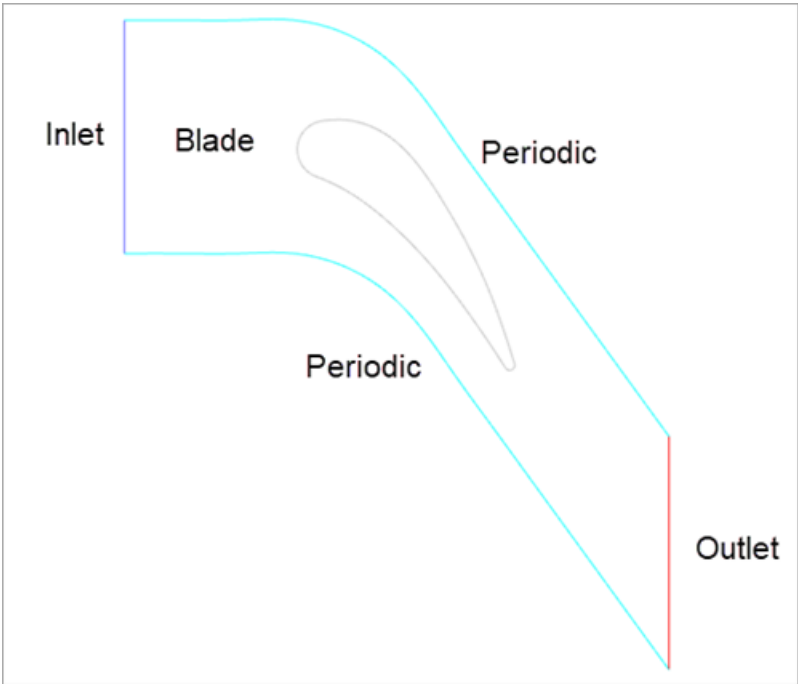
Overview

Reference	Goldman, L.G., & McLallin, K.L. (1977). <i>Cold-Air Annular Cascade Investigation of Aerodynamic Performance of Core-Engine-Cooled Turbine Vanes: In Solid Vane Performance and Facility Description</i> , NASA Technical Memorandum X-3224.
Solver	Ansys Fluent
Physics/Models	Turbomachinery
Input File	VMFL071_FLUENT . cas . h5
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Airflow over a Goldman stator blade at the mid-span is considered. This 2D analysis provides insight into Ansys Fluent's ability to predict the complicated flow features typical of turbomachinery applications. The geometry of the domain under consideration is shown in the figure below. The inlet and the outlet are located approximately 0.03 m away from the blade's leading and trailing edges, respectively. They are located where their presence does not affect predictions near the blade.

Figure .71.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Fluid: Air Density: Ideal gas	Dimensions of the domain: ~ 0.1 m x 0.12 m	Inlet velocity: 71.75 m/s Translational periodic boundary condition on the top and bottom

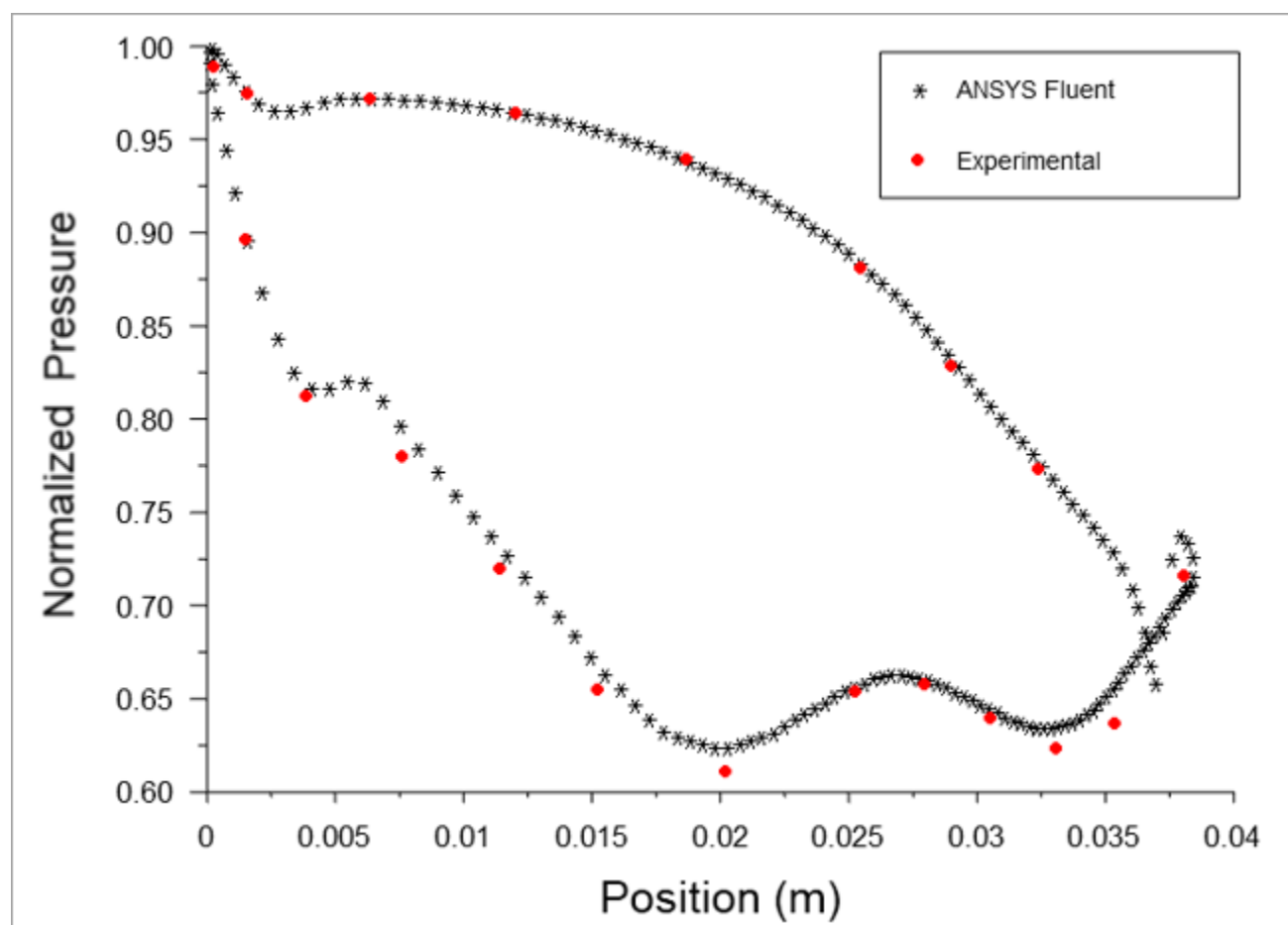
Material Properties	Geometry	Boundary Conditions
Viscosity: 1.78e-05		walls simulate the test section mid-span.

Analysis Assumptions and Modeling Notes

The Reynolds number $Re = \rho * u * h / \mu$ is based on the chord length of the blade, and the free-stream velocity is 500,000. Therefore the flow is turbulent. The inlet Mach number is approximately 0.2 which implies subsonic flow.

Results Comparison for Ansys Fluent

Figure .71.2: Comparison of Pressure Ratio with Experimental Result



VMFL072: Liquid Water Flow Over a Flat Plate Under the Influence of Gravity

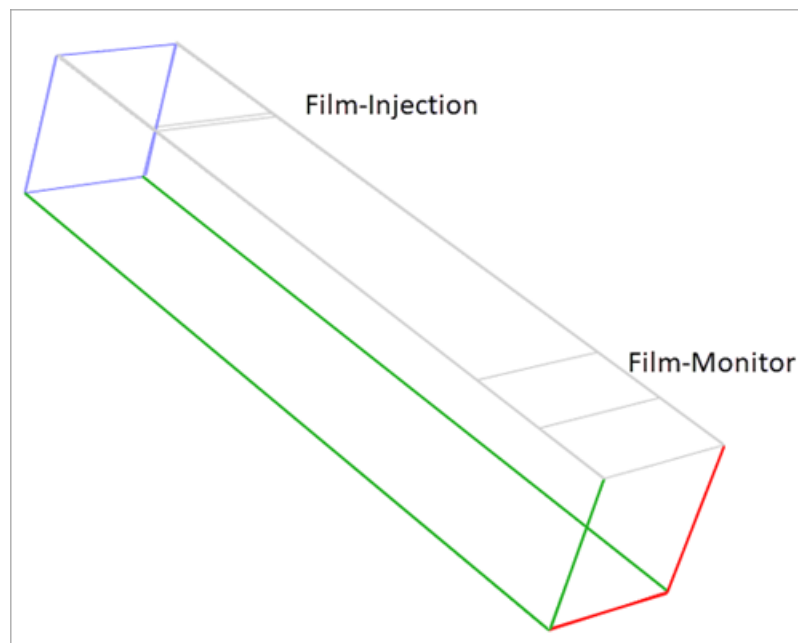
Overview

Reference	Roy, R.P. & Jain, S. (1989). A study of thin water film flow down an inclined plate without and with countercurrent air flow. <i>Experiments in Fluids</i> , (7) 318-328.
Solver	Ansys Fluent
Physics/Models	Eulerian Wall Film
Input File	VMFL072_FLUENT . cas . h5
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A water film Reynolds number of 417 and a plate inclination angle of 40° without counter current airflow are selected for the present study. The film is introduced as a mass flux boundary on the wall patch injection. Film thickness is calculated at a downstream location wall-monitor.

Figure .72.1: Flow Domain



Material Properties	Geometry	Boundary Conditions.
Fluid domain is air while water-liquid is injected at the wall patch.	Dimensions of the domain: 500 mm x 100 mm x 100 mm Injection width: 5 mm Wall-monitor width: 50 mm	Mass Flux of Water-liquid = 76.2 kg/m ² -s X-Gravitational Acceleration = 6.305746 m/s ²

Material Properties	Geometry	Boundary Conditions.
		Y-Gravitational Acceleration = 0 m/s ² Z-Gravitational Acceleration = -7.514896 m/s ²

Analysis Assumptions and Modeling Notes

Air flow is zero, and the film flow is laminar.

Results Comparison for Ansys Fluent

Table .72.1: Comparison of Film Thickness

	Target	Ansys Fluent	Ratio
Film Thickness (mm)	0.555	0.5497	0.99

VMFL073: Turbulent Separated Flow in an Axisymmetric Diffuser

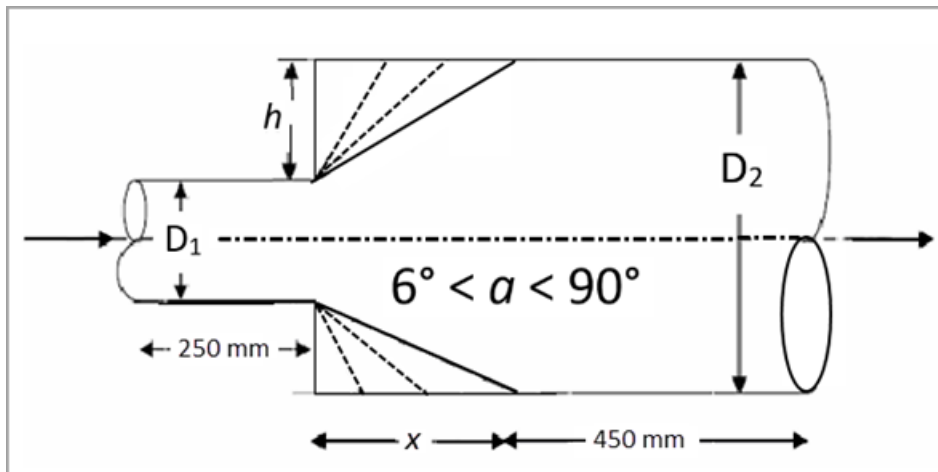
Overview

Reference	Nitsche, Wolfgang & Haberland, Christoph. (1993). <i>On Turbulent Separated Flows in Axisymmetric Diffusers</i> . 10.1007/978-3-663-13986-7_17.
Solver	Ansys Fluent
Physics/Models	Turbulence, Separation
Input File	VMFL073_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A fully developed turbulent flow is modeled in an axisymmetric diffuser. Half angle of 90° is selected for the present study.

Figure .73.1: Flow Domain



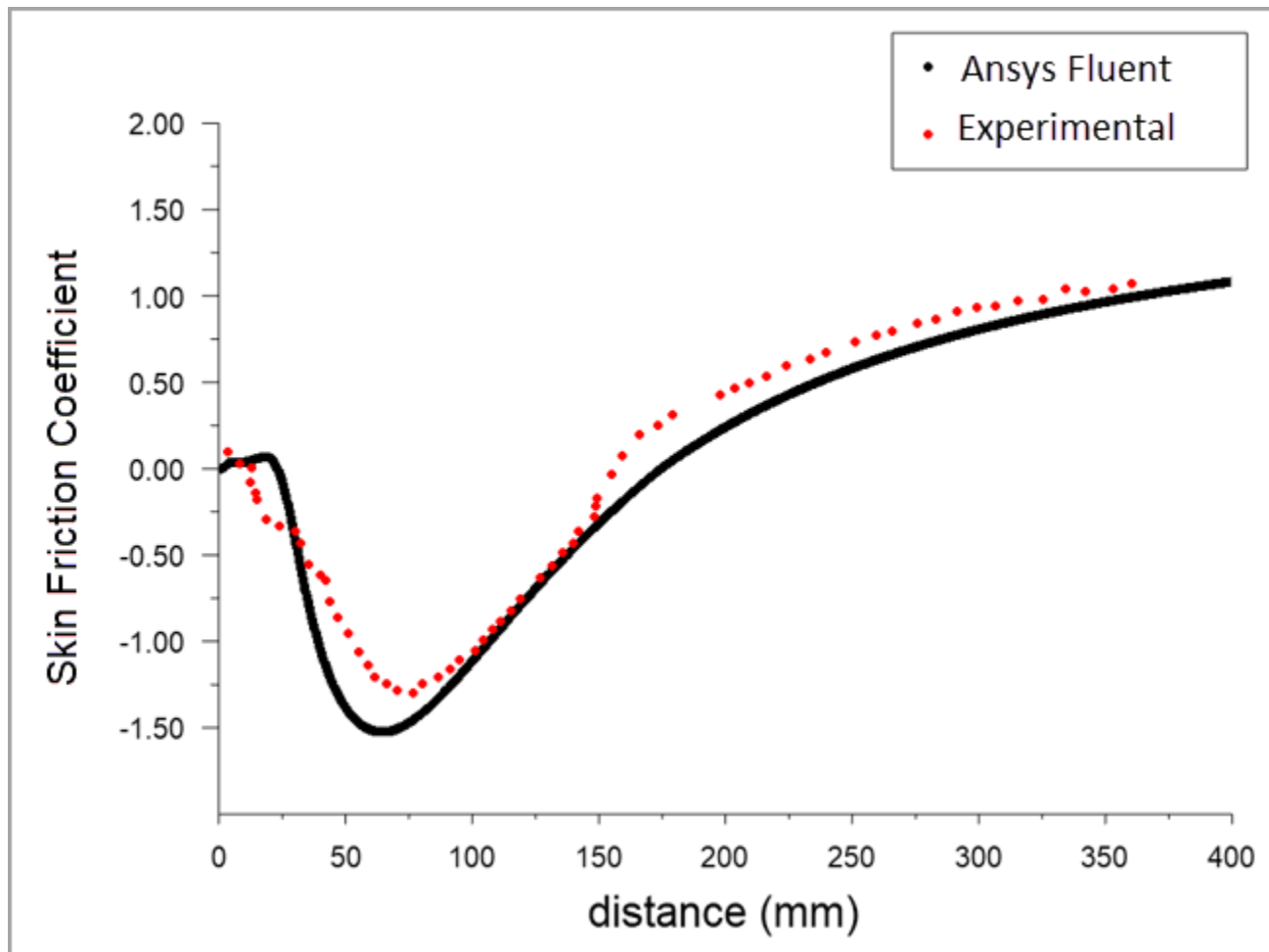
Material Properties	Geometry	Boundary Conditions
Density: 1.22478 kg/m ³ Viscosity: 1.7894e-05 kg/m-s	Half angle = 90 degrees Inlet radius = 0.025 m Outlet radius = 0.04 m	Fully developed turbulent profile at inlet with an average velocity = 14.61 m/s

Analysis Assumptions and Modeling Notes

Steady turbulent flow is assumed. An SST k-w model is used for turbulence.

Results Comparison for Ansys Fluent

Figure .73.2: Comparison of Skin Friction Coefficient along the Diffuser Wall



VMFL074: Modeling of a Plug Flow Atomizer

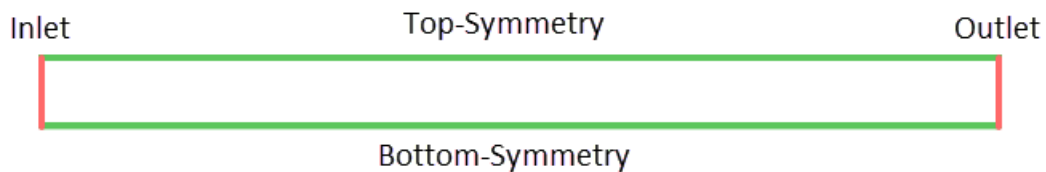
Overview

Reference	Bhamidipati, S., Panchagnula, M., Peddieson, J. (2008, January 3-5). <i>Discrete population balance modeling of a plug flow atomizer</i> [Paper presentation]. Heat and Mass Transfer Conference, Hyderabad, India.
Solver	Ansys Fluent
Physics/Models	Multi-phase, population balance model, turbulence
Input File	VMFL074_FLUENT.cas.h5
Project Files	Link to Project Files Download Page (p. 5)

Test Case

A plug flow atomizer is modeled using a discrete population balance, multi-phase model. The Ramkrishna method is used for the breakage kernel which is defined through a compiled udf.

Figure .74.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Air Phase Density: 1.225 kg/m ³ Viscosity: 1.7894e-05 kg/m-s	Inlet = 0.145 m Outlet = 0.145 m Domain Length = 2 m	Inlet Velocity = 0.01 m/s
Water Phase Density: 998.2 kg/m ³ Viscosity: 0.001003 kg/m-s		

Analysis Assumptions and Modeling Notes

A steady turbulent flow is assumed. Ansys Fluent's simulation results are compared to an analytical solution.

Results Comparison for Ansys Fluent

Figure .74.2: Comparison of Number Density of bin-2 Fraction

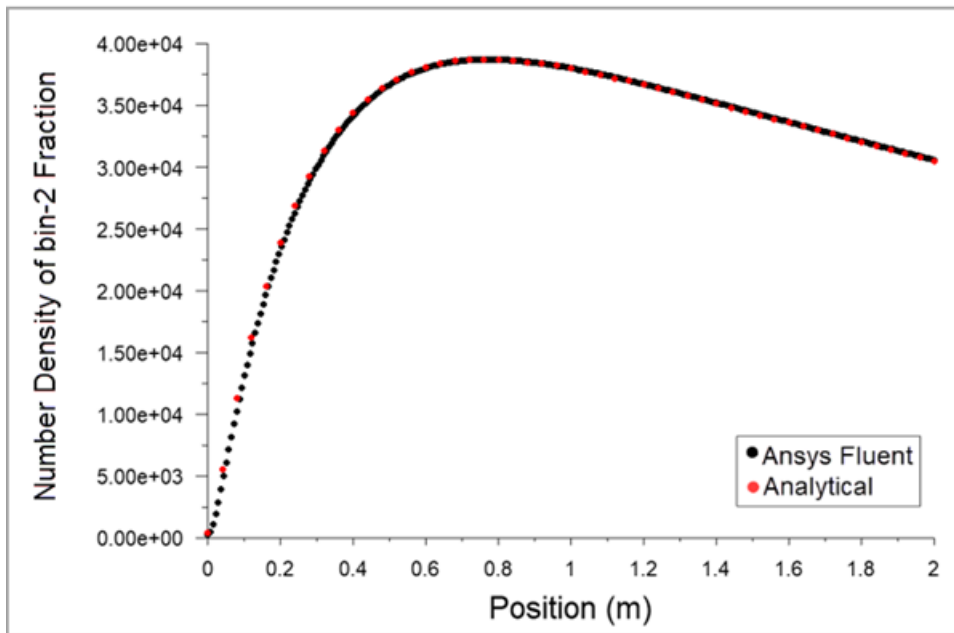
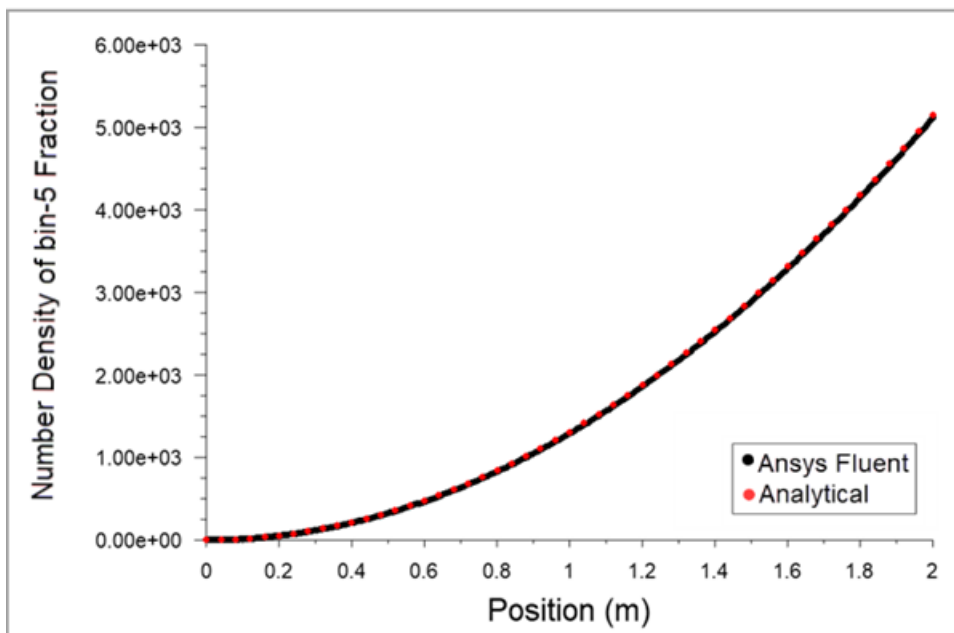


Figure .74.3: Comparison of Number Density of bin-5 Fraction



VMFL075: Supersonic Flow Over a Circular Arc Bump

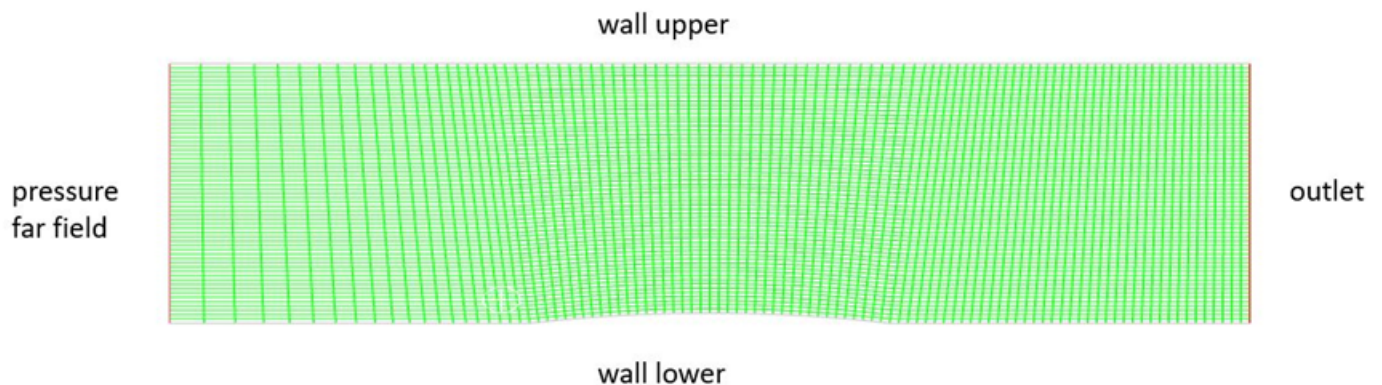
Overview

Reference	Yeng-Yung Tsui and Tian-Cherng Wu, (2008) <i>A Pressure-Based Unstructured-Grid Algorithm Using High-Resolution Schemes for All-Speed Flows</i> , Numerical Heat Transfer, Part B: Fundamentals, 53:1, 75 - 96
Solver	Ansys Fluent
Physics/Models	Inviscid, Compressible, Supersonic
Input File	VMFL075_FLUENT.cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Supersonic ideal gas enters a tube with a circular arc bump inside. The supersonic flow shows compression shockwaves before and after the bump, which reflect off the upper wall of the tube.

Figure .75.1: Flow Domain



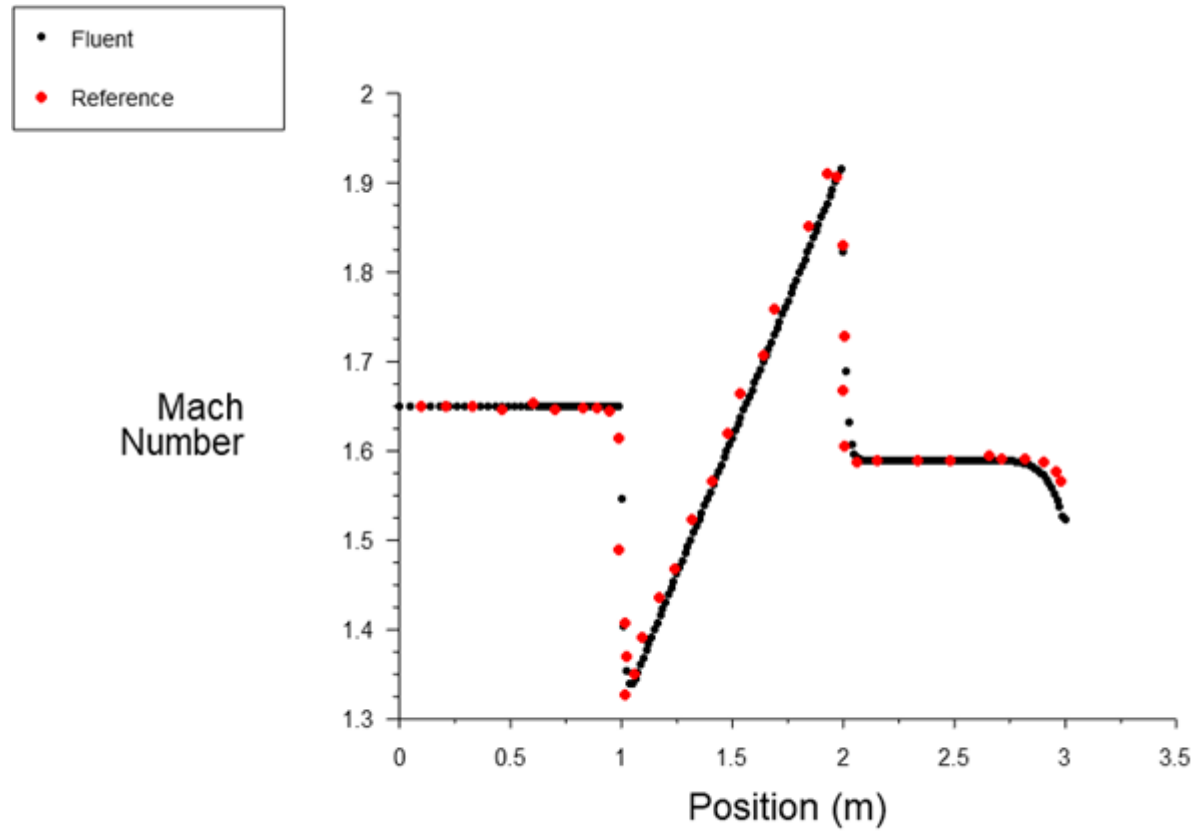
Material Properties	Geometry	Boundary Conditions
Air Density: ideal gas	Rectangular domain Length = 3m Width = 1m Length of the circular arc bump = 1m	Pressure Far Field with Mach Number = 1.65 bar

Analysis Assumptions and Modeling Notes

Inviscid and steady flow. Coupled solver with pseudo-transient is used.

Results Comparison for Ansys Fluent

Figure .75.2: Comparison of Mach Number along the lower wall with published reference



VMFL076: Forced Convection Over a Flat Plate

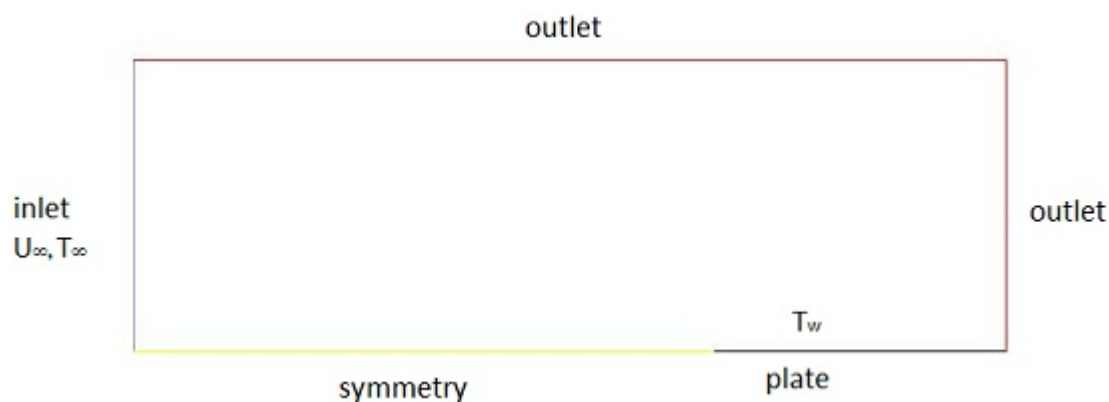
Overview

Reference	NASA Memorandum 02-27-1959, <i>DETAILS OF EXACT LOW PRANDTL NUMBER BOUNDARY-LAYER SOLUTIONS FOR FORCED AND FOR FREE CONVECTION</i> , By E. M. Sparrow and J. L. Gregg, Lewis Research Center, Cleveland, Ohio
Solver	Ansys Fluent
Physics/Models	Laminar, Convection
Input File	VMFL076_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Analysis of forced convection heat transfer with very low Prandtl numbers, typically encountered in liquid metal flows.

Figure .76.1: Flow Domain



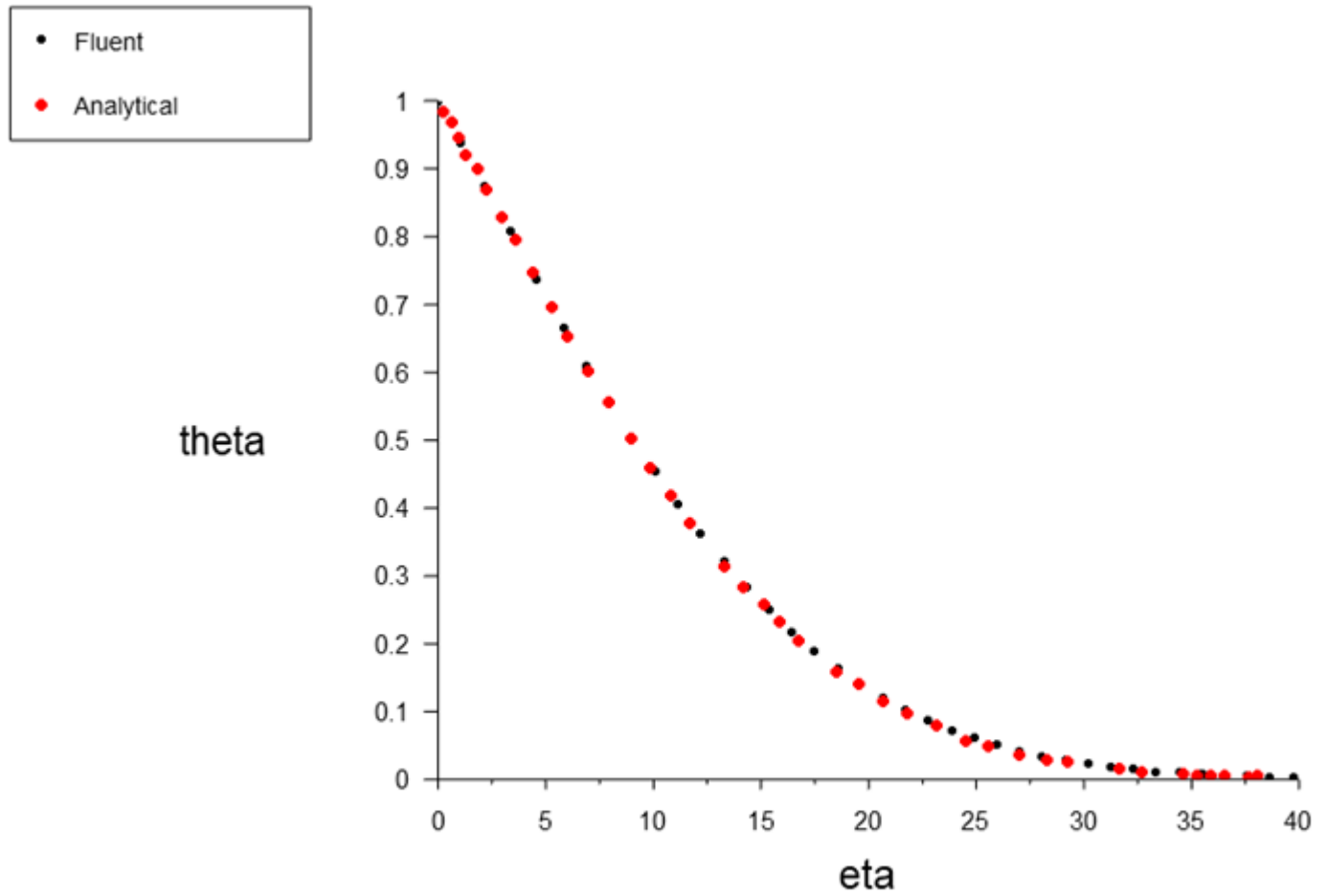
Material Properties	Geometry	Boundary Conditions
Density= 900 kg/m ³ Viscosity: 0.01 kg/m-s Specific Heat = 42.6 J/kg-K Thermal Conductivity = 142 W/m-K	Flat Plate Length = 1 m	Inlet Velocity = 1 m/s Inlet temperature = 300 K temperature of the plate = 350 K

Analysis Assumptions and Modeling Notes

Laminar steady flow. Pressure based coupled solver with pseudo-transient is used.

Results Comparison for Ansys Fluent

Figure .76.2: Comparison of Normalized Temperature with analytical results



VMFL077: Free Surface Flow Around a Ship

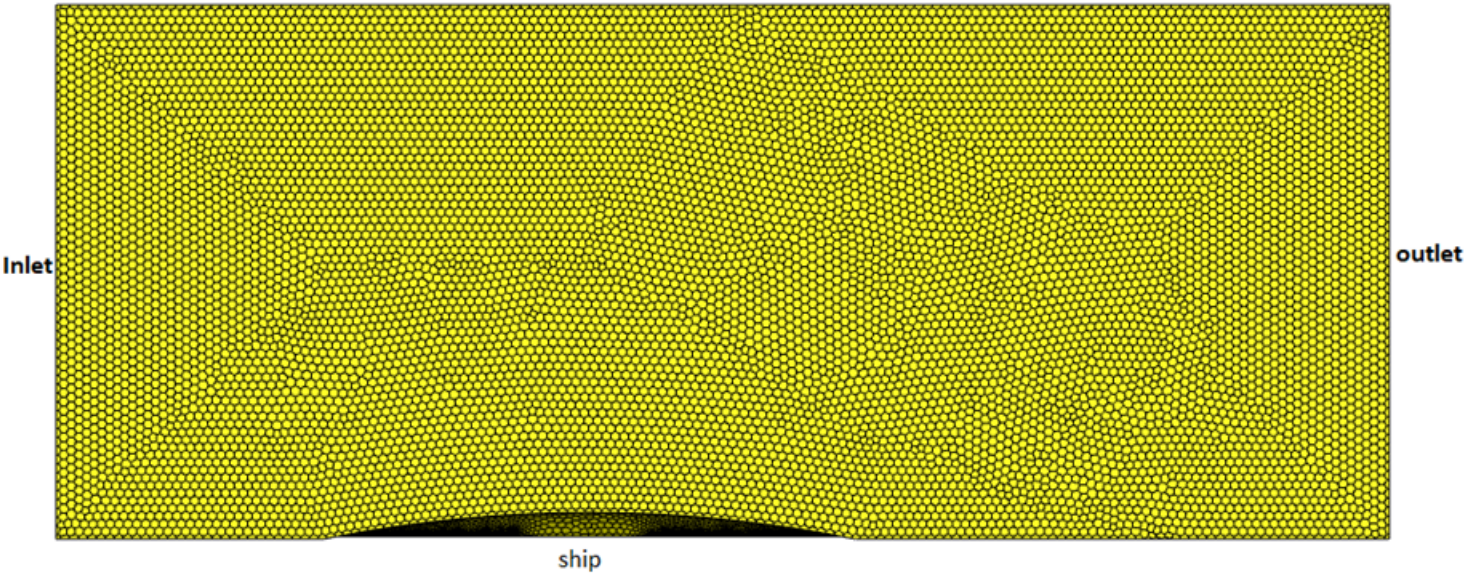
Overview

Reference	Sarda, O. P. <i>Turbulent Flows Past Ship Hulls – An Experimental and Computational Study</i> , PhD thesis, University of Iowa, 1986.
Solver	Ansys Fluent
Physics/Models	Turbulent, Free Surface Flow
Input File	VMFL077_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

Free surface flow around a ship. In this multiphase open channel flow problem, the profile of free surface at ship hull is compared with experimental results.

Figure .77.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Water Density : 997 kg/m ³	The hull geometry has a parabolic shape	Pressure inlet and pressure outlet with multiphase open

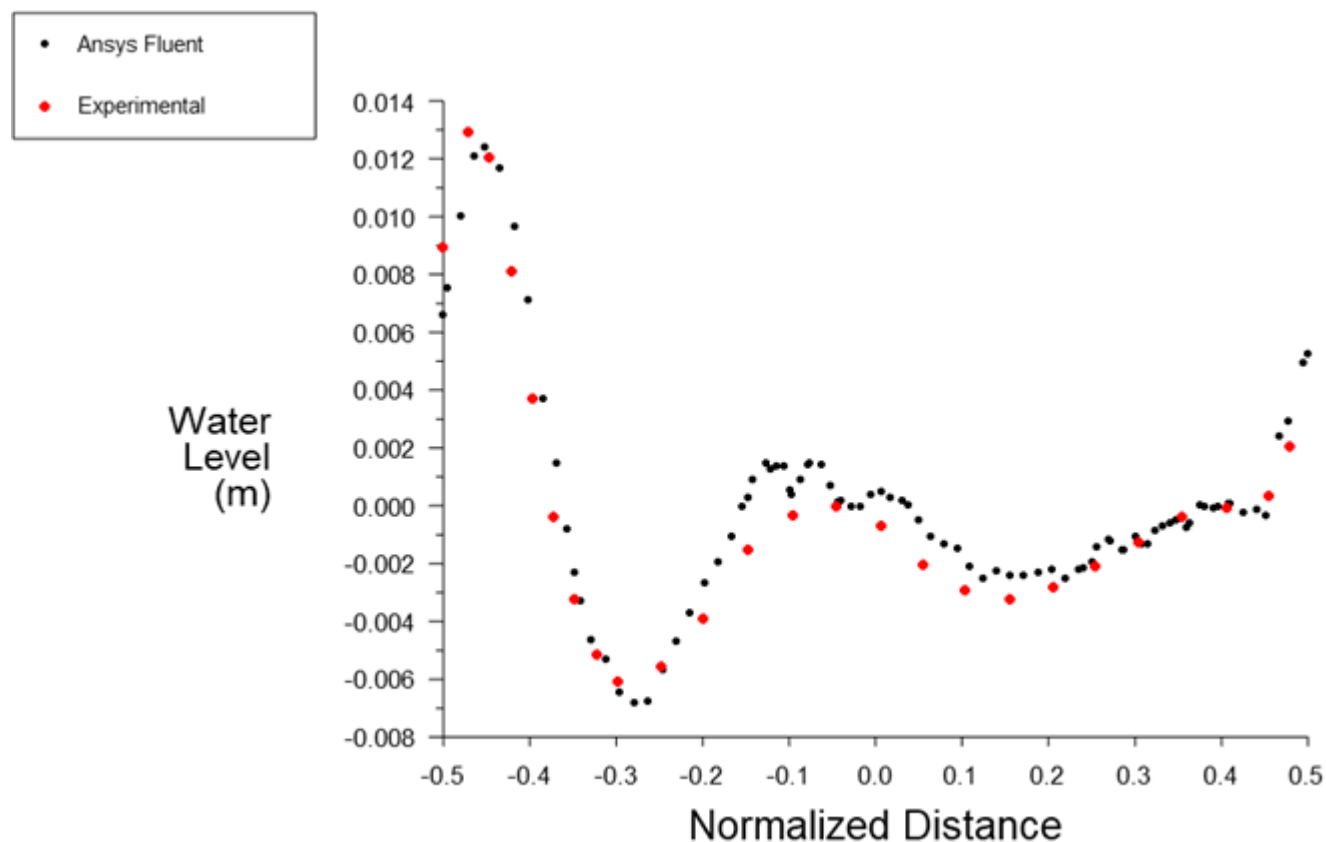
Material Properties	Geometry	Boundary Conditions
Water Viscosity: 0.008899 kg/m-s	with a hull length = 3.014 m.	channel boundary. Reynold's number=4900000 at inlet.

Analysis Assumptions and Modeling Notes

The flow is steady. SST k-w model is used for turbulence modeling. Polyhedral mesh of 921470 cells is used to discretize the domain.

Results Comparison for Ansys Fluent

Figure .77.2: Comparison of water level on the hull surface with experimental data



VMFL078: Polyhedral Mesh Accuracy

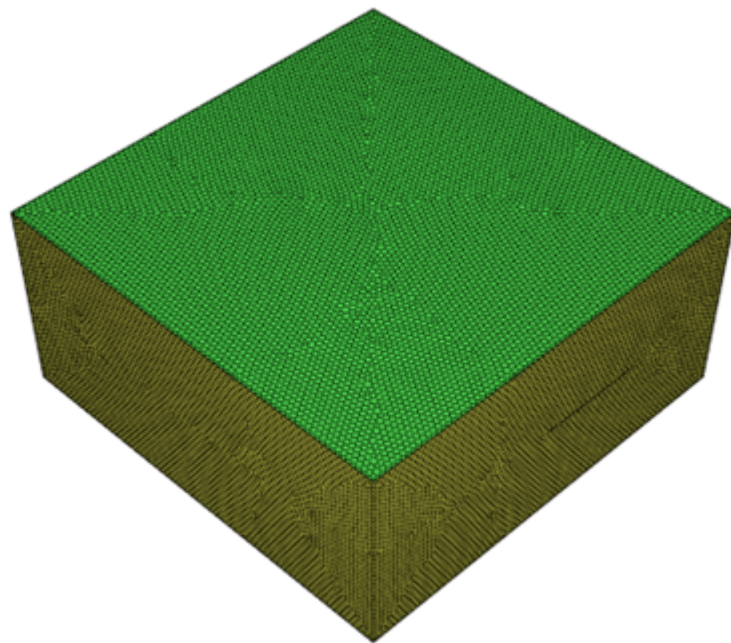
Overview

Reference	Parallel Simulation of 3D Lid-driven Cubic Cavity Flows by Finite Element Method, Jifei Wang and Decheng Wan. Proceedings of the Twenty-first (2011) International Offshore and Polar Engineering Conference, Maui, Hawaii, USA, June 19-24, 2011.
Solver	Ansys Fluent
Physics/Models	Laminar Flow, driven by a moving wall
Input File	VMFL078_FLUENT . cas
Project Files	Link to Project Files Download Page (p. 5)

Test Case

The laminar incompressible flow in 3D driven cavity is solved at Reynolds number = 1000. The study demonstrates accuracy of polyhedral mesh.

Figure .78.1: Flow Domain



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Material Properties	Geometry	Boundary Conditions
Density: 1 kg/m ³ Viscosity: 0.001 kg/m-s	A half domain is modeled with height = 1 m, length = 1 m and breadth = 0.5 m	Moving boundary condition is at the top of the domain with a velocity= 1 m/s All other walls are stationary

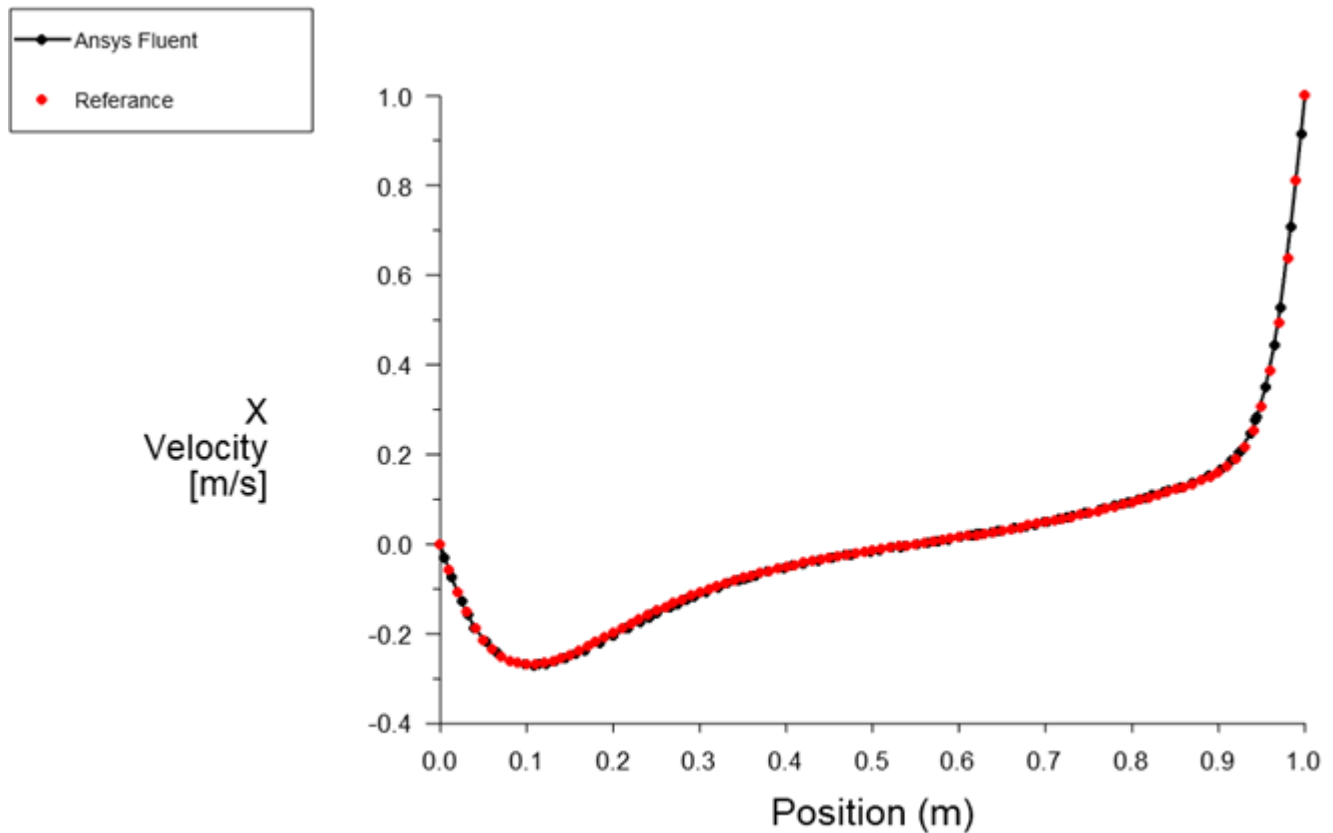
Material Properties	Geometry	Boundary Conditions
	using symmetry boundary condition.	

Analysis Assumptions and Modeling Notes

The flow is steady. Polyhedral mesh of 279894 cells is used to discretize the domain.

Results Comparison for Ansys Fluent

Figure .78.2: Comparison of X-Velocity along the vertical centerline in the symmetry plane



VMFLGPU001: Flow Between Rotating and Stationary Concentric Cylinders

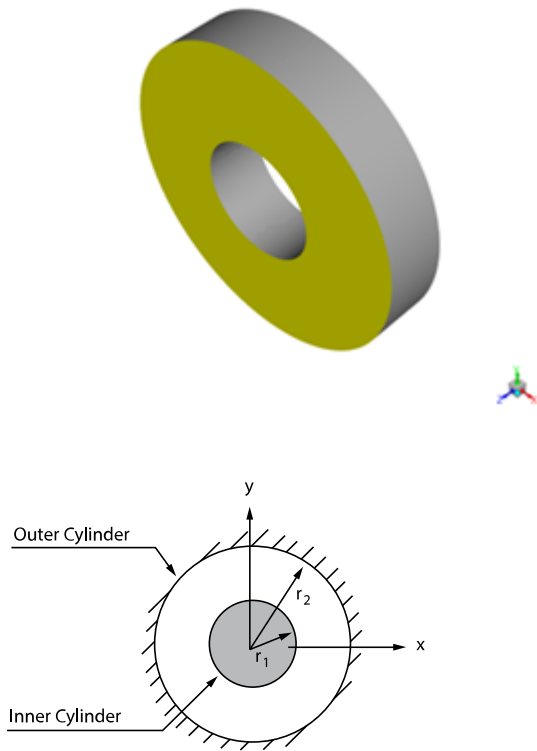
Overview

Reference	F. M. White. <i>Viscous Fluid Flow</i> . Section 3-2.3. McGraw-Hill Book Co., Inc.. New York, NY. 1991.
Solver	Ansys Fluent GPU
Physics/Models	Laminar flow, rotating wall
Input Files	vt001.msh, vmfl001.jou for Ansys Fluent GPU

Test Case

Steady laminar flow between two concentric cylinders is modeled. The flow is induced by rotation of the inner cylinder with a constant angular velocity, while the outer cylinder is held stationary. Due to periodicity only a section of the domain needs to be modeled. In the present simulation a 180° segment (half of the domain shown in [Figure .gpu001.1: Flow Domain \(p. 225\)](#)) is modeled. The sketch is not to scale.

Figure .gpu001.1: Flow Domain



Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³ Viscosity = 0.0002 kg/m-s	Radius of the Inner Cylinder = 17.8 mm Radius of the Outer Cylinder = 46.28 mm	Angular velocity of the inner wall = 1 rad/s

Analysis Assumptions and Modeling Notes

The flow is steady. The tangential velocity at various sections can be calculated using analytical equations for laminar flow. These values are used for comparison with simulation results.

Results Comparison for Ansys Fluent GPU

Table .gpu001.1: Comparison of Tangential Velocity in the Annulus at Various Radial Locations

Tangential Velocity at	Target, m/s	Ansys Fluent GPU, m/s	Ratio
r = 20 mm	0.0151	0.0152	1.006
r = 25 mm	0.0105	0.0105	1.000
r = 30 mm	0.0072	0.0072	1.000
r = 35 mm	0.0046	0.0045	0.978

VMFLGPU002: Laminar Flow in a 90° Tee-Junction.

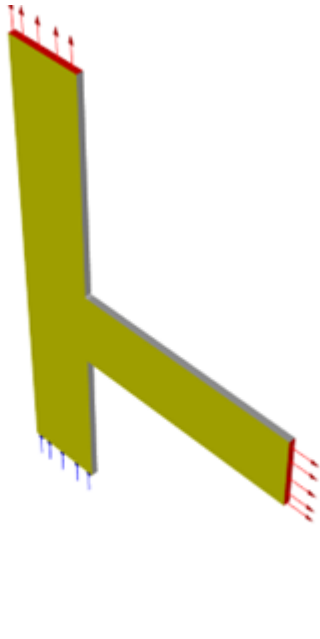
Overview

Reference	R.E. Hayes, K. Nandkumar, H. Nasr-El-Din, "Steady Laminar Flow in a 90 Degree Planar Branch". <i>Computers and Fluids</i> , Vol 17, pp. 537-553, 1989.
Solver	Ansys Fluent GPU
Physics/Models	Laminar flow
Input File	vt002.msh.h5, vel-inlet.prof, vmfl002.jou for Ansys Fluent GPU

Test Case

The purpose of this test is to compare prediction of the fractional flow in a dividing tee-junction with experimental results. The fluid enters through the bottom branch and divides into the two channels whose exit planes are held at the same static pressure.

Figure .gpu002.1: Flow Domain



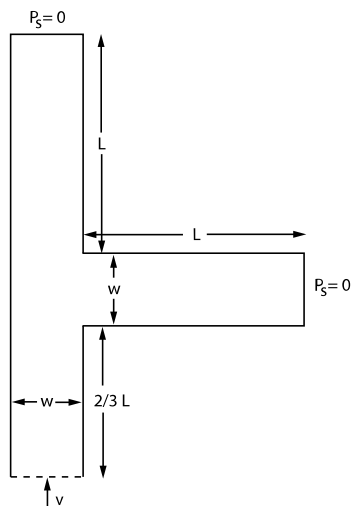


Table .gpu002.1: Comparison of Flow Split from Tee

Material Properties	Geometry	Boundary Conditions
Fluid: Air	L=3.0 m	Fully developed inlet velocity profile for:
Density : 1 kg/m ³	W=1.0 m	$Re = \frac{\rho V_c W}{\mu} = 300$ where V_c is the inlet centerline velocity.
Viscosity: 0.003333 kg/m-s		$P_s = 0$

Analysis Assumptions and Modeling Notes

The flow is steady and incompressible. Pressure based solver is used. It is seen that with increasing flow rate in the main channel, less fluid escapes through the secondary (right) branch. For analysis of results, we calculate and compare the fractional flow in the upper branch.

Results Comparison for Ansys Fluent GPU

Table .gpu002.2: Comparison of Flow Split from Tee

	Target	Ansys Fluent GPU	Ratio
Flow split	0.887	0.884	0.997

VMFLGPU003: Laminar Flow in a Triangular Cavity

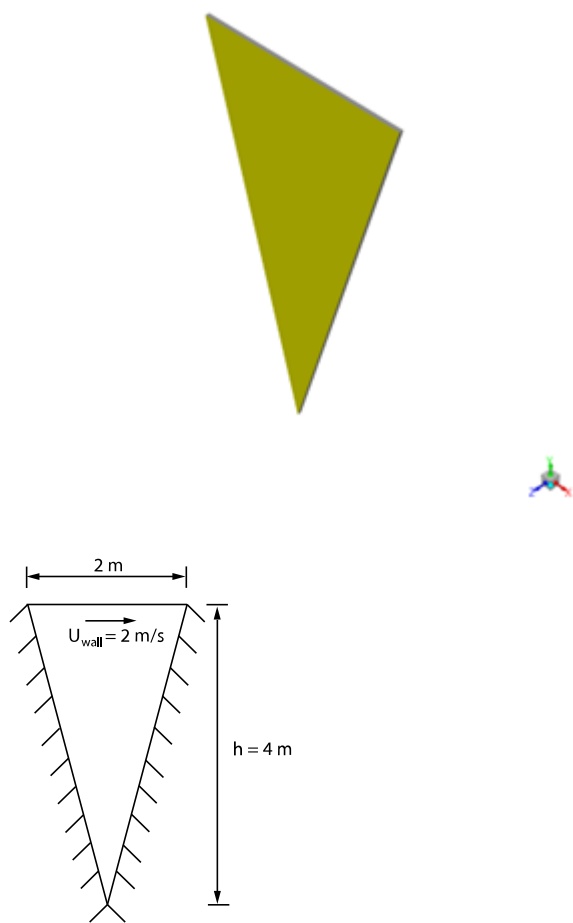
Overview

Reference	R. Jyotsna, S.P. Vanka. "Multigrid Calculation of Steady, Viscous Flow in a Triangular Cavity". <i>J. Comp. Phys.</i> , Vol 122, pp. 107-117, 1995.
Solver	Ansys Fluent GPU
Physics/Models	Viscous flow, driven by a moving wall
Input Files	vt003.msh.h5 , vmfl003.jou for Ansys FluentGPU

Test Case

Laminar flow induced by the motion of the top wall of a triangular cavity (Figure .gpu003.1: Flow Domain (p. 229)). The side walls are stationary.

Figure .gpu003.1: Flow Domain



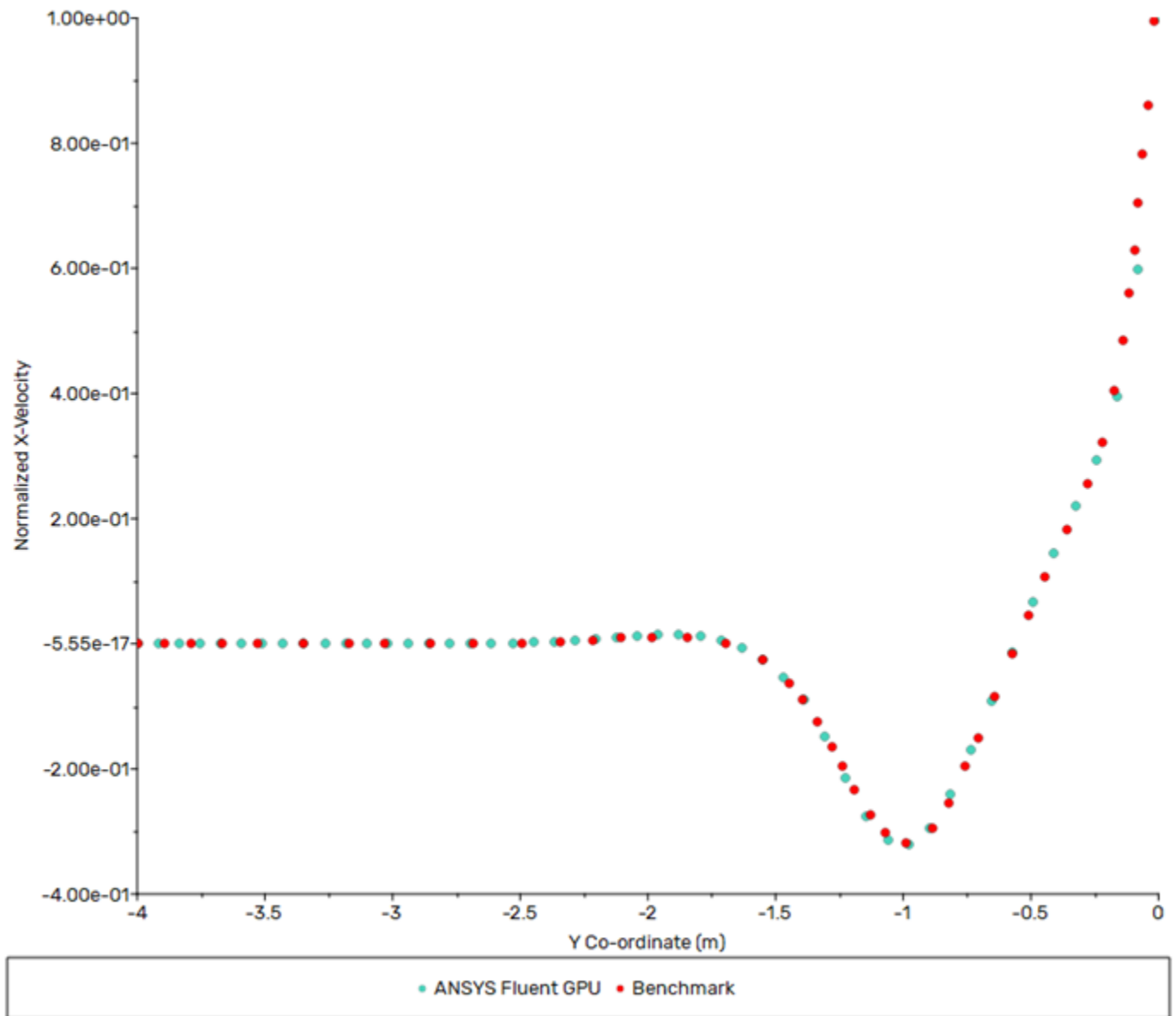
Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³	Height of the triangular cavity = 4 m	Velocity of the top (base) wall = 2 m/s
Viscosity = 0.01 kg/m-s	Width of the base = 2 m	Other walls are stationary

Analysis Assumptions and Modeling Notes

The flow is steady. Pressure based solver is used. A hybrid mesh with tetrahedral and hexahedral cells is used to discretize the domain.

Results Comparison for FLUENT GPU

Figure .gpu003.2: Comparison of Distribution of Normalized X-Velocity Along a Vertical Line that Bisects the Base of the Cavity



In this figure, X-velocity is normalized by the velocity of the moving wall.

VMFLGPU004: Anisotropic Conduction Heat Transfer

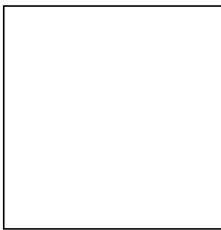
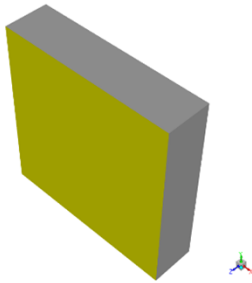
Overview

Reference	
Solver	Ansys Fluent GPU
Physics/Models	Heat conduction, anisotropic conductivity
Input File	vt004.msh.h5, vmfl004.jou, 3d.csv

Test Case

Heat conduction in a solid with anisotropic thermal conductivity is modeled. Two opposite walls are maintained at uniform temperatures. Conductivity of the solid material is specified using matrix components to account for the anisotropy. The simulation results are compared with analytical solution for temperature distribution.

Figure .gpu004.1: Domain



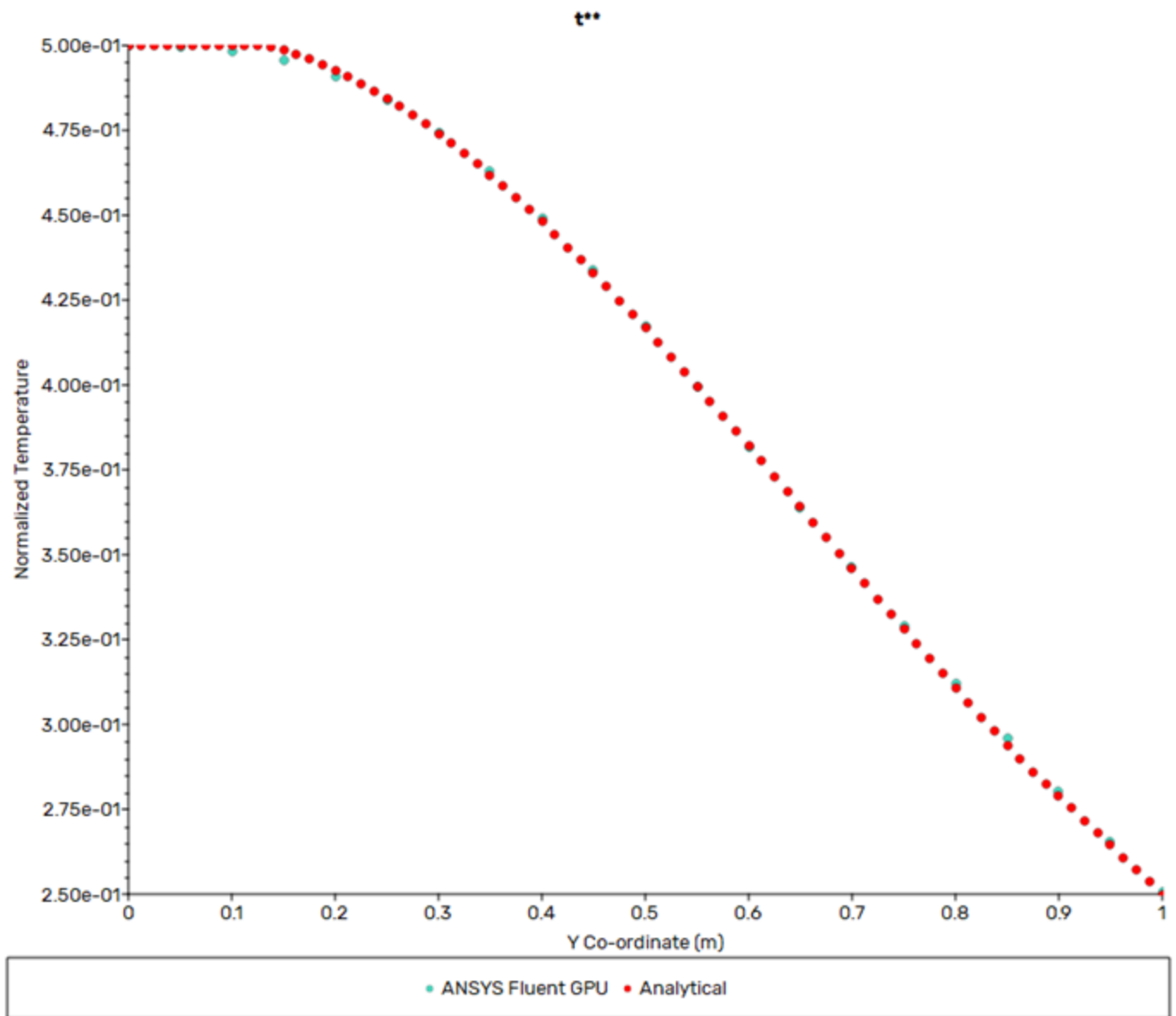
Material Properties	Geometry	Boundary Conditions
Density of solid = 2719 kg/m^3 Specific heat = 871 J/kg-K Thermal conductivity: Anisotropic	Dimensions of the domain: $1 \text{ m} \times 1 \text{ m}$	Fixed wall temperatures = 100 K and 200 K respectively User-defined profile for temperature distribution on the other two walls

Analysis Assumptions and Modeling Notes

Steady state conduction. Anisotropic conductivity modeled by specified matrix components for the solid conductivity.

Results Comparison

Figure .gpu004.2: Comparison of Normalized Temperature Distribution at X = 0.5 m



VMFLGPU005: Turbulent Natural Convection Inside a Tall Cavity

Overview

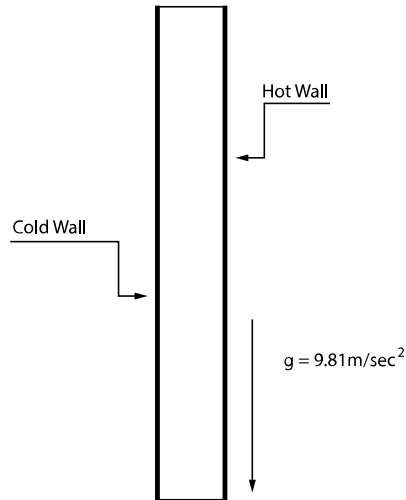
Reference	P.L. Betts, I.H. Bokhari. "Experiments on turbulent natural convection in an enclosed tall cavity". <i>International Journal of Heat and Fluid Flow</i> , Vol. 21, pp. 675-683, 2000
Solver	Ansys Fluent GPU
Physics/Models	Turbulent flow, buoyancy effects, Boussinesq approximation
Input File	vt005.msh, vmfl005.jou

Test Case

Natural convection in the turbulent flow field of an enclosed cavity with a length-to-width ratio of 28.6 is modeled. The Rayleigh number is in the turbulent range. The two vertical walls are kept at different temperatures, while the horizontal walls are adiabatic.

Figure .gpu005.1: Flow Domain (not to scale)





Material Properties	Geometry	Boundary Conditions
Density : Boussinesq approximation	Length of the cavity = 2.18 m	Temperature of Cold wall = 288.25 K
Specific Heat = 1005 J/kg-K	Width of the cavity = 0.0762 m	Temperature of Hot wall = 307.85 K
Viscosity = 1.81×10^{-5} kg/m-sec		Top and bottom walls are adiabatic
Molecular weight = 28.966		

Analysis Assumptions and Modeling Notes

The flow is steady and is induced by natural convective heat transfer.

Results Comparison for Ansys Fluent GPU

Figure .gpu005.2: Comparison of Vertical Velocity at $Y/h = 0.05$

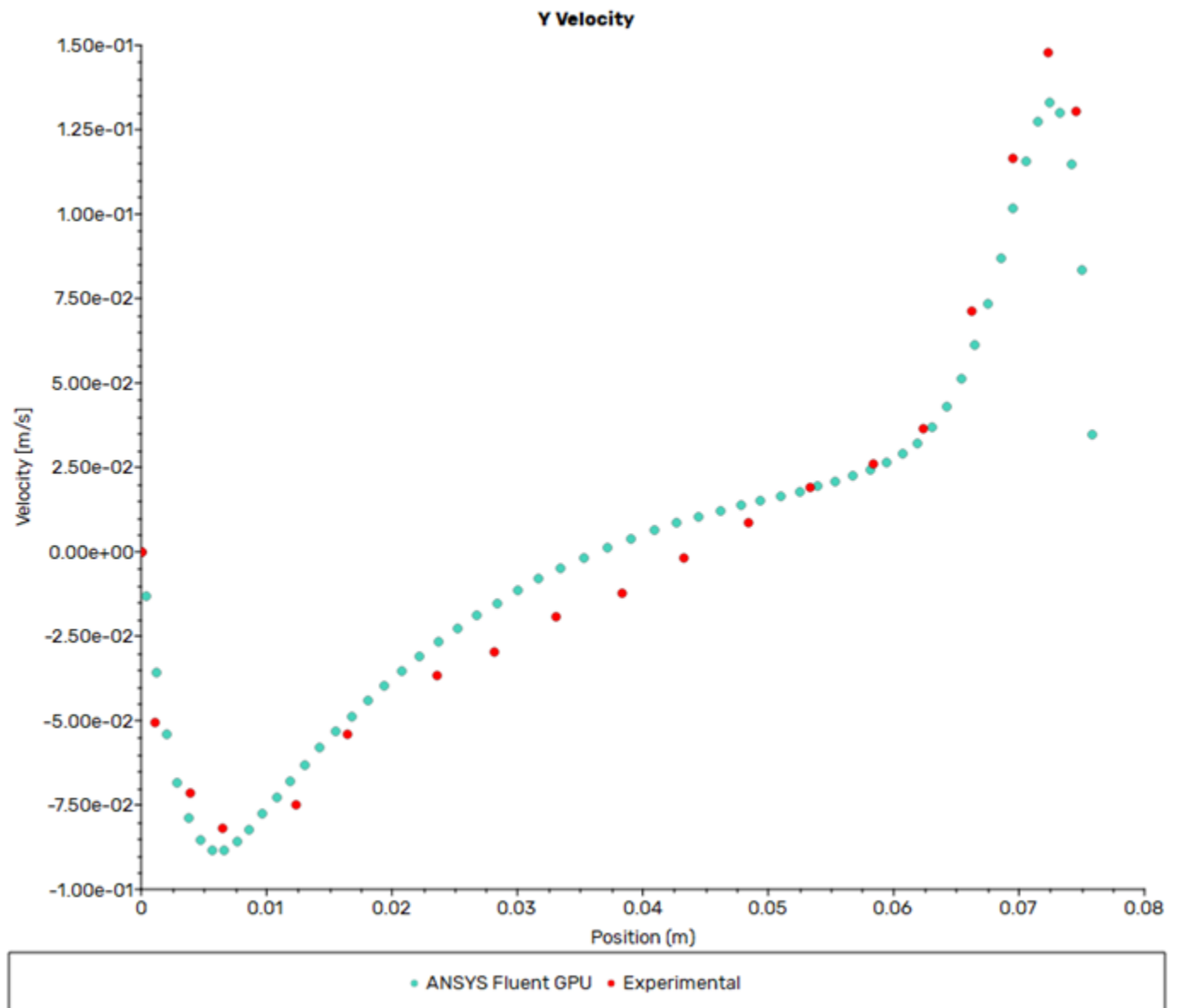
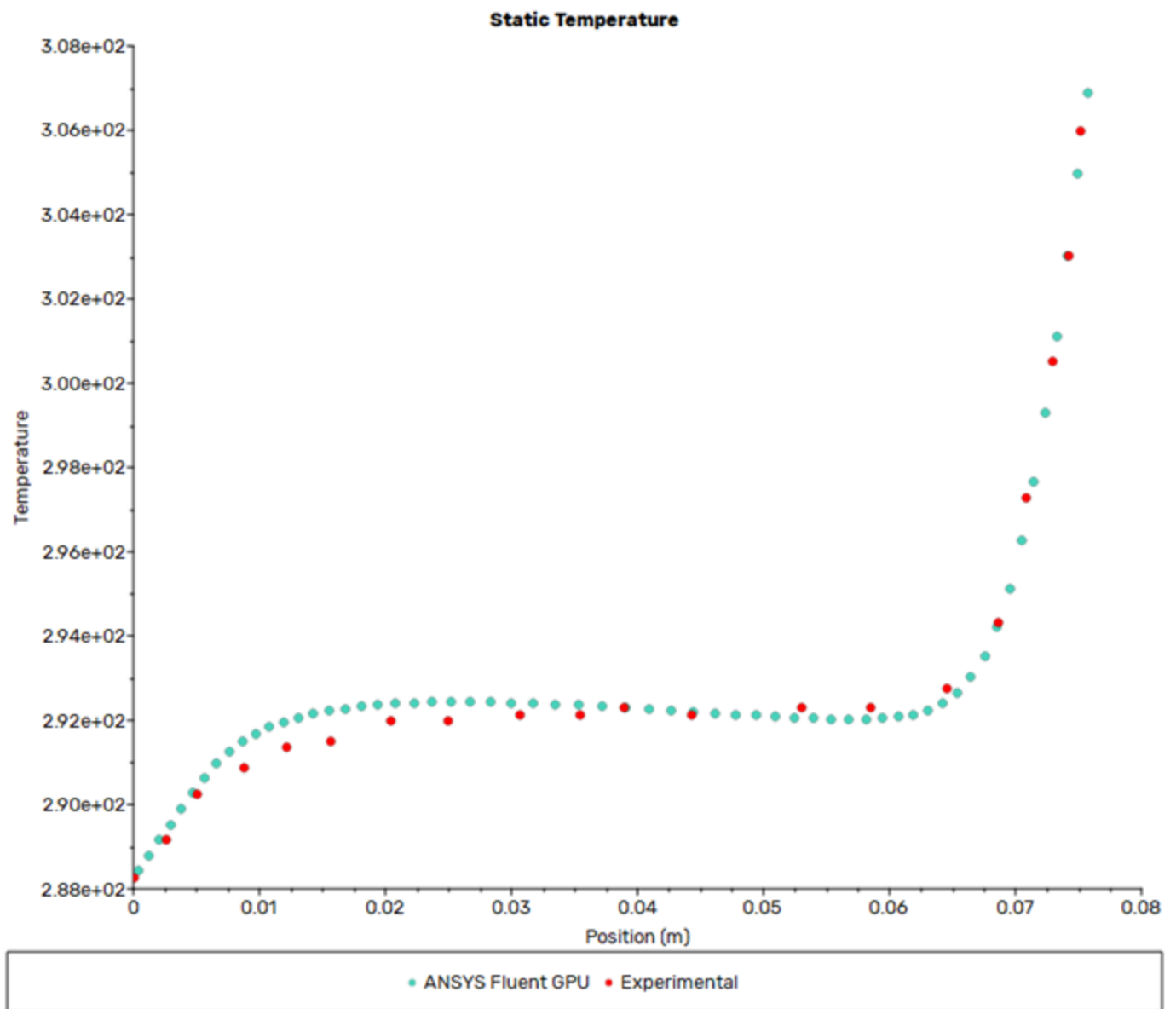


Figure .gpu005.3: Comparison of Temperature at $Y/h = 0.05$ 

VMFLGPU006: Mid-Span Flow Over a Goldman Stator Blade

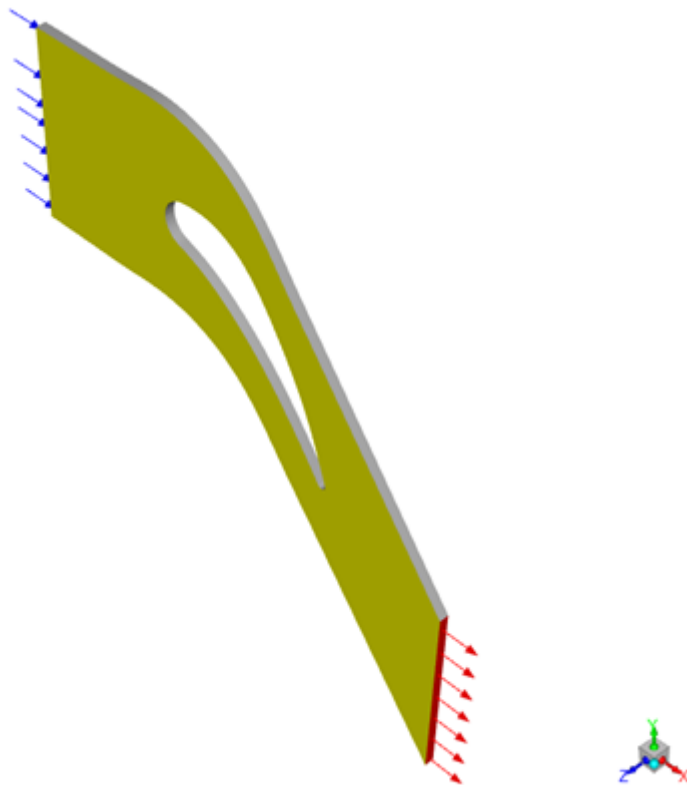
Overview

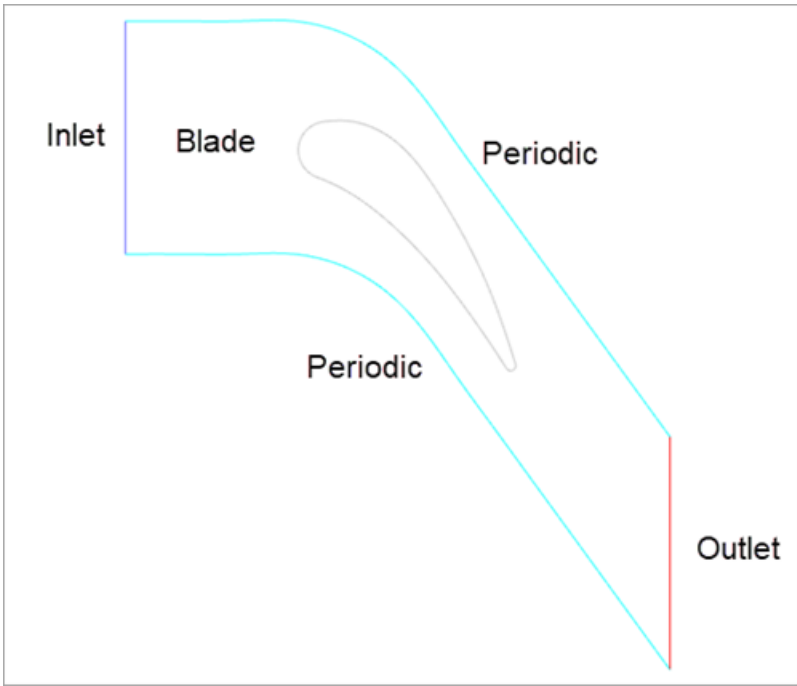
Reference	Goldman, L.G., & McLallin, K.L. (1977). <i>Cold-Air Annular Cascade Investigation of Aerodynamic Performance of Core-Engine-Cooled Turbine Vanes: In Solid Vane Performance and Facility Description</i> , NASA Technical Memorandum X-3224.
Solver	Ansys Fluent GPU
Physics/Models	Turbomachinery
Input File	vt006.msh.h5, vmfl006.jou

Test Case

Airflow over a Goldman stator blade at the mid-span is considered. This 2.5D analysis provides insight into Ansys Fluent's ability to predict the complicated flow features typical of turbomachinery applications. The geometry of the domain under consideration is shown in the figure below. The inlet and the outlet are located approximately 0.03 m away from the blade's leading and trailing edges, respectively. They are located where their presence does not affect predictions near the blade.

Figure .gpu006.1: Flow Domain





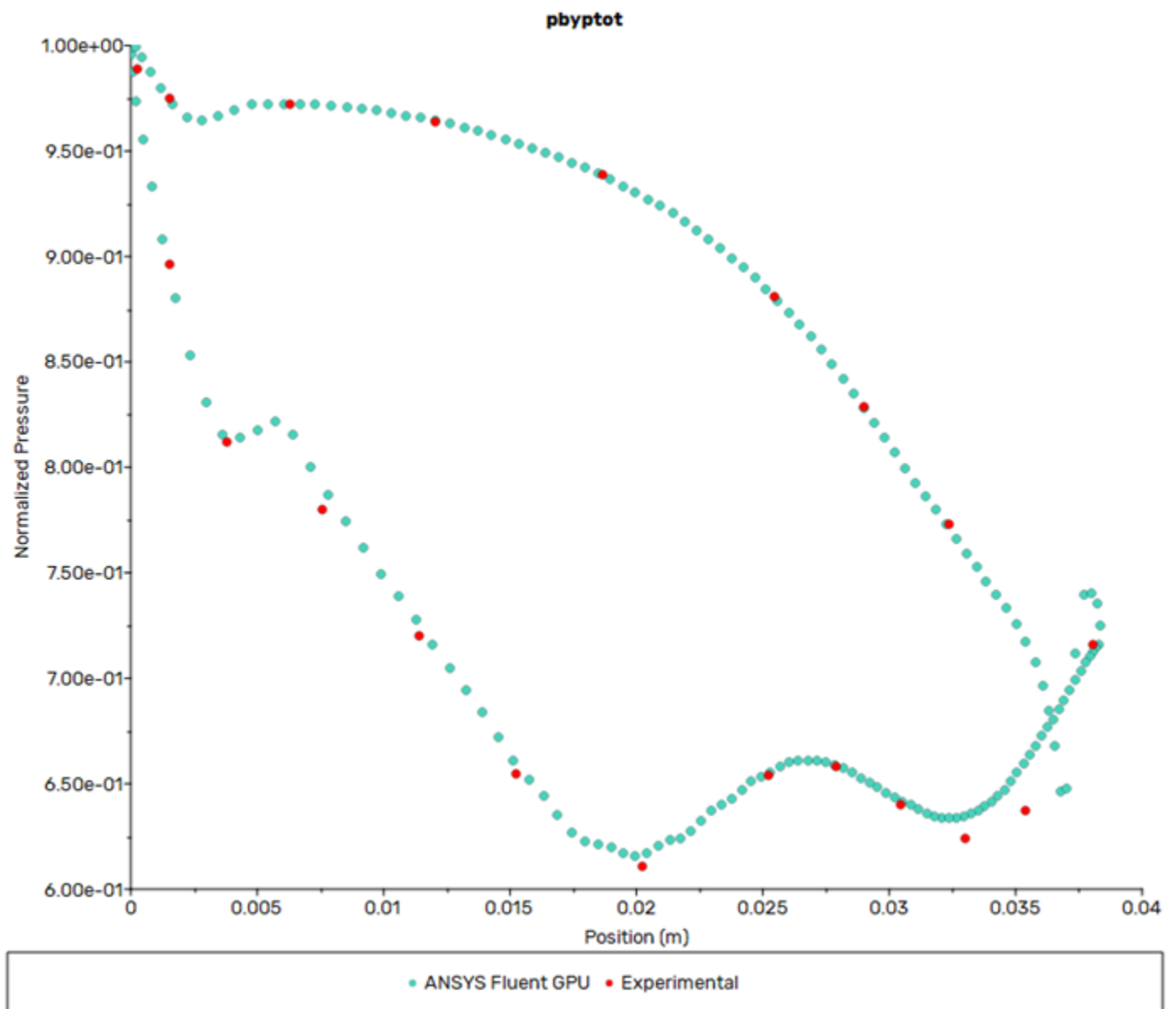
Material Properties	Geometry	Boundary Conditions
Fluid: Air Density: Ideal gas Viscosity: 1.78e-05	Dimensions of the domain: ~ 0.1 m x 0.12 m	Inlet velocity: 71.75 m/s Translational periodic boundary condition on the top and bottom walls simulate the test section mid-span.

Analysis Assumptions and Modeling Notes

The Reynolds number $Re = \rho * u * h / \mu$ is based on the chord length of the blade, and the free-stream velocity is 500,000. Therefore the flow is turbulent. The inlet Mach number is approximately 0.2 which implies subsonic flow.

Results Comparison for Ansys Fluent GPU

Figure .gpu006.2: Comparison of Pressure Ratio with Experimental Result



VMFLGPU007: Turbulent Flow with Heat Transfer in a Backward-Facing Step

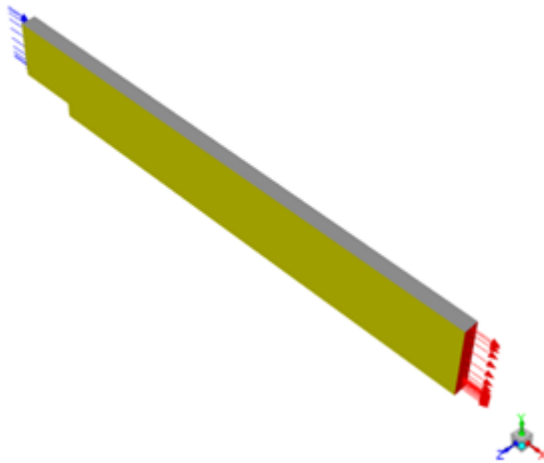
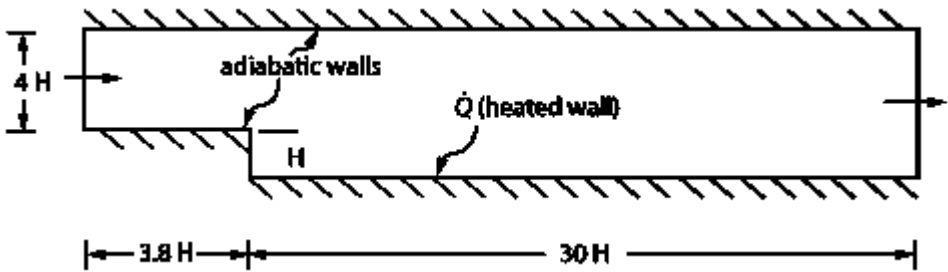
Overview

Reference	J.C. Vogel, J.K. Eaton, "Combined Heat Transfer and Fluid Dynamic Measurements Downstream of a Backward-Facing Step". <i>Journal of Heat Transfer</i> , Vol. 107, pp. 922-929, 1985.
Solver	Ansys Fluent GPU
Physics/Models	Incompressible, turbulent flow with heat convection and reattachment.
Input File	vt007.msh, vmfl007.jou

Test Case

Airflow over a Goldman stator blade at the mid-span is considered. This 2.5D analysis provides insight into Ansys Fluent's ability to predict the complicated flow features typical of turbomachinery applications. The geometry of the domain under consideration is shown in the figure below. The inlet and the outlet are located approximately 0.03 m away from the blade's leading and trailing edges, respectively. They are located where their presence does not affect predictions near the blade.

Figure .gpu007.1: Flow Domain



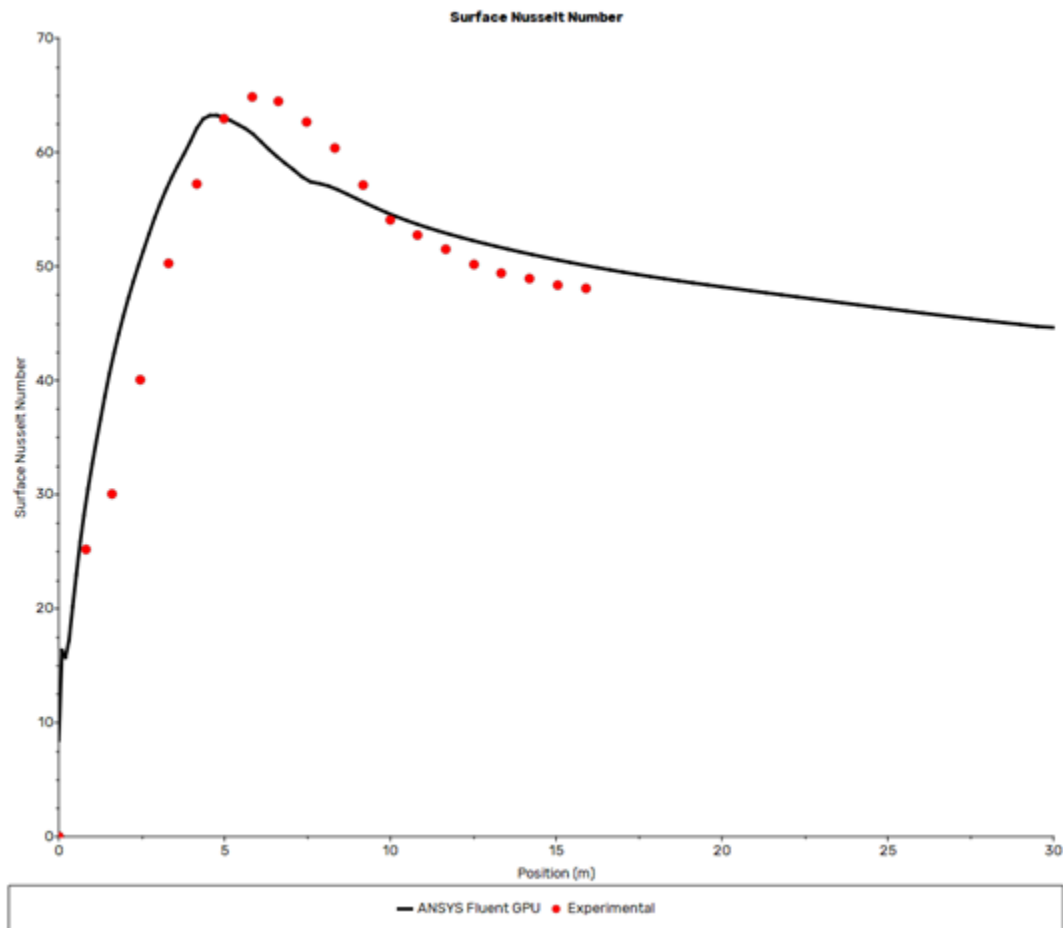
Material Properties	Geometry	Boundary Conditions
Density = 1 kg/m ³ Viscosity = 0.0001 kg/m-s Conductivity = 1.408 W/m-K Specific Heat = 10,000 J/kg-K	H = 1m	I Velocity profile at inlet corresponding to $Re_H = 28,000$ Wall heat transfer, $Q' = 1,000$ W/m ²

Analysis Assumptions and Modeling Notes

The flow is steady and incompressible. Fluid properties are considered constant. Pressure based solver is used. The inlet boundary conditions are specified using the fully developed profiles for the velocity, k , and ϵ . The incoming boundary layer thickness is 1.1 H . Under the given pressure conditions, the Reynolds number, Re_H is about 28,000. The standard $k-\epsilon$ model with standard wall functions is used for accounting turbulence.

Results Comparison for Ansys Fluent GPU

Figure .gpu007.2: Comparison of Surface Nusselt Number with Experimental Result



VMFLGPU008: Radiative Heat Transfer in a Rectangular Enclosure with Participating Medium

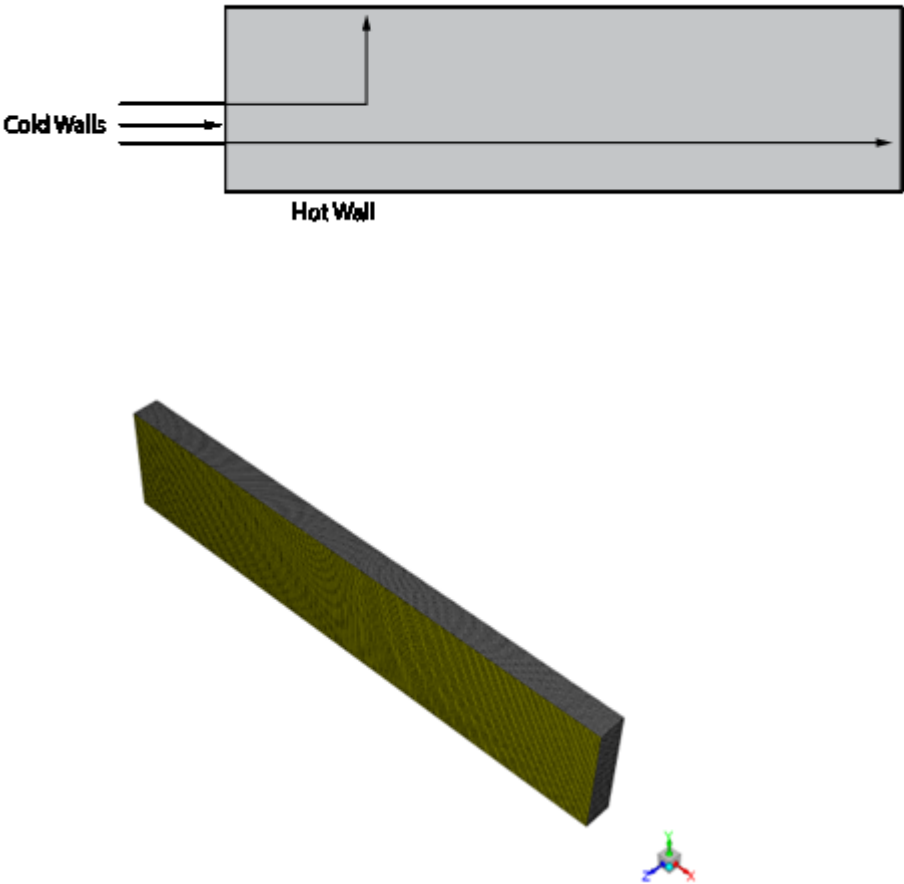
Overview

Reference	G.D Raithby, E.H. Chui. "A Finite Volume Method for Predicting a Radiant Heat Transfer in Enclosures with Participating Media". <i>Journal of Heat Transfer</i> . Volume 112, pp. 415-423, 1990
Solver	Ansys Fluent GPU
Physics/Models	Radiation modeling, discrete ordinate model
Input File	vt008.cas.h5, vmfl008.jou

Test Case

Radiative heat transfer in an enclosure with one hot wall and three cold walls at equal temperature is modeled. The enclosure is a cavity with a length-to-width ratio of 5. For the problem being considered, $\sigma_s L_y = 1.0$, where σ_s is the scattering coefficient and L_y is the normal distance between the hot wall and the cold wall opposite to it.

Figure .gpu008.1: Flow Domain



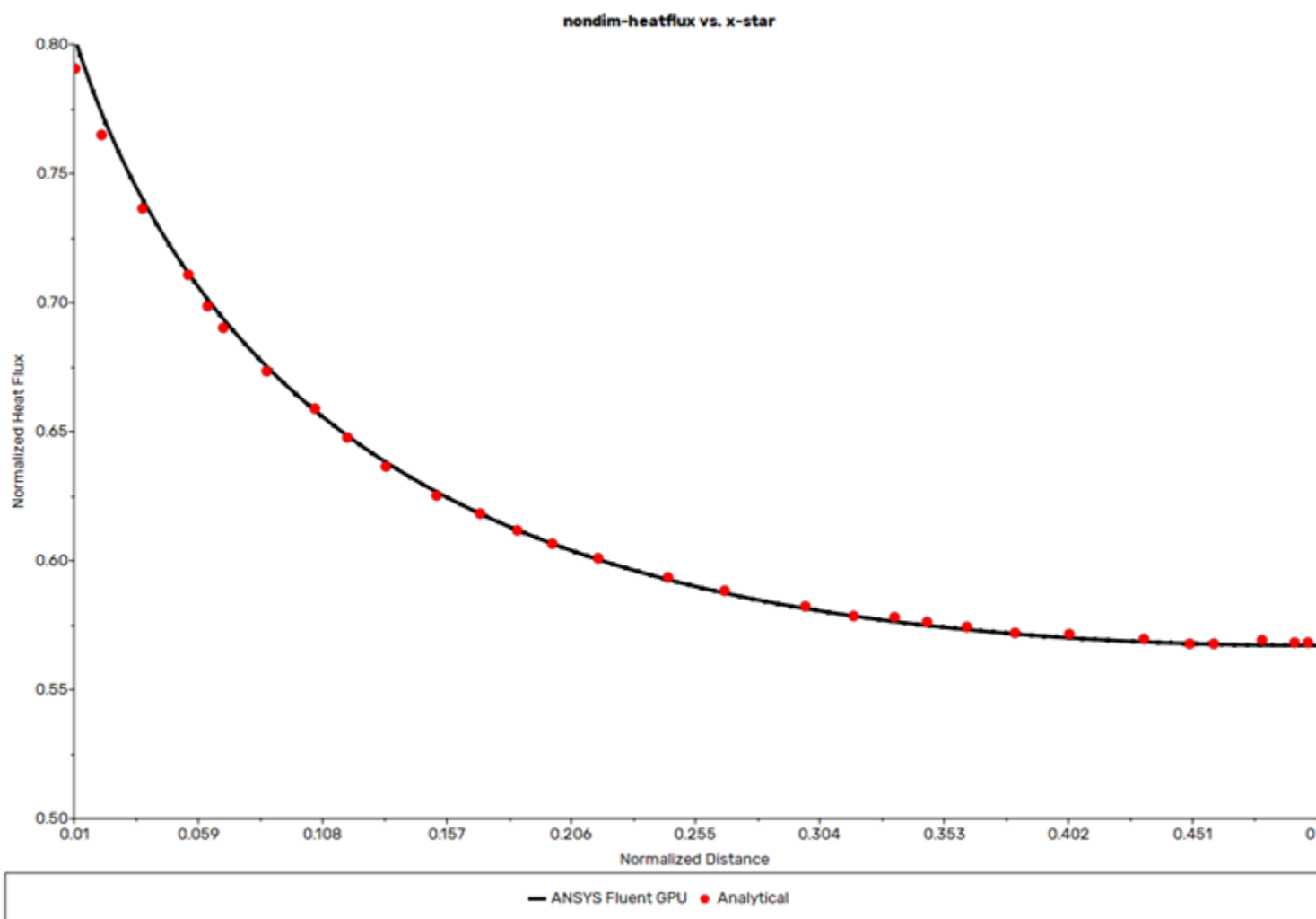
Material Properties	Geometry	Boundary Conditions
Scattering coefficient = 0.5/m	Dimensions of the domain: 10 m x 2 m x 0.5 m	Temperature of the hot wall = 200 K Temperature of the cold walls = 100 K

Analysis Assumptions and Modeling Notes

Isotropic scattering and radiative equilibrium are assumed.

Results Comparison for Ansys Fluent GPU

Figure .gpu008.2: Comparison of Non-Dimensional Heat Flux vs x-Star compared to Analytic Solution



VMFLGPU009: Two Phase Poiseuille Flow

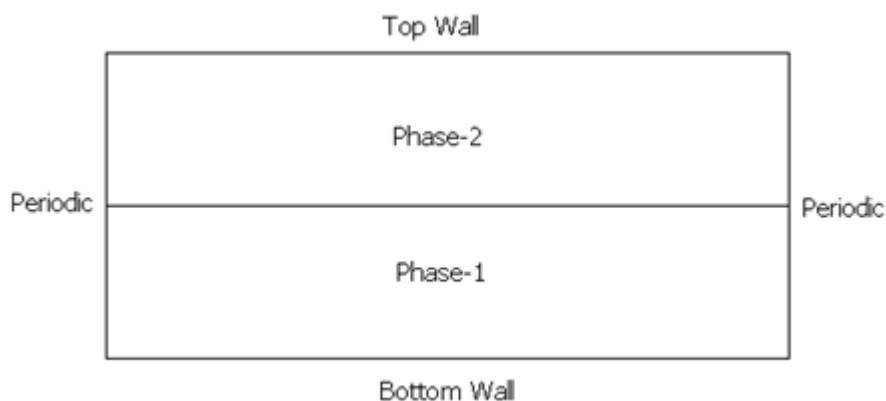
Overview

Reference	E. Marchandise, J.F. Remacle, "A stabilized finite element method using a discontinuous level set approach for solving two phase incompressible flows", <i>Journal of Computational Physics</i> . Volume 219, pp. 180-800, 2006
Solver	Ansys Fluent GPU
Physics/Models	Steady flow, laminar, two phase
Input File	vt009.msh.h5, vmfl009.jou

Test Case

This test case considers the horizontal stratified Poiseuille flow of two fluids between parallel walls. The interface between the two phases is located at half of the height of the channel.

Figure .gpu009.1: Flow Domain



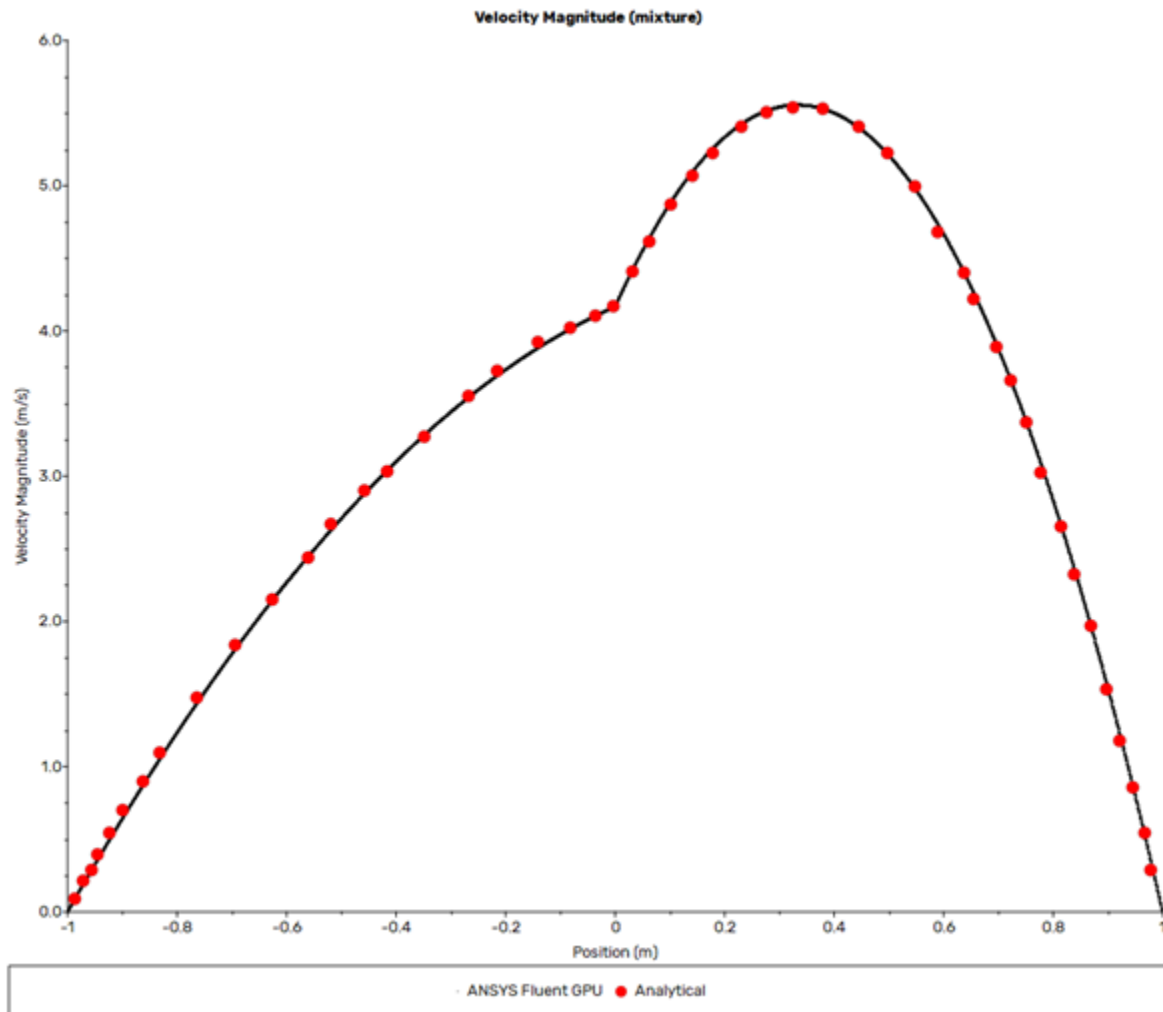
Material Properties	Geometry	Boundary Conditions
The two fluids have the same density. Kinematic Viscosity Fluid-1 = 0.1 Fluid-2 = 0.02	Dimensions of the domain: 2 m X 4 m	Periodic Boundary is used with a Pressure Gradient = -0.5 Pa/m.

Analysis Assumptions and Modeling Notes

The flow is steady. Deformation of the interface is not modeled.

Results Comparison for Ansys Fluent GPU

Figure .gpu009.2: Comparison of Velocity Magnitude vs Position



VMFLGPU010: Surface to Surface Radiative Heat Transfer Between Two Concentric Cylinders

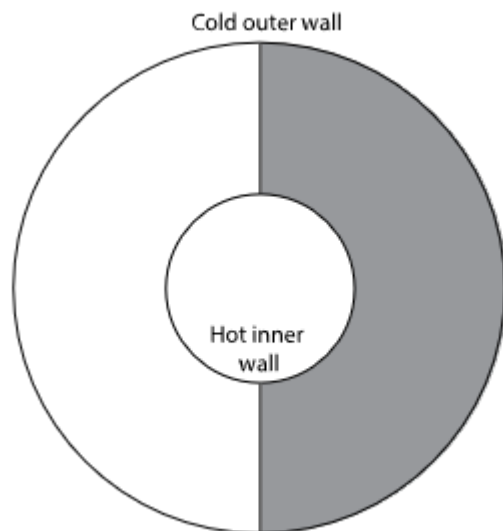
Overview

Reference	F.P. Incropera, D.P. Dewitt. <i>Fundamentals of Heat and Mass Transfer</i> . 4th Edition, John Wiley & Sons, Inc., New York City, New York, 1996.
Solver	Ansys Fluent GPU
Physics/Models	Radiation Modeling (S2S Model)
Input File	vt010.msh.h5, vmfl010.jou

Test Case

Radiative heat transfer between two cylindrical surfaces forming a concentric annulus is modeled. There is no participating medium. Due to symmetry only the shaded portion of the domain is modeled.

Figure .gpu010.1: Flow Domain



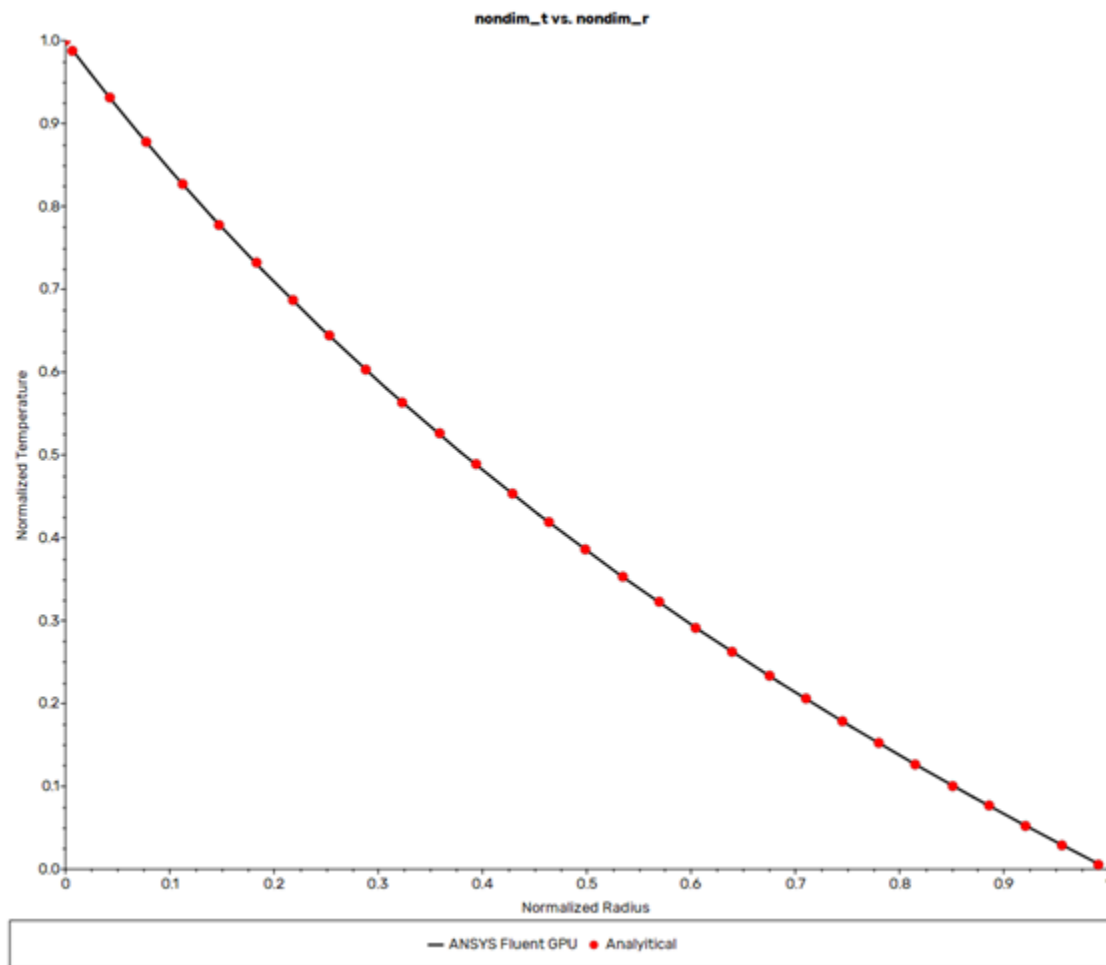
Geometry	Boundary Conditions
Radius of outer wall = 0.04625 m	Temperature of inner wall = 700 K
Radius of inner wall = 0.0178 m	Temperature of outer wall = 300 K

Analysis Assumptions and Modeling Notes

There is no flow, only the energy equation is solved. Surface to Surface Radiation model is used for the simulation. Heat transfer is purely due to radiation between the two surfaces.

Results Comparison for Ansys Fluent GPU

Figure .gpu010.2: Comparison of Non-Dimensional Temperature vs Normalized Radius



Part 3: Ansys Forte Test Cases

VMFRT001: Large Eddy Simulation in Internal Combustion Engine Case

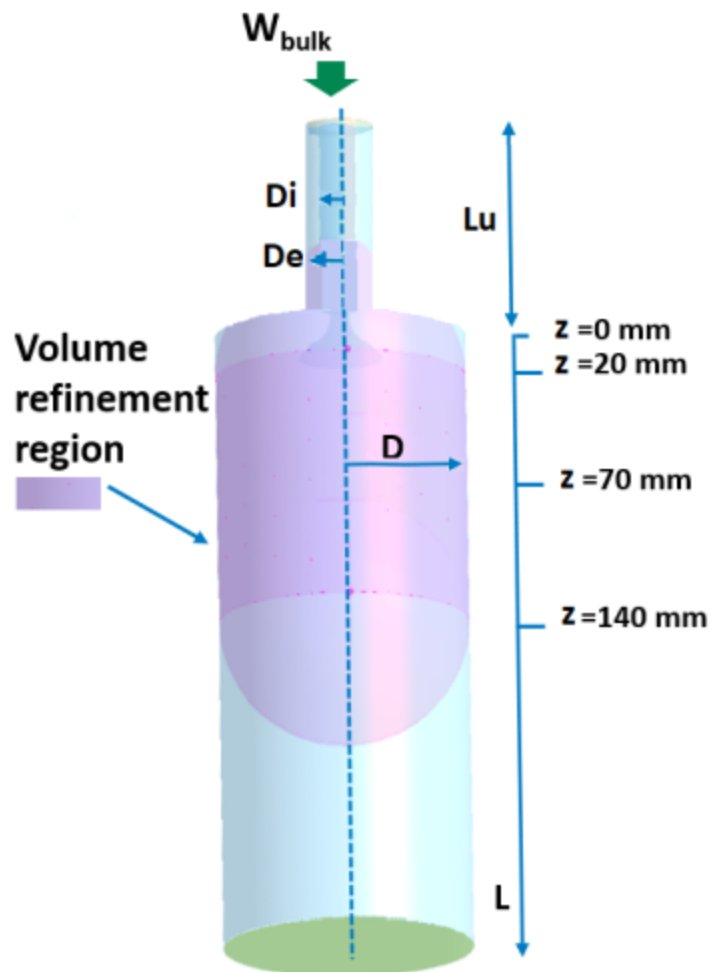
Overview

Reference	Thobois, L., Rymer, G., Soulères, T., and Poinso, T., "Large-Eddy Simulation in IC Engine Geometries, SAE Paper 2004-01-1854," SAE Technical Paper Series, 2004-01-1854 , 2004.
Solver	Ansys Forte
Physics/Models	Non-reacting flow, gas-phase only
Input Files	VM_LES_initialization.ftsim
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test case simulates a simplified IC engine geometry with a nonreacting steady-state flow. See [Figure .01.1: Sketch of the geometry with the static volume refinement region \(p. 256\)](#). The valve is fixed in the open position and has a seat of 4 mm, the expansion ratio is 3.5 and there is no swirl. A static volume refinement is present around the jet region to better capture the recirculation, while Solution Adaptive Mesh refinements (SAM) are activated based on the velocity gradient (see the table below for details). The turbulence model settings are the default for all parameters for both LES and Standard k- ϵ while the LES simulation is initialized using the RANS solution. The resulting velocity profiles and their fluctuations in [Figure .01.2: Normalized axial velocity at \$z = 20\$ mm \(p. 257\)](#), [Figure .01.3: Normalized axial velocity fluctuation at \$z = 20\$ mm \(p. 257\)](#), and [Figure .01.4: Normalized axial velocity at \$z = 70\$ mm \(p. 258\)](#) are compared to the laser doppler anemometry experimental measurements (Thobois 2004 (p. 255)).

Figure .01.1: Sketch of the geometry with the static volume refinement region



Test case properties										
Fluid	Reynolds	Q_m	W_{bulk}	P_{outlet}	Valve lift	L	L_u	D	D_i	D_e
N_2	30000	0.055 kg/s	65 m/s	101.3 kPa	10 mm	300 mm	104 mm	60 mm	17 mm	34 mm
Simulation Settings										
LES model		RANS model		SAM on ∇Vel		Mesh size _{MAX}		Refinement level		Tot n. cells
Dynamic Smagorinsky		Std-k		0.5 mm		4 mm		0.5 mm		3.3 M

Results Comparison

Figure .01.2: Normalized axial velocity at $z = 20$ mm

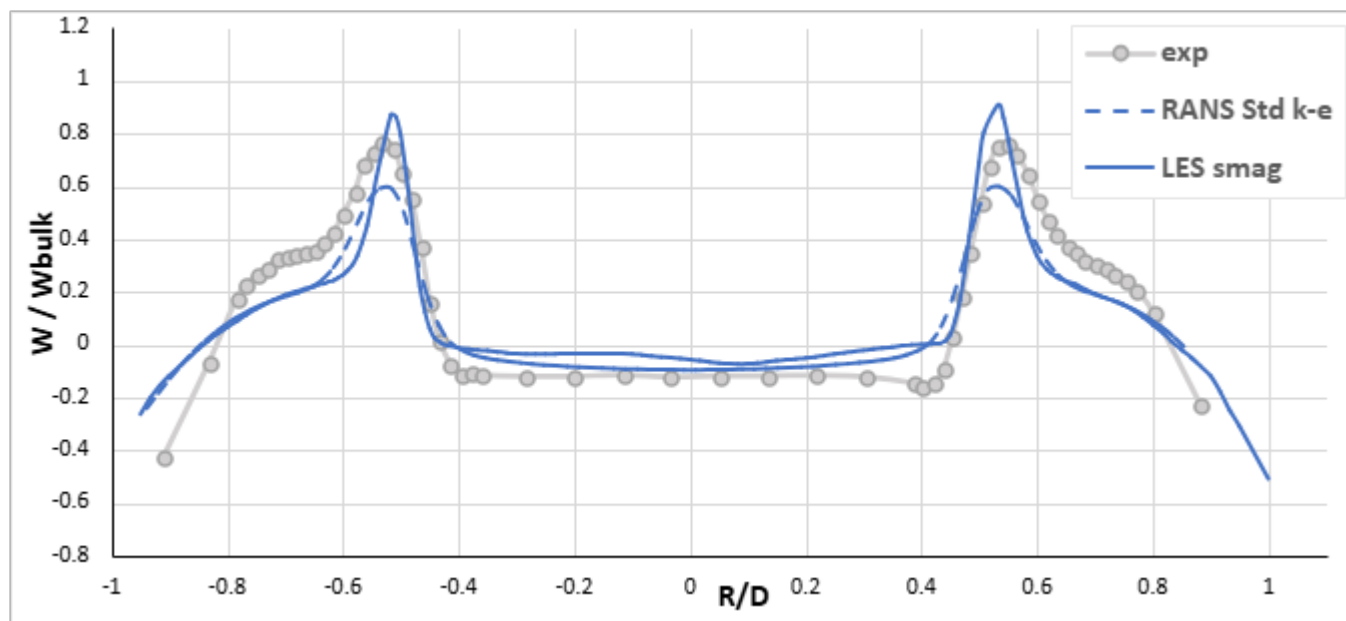


Figure .01.3: Normalized axial velocity fluctuation at $z = 20$ mm

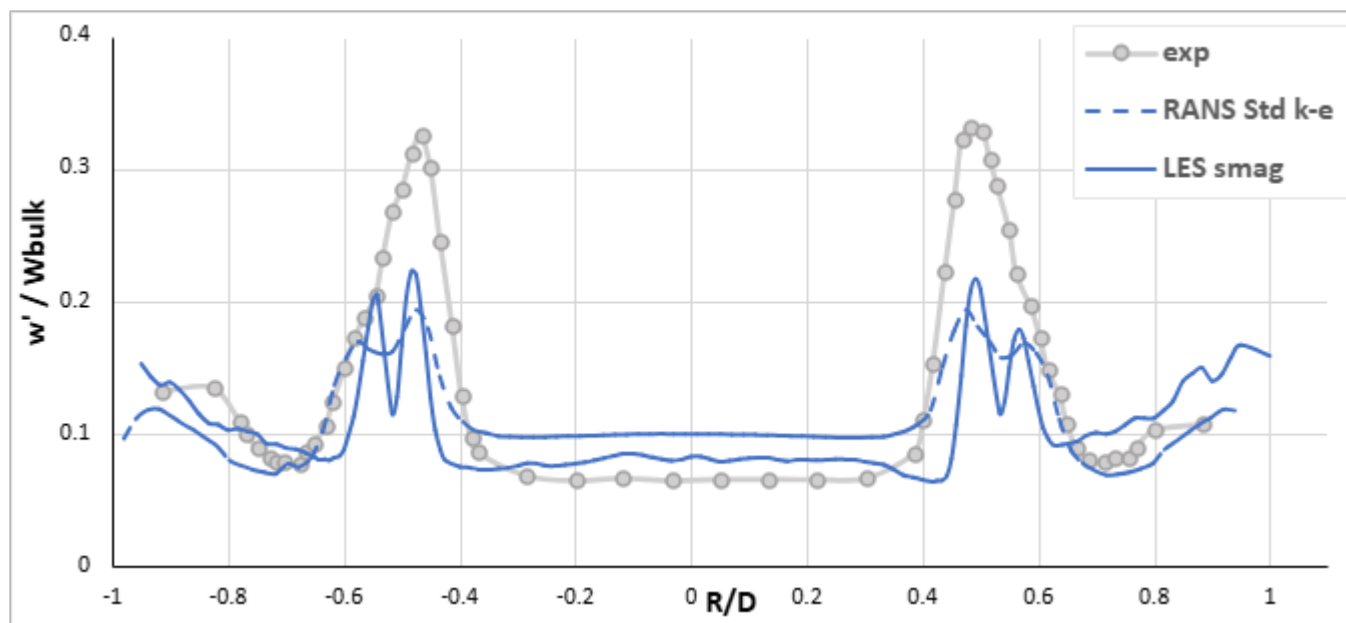
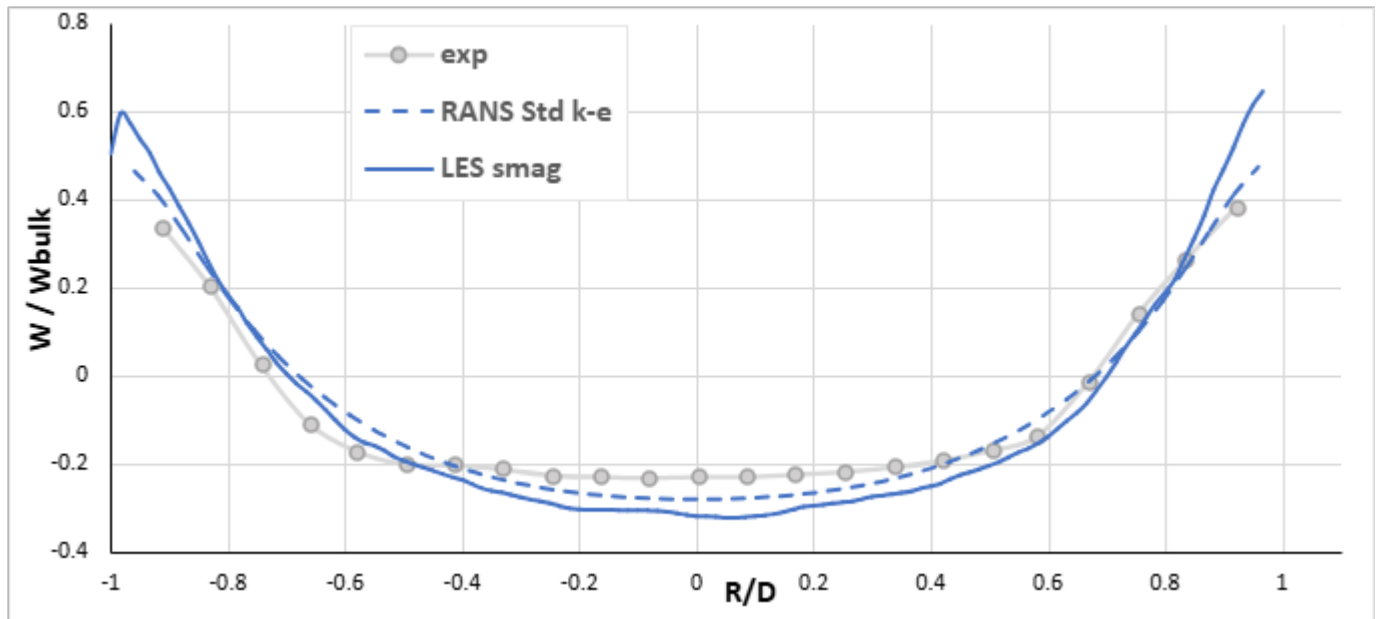
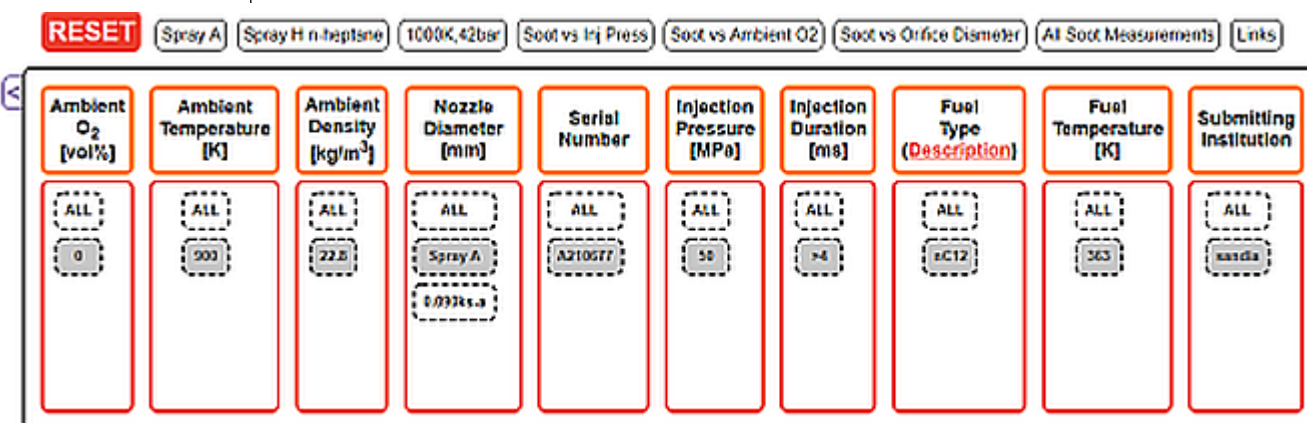


Figure .01.4: Normalized axial velocity at $z = 70$ mm

VMFRT002: Engine Combustion Network Nonreacting Flow Case - bklraAL4

Overview

Reference	Retrieved from Engine Combustion Network (ECN) data search utility: https://ecn.sandia.gov/ecn-data-search/ Accessed on April 7, 2020. Find relevant data and information by choosing the parameters highlighted in gray below on the ECN data search utility.
	
Solver	Ansys Forte
Physics/Models	Engine Combustion Network (ECN) Nonreacting Flow Case bklraAL4
Input Files	P_50MPa_NR.fts.sim
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test recreates the Engine Combustion Network (ECN) Spray A target case characterized by high-pressure liquid fuel injection in conditions relevant to diesel engines. In this case, the combustion chamber has a cubical shape (length = 10.8 cm), the oxygen content is zero, and the difference between the fuel injector pressure and ambient gas pressure is 50 MPa. The fuel entering the combustion chamber through the spray injector is *n*-dodecane (nC12). The model includes two refined regions of respectively 2 mm and 1 mm around the jet. The table below lists the setup parameters for the experiment and simulation. Results are compared against the experimental data from the ECN data search utility, 2019.

Additional test details can be found on the ECN web site. The experimental data for the vapor and liquid penetration lengths can be found in the form of text files on the ECN site at the following links:

- Vapor: bklraAL4-pen.txt
(<https://ecn.sandia.gov/cvdata/assets/datafiles/pen/bklraAL4-pen.txt>)
- Liquid: bklraAL4-liq.txt
(<https://ecn.sandia.gov/cvdata/assets/datafiles/liq/bklraAL4-liq.txt>)

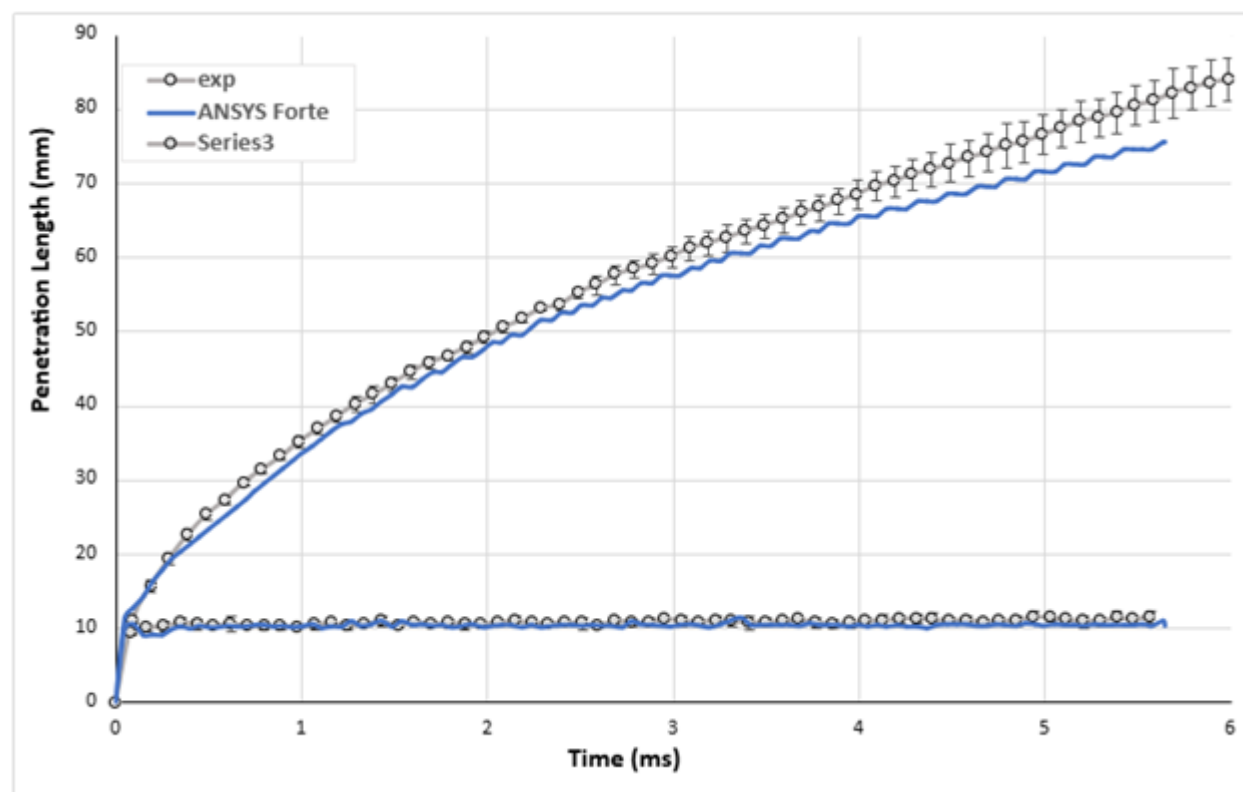
The modeling predictions for vapor and liquid penetration lengths are compared with the experimental results including error bars in [Figure .02.1: Vapor and liquid penetration length \(p. 260\)](#).

Test case properties									
P_{inj}	M_{inj}	$Duration_{inj}$	V_{amb}	ρ_{amb}	$Composition_{amb}$	P_{amb}	T_{amb}	T_{fuel}	D_{nozzle}
50 MPa	7.4 mg	5.65 ms	20 cm/s	22.8 kg/m ³	N ₂ = 89.71% CO ₂ = 6.52% H ₂ O = 3.77%	6.07 MPa	900 K	373 K	0.084 mm
Simulation Settings									
RT_{dist}	v_{inj} profile	Spray initialization model	L/D	R/D	SAM on $\nabla T \nabla V e l \nabla Y^*[a]$	Turbulence model	Mesh size _{MAX}	$dt_{initial}$	
2.2	Flat	Nozzle flow	11.9	0.238	0.5 mm	RANS RNG-k	4 mm	1E-7 s	

[a] Y^* =Fuel Vapor Mass Fraction

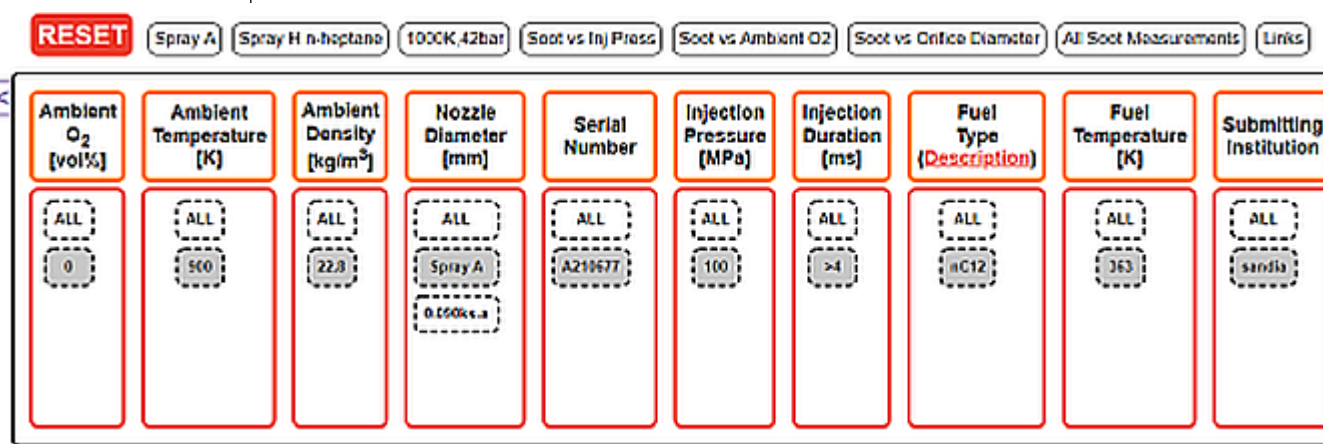
Results Comparison

Figure .02.1: Vapor and liquid penetration length



VMFRT003: Engine Combustion Network Nonreacting Flow Case - bklfaAL4

Overview

Reference	Retrieved from Engine Combustion Network data search utility: https://ecn.sandia.gov/ecn-data-search/ Accessed on April 7, 2020. Find relevant data and information by choosing the parameters highlighted in gray below on the ECN data search utility.
	
Solver	Ansys Forte
Physics/Models	Engine Combustion Network Nonreacting Flow Case - bklfaAL4
Input Files	P_100MPa_NR.fts _{sim}
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test recreates the Engine Combustion Network (ECN) Spray A target case characterized by high-pressure liquid fuel injection in conditions relevant to diesel engines. In this case, the combustion chamber has a cubical shape (length = 10.8 cm), the oxygen content is zero, and the difference between the fuel injector pressure and ambient gas pressure is 100 MPa. The fuel entering the combustion chamber through the spray injector is *n*-dodecane (nC12). The model includes two refined regions of respectively 2 mm and 1 mm around the jet. The table below lists the setup parameters for the experiment and simulation.

Additional test details can be found on the ECN web site. The experimental data for the vapor and liquid penetration lengths can be found in the form of text files on the ECN site at the following links:

- Vapor: [bklfaAL4-pen.txt](#)
(<https://ecn.sandia.gov/cvdata/assets/datafiles/pen/bklfaAL4-pen.txt>)
- Liquid: [bklfaAL4-liq.txt](#)
()

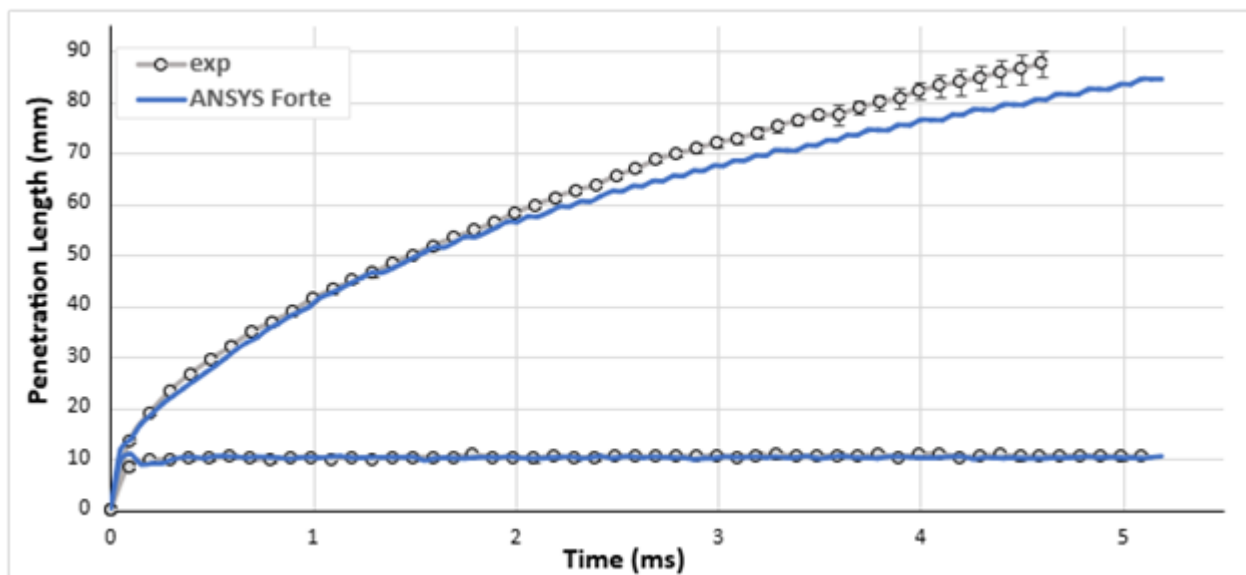
The modeling predictions for vapor and liquid penetration lengths are compared with the experimental results including error bars in [Figure .03.1: Vapor and liquid penetration length \(p. 262\)](#).

Test case properties									
P_{inj}	M_{inj}	$Duration_{inj}$	V_{amb}	ρ_{amb}	Composition _{amb}	P_{amb}	T_{amb}	T_{fuel}	D_{nozzle}
100 MPa	9.672 mg	5.2 ms	20 cm/s	22.8 kg/m ³	N ₂ = 89.71% CO ₂ = 6.52% H ₂ O = 3.77%	6.07 MPa	900 K	373 K	0.084 mm
Simulation Settings									
RT_{dist}	v_{inj} profile	Spray initialization model	L/D	R/D	SAM on $\nabla T \nabla Vel \nabla Y^{[a]}$	Turbulence model	Mesh size _{MAX}	$dt_{initial}$	
2.2	Flat	Nozzle flow	11.9	0.238	0.5 mm	RANS RNG-k	4 mm	1E-7 s	

[a] Y^* =Fuel Vapor Mass Fraction

Results Comparison

Figure .03.1: Vapor and liquid penetration length



VMFRT004: Engine Combustion Network Nonreacting Flow Case - bkldaAL4

Overview

Reference	Retrieved from Engine Combustion Network data search utility: https://ecn.sandia.gov/ecn-data-search/ Accessed on April 7, 2020. Find relevant data and information by choosing the parameters highlighted in gray below on the ECN data search utility.
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RESET

Spray A

Spray H n-heptane

1000K, 42bar

Soot vs Inj Press

Soot vs Ambient O2

Soot vs Orifice Diameter

All Soot Measurements

Links

Ambient O ₂ [vol%]	Ambient Temperature [K]	Ambient Density [kg/m ³]	Nozzle Diameter [mm]	Serial Number	Injection Pressure [MPa]	Injection Duration [ms]	Fuel Type (Description)	Fuel Temperature [K]	Submitting Institution
ALL 0	ALL 500	ALL 22.8	ALL Spray A 0.090ks-a	ALL A210677	ALL 150	ALL >4	ALL nC12	ALL 363	ALL sandia

Solver	Ansys Forte
Physics/Models	Engine Combustion Network (ECN) Nonreacting Flow Case bkldaAL4
Input Files	P_150MPa_NR.ftsims
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test recreates the Engine Combustion Network (ECN) Spray A target case characterized by high-pressure liquid fuel injection in conditions relevant to diesel engines. In this case, the combustion chamber has a cubical shape (length = 10.8 cm), the oxygen content is zero, and the difference between the fuel injector pressure and ambient gas pressure is 150 MPa. The fuel entering the combustion chamber through the spray injector is *n*-dodecane (nC12). The model includes two refined regions of respectively, 2 mm and 1 mm, around the jet. The table below lists the setup parameters for the experiment and simulation.

Additional test details can be found on the ECN web site. The experimental data for the vapor and liquid penetration lengths can be found in the form of text files on the ECN site at the following links:

- Vapor: [bkldaAL4-pen.txt](#)
(<https://ecn.sandia.gov/cvdata/assets/datafiles/pen/bkldaAL4-pen.txt>)

- Liquid: [bkldaAL4-liq.txt](#)

()

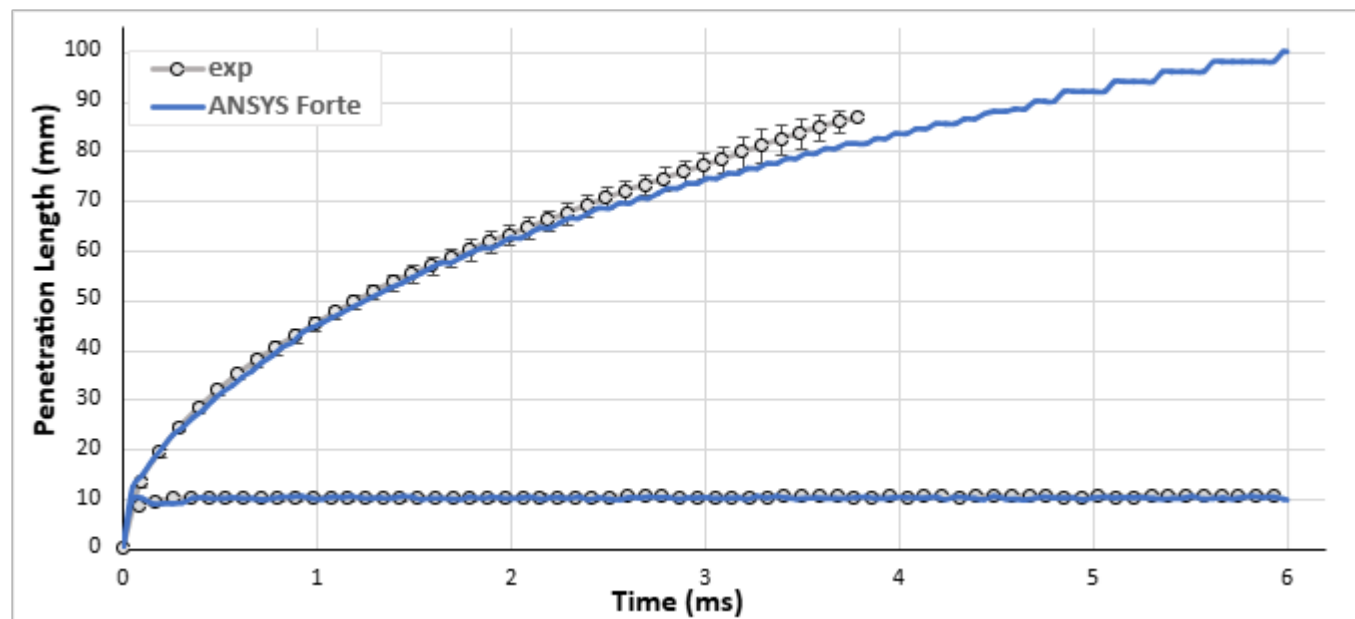
The modeling predictions for vapor and liquid penetration lengths are compared with the experimental results including error bars in [Figure .04.1: Vapor and liquid penetration length \(p. 264\)](#).

Test case properties									
P_{inj}	M_{inj}	$Duration_{inj}$	V_{amb}	ρ_{amb}	Composition _{amb}	P_{amb}	T_{amb}	T_{fuel}	D_{nozzle}
150 MPa	13.77 mg	6 ms	20 cm/s	22.8 kg/m ³	N ₂ = 89.71% CO ₂ = 6.52% H ₂ O = 3.77%	6.05 MPa	900 K	373 K	0.084 mm
Simulation Settings									
RT_{dist}	v_{inj} profile	Spray initialization model	L/D	R/D	SAM on $\nabla T \nabla Vel \nabla Y^*_{[a]}$	Turbulence model	Mesh size _{MAX}	$dt_{initial}$	
2.2	Flat	Nozzle flow	11.9	0.238	0.5 mm	RANS RNG-k	4 mm	1E-7 s	

[a] Y^* =Fuel Vapor Mass Fraction

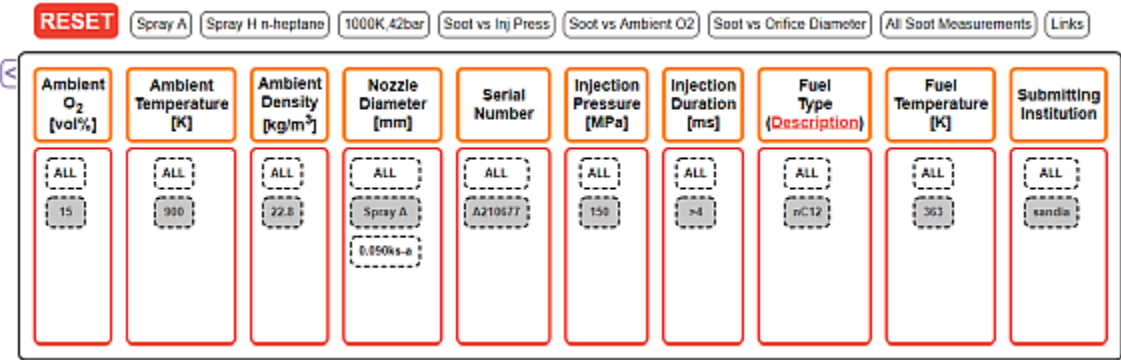
Results Comparison

Figure .04.1: Vapor and liquid penetration length



VMFRT005: Engine Combustion Network Reacting Flow Case *jkldaAL4*

Overview

Reference	Retrieved from Engine Combustion Network data search utility: https://ecn.sandia.gov/ecn-data-search/ Accessed on April 7, 2020. Find relevant data and information by choosing the parameters highlighted in gray below on the ECN data search utility. 
Solver	Ansys Forte
Physics/Models	Engine Combustion Network (ECN) Nonreacting Flow Case <i>jkldaAL4</i>
Input Files	P_150MPa_R.fts im
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test recreates the Engine Combustion Network (ECN) Spray A target case characterized by high-pressure liquid fuel injection in conditions relevant to diesel engines. In this case, the combustion chamber has a cubical shape (length = 10.8 cm), the oxygen content is 15%, and the difference between the fuel injector pressure and ambient gas pressure is 150 MPa. The fuel entering the combustion chamber through the spray injector is *n*-dodecane (nC12).

The model includes two refined regions of respectively 2 mm and 1 mm around the jet. The table below lists the setup parameters for the experiment and simulation. No liquid penetration length measurements were found on the ECN repository for this case, but experimental data for the vapor penetration length can be found on the ECN site in the form of a text file:

- Vapor: [jkldaAL4-pen.txt](#)

(<https://ecn.sandia.gov/cvdata/assets/datafiles/pen/jkldaAL4-pen.txt>)

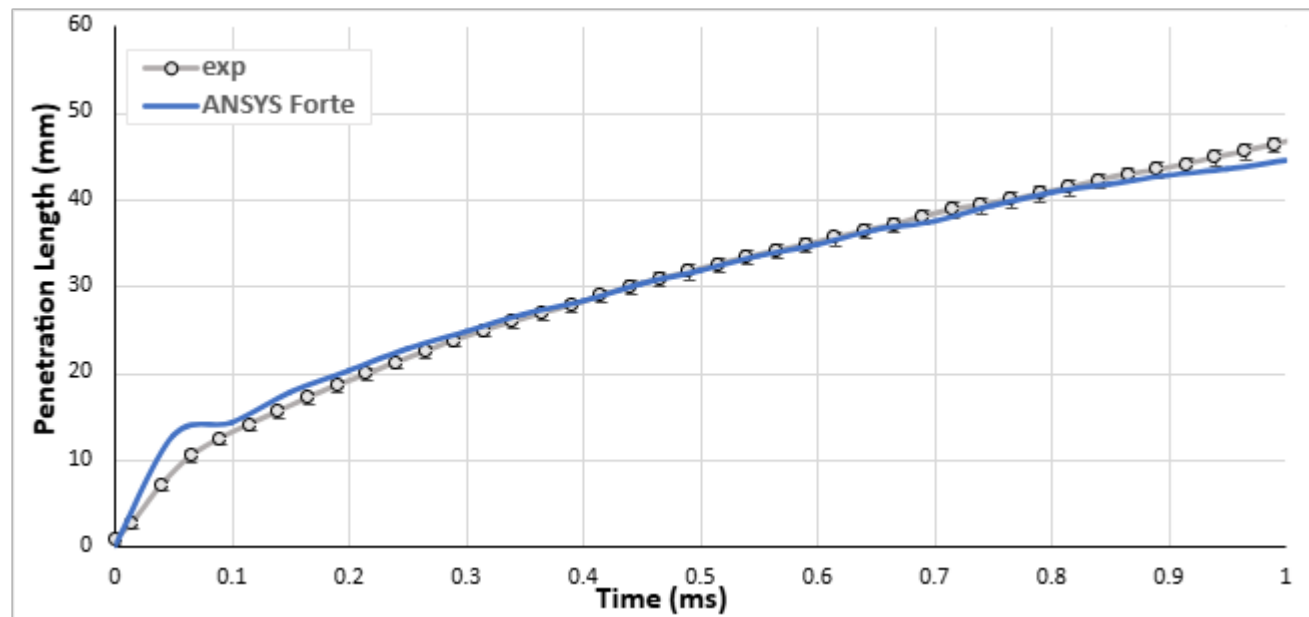
The modeling prediction for vapor penetration length is compared with the experimental results including error bars in [Figure .05.1: Vapor and liquid penetration length \(p. 266\)](#). Additional measurements on the ECN data search utility report the lift-off length measurement and the ignition delay time. The slight overprediction is likely due to imperfections in the low temperature chemical kinetics of the reacting fuel and air mixture.

Test case properties									
P_{inj}	M_{inj}	$Duration_{inj}$	V_{amb}	ρ_{amb}	Composition	P_{amb}	T_{amb}	T_{fuel}	D_{nozzle}
150 MPa	14 mg	6 ms	20 cm/s	22.8 kg/m ³	$N_2 = 75.15\%$ $CO_2 = 6.22\%$ $H_2O = 3.62\%$ $O_2 = 15.00\%$	5.94 MPa	900 K	373 K	0.084 mm
Simulation Settings									
RT_{dist}	v_{inj} profile	Spray initialization model	L/D	R/D	SAM on $\nabla T \nabla Vel \nabla Y^{[a]}$	Turbulence model	Mesh size _{MAX}	$dt_{initial}$	
2.2	Flat	Nozzle flow	11.9	0.238	0.5 mm	RANS RNG-ke	4 mm	1E-7 s	

[a] Y^* =Fuel Vapor Mass Fraction

Results Comparison for Ansys Forte

Figure .05.1: Vapor and liquid penetration length



Target	Predicted
Lift-off length	

Target	Predicted
16.7 mm	18.0 mm
Ignition delay time	
0.41 ms	0.7 ms

VMFRT006: Adiabatic Compression of Air in Cylinder by a Reciprocating Piston

Overview

Reference	Russell, L.D., Adebisi, G.A. <i>Classical Thermodynamics</i> , Saunders College Publishing, Philadelphia, PA, 1993.
Solver	Ansys Forte
Physics/Models	Dynamic Mesh, Transient flow with ideal gas effects
Input Files	VMFRT006.fts
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test simulates air undergoing an adiabatic compression due to the upward movement of a piston inside a cylinder. Initially the air inside the cylinder is at atmospheric conditions and the piston is located at the Bottom Dead Center (BDC). The piston reaches the Top Dead Center (TDC) when at the 360° crank angle. The compression is assumed to be adiabatic. Results in [Figure .06.1: Pressure \(p. 270\)](#) and [Figure .06.2: Temperature \(p. 271\)](#) are compared against the 2-D analytical solution mentioned in the Reference.

Test case properties					
RPM	Stroke	Bore _{diameter}	Composition _{amb}	P _{initial}	T _{initial}
10	8 m	8 m	O2=21% N2=79%	1 atm	300 K
Simulation Settings					
Initial CA	Final CA	Cycle type	Wall refinements	Mesh size _{max}	dt _{MAX}
180 deg	540 deg	4-stroke	Head=0.5 m Liner=0.5 m Piston=0.25 m	1 m	5 ms

Results Comparison for Ansys Forte

Figure .06.1: Pressure

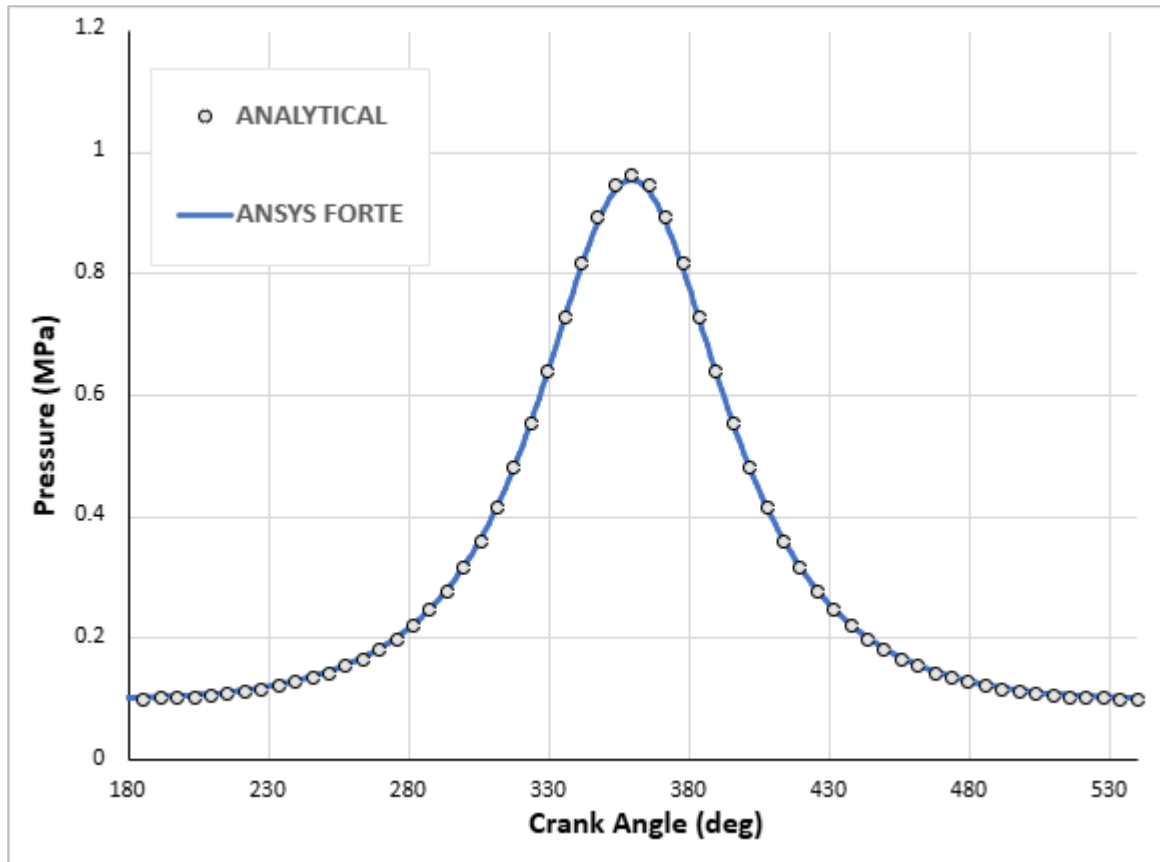
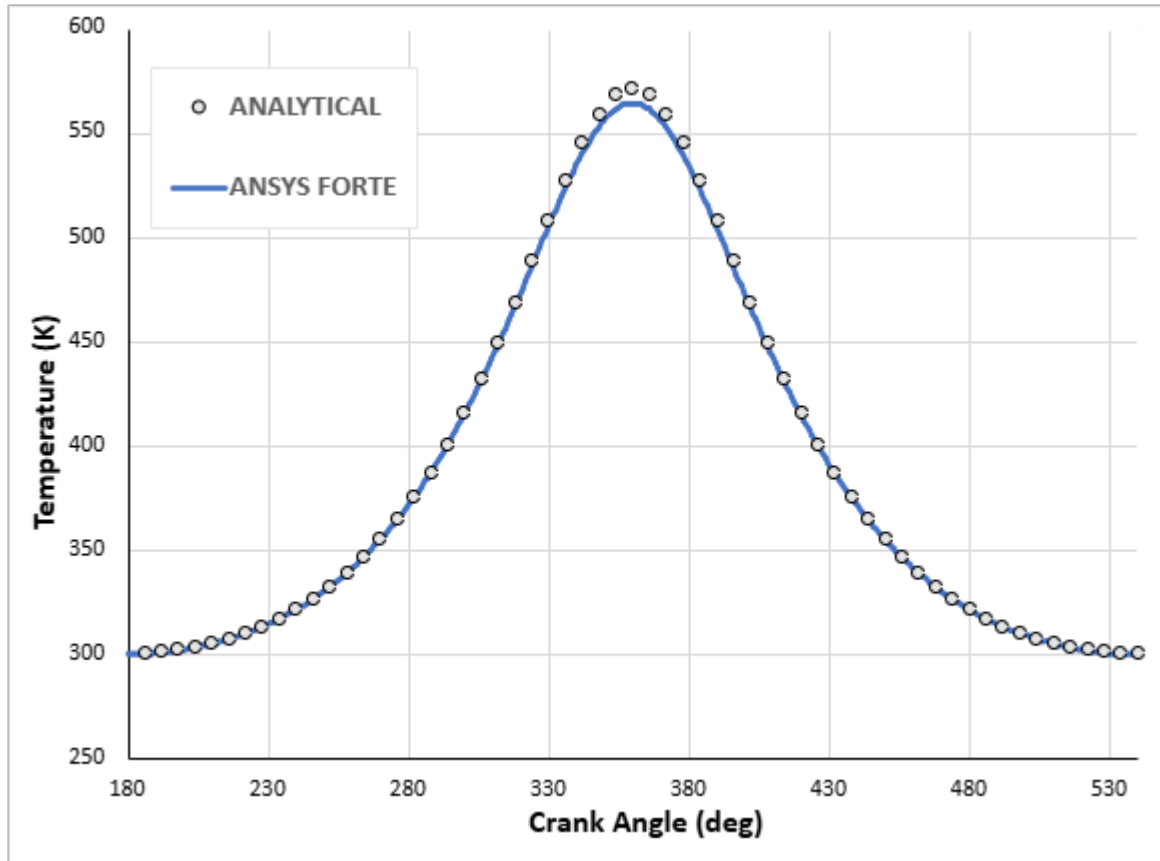


Figure .06.2: Temperature

VMFRT007: Small-Bore Direct Injection Diesel Engine

Overview

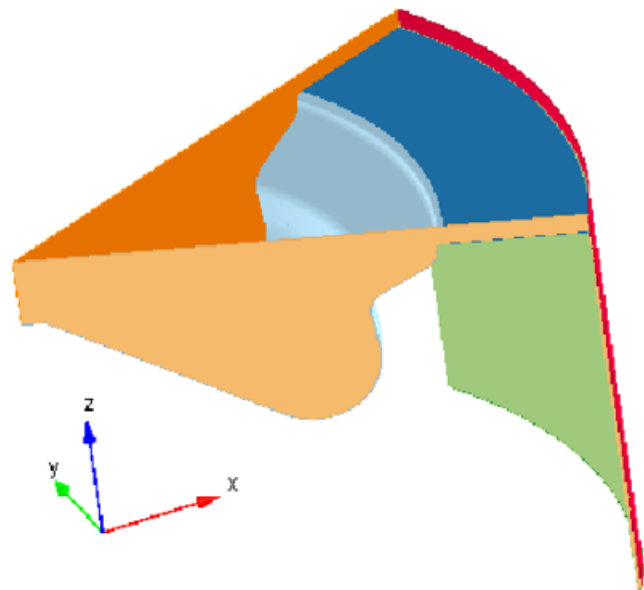
Reference	Retrieved from Engine Combustion Network website: https://ecn.sandia.gov/Small-BoreDieselData/CFD/Engine_Geometry/SmallBore_Full_Engine_SL_Piston.stl
Solver	Ansys Forte
Physics/Models	Engine Combustion Network (ECN) - Light duty engine combustion
Input Files	SmallBore.fts
Project Files	Link to Project Files Download Page (p. 5)

Test Case

This test simulates the closed cycle combustion in a small-bore direct injection diesel engine. The engine has a 7-hole injector and here it is modeled as a (360/7) sector with a stepped-lip piston bowl and no valve cutouts as shown in [Figure .07.1: Geometry sector and dimensions \(p. 273\)](#).

Figure .07.1: Geometry sector and dimensions

Bore	82 mm
Stroke	90.4 mm
Pin offset	1.5875 mm
Squish	1.36 mm
CR	15.6

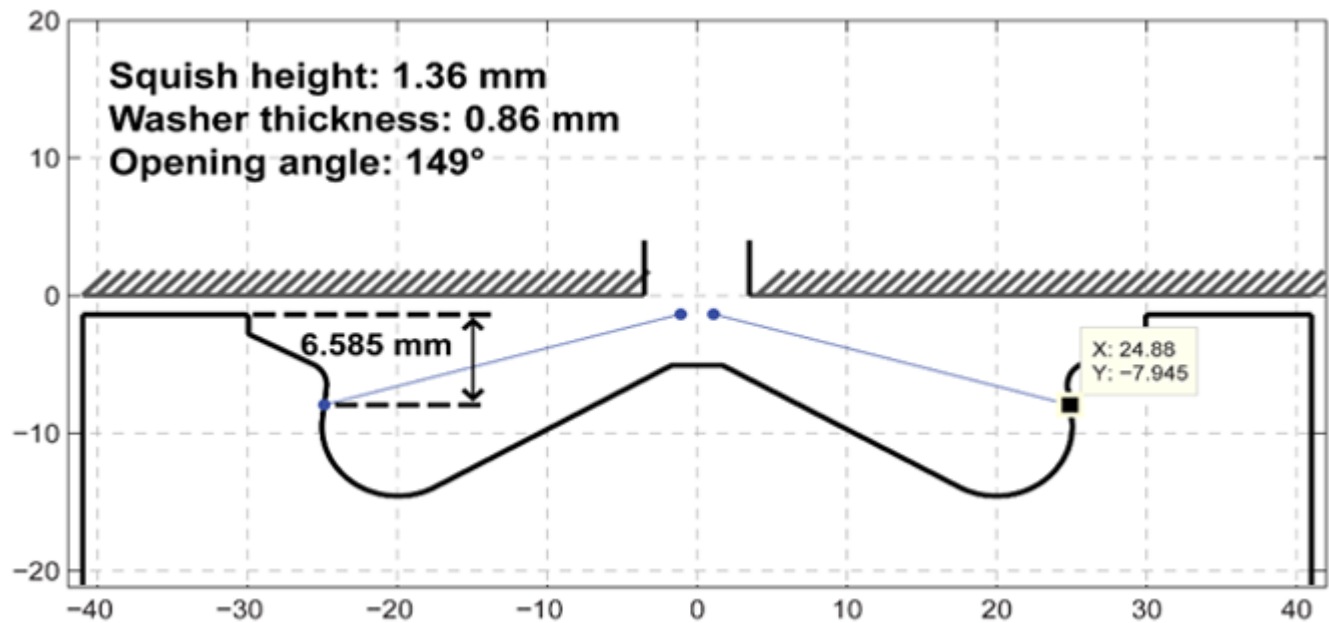


The case is labeled on the ECN website as *conventional diesel combustion* at part load (9bar IMEPg), CDC9_SL_SSEp17b. It operates with a pilot-main injection strategy applied with a single profile and the spray targeting data, shown in [Figure .07.2: Spray targeting data \(p. 274\)](#), can be found on the ECN site at this link:

- [Spray Targeting](#)

The file contains both the re-entrant and the stepped-lip piston bowl data.

Figure .07.2: Spray targeting data



The table below lists the setup parameters, and results are compared against the experimental data. Data for the pressure trace can be found in the form of a Microsoft Excel file on the ECN site at the following link:

• [Pressure](#)

The file is named CDC9 InjRate Pcyl PIntake Fired and the worksheet it references is named CDC9_SL_SSEp17b. Notice that the engine has a piston pin offset of 1.5875 mm and the minimum volume occurs at -0.43 ATDC. Forte’s convention is to define piston TDC as crank angle= 0 , therefore the experimental curve is shifted by +0.43 degrees forward for a consistent comparison (see [Slider Crank Motion in the Ansys Forte User's Guide](#)). In addition, it is presented as the apparent heat release rate with constant gamma (=1.35) derived from the pressure trace (see [Spatially averaged variables that are always output to the .csv files in the Ansys Forte User's Guide](#)) according to:

$$\frac{dQ}{dt} = \frac{\gamma}{\gamma-1} p \frac{dV}{dt} + \frac{1}{\gamma-1} V \frac{dp}{dt}$$

Test case properties								
P _{initial}	T _{initial}	Tot M _{inj}	SOI	Duration _{inj}	Composition _{fuel}	CR	RPM	D _{nozzle}
1.74	440	24.24	-13.4 deg	23.5	hmn =58,7%	15.6	1500	0.139
bar	K	mg		deg	nc16h34 =41,3%			mm
Simulation Settings								
T _{droplet}			Cd	Spray angle	SAM on ∇Vel	IVC	Mesh size _{MAX}	dt _{initial}

Test case properties						
400 K	0.8	12 deg	$\nabla T = 1 \text{ mm } \nabla \text{Vel},$ $\nabla Y^{[a]}$	-138 deg	2 mm	5e-7 sec

[a] Y=Fuel Vapor Mass Fraction

Results Comparison for Ansys Forte

Figure .07.3: Pressure and apparent heat release rate

